

Supplementary file 1 - The further analysis for "The Functional Regression with Reconstructed Functions from Hybrid Principal Components Analysis: With EEG-fMRI Application"

Mohammad Fayaz^{1,*}, Alireza Abadi^{2,3}, Soheila Khodakarim⁴

¹ PhD in Biostatistics, Department of Biostatistics, School of Allied Medical Sciences, Shahid Beheshti University of Medical Sciences, Tehran, Iran. (<https://orcid.org/0000-0002-5643-9763>). ² Professor of Biostatistics, Department of Community Medicine, Faculty of medicine, Shahid Beheshti University of Medical Sciences, Tehran, Iran. (<http://orcid.org/0000-0003-2653-6623>). ³ Social Determinants of Health Research Center, Shahid Beheshti University of Medical Sciences, Tehran, Iran. ⁴ Associate Professor, School of Allied Medical Sciences, School of Public Health and Safety, Shahid Beheshti University of Medical Sciences, Tehran, Iran. (<https://orcid.org/0000-0002-5473-999X>)

Abstract It is a supplementary file for "The Functional Regression with Reconstructed Functions from Hybrid Principal Components Analysis: With EEG-fMRI Application". In this supplementary, we analyze four datasets including 17 participants. The Electroencephalography (EEG) curves are response functions and the region of interests (ROI) curves of functional Magnetic Resonance Images (fMRI) for all Broadmann Areas' regions are the functional covariates. There are two oddball tasks: 1) Auditory and 2) Visual. Each of them has two stimuli: 1) Standard and 2) Target. Therefore, there are four datasets including 1) Auditory Oddball task, standard stimuli, 2) Auditory Oddball task, target stimuli, 3) Visual Oddball task, standard stimuli, and 4) Visual Oddball task, target stimuli. We divide the 36 electrodes into 7 groups: 1) Whole Brain, 2) Frontal, 3) Central, 4) Parietal, 5) Occipital, 6) Right Temporal, and 7) Left Temporal Regions. The process for each dataset is: 1) It decomposes the EEG into functional, longitudinal, and regional dimensions with hybrid principal components analysis (HPCA), 2) It reconstruct the EEG curves based on the first to KLM HPCAs and it fits the signal-compression function-on-function regression with multiple ROI curves (train/test), 3) the correlation and mean squared prediction error (MSPE) between reconstructed EEG and predicted values are calculated and finally, the results are procudes in the graphs. The best number of HPCA based on the MSPE, correlation and elapsed times group by region of brain are chosen.

Keywords Functional Data Analysis, Hybrid Principal Component, Multiple Functional Covariates, Functional Regression, EEG, fMRI

AMS 2010 subject classifications 62M10 , 62F15

DOI: 10.19139/soic-2310-5070-1437

*Correspondence to: Mohammad Fayaz (Email: Mohammad.Fayaz.89@gmail.com). PhD in Biostatistics, Department of Biostatistics, School of Allied Medical Sciences, Shahid Beheshti University of Medical Sciences, Tehran, Iran. (<https://orcid.org/0000-0002-5643-9763>)

Contents

1	Introduction	2
2	The Auditory Odd Ball Task	3
2.1	Standard Stimuli	3
2.1.1	The HPCA Decomposition	3
2.1.2	The Reconstructed of the Response	9
2.1.3	The function-on-function result	16
2.1.4	Conclusion	40
2.2	Target Stimuli	41
2.2.1	The HPCA Decomposition	41
2.2.2	The Reconstructed of the Response	47
2.2.3	The function-on-function result	54
2.2.4	Conclusion	85
3	The Visual Odd Ball Task	86
3.1	Standard Stimuli	86
3.1.1	The HPCA Decomposition	86
3.1.2	The Reconstructed of the Response	92
3.1.3	The function-on-function result	99
3.1.4	Conclusion	123
3.2	Target Stimuli	124
3.2.1	The HPCA Decomposition	124
3.2.2	The Reconstructed of the Response	130
3.2.3	The function-on-function result	137
3.2.4	Conclusion	161
4	Bayesian Simulation	162
4.1	Without HPCA	163
4.2	First HPCA	165
4.3	All HPCA	167

1. Introduction

There are four datasets in this analysis for 17 participant with Electroencephalography (EEG) curves as response functions and the region of interests (ROI) curves of functional Magnetic Resonance Images (fMRI) as functional covariates ([1]):

1. Auditory Oddball task, standard stimuli
2. Auditory Oddball task, target stimuli
3. Visual Oddball task, standard stimuli
4. Visual Oddball task, target stimuli

In each dataset, following analysis were done:

1. The Hybrid Principal Component Analysis (HPCA) of EEG curves (response function)
2. Reconstructing the EEG curves from HPCA = 1 to HPCA=KLM, separately for different region of the brains:
 - Whole brain (considering all 36 electrodes)
 - Frontal Region (Fp1 – Fp2 – AF3 – AF4- Fz)
 - Central Region (FC1 – FC2 – C3 – Cz – C4)
 - Parietal Region (CP1 – CP2 – P3 – Pz – P4)
 - Occipital Region (PO3 – PO4 – PO7 – PO8 – O1 – Oz – O2)
 - Right Temporal Region (F4 – F8 – FC6 – T8 – RM – CP6 – P8)
 - Left Temporal Region (F3 – F7 – FC5 – T7 – LM – CP5 – P7)
3. The signal compression function-on-function regression with reconstructed EEG as response function and ROIs of fMRI as functional covariates were done for each number of HPCA and each region of the brain. The results are shown in the:
 - Train/test with 10 iterations and two indices:
 - Mean Squared Prediction Error (MSPE).
 - Correlation.
 - Running Time with 10 iterations:
 - User
 - System
 - Elapsed
 - The model fit indices in 10 iterations
4. Finally, the concluding figures were produced:
 - The MSPE over number of HPCA group by train/test by regions.
 - The correlation over number of HPCA group by train/test by regions.
 - The elapsed time over number of HPCA group by train/test by regions.

2. The Auditory Odd Ball Task

2.1. Standard Stimuli

2.1.1. The HPCA Decomposition

2.1.1.1 The Functional dimension

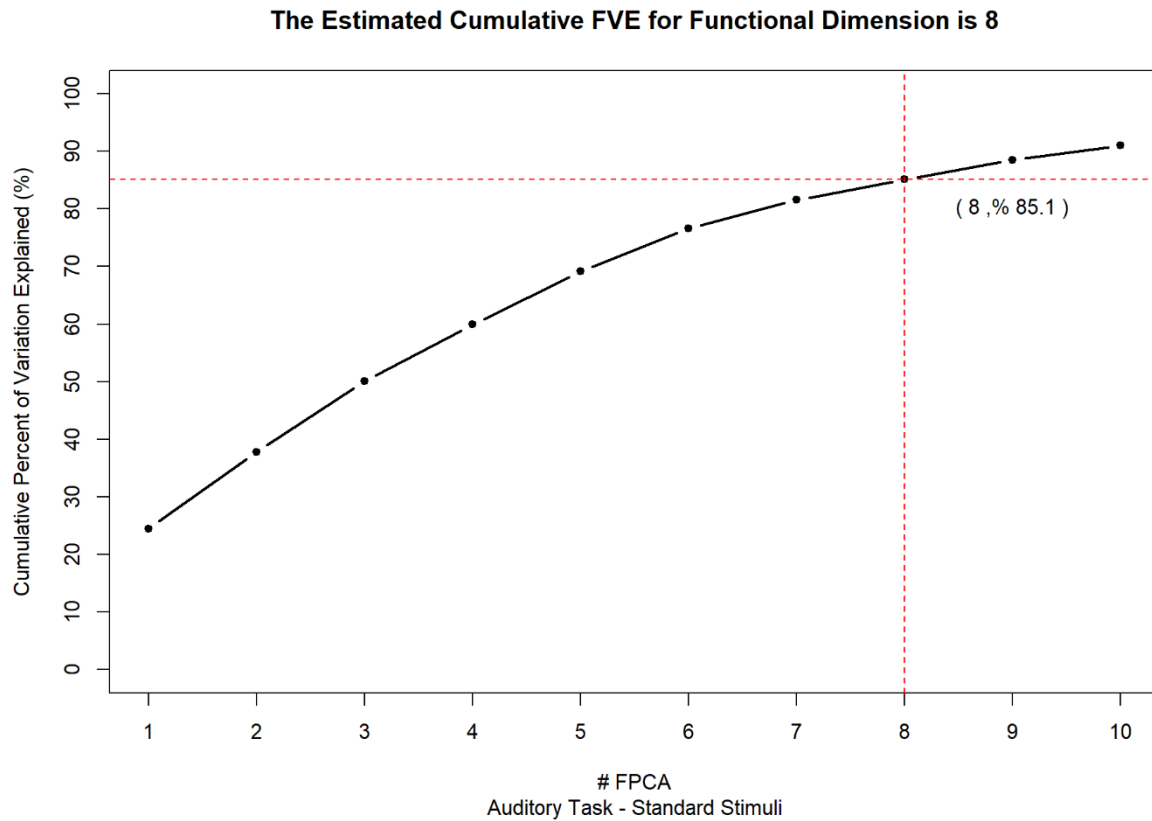


Figure 1. The estimated cumulative fraction of variations (FVE) for functional dimension.

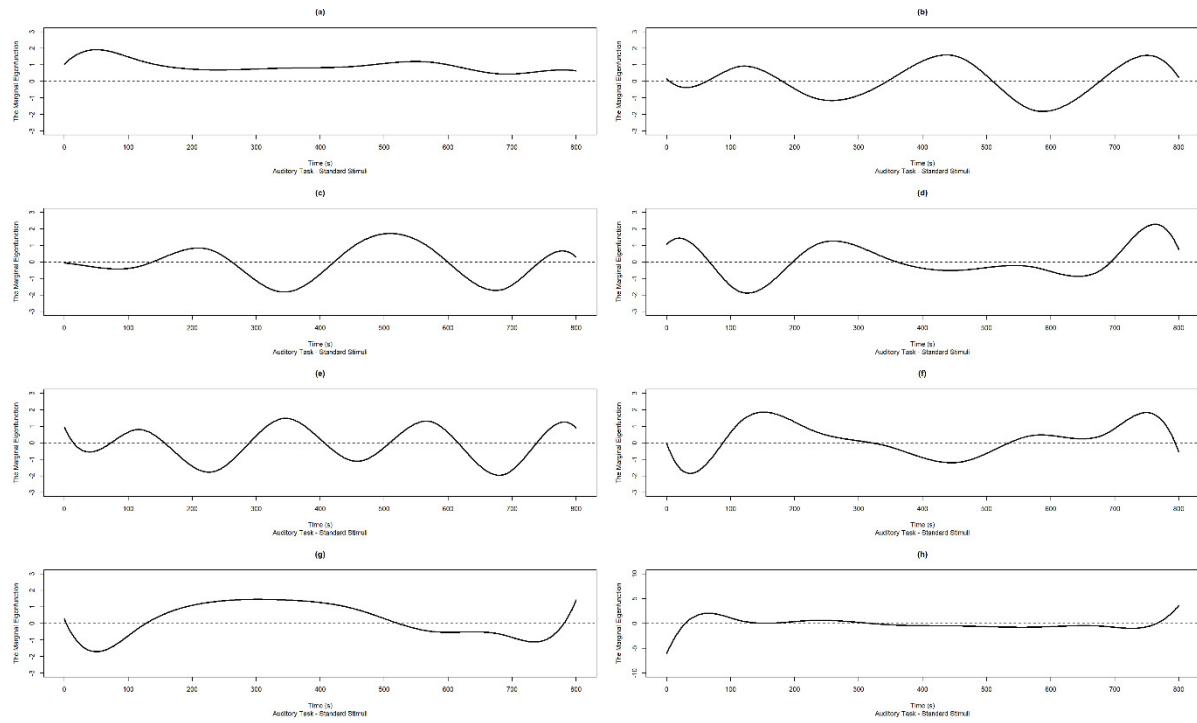


Figure 2. (a, h) Estimated first to eight leading functional marginal eigenfunctions.

According to the [Figure 1](#), the first eight marginal functional principal component analysis (FPCA) capture at least 85% of variation in this dimension. The [Figure 2](#) plots estimated first to eight leading functional marginal eigenfunctions on the x-axis (time).

2.1.1.2 The Longitudinal dimension

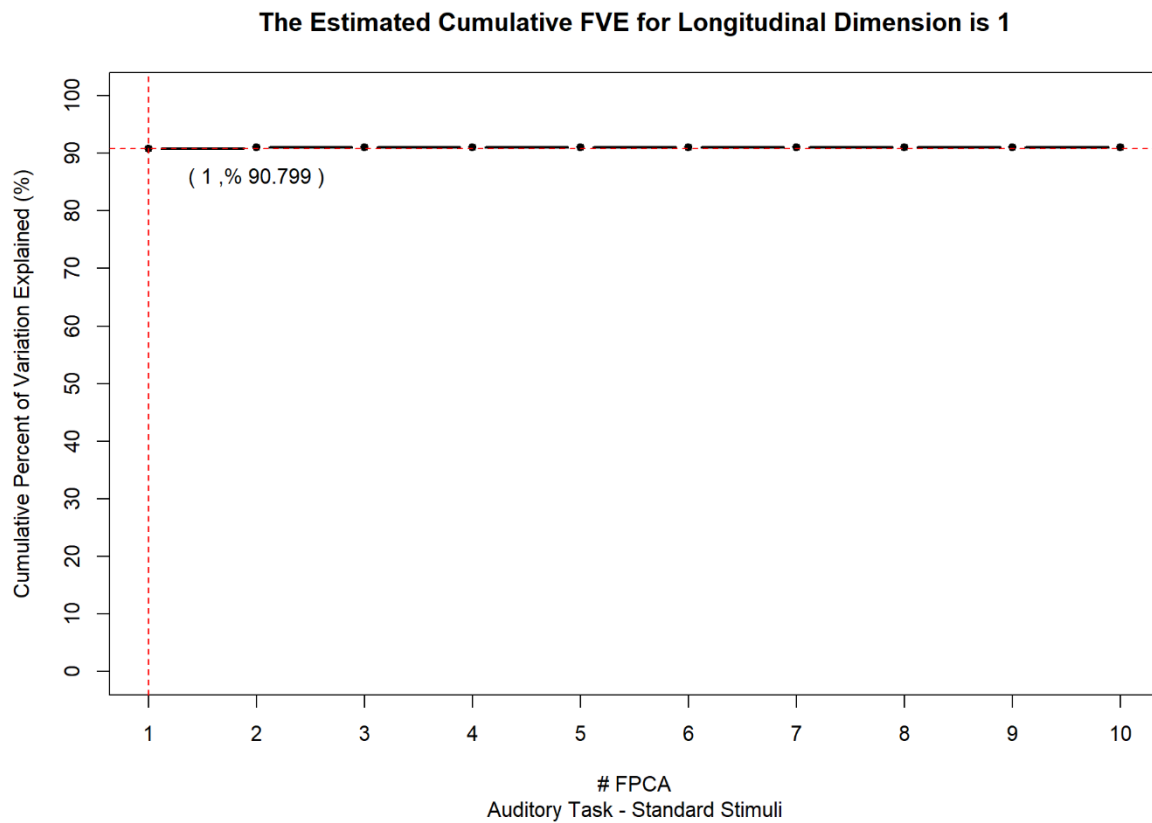


Figure 3. The estimated cumulative fraction of variations (FVE) for longitudinal dimension.

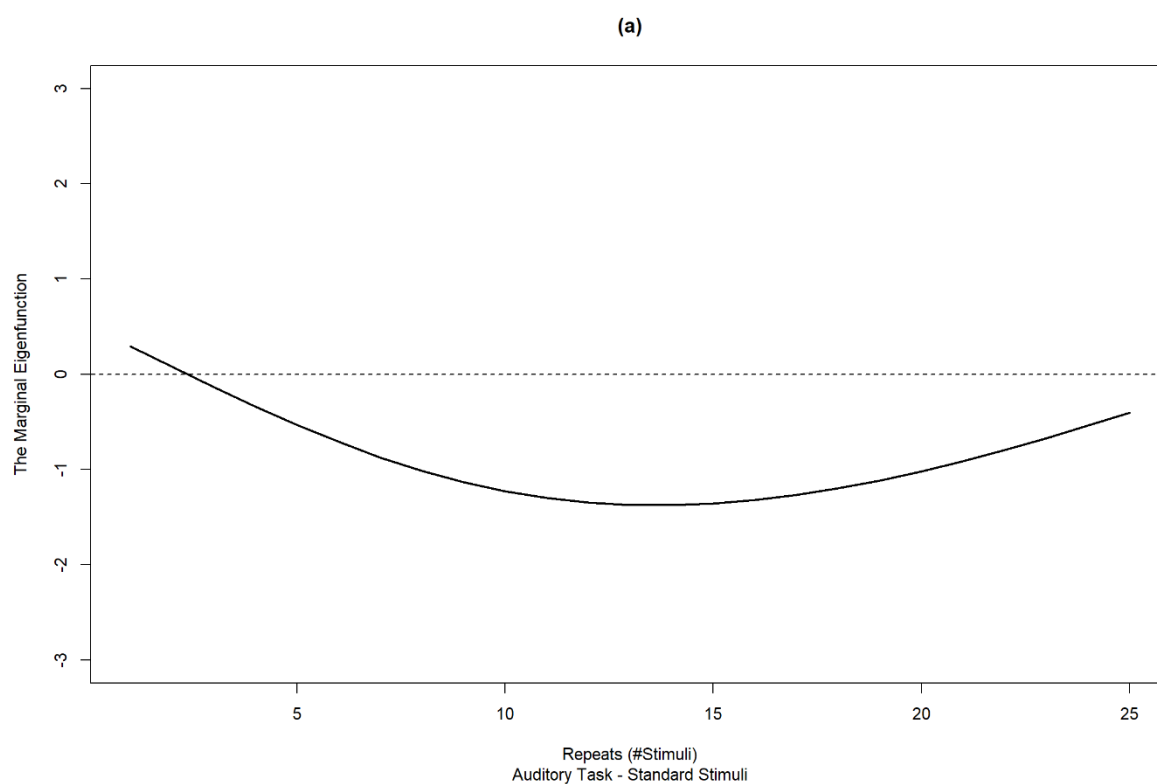


Figure 4. Estimated first leading longitudinal marginal eigenfunctions.

According to the [Figure 3](#), the first marginal functional principal component analysis (FPCA) capture at least 90% of variation in this dimension. The [Figure 4](#) plots estimated first leading longitudinal marginal eigenfunctions.

2.1.1.3 The Regional dimension

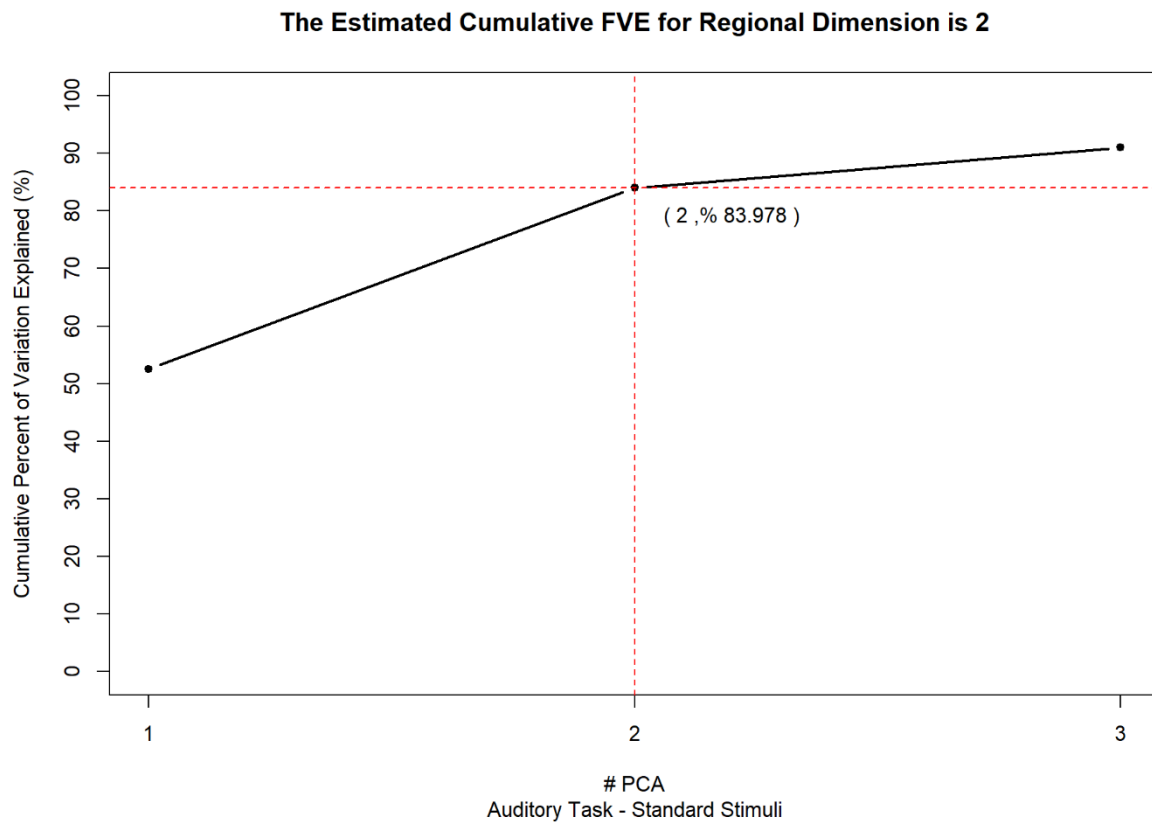


Figure 5. The estimated cumulative fraction of variations (FVE) for regional dimension.

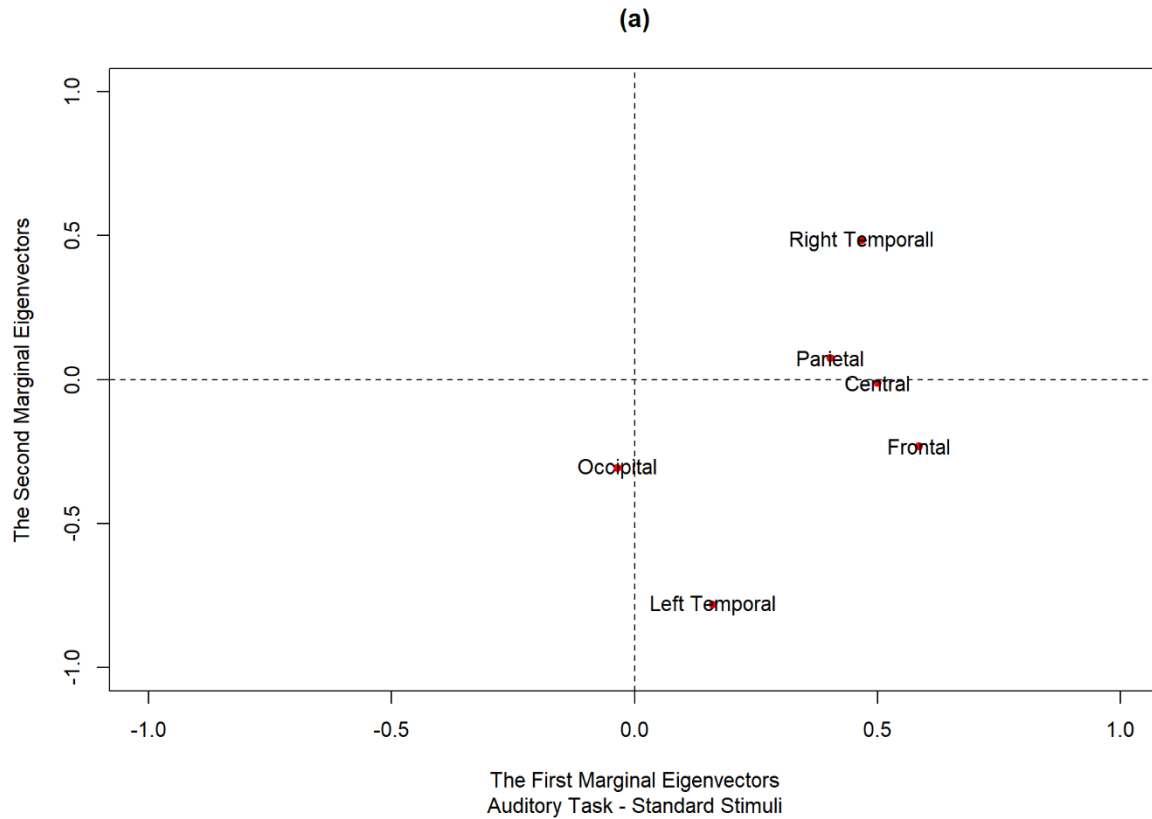


Figure 6. Estimated first and second leading regional marginal eigenvectors.

According to the Figure 5, the first and second marginal principal component analysis (PCA) capture at least nearly 85% of variation in this dimension. The Figure 6 plots estimated first and second leading regional marginal eigenvectors.

2.1.2. The Reconstructed of the Response .

The fof regression were conducted group by the number of HPCA and the regions. In the following part, we show the reconstructed ERPs for each subject (in gray) based on the number of HPCA. In this trial, we have 16 HPCA. We also show the point-wise mean and mean $\pm$ standard deviation. (Whole Brain: **Figure 7**, Frontal region (1): **Figure 8**, Central Region (2): **Figure 9**, Occipital Region (3): **Figure 10**, Parietal Region (4): **Figure 11**, Left Temporal (5): **Figure 12**, Right Temporal (6): **Figure 13**).

2.1.2.1 Whole Brain

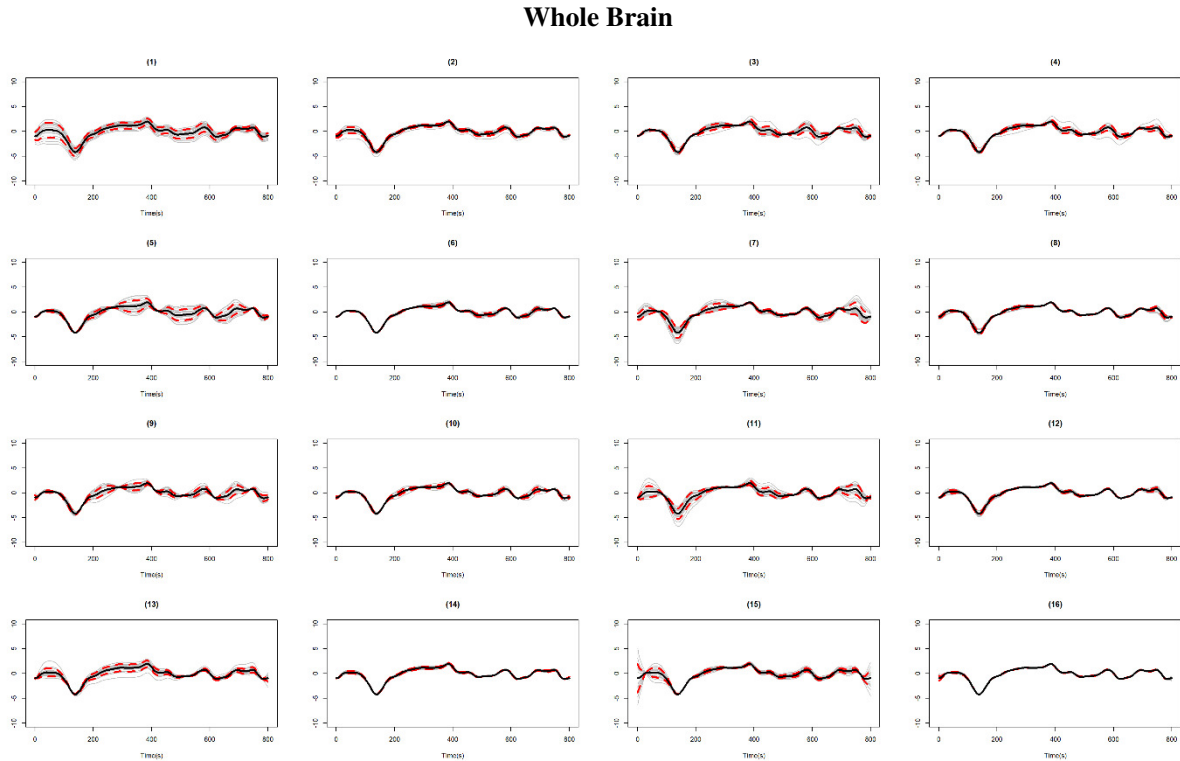


Figure 7. (1-16) Reconstructed ERPs for each subject (Gray line), point-wise mean (solid black line), and mean $\pm$ standard deviation (dotted red line) for whole Brain.

2.1.2.2 Frontal Region (1)

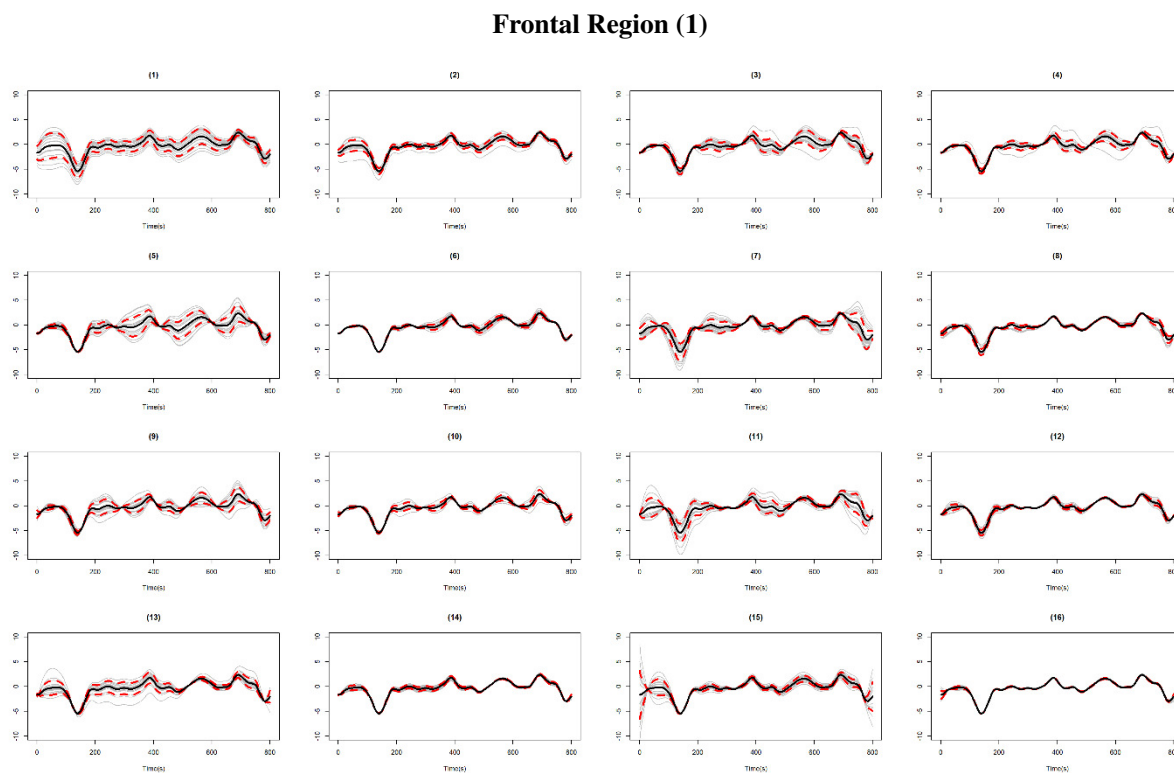


Figure 8. (1-16) Reconstructed ERPs for each subject (Gray line), point-wise mean (solid black line), and mean $\pm$ standard deviation (dotted red line) for frontal region of the Brain.

2.1.2.3 Central Region (2)

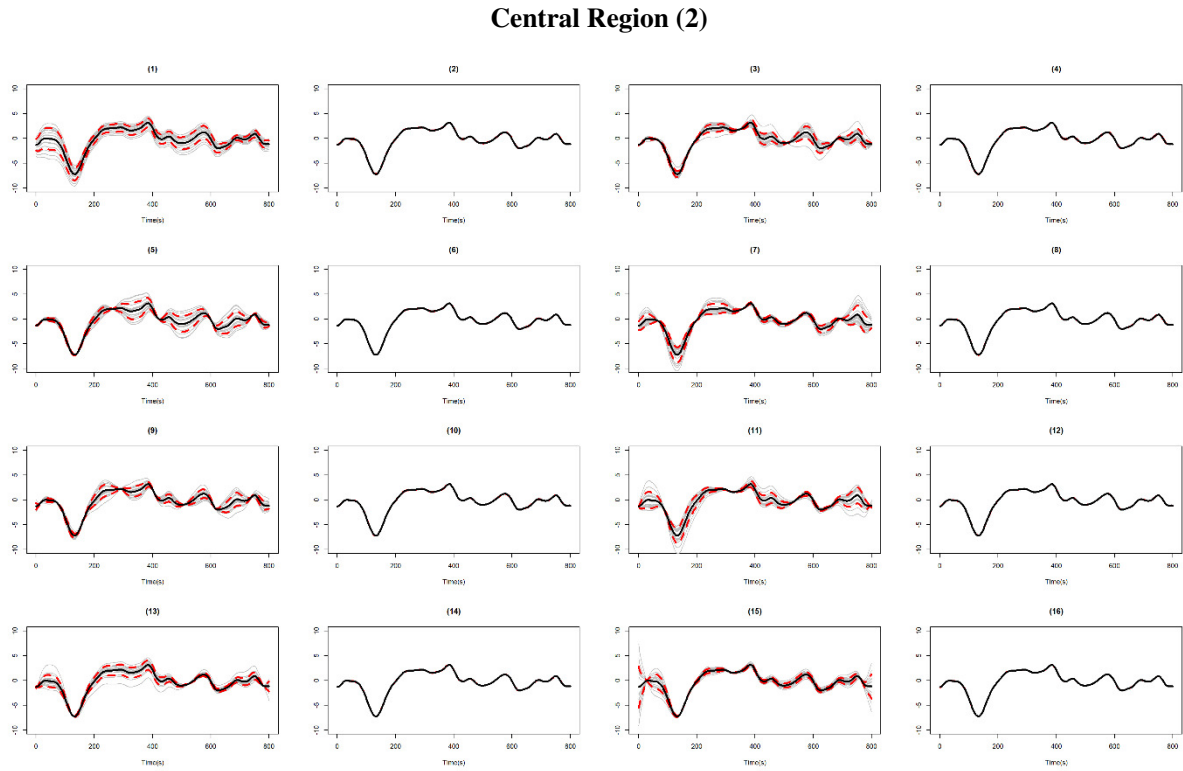


Figure 9. (1-16) Reconstructed ERPs for each subject (Gray line), point-wise mean (solid black line), and mean $\pm$ standard deviation (dotted red line) for central region of the Brain.

2.1.2.4 Occipital Region (3)

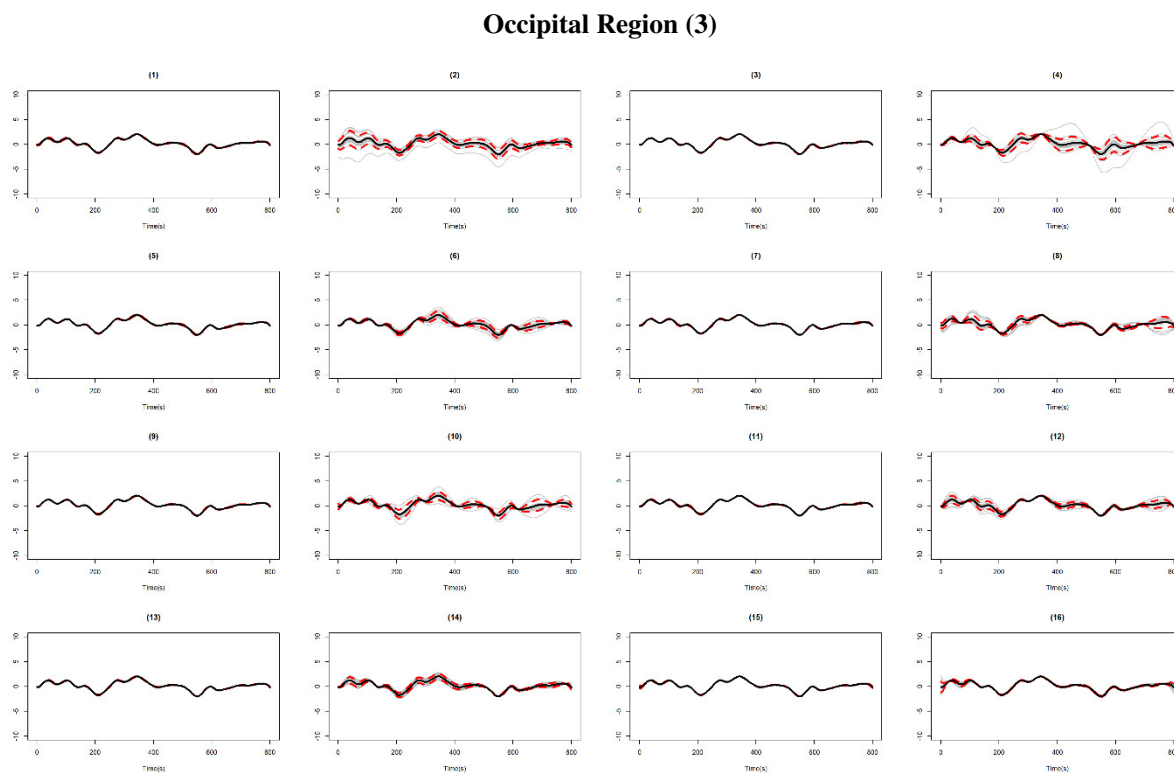


Figure 10. (1-16) Reconstructed ERPs for each subject (Gray line), point-wise mean (solid black line), and mean $\pm$ standard deviation (dotted red line) for occipital region of the Brain.

2.1.2.5 Parietal Region (4)

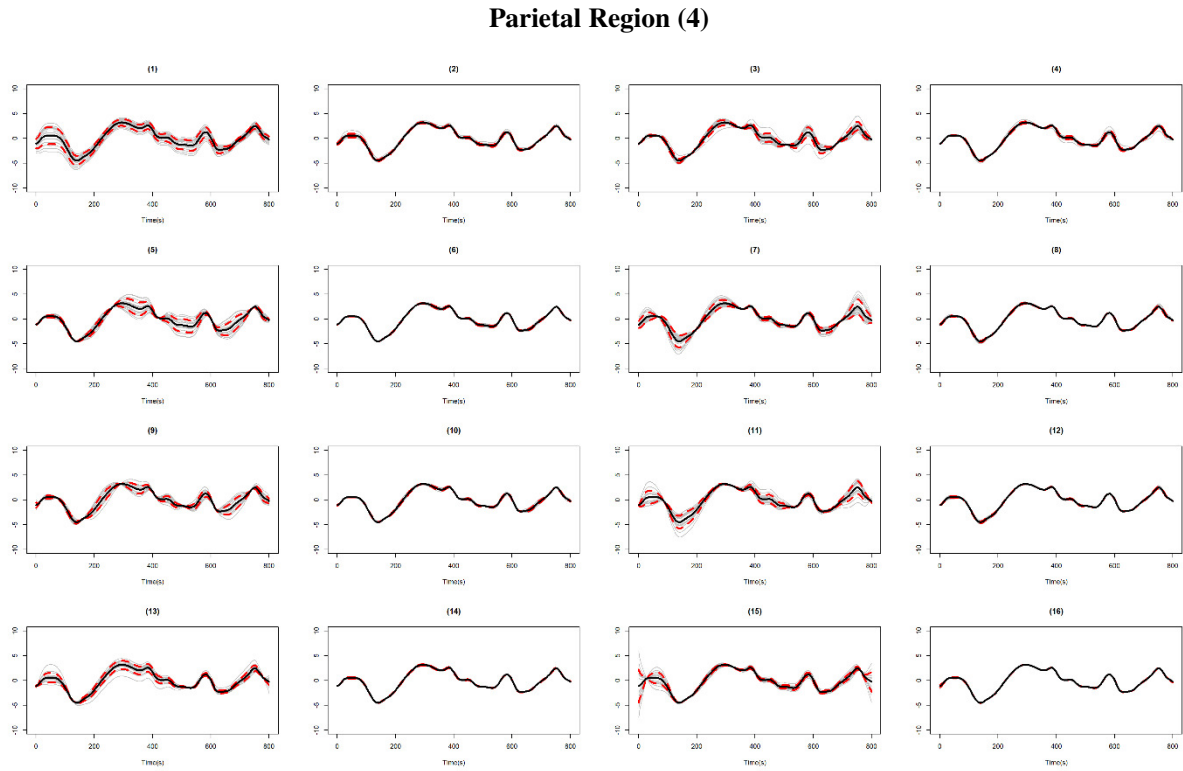


Figure 11. (1-16) Reconstructed ERPs for each subject (Gray line), point-wise mean (solid black line), and mean $\pm$ standard deviation (dotted red line) for parietal region of the Brain.

2.1.2.6 Left Temporal (5)

Left Temporal (5)

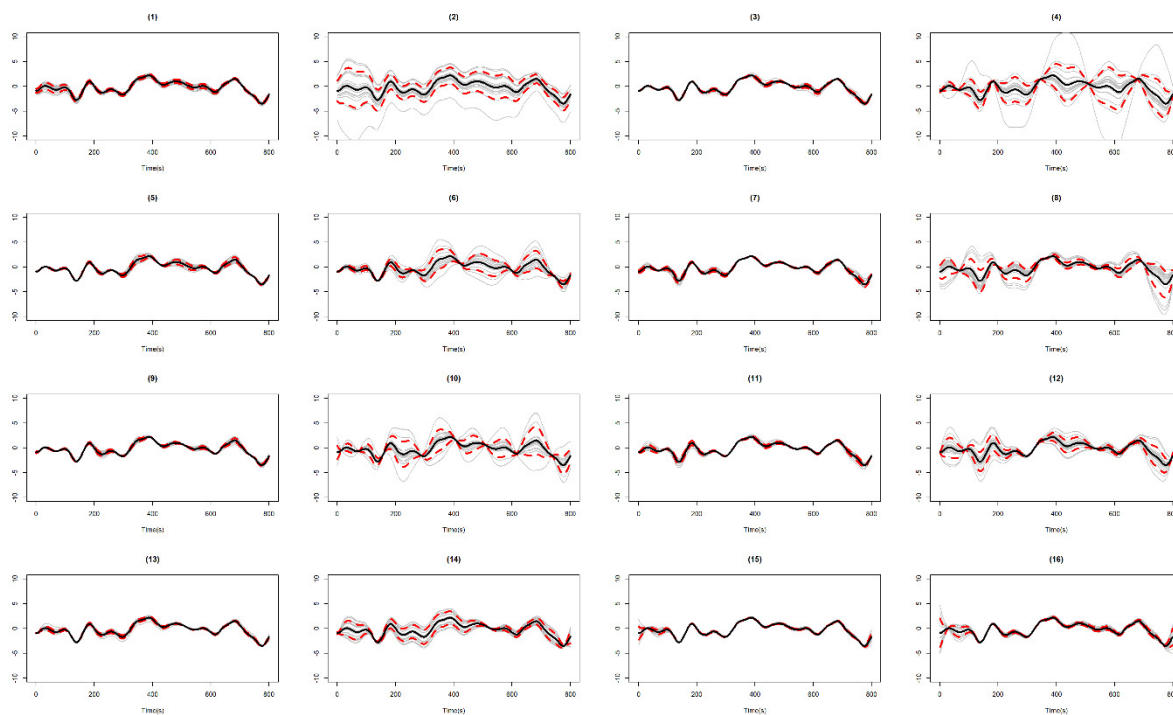


Figure 12. (1-16) Reconstructed ERPs for each subject (Gray line), point-wise mean (solid black line), and mean $\pm$ standard deviation (dotted red line) for left temporal region of the Brain.

2.1.2.7 Right Temporal (6)

Right Temporal (6)

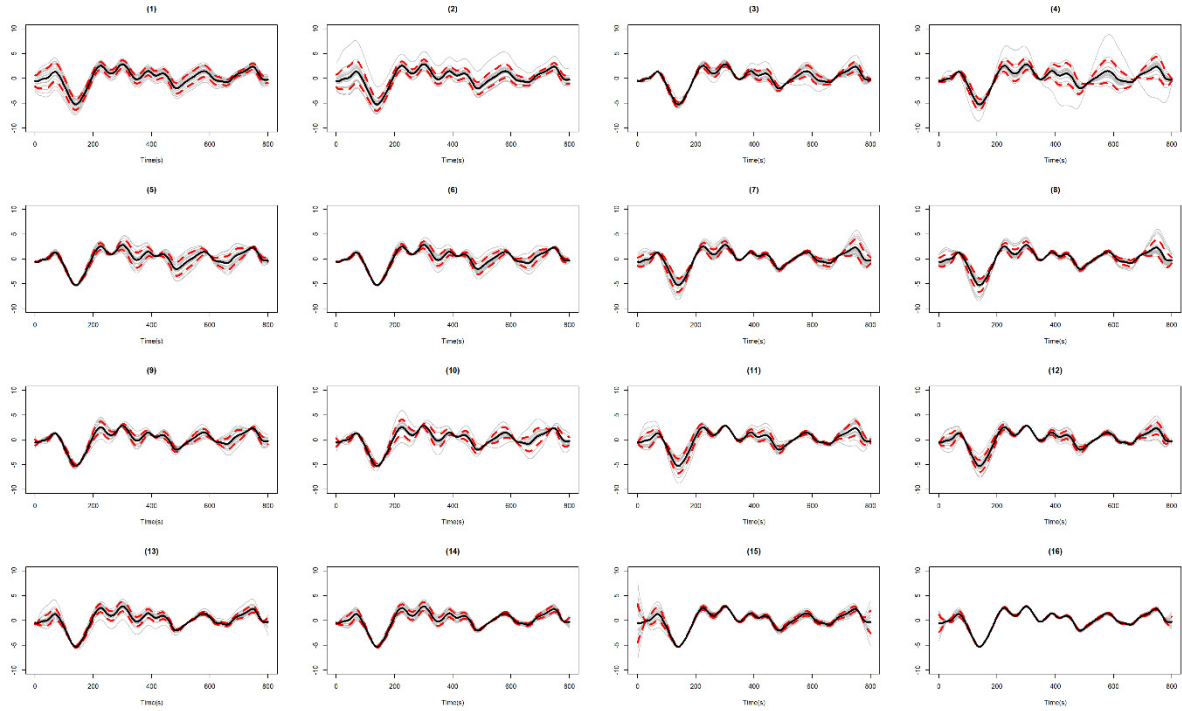


Figure 13. (1-16) Reconstructed ERPs for each subject (Gray line), point-wise mean (solid black line), and mean $\pm$ standard deviation (dotted red line) for right temporal region of the Brain.

2.1.3. *The function-on-function result* .

The fof regression result group by the region of the brain and the number of hybrid principal component analysis (HPCA) were reported.

2.1.3.1 **Whole Brain**

The MSPE and Correlation group by number of HPCA, Train and Test ([Table 1](#))

The User, System and Elapsed Time group by number of HPCA ([Table 2](#)): The median (first quantile – third quantile) of the Time (user-system-elapsed) in 10 iterations group by the number of hybrid principal component analysis (HPCA) for whole brain.

The Model fitting parameters group by number of HPCA ([Table 3](#)): The median (first quantile – third quantile) of the model fit indices in 10 iterations group by the number of hybrid principal component analysis (HPCA) for whole brain.

HPCA	MSPE		Correlation	
	Train	Test	Train	Test
1	0(0,0.002)	0.071(0.071,0.08)	1(0.999,1)	0.964(0.96,0.964)
2	0(0,0.003)	0.033(0.033,0.044)	1(0.999,1)	0.983(0.978,0.983)
3	0.094(0.013,0.094)	0.249(0.242,0.249)	0.952(0.952,0.994)	0.87(0.87,0.873)
4	0.108(0.108,0.108)	0.218(0.218,0.218)	0.944(0.944,0.944)	0.884(0.884,0.884)
5	0.12(0.12,0.12)	0.563(0.563,0.563)	0.938(0.938,0.938)	0.692(0.692,0.692)
6	0.121(0.121,0.121)	0.746(0.746,0.746)	0.937(0.937,0.937)	0.591(0.591,0.591)
7	0.192(0.192,0.192)	0.737(0.719,0.737)	0.899(0.899,0.899)	0.578(0.578,0.586)
8	0.176(0.176,0.176)	0.76(0.76,0.76)	0.908(0.907,0.908)	0.566(0.566,0.566)
9	0.221(0.214,0.236)	0.776(0.677,0.778)	0.882(0.874,0.886)	0.559(0.558,0.622)
10	0.254(0.227,0.265)	0.666(0.646,0.709)	0.864(0.857,0.879)	0.637(0.597,0.644)
11	0.319(0.316,0.327)	0.751(0.732,0.751)	0.825(0.82,0.827)	0.568(0.568,0.58)
12	0.288(0.258,0.318)	0.739(0.726,0.744)	0.844(0.826,0.861)	0.579(0.575,0.592)
13	0.29(0.284,0.333)	0.842(0.779,0.866)	0.842(0.816,0.846)	0.513(0.493,0.543)
14	0.347(0.341,0.352)	0.877(0.877,0.884)	0.808(0.804,0.811)	0.487(0.487,0.488)
15	0.364(0.339,0.405)	0.918(0.879,0.95)	0.797(0.771,0.813)	0.451(0.448,0.453)
16	0.341(0.219,0.352)	1.037(0.938,1.054)	0.812(0.805,0.884)	0.424(0.358,0.436)

Table 1. The median (first quantile – third quantile) of the Mean Square Prediction Error (MSPE) and Correlation in 10 iterations group by Train and Test and the number of hybrid principal component analysis (HPCA) for whole brain.

HPCA	Time (s)		
	User	System	Elapsed
1	72.645(71.585,73.463)	0.115(0.05,0.222)	72.805(72.295,73.778)
2	69.75(67.78,70.74)	0.045(0.023,0.06)	69.77(67.833,70.76)
3	136.82(136.033,139.845)	0.06(0.03,0.11)	136.87(136.243,139.985)
4	144.04(142.655,145.29)	0.11(0.087,0.125)	144.16(142.733,145.543)
5	184.945(182.778,185.485)	0.085(0.063,0.128)	185.02(182.86,186.23)
6	182.69(180.285,186.288)	0.085(0.073,0.097)	182.82(180.44,186.363)
7	223.605(220.992,226.475)	0.11(0.11,0.133)	223.77(221.155,226.655)
8	229.2(225.528,231.83)	0.11(0.075,0.125)	229.315(225.71,231.87)
9	266.25(262.32,269.9)	0.11(0.09,0.153)	266.44(262.48,270.09)
10	265.565(262.665,267.167)	0.12(0.112,0.14)	265.78(262.802,267.32)
11	280.89(279.818,286.23)	0.085(0.08,0.14)	281.015(280.05,286.363)
12	286.305(283.575,290.59)	0.14(0.123,0.17)	286.545(283.8,290.857)
13	293.635(288.247,295.518)	0.14(0.1,0.18)	293.81(288.548,295.685)
14	294.41(287.753,300.593)	0.125(0.103,0.14)	294.735(287.867,300.715)
15	312.465(309.808,315.228)	0.135(0.09,0.178)	312.77(309.968,315.358)
16	314.9(311.79,318.95)	0.16(0.115,0.222)	315.09(312.093,319.1)

Table 2. The median (first quantile – third quantile) of the Time (user-system-elapsed) in 10 iterations group by the number of hybrid principal component analysis (HPCA) for whole brain.

HPCA	Min.Error	Optimum					
		index	K	Tau	Lambda	Eta	Kappa
1	0.692(0.625,0.792)	5(5,8)	2(1,2.75)	0.1(0.1,100)	0.1(0.1,0.4)	0(0,0)	0(0,0)
2	0.884(0.664,0.906)	5(4,5)	1(1,2)	0.1(0.1,100)	0.1(0.1,0.4)	0(0,0)	0(0,0)
3	2.998(2.531,3.429)	12(2,12)	1(1,1.75)	100(0.1,100)	0.4(0.1,0.4)	0.01(0,0.01)	0(0,0)
4	3.573(3.503,3.7)	12(12,12)	1(1,1)	100(100,100)	0.4(0.4,0.4)	0.01(0.01,0.01)	0(0,0)
5	4.55(4.309,5.022)	12(12,12)	2(2,2)	100(100,100)	0.4(0.4,0.4)	0.01(0.01,0.01)	0(0,0)
6	4.488(4.286,4.869)	12(12,12)	2(2,2)	100(100,100)	0.4(0.4,0.4)	0.01(0.01,0.01)	0(0,0)
7	6.261(5.898,6.895)	12(12,12)	2(2,2)	100(100,100)	0.4(0.4,0.4)	0.01(0.01,0.01)	0(0,0)
8	5.951(5.51,6.22)	12(12,12)	2(2,2)	100(100,100)	0.4(0.4,0.4)	0.01(0.01,0.01)	0(0,0)
9	7.417(6.954,7.844)	6(1.25,12)	2(1,2)	100(0.325,100)	0.4(0.125,0.4)	0(0,0.01)	0(0,0)
10	7.453(6.825,8.711)	4.5(4,11)	1(1,1.75)	100(100,100)	0.4(0.4,0.4)	0(0,0.008)	0(0,0)
11	7.81(7.412,7.924)	12(5,12)	1(1,1)	100(100,100)	0.4(0.4,0.4)	0.01(0,0.01)	0(0,0)
12	7.575(7.539,7.698)	8(3.25,12)	1(1,1)	100(32.5,100)	0.4(0.325,0.4)	0(0,0.01)	0(0,0)
13	8.578(8.148,8.772)	4(1.75,7.5)	1(1,1.75)	100(0.325,100)	0.4(0.125,0.4)	0(0,0)	0(0,0)
14	8.463(7.949,8.823)	9(4,12)	1(1,1)	100(100,100)	0.4(0.4,0.4)	0.005(0,0.01)	0(0,0)
15	8.762(8.591,9.078)	4(2.25,7)	1(1,1)	100(3.25,100)	0.4(0.225,0.4)	0(0,0)	0(0,0)
16	8.6(8.37,8.869)	4(3,4)	1(1,3)	100(10,100)	0.4(0.3,0.4)	0(0,0)	0(0,0)

Table 3. The median (first quantile – third quantile) of the model fit indices in 10 iterations group by the number of hybrid principal component analysis (HPCA) for whole brain.

2.1.3.2 Frontal Region (1)

The MSPE and Correlation group by number of HPCA, Train and Test (**Table 4**)

The User, System and Elapsed Time group by number of HPCA (**Table 5**): The median (first quantile – third quantile) of the Time (user-system-elapsed) in 10 iterations group by the number of hybrid principal component analysis (HPCA) for frontal region of brain.

The Model fitting parameters group by number of HPCA (**Table 6**): The median (first quantile – third quantile) of the model fit indices in 10 iterations group by the number of hybrid principal component analysis (HPCA) for whole brain.

HCPA	MSPE		Correlation	
	Train	Test	Train	Test
1	0(0,0.001)	0.126(0.126,0.126)	1(0.999,1)	0.936(0.935,0.936)
2	0.002(0,0.006)	0.063(0.057,0.095)	0.999(0.997,1)	0.968(0.952,0.971)
3	0.034(0.011,0.056)	0.366(0.36,0.371)	0.983(0.972,0.994)	0.801(0.798,0.807)
4	0.151(0.096,0.158)	0.316(0.301,0.316)	0.921(0.918,0.951)	0.827(0.827,0.836)
5	0.175(0.175,0.175)	0.727(0.727,0.727)	0.908(0.908,0.908)	0.585(0.585,0.585)
6	0.174(0.174,0.174)	0.943(0.943,0.943)	0.909(0.909,0.909)	0.462(0.462,0.462)
7	0.328(0.301,0.328)	0.777(0.756,0.866)	0.819(0.819,0.84)	0.555(0.487,0.566)
8	0.302(0.271,0.314)	0.84(0.818,0.917)	0.835(0.828,0.854)	0.528(0.454,0.539)
9	0.345(0.329,0.388)	0.984(0.973,1.022)	0.809(0.782,0.819)	0.439(0.4,0.444)
10	0.376(0.353,0.386)	0.807(0.804,0.81)	0.79(0.783,0.804)	0.535(0.532,0.539)
11	0.45(0.441,0.601)	0.931(0.93,0.947)	0.741(0.639,0.748)	0.43(0.319,0.435)
12	0.403(0.37,0.431)	0.907(0.901,0.942)	0.772(0.754,0.793)	0.452(0.419,0.461)
13	0.453(0.413,0.486)	0.89(0.814,0.895)	0.739(0.717,0.766)	0.454(0.451,0.491)
14	0.427(0.413,0.476)	0.918(0.833,0.919)	0.757(0.723,0.766)	0.444(0.435,0.479)
15	0.474(0.446,0.509)	1.035(0.945,1.085)	0.725(0.7,0.744)	0.372(0.309,0.385)
16	0.491(0.374,0.507)	1.099(1.027,1.162)	0.713(0.701,0.79)	0.349(0.341,0.37)

Table 4. The median (first quantile – third quantile) of the Mean Square Prediction Error (MSPE) and Correlation in 10 iterations group by Train and Test and the number of hybrid principal component analysis (HPCA) for frontal region brain.

HPCA	Time (s)		
	User	System	Elapsed
1	95.55(94.568,96.358)	0.08(0.06,0.097)	95.78(94.678,96.447)
2	95.51(94.533,97.14)	0.07(0.045,0.087)	95.565(94.612,97.222)
3	143.905(143.158,145.185)	0.08(0.063,0.102)	144.015(143.225,145.305)
4	156.825(154.452,163.548)	0.115(0.087,0.14)	156.89(154.563,164.062)
5	187.61(184.643,190.002)	0.075(0.063,0.1)	187.705(184.955,190.142)
6	190.755(188.99,191.795)	0.08(0.053,0.135)	190.835(189.088,191.923)
7	220.5(216.895,222.17)	0.135(0.093,0.183)	220.635(217.065,222.34)
8	225.53(224.603,228.035)	0.095(0.07,0.13)	225.715(224.675,228.185)
9	269.96(268.198,272.765)	0.135(0.112,0.165)	270.145(268.365,272.968)
10	270.205(268.855,274.842)	0.125(0.105,0.155)	270.445(268.973,275.013)
11	299.08(295.595,305.542)	0.125(0.09,0.147)	299.3(295.74,305.75)
12	304.335(298.033,310.727)	0.135(0.08,0.158)	304.46(298.25,310.942)
13	305.765(302.532,309.892)	0.135(0.11,0.155)	305.92(302.957,310.098)
14	302.06(298.79,303.965)	0.115(0.087,0.14)	302.225(299.035,304.17)
15	332.72(330.39,335.655)	0.165(0.135,0.19)	332.96(330.552,335.84)
16	326.68(325.358,329.56)	0.155(0.15,0.198)	326.91(325.505,329.818)

Table 5. The median (first quantile – third quantile) of the Time (user-system-elapsed) in 10 iterations group by the number of hybrid principal component analysis (HPCA) for frontal region of brain.

HPCA	Min_Error	Optimum					
		index	K	Tau	Lambda	Eta	Kappa
1	1.344(0.987,1.355)	5(5,5)	3(1.25,3)	0.1(0.1,0.1)	0.1(0.1,0.1)	0(0,0)	0(0,0)
2	1.468(1.183,1.608)	6(5,8)	1.5(1,2.75)	5.05(0.1,100)	0.2(0.1,0.4)	0(0,0)	0(0,0)
3	4.932(4.617,5.082)	1(1,1)	1.5(1,2)	0.1(0.1,0.1)	0.1(0.1,0.1)	0(0,0)	0(0,0)
4	5.02(4.847,5.263)	12(4,12)	1(1,1.75)	100(100,100)	0.4(0.4,0.4)	0.01(0,0.01)	0(0,0)
5	5.794(5.709,6.1)	12(12,12)	2(2,2)	100(100,100)	0.4(0.4,0.4)	0.01(0.01,0.01)	0(0,0)
6	5.776(5.336,6.175)	12(12,12)	2(2,2)	100(100,100)	0.4(0.4,0.4)	0.01(0.01,0.01)	0(0,0)
7	9.085(8.691,9.351)	4(4,12)	1(1,1.75)	100(100,100)	0.4(0.4,0.4)	0(0,0.01)	0(0,0)
8	8.048(7.4,8.769)	12(6,12)	1(1,2)	100(100,100)	0.4(0.4,0.4)	0.01(0.003,0.01)	0(0,0)
9	10.122(9.518,10.319)	8(5,11.75)	1(1,1.75)	100(32.5,100)	0.4(0.325,0.4)	0(0,0.01)	0(0,0)
10	9.243(9.078,9.417)	7.5(4,12)	1(1,1)	100(100,100)	0.4(0.4,0.4)	0.005(0,0.01)	0(0,0)
11	10.668(9.741,11.998)	6(4,12)	1(1,1)	100(100,100)	0.4(0.4,0.4)	0(0,0.01)	0(0,0.008)
12	10.457(10.265,10.997)	8(4,12)	1(1,2)	100(100,100)	0.4(0.4,0.4)	0.005(0,0.01)	0(0,0)
13	11.347(11.254,11.841)	3.5(1.25,8)	1(1,1)	55(0.325,100)	0.35(0.125,0.4)	0(0,0)	0(0,0)
14	11.283(11.014,11.597)	3.5(1.5,6.25)	1(1,1)	10(2.575,100)	0.3(0.15,0.4)	0(0,0)	0(0,0)
15	11.756(11.404,12.145)	5.5(1.5,8)	1(1,1.75)	55(2.575,100)	0.35(0.15,0.4)	0(0,0)	0(0,0)
16	11.283(11.045,11.405)	5.5(4,10.25)	1(1,2.5)	100(32.5,100)	0.4(0.325,0.4)	0(0,0.008)	0(0,0)

Table 6. The median (first quantile – third quantile) of the model fit indices in 10 iterations group by the number of hybrid principal component analysis (HPCA) for frontal region of brain.

2.1.3.3 Central Region (2)

The MSPE and Correlation group by number of HPCA, Train and Test (**Table 7**)

The User, System and Elapsed Time group by number of HPCA (**Table 8**): The median (first quantile – third quantile) of the Time (user-system-elapsed) in 10 iterations group by the number of hybrid principal component analysis (HPCA) for whole brain.

The Model fitting parameters group by number of HPCA (**Table 9**): The median (first quantile – third quantile) of the model fit indices in 10 iterations group by the number of hybrid principal component analysis (HPCA) for whole brain.

HPCA	MSPE		Correlation	
	Train	Test	Train	Test
1	0(0,0.001)	0.046(0.046,0.053)	1(1,1)	0.977(0.973,0.977)
2	0(0,0.001)	0.044(0.044,0.049)	1(0.999,1)	0.978(0.975,0.978)
3	0.027(0.009,0.068)	0.171(0.168,0.221)	0.986(0.965,0.995)	0.911(0.885,0.913)
4	0.048(0.027,0.069)	0.169(0.169,0.228)	0.976(0.965,0.986)	0.913(0.882,0.913)
5	0.086(0.086,0.086)	0.402(0.402,0.402)	0.956(0.956,0.956)	0.788(0.788,0.788)
6	0.086(0.086,0.086)	0.41(0.41,0.41)	0.956(0.956,0.956)	0.784(0.784,0.784)
7	0.137(0.137,0.141)	0.441(0.424,0.441)	0.929(0.927,0.929)	0.761(0.761,0.779)
8	0.136(0.136,0.164)	0.42(0.39,0.44)	0.929(0.914,0.929)	0.772(0.761,0.792)
9	0.197(0.185,0.217)	0.415(0.412,0.45)	0.896(0.884,0.903)	0.775(0.763,0.777)
10	0.193(0.189,0.197)	0.449(0.417,0.461)	0.898(0.896,0.901)	0.764(0.759,0.775)
11	0.258(0.218,0.263)	0.483(0.473,0.504)	0.861(0.858,0.884)	0.731(0.718,0.737)
12	0.213(0.211,0.258)	0.494(0.471,0.509)	0.887(0.861,0.888)	0.725(0.715,0.739)
13	0.24(0.189,0.257)	0.618(0.608,0.641)	0.871(0.862,0.9)	0.654(0.638,0.667)
14	0.257(0.241,0.289)	0.616(0.606,0.648)	0.862(0.843,0.871)	0.661(0.634,0.665)
15	0.242(0.185,0.294)	0.702(0.635,0.715)	0.87(0.84,0.902)	0.61(0.589,0.636)
16	0.294(0.224,0.327)	0.661(0.63,0.695)	0.84(0.82,0.88)	0.629(0.622,0.633)

Table 7. The median (first quantile – third quantile) of the Mean Square Prediction Error (MSPE) and Correlation in 10 iterations group by Train and Test and the number of hybrid principal component analysis (HPCA) for central region of the brain.

HPCA	Time (s)		
	User	System	Elapsed
1	70.32(68.812,71.565)	0.045(0.023,0.08)	70.42(69.155,71.75)
2	68.69(67.558,69.393)	0.03(0.02,0.045)	68.785(67.597,69.475)
3	146.825(145.66,147.255)	0.11(0.065,0.118)	147.065(145.9,147.32)
4	142.945(141.553,145.535)	0.07(0.045,0.08)	143.03(141.645,145.835)
5	180.79(179.892,182.508)	0.11(0.07,0.138)	180.885(179.943,182.69)
6	176.195(175.48,177.915)	0.09(0.08,0.097)	176.345(175.608,178.015)
7	214.72(214.553,215.67)	0.1(0.08,0.11)	214.93(214.642,215.763)
8	216.76(214.187,218.365)	0.095(0.08,0.17)	216.855(214.357,218.458)
9	262.21(259.643,265.375)	0.12(0.112,0.16)	262.32(259.89,265.573)
10	263.77(259.072,266.665)	0.105(0.1,0.118)	263.91(259.205,266.798)
11	291.36(286.702,295.69)	0.16(0.125,0.2)	291.66(286.815,296.205)
12	291.295(287.503,297.415)	0.13(0.11,0.182)	291.485(287.67,297.73)
13	287.85(284.567,290.38)	0.14(0.115,0.198)	288.1(284.76,290.535)
14	284.445(281.037,286.245)	0.135(0.115,0.165)	284.68(281.22,286.41)
15	315.955(314.29,319.553)	0.145(0.13,0.19)	316.19(314.48,319.848)
16	313.405(311.62,315.738)	0.15(0.13,0.167)	313.805(311.913,315.9)

Table 8. The median (first quantile – third quantile) of the Time (user-system-elapsed) in 10 iterations group by the number of hybrid principal component analysis (HPCA) for central region of the brain.

HPCA	Min_Error	Optimum					
		index	K	Tau	Lambda	Eta	Kappa
1	0.524(0.452,0.553)	5(5,7.5)	2(1,2.75)	0.1(0.1,75.25)	0.1(0.1,0.35)	0(0,0)	0(0,0)
2	0.505(0.442,0.516)	5(5,8)	2(1,2.75)	0.1(0.1,100)	0.1(0.1,0.4)	0(0,0)	0(0,0)
3	1.96(1.8,2.215)	7(2,12)	1(1,1.75)	0.1(0.1,100)	0.1(0.1,0.4)	0.005(0,0.01)	0(0,0)
4	2.007(1.841,2.151)	10.5(5,12)	1(1,1)	50.05(0.1,100)	0.25(0.1,0.4)	0.01(0,0.01)	0(0,0)
5	2.913(2.67,3.043)	12(12,12)	2(2,2)	100(100,100)	0.4(0.4,0.4)	0.01(0.01,0.01)	0(0,0)
6	3.043(2.753,3.261)	12(12,12)	2(2,2)	100(100,100)	0.4(0.4,0.4)	0.01(0.01,0.01)	0(0,0)
7	4.263(4.016,4.351)	12(11.25,12)	2(1.25,2)	100(100,100)	0.4(0.4,0.4)	0.01(0.01,0.01)	0(0,0)
8	4.202(4.084,4.412)	12(12,12)	2(1,2)	100(100,100)	0.4(0.4,0.4)	0.01(0.01,0.01)	0(0,0)
9	5.339(4.879,5.592)	6(4,11)	1(1,1)	100(10,100)	0.4(0.3,0.4)	0(0,0.01)	0(0,0)
10	5.185(4.983,5.606)	9.5(3,11)	1(1,1)	10(10,77.5)	0.3(0.3,0.375)	0.005(0,0.01)	0(0,0)
11	6.182(5.968,6.599)	12(11.25,12)	1(1,1.75)	100(100,100)	0.4(0.4,0.4)	0.01(0.01,0.01)	0(0,0)
12	6.156(6.047,6.363)	10(4,12)	1.5(1,2)	100(100,100)	0.4(0.4,0.4)	0.005(0,0.01)	0(0,0)
13	6.701(6.504,6.84)	10(5,11.75)	1(1,2.75)	100(10,100)	0.4(0.3,0.4)	0.01(0,0.01)	0(0,0)
14	6.876(6.329,7.133)	11.5(5.75,12)	1(1,1)	100(10,100)	0.4(0.3,0.4)	0.01(0.003,0.01)	0(0,0)
15	7.107(6.927,7.172)	6(4,9.5)	2(1,3)	100(32.5,100)	0.4(0.325,0.4)	0(0,0.008)	0(0,0)
16	7.239(6.893,7.649)	11(8,12)	1(1,2.5)	100(10,100)	0.4(0.3,0.4)	0.01(0,0.01)	0(0,0)

Table 9. The median (first quantile – third quantile) of the model fit indices in 10 iterations group by the number of hybrid principal component analysis (HPCA) for central region of the brain..

2.1.3.4 Occipital Region (3)

The MSPE and Correlation group by number of HPCA, Train and Test (Table 10)

The User, System and Elapsed Time group by number of HPCA (Table 11): The median (first quantile – third quantile) of the Time (user-system-elapsed) in 10 iterations group by the number of hybrid principal component analysis (HPCA) for whole brain.

The Model fitting parameters group by number of HPCA (Table 12): The median (first quantile – third quantile) of the model fit indices in 10 iterations group by the number of hybrid principal component analysis (HPCA) for whole brain.

HCPA	MSPE		Correlation	
	Train	Test	Train	Test
1	0(0,0)	0.002(0.001,0.002)	1(1,1)	0.999(0.999,0.999)
2	0.01(0.009,0.012)	0.281(0.253,0.31)	0.995(0.994,0.996)	0.86(0.85,0.87)
3	0.012(0.01,0.014)	0.285(0.245,0.316)	0.994(0.993,0.995)	0.858(0.847,0.877)
4	0.078(0.057,0.078)	0.4(0.256,0.429)	0.96(0.96,0.971)	0.781(0.768,0.867)
5	0.069(0.036,0.08)	0.353(0.273,0.42)	0.965(0.959,0.982)	0.806(0.773,0.858)
6	0.092(0.092,0.111)	0.496(0.487,0.619)	0.953(0.943,0.953)	0.74(0.664,0.742)
7	0.106(0.096,0.115)	0.501(0.489,0.554)	0.945(0.941,0.951)	0.738(0.708,0.741)
8	0.185(0.185,0.202)	0.688(0.685,0.715)	0.902(0.893,0.903)	0.6(0.597,0.604)
9	0.187(0.187,0.208)	0.684(0.684,0.713)	0.901(0.89,0.901)	0.601(0.601,0.601)
10	0.371(0.362,0.378)	0.844(0.765,0.851)	0.793(0.788,0.798)	0.433(0.432,0.519)
11	0.354(0.354,0.375)	0.821(0.821,0.842)	0.803(0.79,0.803)	0.448(0.436,0.448)
12	0.367(0.367,0.373)	0.902(0.902,0.908)	0.795(0.791,0.795)	0.38(0.376,0.38)
13	0.382(0.368,0.388)	0.913(0.901,0.918)	0.786(0.782,0.795)	0.376(0.375,0.382)
14	0.388(0.388,0.422)	1.008(1,1.009)	0.782(0.76,0.782)	0.334(0.328,0.35)
15	0.43(0.387,0.434)	1.02(1.013,1.043)	0.755(0.752,0.783)	0.333(0.324,0.347)
16	0.428(0.41,0.45)	0.985(0.973,1.012)	0.756(0.741,0.768)	0.339(0.33,0.343)

Table 10. The median (first quantile – third quantile) of the Mean Square Prediction Error (MSPE) and Correlation in 10 iterations group by Train and Test and the number of hybrid principal component analysis (HPCA) for occipital region of the brain.

HPCA	Time (s)		
	User	System	Elapsed
1	73.99(73.703,74.662)	0.045(0.033,0.09)	74.19(73.82,74.735)
2	97.26(95.125,98.603)	0.05(0.03,0.075)	97.335(95.252,98.625)
3	121.24(120.955,122.175)	0.075(0.06,0.105)	121.36(121.055,122.27)
4	158.97(155.26,161.55)	0.085(0.065,0.107)	159.095(155.685,161.63)
5	153.09(149.975,155.527)	0.085(0.063,0.105)	153.215(150.053,155.633)
6	188.6(185.438,192.718)	0.1(0.083,0.11)	188.7(185.55,192.83)
7	189.01(184.708,190.642)	0.11(0.072,0.155)	189.12(185.083,190.73)
8	215.275(212.437,220.477)	0.125(0.093,0.16)	215.495(212.597,220.617)
9	206.62(205.715,207.52)	0.095(0.09,0.15)	206.81(205.823,207.633)
10	237.12(231.073,238.997)	0.1(0.075,0.12)	237.195(231.22,239.125)
11	218.29(215.97,219.05)	0.095(0.082,0.133)	218.4(216.185,219.18)
12	268.29(265.86,270.997)	0.13(0.08,0.182)	268.43(265.982,271.165)
13	248.595(245.99,252.612)	0.11(0.087,0.127)	248.725(246.07,252.783)
14	274.22(271.423,295.435)	0.125(0.103,0.22)	274.64(271.897,297.185)
15	464.355(400.745,538.865)	0.255(0.2,0.37)	469.43(402.81,540.52)
16	584.05(580.36,591.583)	0.285(0.245,0.36)	585.795(581.923,593.19)

Table 11. The median (first quantile – third quantile) of the Time (user-system-elapsed) in 10 iterations group by the number of hybrid principal component analysis (HPCA) for occipital region of the brain.

HPCA	Min_Error	Optimum					
		index	K	Tau	Lambda	Eta	Kappa
1	0.019(0.017,0.02)	5(2,5)	1(1,2)	0.1(0.1,0.1)	0.1(0.1,0.1)	0(0,0)	0(0,0)
2	0.969(0.852,1.128)	12(12,12)	2(2,2.75)	100(100,100)	0.4(0.4,0.4)	0.01(0.01,0.01)	0(0,0)
3	1.086(0.972,1.241)	12(6.75,12)	2(2,2)	100(100,100)	0.4(0.4,0.4)	0.01(0.003,0.01)	0(0,0)
4	4.553(4.108,5.357)	12(7.25,12)	2(1,2)	100(3.25,100)	0.4(0.225,0.4)	0.01(0.003,0.01)	0(0,0)
5	4.802(4.33,5.421)	8.5(4,12)	2(1,3.5)	100(32.5,100)	0.4(0.325,0.4)	0.005(0,0.01)	0(0,0)
6	4.872(4.723,5.013)	7.5(2,11.75)	1(1,1.75)	1(1,77.5)	0.2(0.2,0.375)	0.005(0,0.01)	0(0,0)
7	5.043(4.636,5.665)	10.5(2,11)	1(1,1)	5.5(1,10)	0.25(0.2,0.3)	0.01(0,0.01)	0(0,0)
8	6.641(6.351,6.978)	5(3.5,9.75)	1(1,1)	1(0.1,10)	0.2(0.1,0.3)	0(0,0.008)	0(0,0)
9	7.152(6.466,7.643)	1(1,11)	1(1,1)	0.1(0.1,10)	0.1(0.1,0.3)	0(0,0.01)	0(0,0)
10	9.18(8.982,9.368)	12(9,12)	1(1,1)	100(100,100)	0.4(0.4,0.4)	0.01(0.003,0.01)	0(0,0)
11	9.243(9.091,9.746)	4(4,11)	1(1,1)	100(100,100)	0.4(0.4,0.4)	0(0,0.008)	0(0,0)
12	9.249(8.845,9.442)	4(4,7)	1(1,1)	100(100,100)	0.4(0.4,0.4)	0(0,0)	0(0,0)
13	9.176(8.668,9.459)	12(5,12)	1(1,1)	100(100,100)	0.4(0.4,0.4)	0.01(0,0.01)	0(0,0)
14	9.803(9.531,10.297)	6(4,12)	1(1,1)	100(100,100)	0.4(0.4,0.4)	0(0,0.01)	0(0,0)
15	10.252(9.911,10.387)	12(4,12)	1(1,1)	100(100,100)	0.4(0.4,0.4)	0.01(0,0.01)	0(0,0)
16	10.175(9.962,10.293)	10(8,12)	1(1,1)	100(100,100)	0.4(0.4,0.4)	0.005(0,0.01)	0(0,0)

Table 12. The median (first quantile – third quantile) of the model fit indices in 10 iterations group by the number of hybrid principal component analysis (HPCA) for occipital region of the brain.

2.1.3.5 Parietal (4)

The MSPE and Correlation group by number of HPCA, Train and Test ([Table 13](#))

The User, System and Elapsed Time group by number of HPCA ([Table 14](#)): The median (first quantile – third quantile) of the Time (user-system-elapsed) in 10 iterations group by the number of hybrid principal component analysis (HPCA) for whole brain.

The Model fitting parameters group by number of HPCA ([Table 15](#)): The median (first quantile – third quantile) of the model fit indices in 10 iterations group by the number of hybrid principal component analysis (HPCA) for whole brain.

HCPA	MSPE		Correlation	
	Train	Test	Train	Test
1	0(0,0)	0.038(0.037,0.038)	1(1,1)	0.981(0.981,0.981)
2	0(0,0)	0.059(0.059,0.059)	1(1,1)	0.97(0.97,0.97)
3	0.06(0.016,0.06)	0.125(0.125,0.152)	0.969(0.969,0.992)	0.936(0.922,0.936)
4	0.027(0.001,0.027)	0.187(0.145,0.219)	0.986(0.986,1)	0.905(0.889,0.926)
5	0.082(0.077,0.085)	0.34(0.33,0.392)	0.958(0.956,0.961)	0.826(0.798,0.828)
6	0.091(0.079,0.097)	0.314(0.305,0.325)	0.954(0.95,0.959)	0.838(0.832,0.842)
7	0.125(0.118,0.135)	0.308(0.302,0.314)	0.935(0.93,0.939)	0.838(0.836,0.84)
8	0.123(0.112,0.139)	0.301(0.289,0.321)	0.936(0.928,0.942)	0.842(0.831,0.847)
9	0.164(0.148,0.191)	0.321(0.317,0.33)	0.914(0.899,0.923)	0.827(0.822,0.829)
10	0.159(0.143,0.188)	0.364(0.362,0.365)	0.917(0.901,0.925)	0.803(0.803,0.806)
11	0.207(0.186,0.224)	0.436(0.336,0.454)	0.891(0.881,0.902)	0.759(0.749,0.815)
12	0.232(0.201,0.24)	0.443(0.374,0.471)	0.876(0.871,0.894)	0.755(0.739,0.793)
13	0.237(0.161,0.267)	0.539(0.516,0.553)	0.873(0.856,0.916)	0.701(0.689,0.711)
14	0.213(0.161,0.242)	0.524(0.443,0.534)	0.887(0.87,0.916)	0.714(0.702,0.752)
15	0.26(0.188,0.292)	0.572(0.522,0.589)	0.86(0.841,0.901)	0.676(0.668,0.71)
16	0.255(0.239,0.262)	0.526(0.494,0.569)	0.863(0.859,0.872)	0.706(0.678,0.727)

Table 13. The median (first quantile – third quantile) of the Mean Square Prediction Error (MSPE) and Correlation in 10 iterations group by Train and Test and the number of hybrid principal component analysis (HPCA) for parietal region of the brain.

HPCA	Time (s)		
	User	System	Elapsed
1	121.48(119.882,124.642)	0.09(0.083,0.148)	121.975(120.285,125.055)
2	119.41(117.62,120.535)	0.12(0.078,0.17)	119.875(118.11,121.197)
3	274.445(272.063,277.975)	0.175(0.063,0.24)	275.035(272.765,278.803)
4	200.72(199.383,203.535)	0.125(0.102,0.178)	201.225(200.232,203.927)
5	357.05(354.743,359.603)	0.17(0.145,0.218)	357.88(355.452,360.51)
6	349.665(345.548,353.182)	0.185(0.173,0.208)	350.275(346.478,353.978)
7	432.955(426.86,439.942)	0.215(0.17,0.258)	433.79(427.92,440.813)
8	429.29(421.5,438.21)	0.23(0.197,0.265)	430.285(422.19,439.267)
9	533.27(517.603,534.82)	0.27(0.197,0.278)	534.58(518.293,536.305)
10	519.435(514.43,527.802)	0.215(0.175,0.337)	520.58(515.74,528.692)
11	563.67(561.878,573.67)	0.24(0.162,0.25)	564.365(562.575,574.41)
12	567.975(559.682,575.592)	0.18(0.142,0.24)	568.585(560.47,576.295)
13	572.335(567.233,574.612)	0.235(0.193,0.335)	573.105(568.095,575.31)
14	585.51(576.208,592.298)	0.195(0.142,0.248)	586.22(576.853,593.08)
15	635.805(633.02,640.603)	0.275(0.22,0.327)	636.56(633.828,641.463)
16	639.4(630.075,656.25)	0.215(0.193,0.273)	640.07(630.952,657.033)

Table 14. The median (first quantile – third quantile) of the Time (user-system-elapsed) in 10 iterations group by the number of hybrid principal component analysis (HPCA) for parietal region of the brain.

HPCA	Min_Error	Optimum					
		index	K	Tau	Lambda	Eta	Kappa
1	0.432(0.404,0.513)	5(5,5)	1.5(1,2)	0.1(0.1,0.1)	0.1(0.1,0.1)	0(0,0)	0(0,0)
2	0.461(0.384,0.481)	5(5,5.75)	1.5(1,2)	0.1(0.1,0.775)	0.1(0.1,0.175)	0(0,0)	0(0,0)
3	1.761(1.713,1.841)	12(6.75,12)	1(1,1.75)	100(25.075,100)	0.4(0.175,0.4)	0.01(0.003,0.01)	0(0,0)
4	1.62(1.537,1.793)	7(2,9)	1(1,2)	0.1(0.1,0.1)	0.1(0.1,0.1)	0.005(0,0.01)	0(0,0)
5	2.504(2.388,2.642)	11(1.75,11.75)	1(1,2)	10(2.575,100)	0.3(0.15,0.4)	0.01(0,0.01)	0(0,0)
6	2.332(2.203,2.486)	12(11,12)	1(1,1)	100(32.5,100)	0.4(0.325,0.4)	0.01(0.01,0.01)	0(0,0)
7	3.463(3.267,3.679)	12(11,12)	1(1,1)	100(32.5,100)	0.4(0.325,0.4)	0.01(0.01,0.01)	0(0,0)
8	3.498(3.418,3.748)	11(8,12)	1(1,2)	100(100,100)	0.4(0.4,0.4)	0.01(0,0.01)	0(0,0)
9	4.553(4.479,4.724)	4(3.25,11)	1(1,2)	100(32.5,100)	0.4(0.325,0.4)	0(0,0.008)	0(0,0)
10	4.552(4.3,4.663)	12(11.25,12)	1.5(1,2)	100(100,100)	0.4(0.4,0.4)	0.01(0.01,0.01)	0(0,0)
11	5.716(5.611,5.842)	10(4,12)	1(1,2)	100(100,100)	0.4(0.4,0.4)	0.005(0,0.01)	0(0,0)
12	5.755(5.55,5.916)	12(12,12)	1(1,1.75)	100(100,100)	0.4(0.4,0.4)	0.01(0.01,0.01)	0(0,0)
13	5.989(5.657,6.092)	8(5,11.75)	1(1,3)	100(100,100)	0.4(0.4,0.4)	0(0,0.01)	0(0,0)
14	6.189(5.866,6.424)	10.5(5,12)	1(1,3)	100(25.75,100)	0.4(0.25,0.4)	0.01(0,0.01)	0(0,0)
15	6.445(6.243,6.692)	8(4.25,10.25)	1(1,2.75)	55(10,100)	0.35(0.3,0.4)	0(0,0.008)	0(0,0)
16	6.471(6.278,6.68)	11(9,11.75)	1(1,1.75)	55(10,100)	0.35(0.3,0.4)	0.01(0.01,0.01)	0(0,0)

Table 15. The median (first quantile – third quantile) of the model fit indices in 10 iterations group by the number of hybrid principal component analysis (HPCA) for parietal region of the brain.

2.1.3.6 Left Temporal (5)

The MSPE and Correlation group by number of HPCA, Train and Test (Table 16)

The User, System and Elapsed Time group by number of HPCA (Table 17): The median (first quantile – third quantile) of the Time (user-system-elapsed) in 10 iterations group by the number of hybrid principal component analysis (HPCA) for whole brain.

The Model fitting parameters group by number of HPCA (Table 18): The median (first quantile – third quantile) of the model fit indices in 10 iterations group by the number of hybrid principal component analysis (HPCA) for whole brain.

HCPA	MSPE		Correlation	
	Train	Test	Train	Test
1	0(0,0)	0.015(0.015,0.015)	1(1,1)	0.992(0.992,0.993)
2	0.016(0.014,0.028)	0.479(0.369,0.533)	0.992(0.986,0.993)	0.759(0.74,0.812)
3	0.031(0.03,0.032)	0.449(0.391,0.503)	0.984(0.984,0.985)	0.772(0.744,0.798)
4	0.071(0.033,0.095)	0.469(0.421,0.52)	0.964(0.951,0.984)	0.746(0.712,0.767)
5	0.104(0.1,0.118)	0.51(0.465,0.516)	0.947(0.939,0.948)	0.709(0.708,0.756)
6	0.166(0.166,0.166)	0.813(0.811,0.813)	0.913(0.913,0.913)	0.566(0.565,0.566)
7	0.185(0.185,0.187)	0.818(0.799,0.818)	0.902(0.902,0.902)	0.561(0.561,0.562)
8	0.268(0.267,0.283)	0.947(0.941,0.973)	0.855(0.847,0.856)	0.433(0.428,0.438)
9	0.279(0.279,0.279)	0.933(0.931,0.946)	0.849(0.849,0.849)	0.44(0.435,0.442)
10	0.37(0.37,0.382)	0.998(0.998,1.009)	0.794(0.786,0.794)	0.342(0.336,0.347)
11	0.386(0.386,0.386)	1.016(1.015,1.025)	0.783(0.783,0.783)	0.329(0.329,0.329)
12	0.419(0.419,0.42)	1.075(1.068,1.077)	0.762(0.761,0.762)	0.271(0.264,0.282)
13	0.423(0.423,0.432)	1.076(1.067,1.079)	0.759(0.753,0.76)	0.271(0.26,0.277)
14	0.516(0.511,0.516)	1.24(1.218,1.241)	0.695(0.695,0.699)	0.19(0.19,0.192)
15	0.516(0.513,0.519)	1.242(1.22,1.243)	0.695(0.693,0.697)	0.188(0.187,0.189)
16	0.524(0.524,0.528)	1.278(1.269,1.28)	0.689(0.686,0.689)	0.151(0.151,0.152)

Table 16. The median (first quantile – third quantile) of the Mean Square Prediction Error (MSPE) and Correlation in 10 iterations group by Train and Test and the number of hybrid principal component analysis (HPCA) for left temporal region of the brain.

HPCA	Time (s)		
	User	System	Elapsed
1	125.38(124.665,126.722)	0.1(0.072,0.188)	125.885(124.92,127.045)
2	202.82(198.693,205.827)	0.15(0.06,0.198)	203.475(199.13,206.4)
3	294.33(287.265,298.83)	0.14(0.115,0.18)	294.92(287.712,299.675)
4	288.405(285.075,292.317)	0.17(0.125,0.203)	289.155(285.585,293.08)
5	349.74(341.647,354.168)	0.175(0.153,0.197)	350.7(342.77,355.352)
6	356.13(351.65,359.147)	0.22(0.143,0.283)	356.575(352.44,359.81)
7	415.24(411.383,418.195)	0.165(0.123,0.23)	415.75(411.952,419.182)
8	450.155(446.803,455.758)	0.235(0.163,0.265)	450.975(447.36,456.193)
9	503.89(494.803,515.692)	0.15(0.135,0.19)	504.495(495.368,516.228)
10	465.605(463.703,472.972)	0.2(0.172,0.243)	466.325(464.46,473.635)
11	503.055(500.398,506.45)	0.24(0.2,0.265)	504.255(501.382,507.693)
12	545.485(539.507,551.778)	0.24(0.21,0.25)	546.1(540.595,552.39)
13	556.815(547.28,562.213)	0.205(0.183,0.218)	557.445(547.935,562.987)
14	585.765(583.23,595.007)	0.215(0.18,0.272)	586.675(583.753,595.878)
15	595.375(588.203,597.305)	0.28(0.19,0.36)	596.115(588.835,598.375)
16	615.175(613.58,618.99)	0.23(0.213,0.25)	615.825(614.15,619.79)

Table 17. The median (first quantile – third quantile) of the Time (user-system-elapsed) in 10 iterations group by the number of hybrid principal component analysis (HPCA) for left temporal region of the brain.

HPCA	Min_Error	Optimum					
		index	K	Tau	Lambda	Eta	Kappa
1	0.18(0.16,0.184)	5(5,5)	1(1,1.75)	0.1(0.1,0.1)	0.1(0.1,0.1)	0(0,0)	0(0,0)
2	2.168(1.813,2.381)	11.5(10.25,12)	2(1,2)	55(3.25,100)	0.35(0.225,0.4)	0.01(0.01,0.01)	0(0,0)
3	2.287(2.083,2.616)	10.5(9,11.75)	1(1,2.75)	5.5(0.325,77.5)	0.25(0.125,0.375)	0.01(0.01,0.01)	0(0,0)
4	6.626(6.03,7.552)	11(11,11.75)	3(1,4)	10(10,77.5)	0.3(0.3,0.375)	0.01(0.01,0.01)	0(0,0)
5	6.311(5.913,8.319)	11.5(11,12)	2(1,3)	55(10,100)	0.35(0.3,0.4)	0.01(0.01,0.01)	0(0,0)
6	8.178(7.658,9.422)	6(6,6)	1(1,1)	1(1,1)	0.2(0.2,0.2)	0(0,0)	0(0,0)
7	8.721(8.465,9.906)	6(6,10.75)	1(1,1)	1(1,7.75)	0.2(0.2,0.275)	0(0,0.01)	0(0,0)
8	9.141(8.698,9.3)	10(5,11)	1(1,1)	1(0.325,10)	0.2(0.125,0.3)	0.01(0,0.01)	0(0,0)
9	9.7(9.188,10.663)	6(5,8.25)	1(1,1)	0.55(0.1,1)	0.15(0.1,0.2)	0(0,0.008)	0(0,0)
10	10.871(10.187,12.159)	7(5,10.75)	1(1,1)	0.1(0.1,7.75)	0.1(0.1,0.275)	0.005(0,0.01)	0(0,0)
11	10.927(10.528,11.208)	5(5,5.75)	1(1,1)	0.1(0.1,0.1)	0.1(0.1,0.1)	0(0,0)	0(0,0)
12	11.039(10.795,11.289)	5.5(2,9.75)	1(1,1)	0.1(0.1,1)	0.1(0.1,0.2)	0(0,0.01)	0(0,0)
13	11.271(10.791,11.696)	6(2.75,10.75)	1(1,1)	1(0.325,7.75)	0.2(0.125,0.275)	0(0,0.01)	0(0,0)
14	12.273(12.061,12.671)	5.5(3,7)	1(1,1)	10(10,10)	0.3(0.3,0.3)	0(0,0)	0(0,0)
15	11.613(11.409,12.138)	4(3,7)	1(1,1)	10(10,10)	0.3(0.3,0.3)	0(0,0)	0(0,0)
16	11.75(11.227,11.929)	3.5(3,6.75)	1(1,1)	10(10,10)	0.3(0.3,0.3)	0(0,0)	0(0,0)

Table 18. The median (first quantile – third quantile) of the model fit indices in 10 iterations group by the number of hybrid principal component analysis (HPCA) for left temporal region of the brain.

2.1.3.7 Right Temporal (6)

The MSPE and Correlation group by number of HPCA, Train and Test ([Table 19](#))

The User, System and Elapsed Time group by number of HPCA ([Table 20](#)): The median (first quantile – third quantile) of the Time (user-system-elapsed) in 10 iterations group by the number of hybrid principal component analysis (HPCA) for whole brain.

The Model fitting parameters group by number of HPCA ([Table 21](#)): The median (first quantile – third quantile) of the model fit indices in 10 iterations group by the number of hybrid principal component analysis (HPCA) for whole brain.

HCPA	MSPE		Correlation	
	Train	Test	Train	Test
1	0(0,0)	0.063(0.063,0.063)	1(1,1)	0.968(0.968,0.968)
2	0.007(0.004,0.013)	0.227(0.212,0.234)	0.997(0.994,0.998)	0.882(0.879,0.89)
3	0.077(0.077,0.088)	0.329(0.327,0.37)	0.961(0.955,0.961)	0.824(0.804,0.825)
4	0.077(0.053,0.077)	0.282(0.257,0.404)	0.96(0.96,0.973)	0.848(0.774,0.862)
5	0.015(0.015,0.015)	0.447(0.447,0.537)	0.993(0.993,0.993)	0.752(0.719,0.756)
6	0.053(0.024,0.149)	0.467(0.444,0.529)	0.973(0.922,0.988)	0.741(0.703,0.759)
7	0.22(0.22,0.22)	0.459(0.459,0.488)	0.883(0.883,0.883)	0.738(0.728,0.738)
8	0.276(0.276,0.296)	0.479(0.472,0.506)	0.85(0.839,0.85)	0.727(0.712,0.727)
9	0.324(0.324,0.343)	0.556(0.506,0.601)	0.822(0.81,0.822)	0.668(0.644,0.703)
10	0.359(0.359,0.359)	0.745(0.642,0.745)	0.801(0.801,0.801)	0.528(0.528,0.605)
11	0.382(0.381,0.466)	0.727(0.637,0.876)	0.786(0.73,0.787)	0.545(0.436,0.604)
12	0.409(0.409,0.486)	0.776(0.722,0.976)	0.768(0.716,0.768)	0.492(0.367,0.537)
13	0.427(0.427,0.502)	0.777(0.738,0.967)	0.757(0.705,0.757)	0.49(0.359,0.524)
14	0.433(0.433,0.433)	0.841(0.841,0.87)	0.753(0.753,0.753)	0.453(0.436,0.453)
15	0.459(0.459,0.459)	0.897(0.883,0.9)	0.735(0.735,0.735)	0.428(0.427,0.428)
16	0.454(0.454,0.454)	0.854(0.821,0.855)	0.739(0.739,0.739)	0.455(0.452,0.468)

Table 19. The median (first quantile – third quantile) of the Mean Square Prediction Error (MSPE) and Correlation in 10 iterations group by Train and Test and the number of hybrid principal component analysis (HPCA) for right temporal region of the brain.

HPCA	Time (s)		
	User	System	Elapsed
1	122.58(120.777,123.17)	0.08(0.065,0.132)	122.805(120.947,123.365)
2	187.405(183.807,189.842)	0.08(0.06,0.095)	187.6(184.075,190.085)
3	316(308.48,321.705)	0.135(0.11,0.19)	316.305(308.993,322.132)
4	292.425(290.21,293.37)	0.13(0.095,0.223)	292.75(290.587,293.828)
5	360.51(354.398,364.865)	0.16(0.12,0.258)	360.86(355.035,365.158)
6	366.595(364.973,368.71)	0.14(0.102,0.173)	367.015(365.413,369.12)
7	475.41(468.057,480.182)	0.19(0.16,0.225)	476.14(468.795,480.968)
8	470.695(464.493,474.79)	0.2(0.167,0.24)	471.39(465.642,475.475)
9	545.57(535.125,560.993)	0.23(0.183,0.265)	546.285(535.743,561.835)
10	559.395(548.497,573.71)	0.19(0.155,0.25)	560.25(549.15,574.675)
11	613.955(605.148,619.005)	0.275(0.195,0.37)	615.19(606.01,620.918)
12	610.25(601.325,616.26)	0.235(0.188,0.267)	611.105(602.45,617.197)
13	613.53(609.288,624.188)	0.25(0.197,0.31)	614.355(610.585,624.97)
14	617.85(608.253,621.952)	0.205(0.193,0.277)	618.555(609.03,623.845)
15	685.05(669.635,694.07)	0.385(0.313,0.45)	687.175(672.178,696.335)
16	664.235(653.742,672.14)	0.265(0.215,0.365)	665.13(654.727,673.798)

Table 20. The median (first quantile – third quantile) of the Time (user-system-elapsed) in 10 iterations group by the number of hybrid principal component analysis (HPCA) for right temporal region of the brain.

HPCA	Min_Error	Optimum					
		index	K	Tau	Lambda	Eta	Kappa
1	0.684(0.612,0.765)	5(5,5)	2(2,2)	0.1(0.1,0.1)	0.1(0.1,0.1)	0(0,0)	0(0,0)
2	1.27(1.194,1.478)	3(3,4)	1(1,2)	10(10,100)	0.3(0.3,0.4)	0(0,0)	0(0,0)
3	2.957(2.904,3.107)	4.5(1.25,10.25)	1(1,1)	5.5(0.1,77.5)	0.25(0.1,0.375)	0(0,0.008)	0(0,0)
4	5.737(5.476,6.624)	1(1,9.5)	1.5(1,2)	0.1(0.1,75.25)	0.1(0.1,0.35)	0(0,0.008)	0(0,0)
5	6.175(5.963,6.379)	1(1,7.25)	3(3,3)	0.1(0.1,75.025)	0.1(0.1,0.325)	0(0,0)	0(0,0)
6	6.972(6.598,7.128)	8(1.75,8.75)	4(1.5,5)	100(0.1,100)	0.4(0.1,0.4)	0(0,0.008)	0(0,0)
7	8.482(8.218,8.681)	1(1,8.5)	1(1,1)	0.1(0.1,7.525)	0.1(0.1,0.25)	0(0,0.008)	0(0,0)
8	9.151(8.727,9.466)	2(1,8.75)	1(1,1)	0.1(0.1,10)	0.1(0.1,0.3)	0(0,0.008)	0(0,0)
9	8.974(8.748,9.273)	9(3,11)	1(1,1)	0.1(0.1,10)	0.1(0.1,0.3)	0.01(0.003,0.01)	0(0,0)
10	9.932(9.286,10.624)	9(2,9)	1(1,1)	0.1(0.1,0.1)	0.1(0.1,0.1)	0.01(0,0.01)	0(0,0)
11	11.162(10.265,11.788)	8(1.25,10)	1(1,1)	1(0.325,100)	0.2(0.125,0.4)	0(0,0.01)	0(0,0)
12	10.628(10.249,11.143)	8.5(1.75,9)	1(1,1)	0.1(0.1,77.5)	0.1(0.1,0.375)	0.005(0,0.01)	0(0,0)
13	10.873(10.734,11.14)	6.5(1.75,8.75)	1(1,1)	0.1(0.1,75.025)	0.1(0.1,0.325)	0(0,0.008)	0(0,0)
14	11.166(10.649,11.326)	1(1,8)	1(1,1)	0.1(0.1,0.1)	0.1(0.1,0.1)	0(0,0.008)	0(0,0)
15	11.089(10.548,11.274)	9(6,9.75)	1(1,1)	1(0.1,1)	0.2(0.1,0.2)	0.01(0,0.01)	0(0,0)
16	11.58(11.08,11.949)	4.5(1.25,8.25)	1(1,1)	0.1(0.1,1)	0.1(0.1,0.2)	0(0,0.008)	0(0,0)

Table 21. The median (first quantile – third quantile) of the model fit indices in 10 iterations group by the number of hybrid principal component analysis (HPCA) for right temporal region of the brain.

2.1.3.8 Figures

The MSPE over Number of HPCA group by Regions: **Figure 14**

The Correlation over Number of HPCA group by Regions: **Figure 15**

The Elapsed Time over Number of HPCA group by Regions: **Figure 16**

The MSPE over Number of HPCA group by Regions

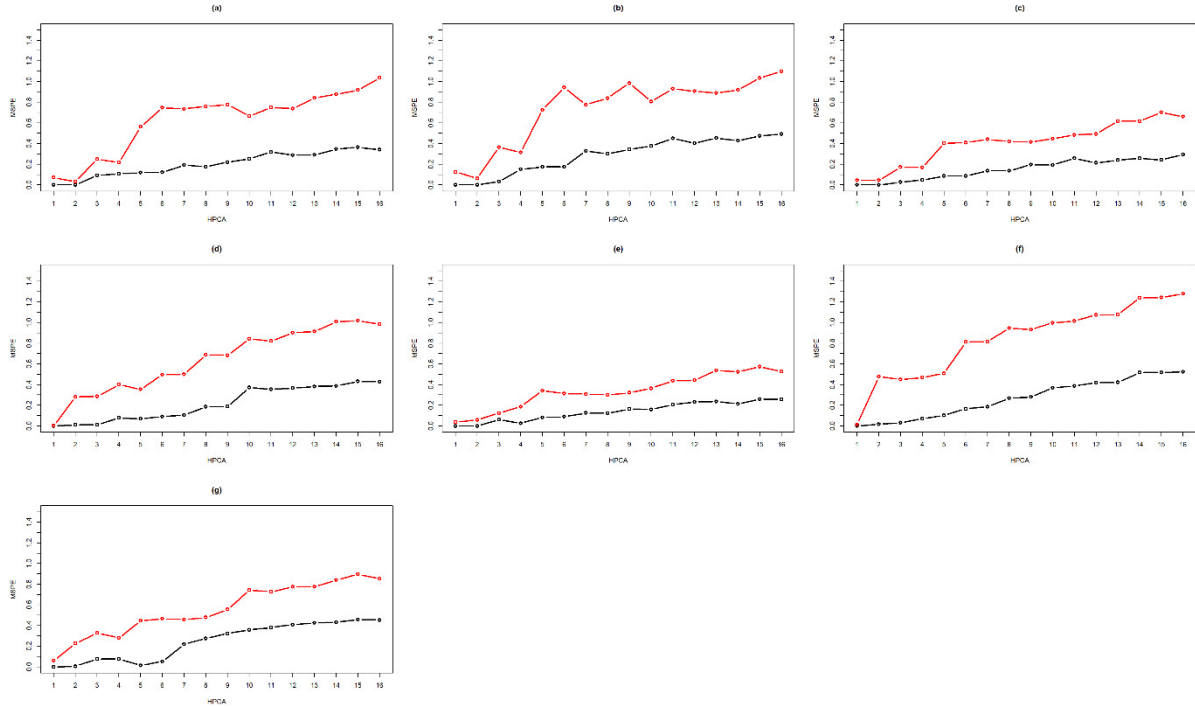


Figure 14. (a-g) Median of mean square prediction error (MSPE) for train (black solid line) and test (red solid line) in 10 iterations of (a) whole, (b) frontal, (c) central, (d) occipital, (e) parietal, (f) left temporal and (g) right temporal regions, respectively.

The Correlation over Number of HPCA group by Regions

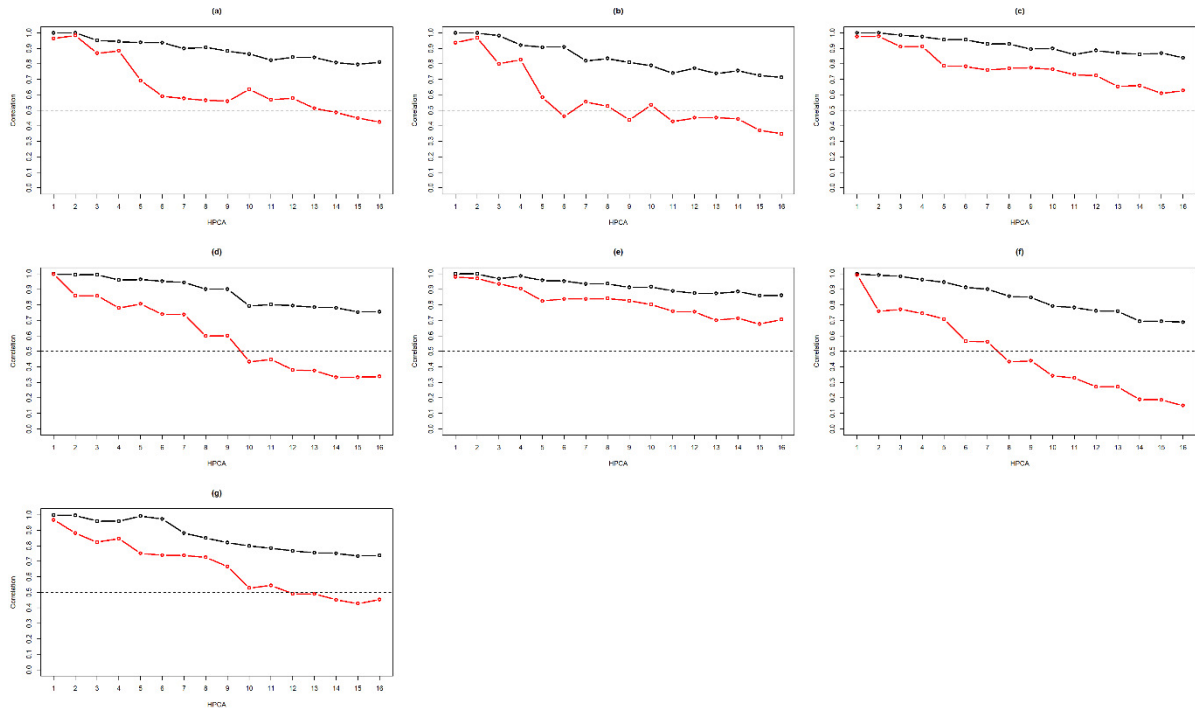


Figure 15. (a-g) Median of correlation for train (black solid line) and test (red solid line) in 10 iterations of (a) whole, (b) frontal, (c) central, (d) occipital, (e) parietal, (f) left temporal and (g) right temporal regions, respectively

The Elapsed Time over Number of HPCA group by Regions

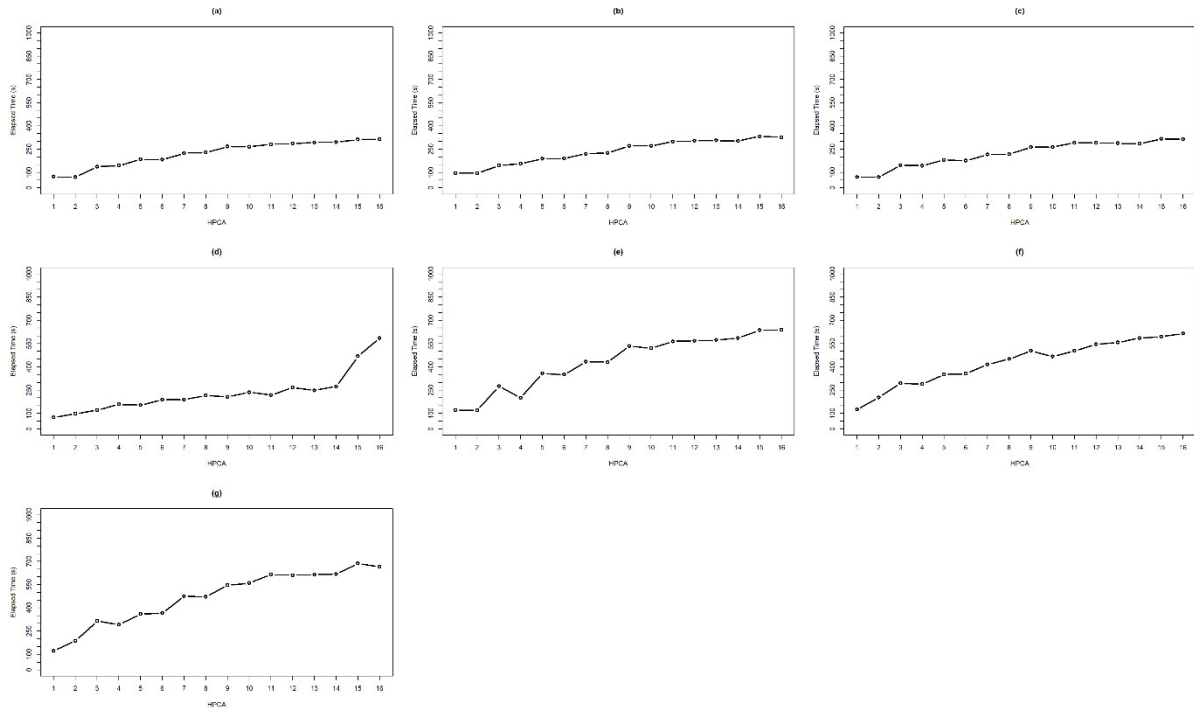


Figure 16. (a-g) Median of elapsed time in 10 iterations of (a) whole, (b) frontal, (c) central, (d) occipital, (e) parietal, (f) left temporal and (g) right temporal regions, respectively

2.1.4. Conclusion .

According to the [Figure 14](#), the MSPE increased by adding HPCA from 1 to KLM in all regions. It also indicates that the difference between training and testing exists. The [Figure 15](#) shows the correlation. The highest correlation is in the first HPCA in all regions fro both training and testing datasets. The elapsed time is also increasing by adding HPCA. ([Figure 16](#))

We conclude that the first HPCA has the best performance based on the MSPE, correlation and elapsed times in all region of the brain.

2.2. Target Stimuli

2.2.1. The HPCA Decomposition

2.2.1.1 The Functional dimension

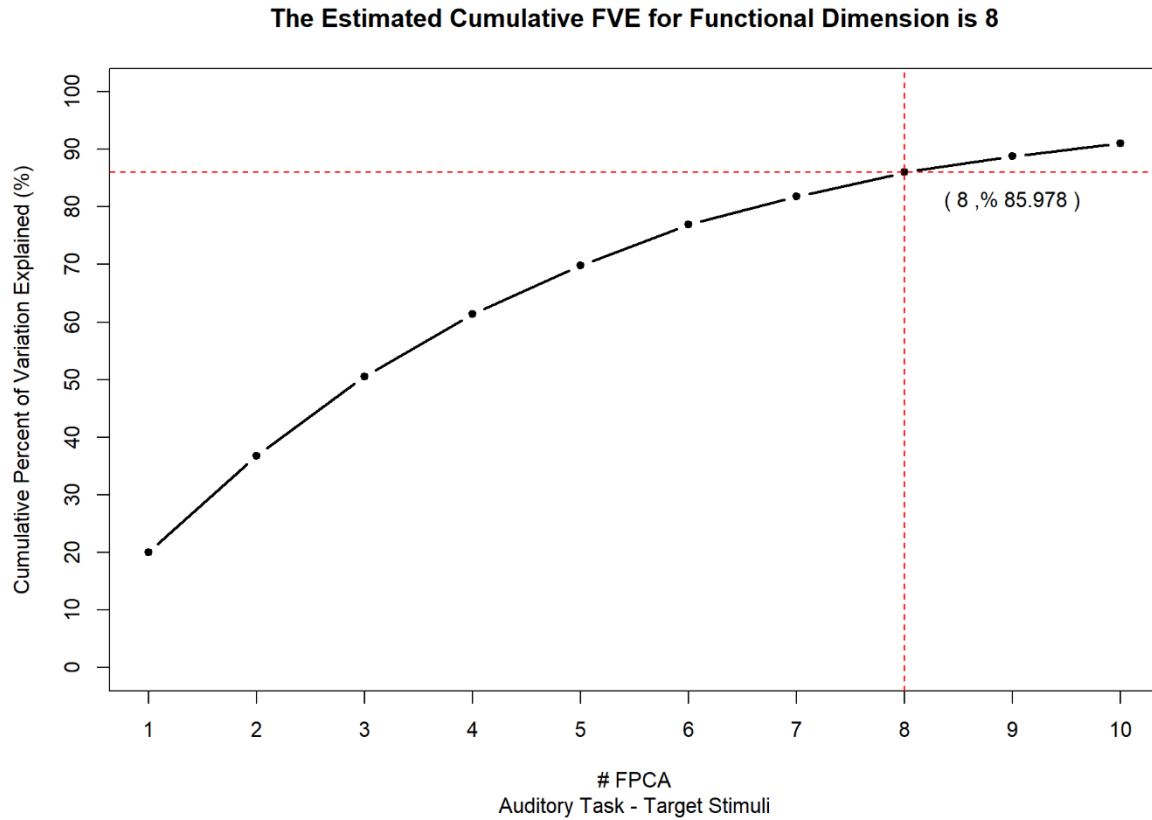


Figure 17. The estimated cumulative fraction of variations (FVE) for functional dimension.

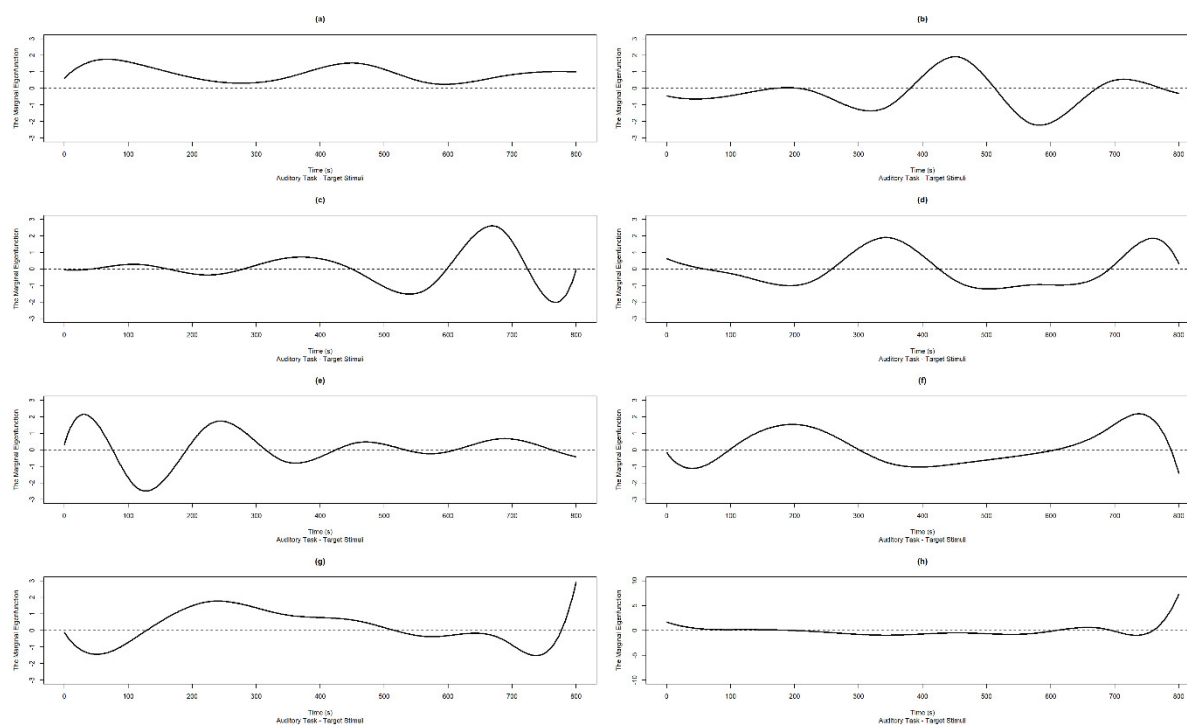


Figure 18. (a, h) Estimated first to eight leading functional marginal eigenfunctions.

According to the [Figure 17](#), the first eight marginal functional principal component analysis (FPCA) capture at least 85% of variation in this dimension. The [Figure 18](#) plots estimated first to eight leading functional marginal eigenfunctions on the x-axis (time).

2.2.1.2 The Longitudinal dimension

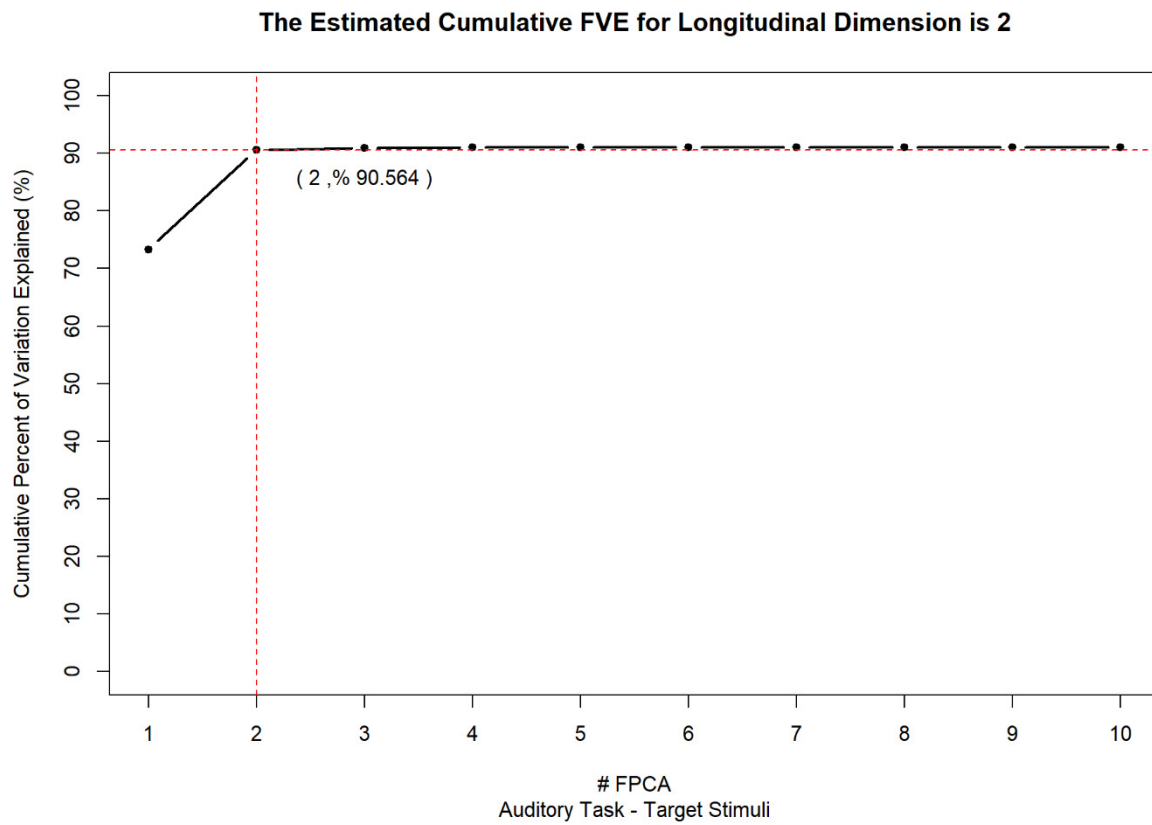


Figure 19. The estimated cumulative fraction of variations (FVE) for longitudinal dimension.

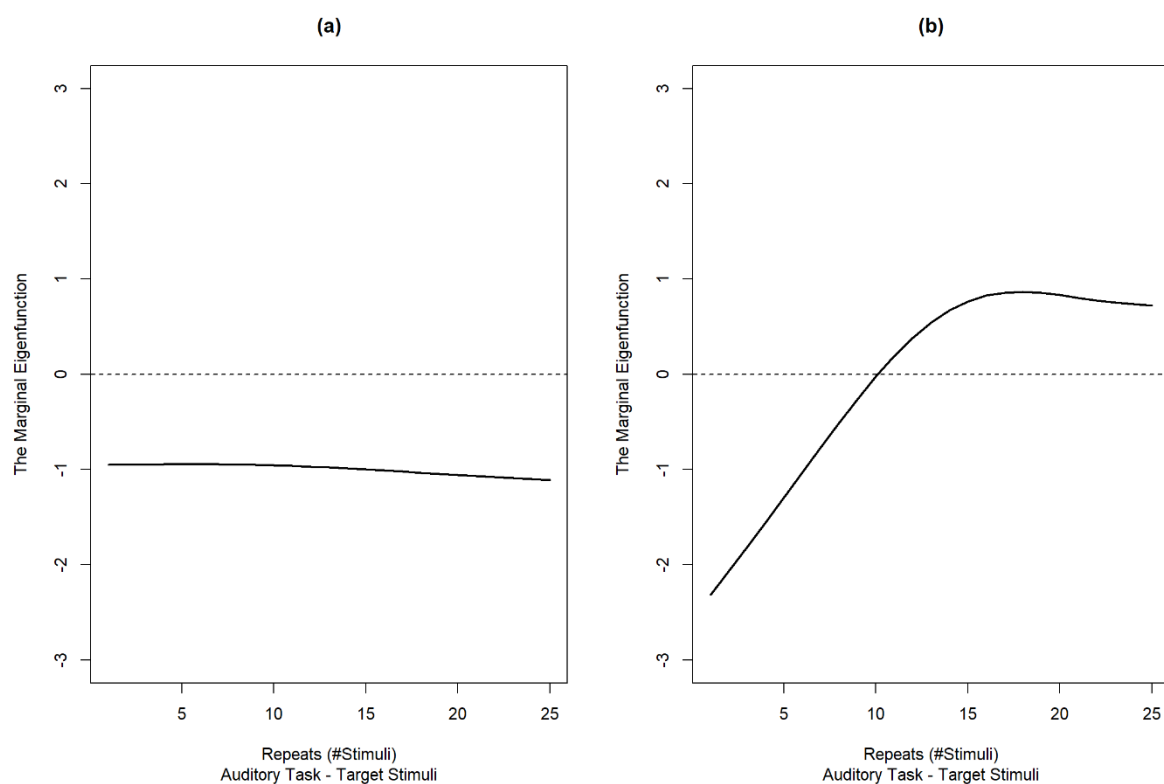


Figure 20. Estimated (a) first and (b) second leading longitudinal marginal eigenfunctions.

According to the [Figure 19](#), the first and second marginal functional principal component analysis (FPCA) capture at least 90% of variation in this dimension. The [Figure 20](#) plots estimated first leading longitudinal marginal eigenfunctions on the receptions.

2.2.1.3 The Regional dimension

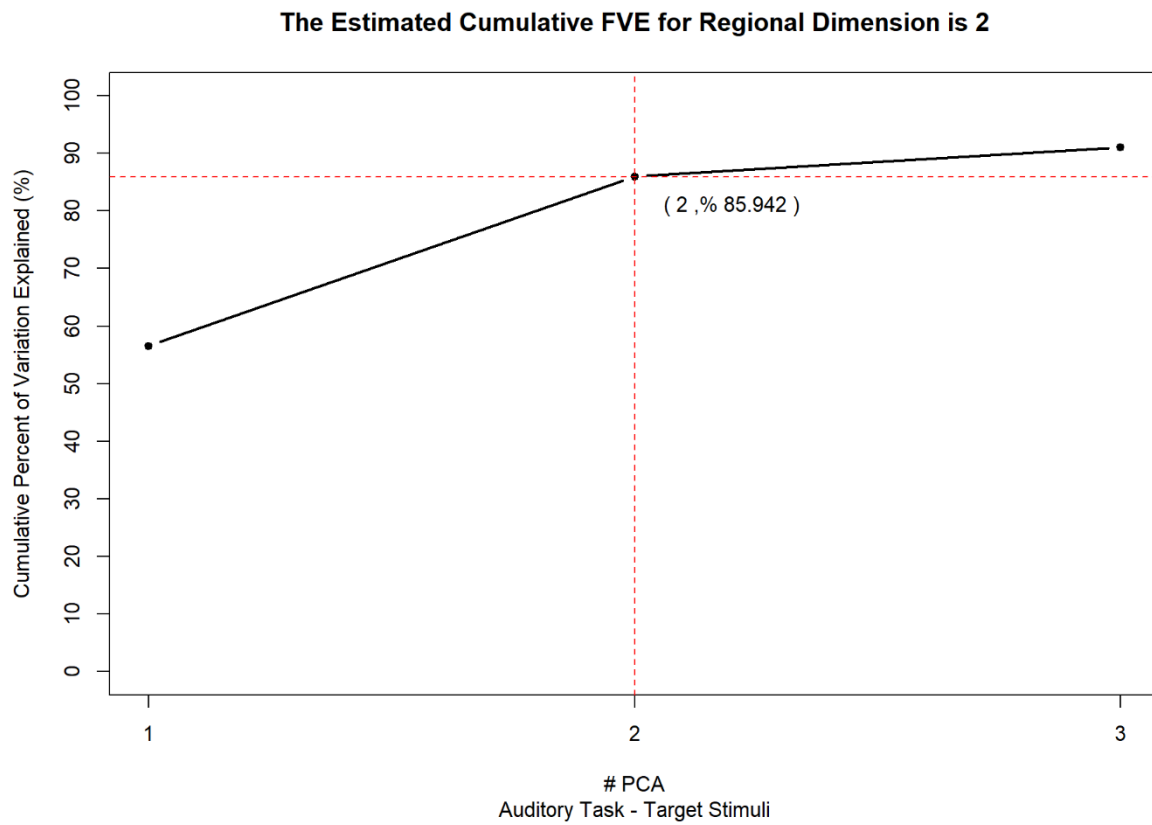


Figure 21. The estimated cumulative fraction of variations (FVE) for regional dimension.

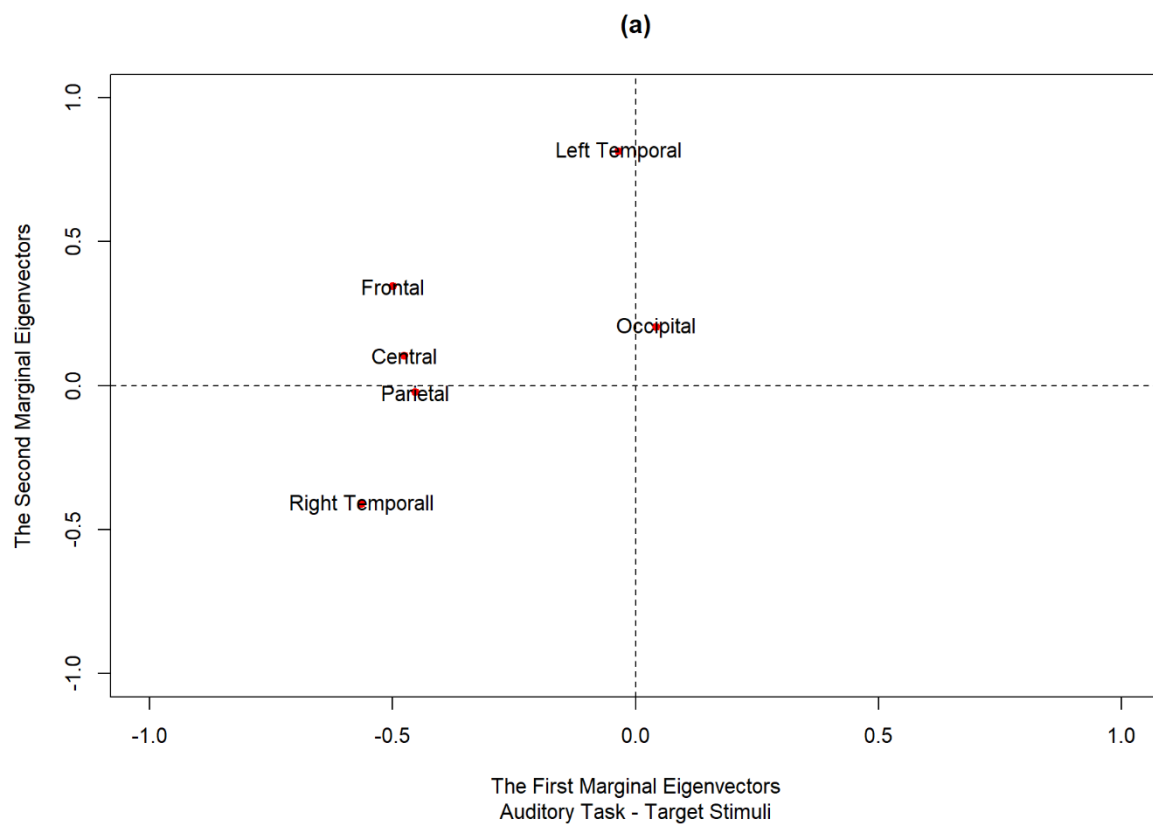


Figure 22. Estimated first and second leading regional marginal eigenvectors.

According to the [Figure 21](#), the first and second marginal principal component analysis (PCA) capture at least nearly 85% of variation in this dimension. The [Figure 22](#) plots estimated first and second leading regional marginal eigenvectors.

2.2.2. The Reconstructed of the Response .

The fof regression were conducted group by the number of HPCA and the regions. In the following part, we show the reconstructed ERPs for each subject (in gray) based on the number of HPCA. In this trial, we have 32 HPCA. We also show the point-wise mean and mean $\pm$ standard deviation. (Whole Brain: **Figure 23**, Frontal region (1): **Figure 24**, Central Region (2): **Figure 25**, Occipital Region (3): **Figure 26**, Parietal Region (4): **Figure 27**, Left Temporal (5): **Figure 28**, Right Temporal (6): **Figure 29**).

2.2.2.1 Whole Brain

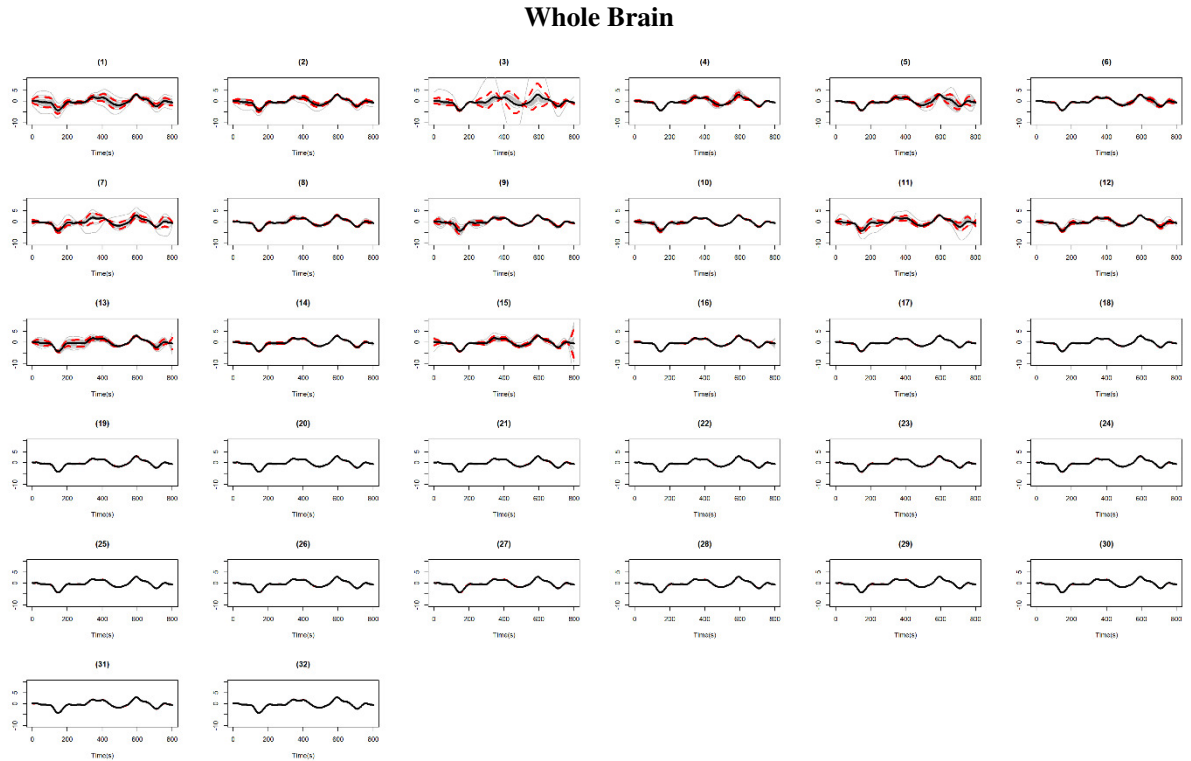


Figure 23. (1-32) Reconstructed ERPs for each subject (Gray line), point-wise mean (solid black line), and mean $\pm$ standard deviation (dotted red line) for whole region of the Brain

2.2.2.2 Frontal Region (1)

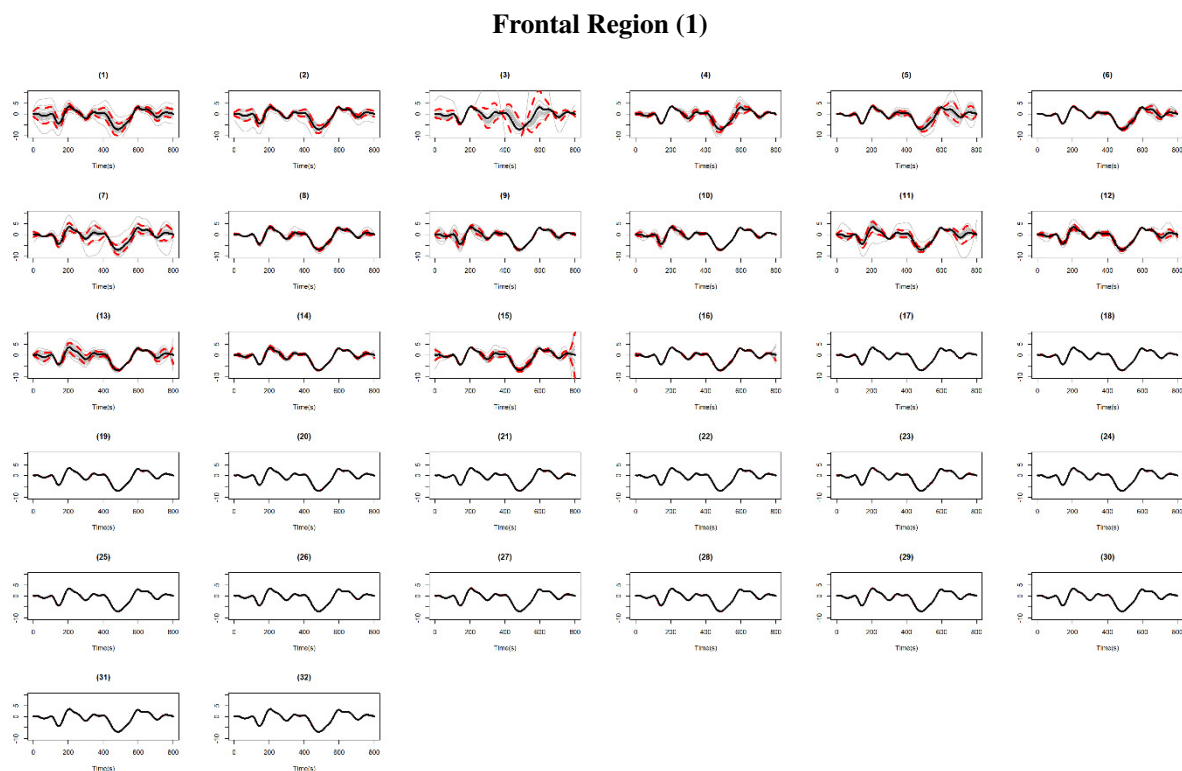


Figure 24. (1-32) Reconstructed ERPs for each subject (Gray line), point-wise mean (solid black line), and mean $\pm$ standard deviation (dotted red line) for frontal region of the Brain.

2.2.2.3 Central Region (2)

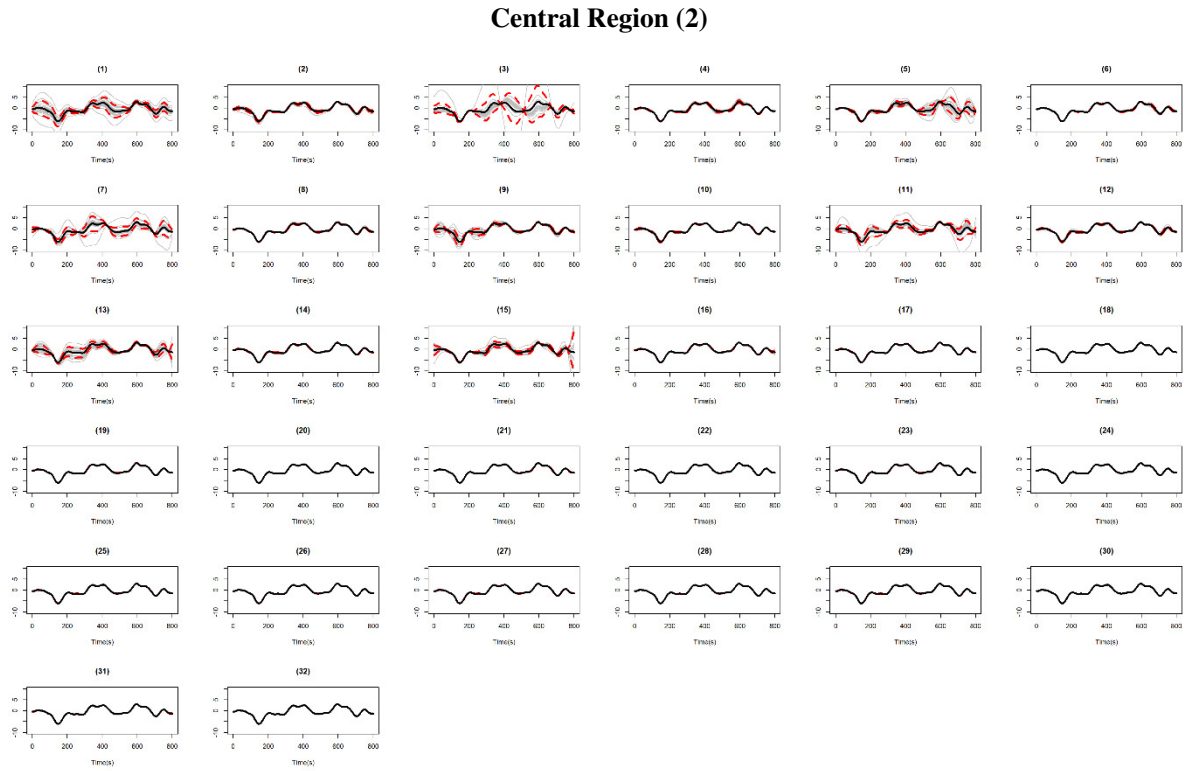


Figure 25. (1-32) Reconstructed ERPs for each subject (Gray line), point-wise mean (solid black line), and mean $\pm$ standard deviation (dotted red line) for central region of the Brain.

2.2.2.4 Occipital Region (3)

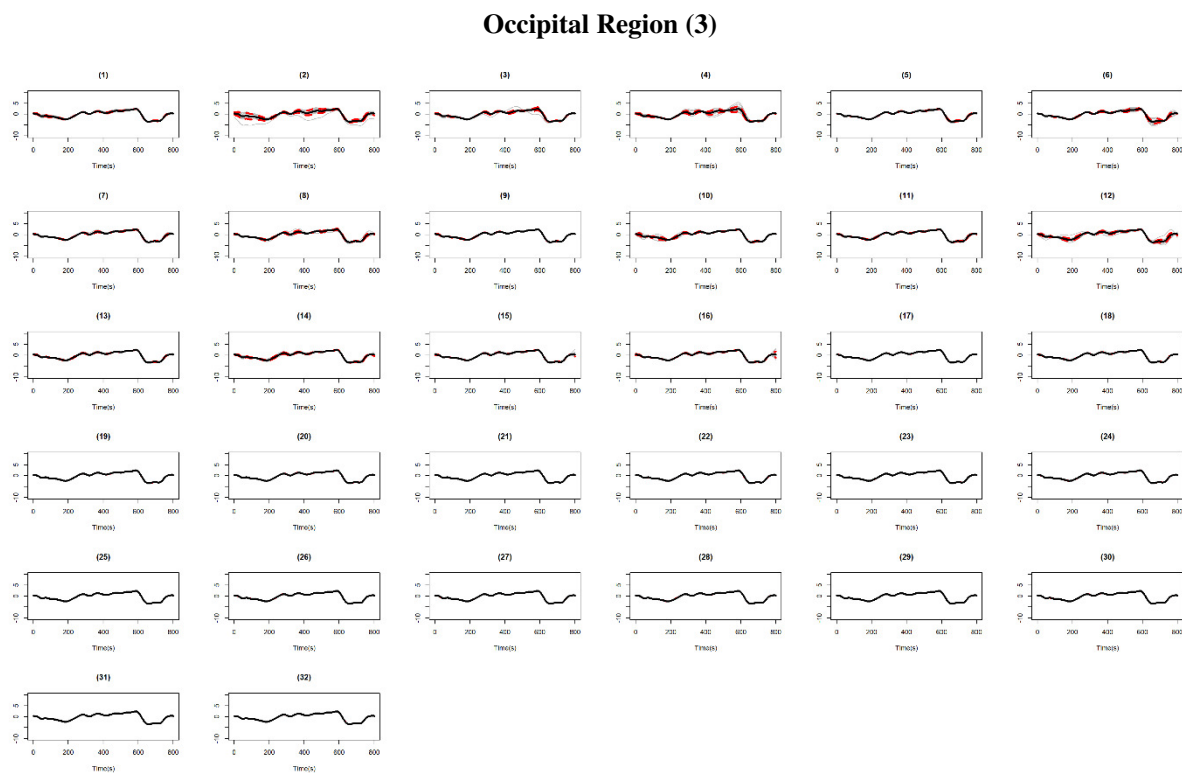


Figure 26. (1-32) Reconstructed ERPs for each subject (Gray line), point-wise mean (solid black line), and mean $\pm$ standard deviation (dotted red line) for occipital region of the Brain.

2.2.2.5 Parietal Region (4)

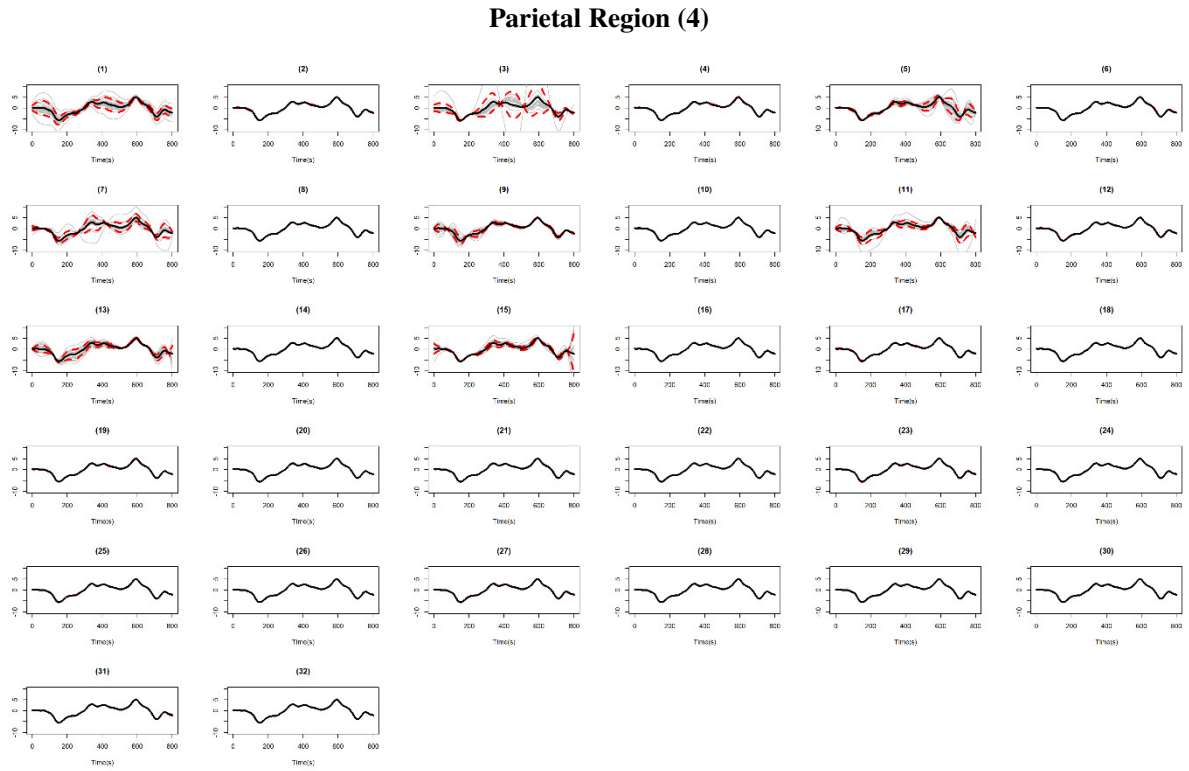


Figure 27. (1-32) Reconstructed ERPs for each subject (Gray line), point-wise mean (solid black line), and mean $\pm$ standard deviation (dotted red line) for parietal region of the Brain.

2.2.2.6 Left Temporal (5)

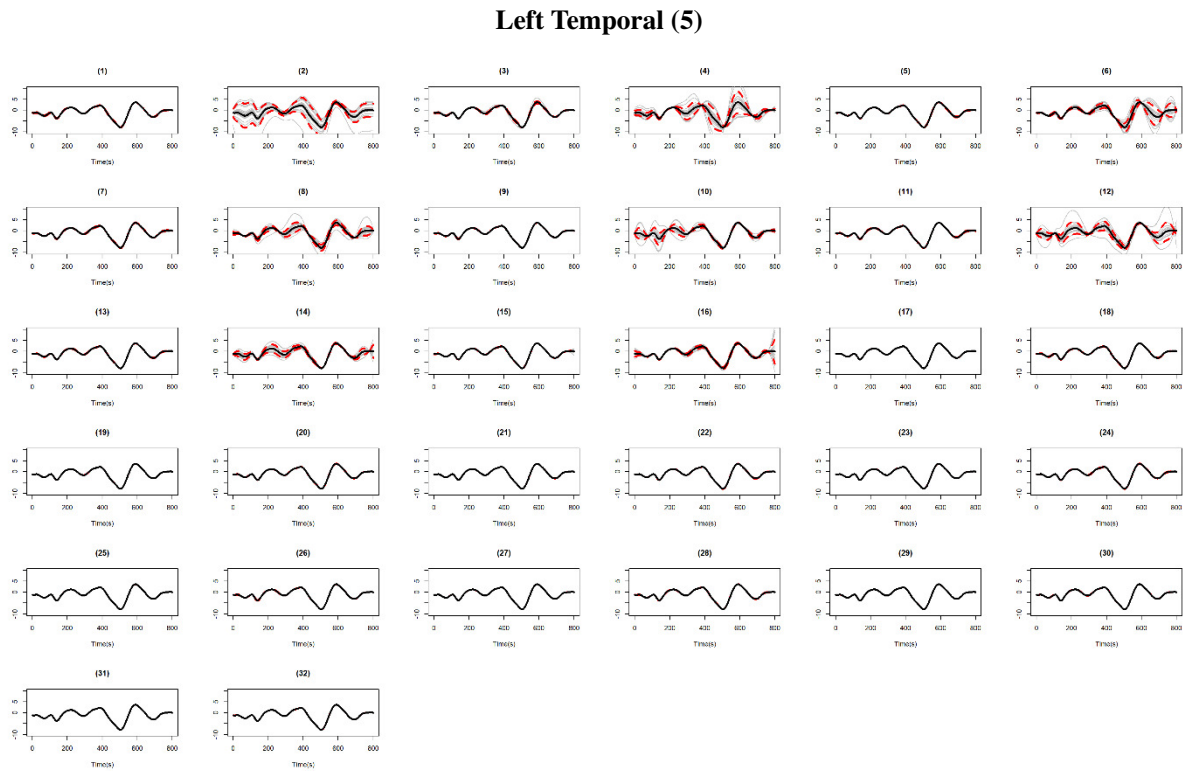


Figure 28. (1-32) Reconstructed ERPs for each subject (Gray line), point-wise mean (solid black line), and mean $\pm$ standard deviation (dotted red line) for left temporal region of the Brain.

2.2.2.7 Right Temporal (6)

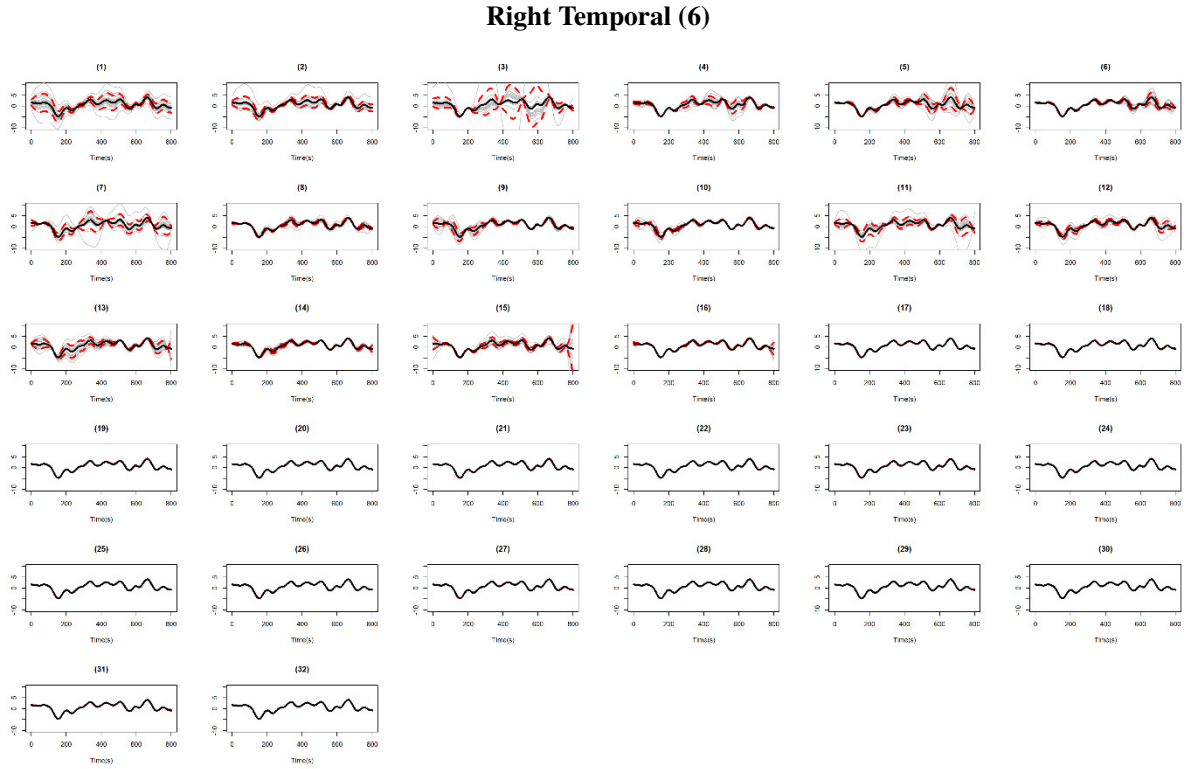


Figure 29. (1-32) Reconstructed ERPs for each subject (Gray line), point-wise mean (solid black line), and mean $\pm$ standard deviation (dotted red line) for right temporal region of the Brain.

2.2.3. *The function-on-function result* .

The fof regression result group by the region of the brain and the number of hybrid principal component analysis (HPCA) were reported.

2.2.3.1 Whole Brain

The MSPE and Correlation group by number of HPCA, Train and Test ([Table 22](#))

The User, System and Elapsed Time group by number of HPCA ([Table 23](#)): The median (first quantile – third quantile) of the Time (user-system-elapsed) in 10 iterations group by the number of hybrid principal component analysis (HPCA) for whole brain.

The Model fitting parameters group by number of HPCA ([Table 24](#)): The median (first quantile – third quantile) of the model fit indices in 10 iterations group by the number of hybrid principal component analysis (HPCA) for whole brain.

HPCA	MSPE		Correlation	
	Train	Test	Train	Test
1	0.013(0.004,0.022)	0.218(0.171,0.244)	0.993(0.989,0.998)	0.888(0.874,0.912)
2	0.003(0.001,0.004)	0.052(0.05,0.056)	0.998(0.998,1)	0.974(0.972,0.975)
3	0.051(0.051,0.111)	0.449(0.417,0.554)	0.974(0.943,0.974)	0.746(0.685,0.768)
4	0.056(0.056,0.056)	0.34(0.34,0.342)	0.972(0.971,0.972)	0.813(0.811,0.813)
5	0.178(0.176,0.178)	0.699(0.699,0.759)	0.907(0.907,0.908)	0.57(0.536,0.57)
6	0.194(0.194,0.261)	0.703(0.671,0.877)	0.898(0.859,0.898)	0.566(0.461,0.591)
7	0.138(0.128,0.138)	0.64(0.64,0.917)	0.929(0.929,0.934)	0.613(0.464,0.613)
8	0.145(0.145,0.161)	0.7(0.693,0.861)	0.925(0.916,0.925)	0.573(0.493,0.586)
9	0.215(0.205,0.222)	0.701(0.647,0.72)	0.886(0.882,0.891)	0.573(0.557,0.611)
10	0.227(0.207,0.29)	0.736(0.736,0.744)	0.879(0.843,0.89)	0.546(0.536,0.546)
11	0.313(0.249,0.43)	0.799(0.762,0.868)	0.843(0.754,0.867)	0.487(0.389,0.52)
12	0.288(0.271,0.374)	0.821(0.737,0.928)	0.843(0.818,0.854)	0.474(0.339,0.535)
13	0.345(0.293,0.482)	0.791(0.753,0.843)	0.826(0.719,0.846)	0.492(0.443,0.506)
14	0.387(0.329,0.474)	0.858(0.754,1.034)	0.8(0.729,0.823)	0.457(0.21,0.504)
15	0.432(0.33,0.508)	0.828(0.78,0.899)	0.761(0.701,0.826)	0.461(0.384,0.475)
16	0.508(0.387,0.569)	0.788(0.753,0.967)	0.701(0.655,0.795)	0.467(0.285,0.507)
17	0.508(0.435,0.587)	0.781(0.747,0.978)	0.701(0.642,0.765)	0.474(0.286,0.506)
18	0.464(0.344,0.567)	0.863(0.789,0.911)	0.739(0.657,0.813)	0.435(0.327,0.471)
19	0.482(0.357,0.567)	0.781(0.747,0.904)	0.726(0.657,0.801)	0.473(0.415,0.512)
20	0.406(0.35,0.568)	0.89(0.767,0.973)	0.778(0.657,0.822)	0.349(0.274,0.506)
21	0.365(0.327,0.506)	0.879(0.792,0.94)	0.804(0.707,0.822)	0.426(0.291,0.479)
22	0.507(0.482,0.588)	0.771(0.737,0.844)	0.701(0.641,0.729)	0.482(0.434,0.514)
23	0.336(0.328,0.508)	0.861(0.781,0.93)	0.817(0.701,0.829)	0.453(0.318,0.472)
24	0.33(0.313,0.507)	0.92(0.848,0.983)	0.828(0.706,0.836)	0.346(0.218,0.441)
25	0.375(0.333,0.487)	0.927(0.8,1.042)	0.806(0.719,0.833)	0.306(0.175,0.469)
26	0.428(0.32,0.519)	0.925(0.868,1.05)	0.77(0.703,0.829)	0.335(0.239,0.434)
27	0.519(0.507,0.572)	0.784(0.777,0.79)	0.693(0.654,0.702)	0.469(0.464,0.476)
28	0.507(0.452,0.521)	0.864(0.778,0.93)	0.701(0.699,0.751)	0.379(0.316,0.474)
29	0.366(0.329,0.519)	0.984(0.865,0.991)	0.803(0.702,0.836)	0.321(0.233,0.427)
30	0.514(0.408,0.572)	0.789(0.746,0.846)	0.697(0.653,0.78)	0.465(0.432,0.505)
31	0.507(0.41,0.527)	0.841(0.77,0.976)	0.701(0.699,0.778)	0.409(0.271,0.483)
32	0.479(0.41,0.59)	0.775(0.742,0.849)	0.721(0.64,0.778)	0.491(0.433,0.518)

Table 22. The median (first quantile – third quantile) of the Mean Square Prediction Error (MSPE) and Correlation in 10 iterations group by Train and Test and the number of hybrid principal component analysis (HPCA) for whole brain.

HPCA	Time		
	User	System	Elapsed
1	192.58(190.075,195.48)	0.195(0.19,0.26)	193.24(190.708,198.12)
2	183.66(180.258,187.238)	0.205(0.118,0.335)	184.295(181.478,189.128)
3	262.67(260.933,267.11)	0.15(0.112,0.18)	263.42(261.817,267.705)
4	283.5(278.993,288.568)	0.155(0.13,0.198)	284.225(279.705,289.563)
5	364.315(361.01,368.888)	0.2(0.183,0.22)	365.3(361.833,369.885)
6	375.095(363.623,380.488)	0.23(0.188,0.248)	376(364.703,381.435)
7	455.57(450.985,461.638)	0.26(0.222,0.315)	456.635(452.08,464.525)
8	468.49(461.908,471.668)	0.22(0.195,0.23)	469.52(463.13,472.93)
9	535.37(532.48,543.223)	0.235(0.2,0.28)	536.74(533.725,544.275)
10	543.99(542.78,553.247)	0.265(0.2,0.325)	545.01(543.935,554.67)
11	558.45(556.647,568.935)	0.205(0.2,0.255)	559.54(557.585,570.247)
12	600.59(591.723,607.72)	0.34(0.3,0.598)	602.03(593.068,609.328)
13	593.775(586.515,599.16)	0.28(0.228,0.375)	594.915(587.697,600.82)
14	586.475(582.255,590.157)	0.215(0.182,0.315)	587.995(583.407,591.29)
15	644.19(637.803,655.495)	0.275(0.235,0.322)	645.255(639.213,657.008)
16	659.475(651.853,664.843)	0.325(0.248,0.352)	661.265(653.25,667.697)
17	648.375(635.745,650.193)	0.325(0.227,0.375)	649.54(636.76,651.035)
18	643.235(640.655,648.33)	0.235(0.198,0.257)	644.695(641.42,649.282)
19	642.445(638.543,649.105)	0.25(0.205,0.34)	643.285(639.2,649.967)
20	642.165(637.233,644.162)	0.22(0.205,0.268)	642.915(637.95,645.01)
21	631.82(626.81,638.77)	0.24(0.17,0.295)	632.515(627.773,639.707)
22	623.94(620.94,629.773)	0.225(0.2,0.273)	624.71(621.885,630.622)
23	637.815(635.733,643.682)	0.245(0.185,0.278)	638.7(636.465,644.61)
24	633.185(625.802,637.128)	0.215(0.155,0.23)	634.03(626.585,637.795)
25	632.16(624.608,648.98)	0.23(0.2,0.25)	632.825(625.285,649.802)
26	647.625(639.403,653.422)	0.275(0.222,0.353)	648.485(640.168,654.23)
27	632.435(629.86,644.05)	0.205(0.165,0.25)	633.265(630.572,644.962)
28	642.86(640.31,653.66)	0.235(0.2,0.265)	643.735(641.103,654.36)
29	653.42(647.357,661.18)	0.285(0.223,0.355)	654.415(648.653,664.15)
30	642.46(635.63,664.748)	0.345(0.247,0.435)	644.125(636.77,667.097)
31	650.96(637.47,655.62)	0.29(0.258,0.33)	652.365(638.997,659.438)
32	642.635(639.618,649.623)	0.305(0.28,0.36)	644.17(641.48,651.642)

Table 23. The median (first quantile – third quantile) of the Time (user-system-elapsed) in 10 iterations group by the number of hybrid principal component analysis (HPCA) for whole brain.

HPCA	Min_Error	Optimum					
		index	K	Tau	Lambda	Eta	Kappa
1	1.068(1.03,1.283)	6.5(4.25,8)	1(1,2)	100(32.5,100)	0.4(0.325,0.4)	0(0,0)	0(0,0)
2	0.849(0.65,1.025)	4(1.75,4.75)	2.5(2,3)	55(0.1,100)	0.35(0.1,0.4)	0(0,0)	0(0,0)
3	4.484(4.082,4.933)	1(1,4)	1(1,1)	0.1(0.1,100)	0.1(0.1,0.4)	0(0,0)	0(0,0)
4	5.04(4.888,5.715)	1(1,3.75)	1(1,1)	0.1(0.1,0.775)	0.1(0.1,0.175)	0(0,0)	0(0,0)
5	6.929(6.747,7.3)	2.5(1,4)	1(1,1.75)	0.55(0.1,100)	0.15(0.1,0.4)	0(0,0)	0(0,0)
6	7.429(7.032,7.553)	4(1,7.75)	1(1,1)	0.55(0.1,100)	0.15(0.1,0.4)	0(0,0.008)	0(0,0)
7	8.553(8.102,9.052)	3(1,7)	2(2,4.25)	5.05(0.1,77.5)	0.2(0.1,0.375)	0(0,0)	0(0,0)
8	8.946(8.747,9.143)	3(1.25,6.25)	2(2,3.5)	10(0.325,77.5)	0.3(0.125,0.375)	0(0,0)	0(0,0)
9	9.145(8.773,9.721)	3(1,3)	2(2,2)	5.5(0.1,10)	0.25(0.1,0.3)	0(0,0)	0(0,0)
10	9.947(9.605,10.124)	1(1,3.75)	2(2,2)	0.1(0.1,10)	0.1(0.1,0.3)	0(0,0)	0(0,0)
11	10.499(10.227,10.829)	3(1.5,3)	2(2,2)	10(2.575,10)	0.3(0.15,0.3)	0(0,0)	0(0,0.008)
12	10.66(10.285,10.858)	3(1.5,9.25)	2(2,3.5)	10(2.575,10)	0.3(0.15,0.3)	0(0,0.008)	0(0,0.008)
13	11.427(11.094,11.679)	3(2,3.75)	2(1,2.75)	10(1,10)	0.3(0.2,0.3)	0(0,0)	0(0,0)
14	11.516(11.288,11.644)	4(3.25,9.25)	2.5(1.25,4.75)	100(10,100)	0.4(0.3,0.4)	0(0,0.008)	0(0,0.01)
15	12.527(11.905,12.637)	2.5(2,3)	2(1,2.75)	5.5(1,10)	0.25(0.2,0.3)	0(0,0)	0(0,0.008)
16	12.613(12.313,12.709)	4(3,4)	1(1,5)	100(10,100)	0.4(0.3,0.4)	0(0,0)	0(0,0.008)
17	12.899(12.623,12.946)	4(2.5,7)	1(1,4)	100(25.75,100)	0.4(0.25,0.4)	0(0,0)	0(0,0.01)
18	12.8(12.696,12.973)	2.5(1,3)	1.5(1,3.5)	5.5(0.1,10)	0.25(0.1,0.3)	0(0,0)	0.005(0,0.01)
19	12.759(11.904,12.837)	3.5(2,4)	1.5(1,2)	55(1,100)	0.35(0.2,0.4)	0(0,0)	0(0,0)
20	12.661(12.399,12.866)	3(1.25,4)	3(1.25,4.75)	10(0.325,100)	0.3(0.125,0.4)	0(0,0)	0.01(0,0.01)
21	12.285(11.792,12.811)	3(1,4)	2.5(1.25,4)	10(0.1,77.5)	0.3(0.1,0.375)	0(0,0)	0(0,0.01)
22	12.466(12.019,12.786)	3(1,4)	1(1,1.75)	5.5(0.1,100)	0.25(0.1,0.4)	0(0,0)	0(0,0)
23	12.493(12.142,12.673)	2.5(1,3)	2(1,4.25)	5.5(0.1,10)	0.25(0.1,0.3)	0(0,0)	0(0,0.008)
24	12.537(12.027,12.729)	2(1,4)	2.5(1.25,4.75)	1(0.1,77.5)	0.2(0.1,0.375)	0(0,0)	0.005(0,0.01)
25	12.6(12.457,12.7)	3(1.25,7)	4.5(1.5,7.75)	50.5(0.325,100)	0.3(0.125,0.4)	0(0,0)	0.01(0,0.01)
26	12.411(11.996,12.768)	3.5(2,4)	2.5(2,4.75)	55(1,100)	0.35(0.2,0.4)	0(0,0)	0.005(0,0.01)
27	12.901(12.341,13.084)	3(1,3.75)	1(1,1)	10(0.1,10)	0.3(0.1,0.3)	0(0,0)	0(0,0)
28	12.838(12.598,13.095)	1.5(1,2.75)	1.5(1,2.75)	0.55(0.1,7.75)	0.15(0.1,0.275)	0(0,0)	0.01(0,0.01)
29	12.357(12.18,12.66)	4(2.5,9.25)	4(2,7.25)	100(3.25,100)	0.4(0.225,0.4)	0(0,0.008)	0.01(0,0.01)
30	12.708(12.493,12.776)	3(2.25,4)	1(1,1.75)	10(3.25,100)	0.3(0.225,0.4)	0(0,0)	0(0,0)
31	12.755(12.508,13.034)	3.5(1,4)	2(1,4.25)	10(0.1,100)	0.3(0.1,0.4)	0(0,0)	0.005(0,0.01)
32	12.755(12.589,12.829)	4(3,4)	1.5(1,2)	100(10,100)	0.4(0.3,0.4)	0(0,0)	0(0,0.008)

Table 24. The median (first quantile – third quantile) of the model fit indices in 10 iterations group by the number of hybrid principal component analysis (HPCA) for whole brain.

2.2.3.2 Frontal Region (1)

The MSPE and Correlation group by number of HPCA, Train and Test ([Table 25](#))

The User, System and Elapsed Time group by number of HPCA ([Table 26](#)): The median (first quantile – third quantile) of the Time (user-system-elapsed) in 10 iterations group by the number of hybrid principal component analysis (HPCA) for frontal region of brain.

The Model fitting parameters group by number of HPCA ([Table 27](#)): The median (first quantile – third quantile) of the model fit indices in 10 iterations group by the number of hybrid principal component analysis (HPCA) for whole brain.

HPCA	MSPE		Correlation	
	Train	Test	Train	Test
1	0.004(0.001,0.01)	0.216(0.184,0.217)	0.998(0.995,1)	0.89(0.889,0.907)
2	0.001(0.001,0.002)	0.032(0.031,0.04)	0.999(0.999,0.999)	0.984(0.98,0.985)
3	0.056(0.047,0.114)	0.34(0.323,0.475)	0.971(0.941,0.976)	0.814(0.736,0.824)
4	0.053(0.051,0.053)	0.33(0.292,0.442)	0.973(0.973,0.974)	0.821(0.749,0.843)
5	0.161(0.161,0.228)	0.616(0.581,0.619)	0.916(0.878,0.916)	0.629(0.627,0.655)
6	0.182(0.182,0.182)	0.685(0.611,0.787)	0.904(0.904,0.904)	0.584(0.522,0.63)
7	0.138(0.096,0.15)	0.567(0.567,0.706)	0.928(0.922,0.951)	0.662(0.573,0.662)
8	0.146(0.146,0.159)	0.645(0.645,0.645)	0.924(0.917,0.924)	0.61(0.61,0.61)
9	0.211(0.202,0.338)	0.667(0.63,0.667)	0.888(0.812,0.893)	0.593(0.593,0.608)
10	0.299(0.212,0.372)	0.656(0.642,0.679)	0.836(0.792,0.888)	0.587(0.582,0.615)
11	0.429(0.353,0.433)	0.742(0.726,0.935)	0.77(0.752,0.836)	0.524(0.292,0.527)
12	0.445(0.445,0.455)	0.757(0.705,0.883)	0.745(0.738,0.745)	0.5(0.357,0.544)
13	0.478(0.428,0.531)	0.734(0.71,0.764)	0.722(0.684,0.756)	0.523(0.491,0.541)
14	0.488(0.346,0.494)	0.772(0.699,0.87)	0.715(0.711,0.812)	0.492(0.406,0.548)
15	0.352(0.326,0.556)	0.91(0.731,0.914)	0.82(0.665,0.837)	0.314(0.309,0.522)
16	0.439(0.346,0.572)	0.932(0.748,0.984)	0.754(0.654,0.823)	0.287(0.152,0.507)
17	0.55(0.387,0.573)	0.851(0.745,1.124)	0.67(0.653,0.796)	0.437(0.157,0.511)
18	0.399(0.318,0.556)	0.917(0.761,0.953)	0.787(0.665,0.841)	0.304(0.278,0.496)
19	0.344(0.318,0.393)	0.964(0.837,1.032)	0.82(0.792,0.839)	0.252(0.189,0.419)
20	0.574(0.424,0.574)	0.734(0.734,0.77)	0.653(0.653,0.767)	0.519(0.489,0.519)
21	0.403(0.338,0.515)	0.936(0.802,1.039)	0.786(0.696,0.829)	0.299(0.183,0.467)
22	0.36(0.307,0.403)	1.069(0.86,1.126)	0.814(0.785,0.848)	0.203(0.125,0.406)
23	0.326(0.318,0.395)	0.944(0.814,0.956)	0.828(0.791,0.841)	0.32(0.247,0.462)
24	0.394(0.337,0.543)	0.861(0.755,1.016)	0.792(0.675,0.822)	0.382(0.222,0.499)
25	0.447(0.319,0.575)	0.779(0.733,0.938)	0.746(0.651,0.84)	0.485(0.279,0.519)
26	0.47(0.379,0.575)	0.835(0.734,1.003)	0.734(0.652,0.801)	0.4(0.232,0.519)
27	0.359(0.319,0.574)	0.853(0.742,0.94)	0.814(0.652,0.84)	0.39(0.277,0.513)
28	0.32(0.3,0.574)	0.84(0.731,1.01)	0.832(0.652,0.853)	0.404(0.288,0.522)
29	0.34(0.319,0.39)	0.94(0.929,1.042)	0.827(0.794,0.841)	0.277(0.207,0.289)
30	0.478(0.32,0.574)	0.859(0.741,0.964)	0.725(0.652,0.84)	0.384(0.249,0.514)
31	0.382(0.322,0.575)	0.946(0.735,1.049)	0.799(0.652,0.839)	0.27(0.184,0.517)
32	0.533(0.387,0.575)	0.759(0.73,0.951)	0.682(0.652,0.796)	0.501(0.264,0.522)

Table 25. The median (first quantile – third quantile) of the Mean Square Prediction Error (MSPE) and Correlation in 10 iterations group by Train and Test and the number of hybrid principal component analysis (HPCA) for frontal region brain.

HPCA	Time		
	User	System	Elapsed
1	178.2(176.198,178.673)	0.205(0.103,0.31)	178.96(176.903,179.585)
2	172.78(169.893,174.69)	0.155(0.13,0.373)	173.415(170.365,175.325)
3	256.885(255.465,259.752)	0.105(0.085,0.145)	257.36(255.873,260.332)
4	270.125(267.97,277.557)	0.18(0.145,0.238)	270.665(268.538,278.173)
5	360.8(357.987,366.925)	0.15(0.105,0.172)	361.645(358.403,367.573)
6	369.625(368.962,370.37)	0.145(0.13,0.197)	370.465(369.29,370.875)
7	454.82(450.585,457.79)	0.185(0.162,0.243)	455.295(452.023,458.378)
8	459.14(455.607,461.945)	0.205(0.19,0.267)	459.74(456.27,462.54)
9	538.2(531.705,540.475)	0.175(0.153,0.198)	538.855(532.462,541.078)
10	540.785(534.775,547.628)	0.24(0.192,0.363)	541.605(535.383,548.17)
11	539.565(534.088,544.357)	0.22(0.192,0.227)	540.75(535.675,545.232)
12	548.82(542.112,552.207)	0.21(0.148,0.238)	550.07(542.968,553.99)
13	596.305(588.862,600.14)	0.265(0.208,0.287)	597.375(589.963,601.415)
14	597.82(587.975,603.365)	0.26(0.21,0.295)	599.59(589.215,604.343)
15	627.21(614.983,633.135)	0.28(0.22,0.337)	627.93(615.8,634.005)
16	625.915(620.675,632.87)	0.17(0.14,0.257)	626.57(621.217,633.425)
17	629.29(622.51,641.743)	0.255(0.185,0.333)	629.945(623.393,642.335)
18	621.12(616.36,624.86)	0.265(0.192,0.345)	621.98(617.175,625.568)
19	625.195(622.09,629.285)	0.175(0.14,0.212)	625.875(622.772,629.96)
20	624.62(618.335,626.585)	0.245(0.192,0.267)	625.36(619.145,627.563)
21	625.935(619.432,629.405)	0.205(0.17,0.22)	626.53(620.058,629.975)
22	626.035(617.165,630.485)	0.22(0.2,0.258)	626.675(617.952,631.012)
23	630.25(617.063,635.045)	0.19(0.155,0.273)	630.925(617.753,635.683)
24	623.45(615.427,631.3)	0.22(0.173,0.243)	624.05(616.072,632.235)
25	618.875(611.662,629.035)	0.225(0.177,0.323)	619.785(612.482,629.59)
26	625.885(614.997,632.513)	0.22(0.2,0.243)	626.545(615.685,633.387)
27	628.15(618.432,634.38)	0.245(0.205,0.285)	628.965(619.19,635.115)
28	632.455(620.033,636.963)	0.25(0.203,0.293)	633.29(620.968,637.69)
29	627.1(617.632,633.415)	0.24(0.212,0.325)	627.995(618.383,634.238)
30	626.73(622.495,634.482)	0.295(0.28,0.35)	627.645(623.48,635.4)
31	627.095(624.605,631.555)	0.225(0.19,0.322)	627.89(625.383,632.597)
32	617.385(611.248,624.925)	0.225(0.22,0.273)	618.28(612.065,626.782)

Table 26. The median (first quantile – third quantile) of the Time (user-system-elapsed) in 10 iterations group by the number of hybrid principal component analysis (HPCA) for frontal region of brain.

HPCA	Min_Error	Optimum					
		index	K	Tau	Lambda	Eta	Kappa
1	0.97(0.791,1.029)	8(5.75,8)	2.5(1.25,5.75)	100(100,100)	0.4(0.4,0.4)	0(0,0)	0(0,0)
2	0.667(0.635,0.728)	3(3,3.75)	2.5(2,4)	10(10,77.5)	0.3(0.3,0.375)	0(0,0)	0(0,0)
3	4.085(3.997,4.481)	3.5(1.5,4)	1(1,1)	55(2.575,100)	0.35(0.15,0.4)	0(0,0)	0(0,0)
4	5.481(4.958,5.96)	4(1,9)	1(1,2.5)	0.1(0.1,7.75)	0.1(0.1,0.275)	0(0,0.01)	0(0,0)
5	6.69(6.482,7.293)	3(1.25,4)	1(1,1)	0.55(0.1,75.25)	0.15(0.1,0.35)	0(0,0)	0(0,0)
6	7.082(6.6,7.413)	3.5(1,9)	1(1,1)	0.1(0.1,1)	0.1(0.1,0.2)	0(0,0.01)	0(0,0)
7	8.2(7.749,8.511)	1(1,3)	2(2,2.75)	0.1(0.1,10)	0.1(0.1,0.3)	0(0,0)	0(0,0)
8	8.683(8.284,8.932)	1(1,1.75)	2(2,2)	0.1(0.1,0.1)	0.1(0.1,0.1)	0(0,0)	0(0,0)
9	9.686(9.345,10.16)	1(1,2.75)	2(1.25,2)	0.1(0.1,0.775)	0.1(0.1,0.175)	0(0,0)	0(0,0)
10	9.718(9.549,9.995)	1(1,3)	1.5(1,2)	0.1(0.1,10)	0.1(0.1,0.3)	0(0,0)	0(0,0)
11	11.155(10.992,11.353)	1(1,1)	1.5(1,4.25)	0.1(0.1,0.1)	0.1(0.1,0.1)	0(0,0)	0(0,0.008)
12	11.092(10.721,11.517)	2(1,3)	1(1,1)	0.1(0.1,10)	0.1(0.1,0.3)	0(0,0)	0(0,0)
13	11.52(11.235,11.955)	4(3,4)	1(1,1.75)	100(10,100)	0.4(0.3,0.4)	0(0,0)	0(0,0)
14	12.076(11.939,12.156)	3(1.5,3)	1(1,2.75)	10(2.575,10)	0.3(0.15,0.3)	0(0,0)	0(0,0)
15	12.793(12.221,13.344)	2(1,3.75)	4(1,5)	5.05(0.1,77.5)	0.2(0.1,0.375)	0(0,0)	0.01(0,0.01)
16	12.42(12.249,13.363)	2.5(1,4)	4.5(1.5,75)	5.05(0.1,100)	0.2(0.1,0.4)	0(0,0)	0.01(0,0.01)
17	12.108(11.381,12.545)	4(4,11.75)	1(1,6.5)	100(100,100)	0.4(0.4,0.4)	0(0,0.01)	0(0,0.01)
18	12.271(11.794,12.662)	3.5(1,4)	4(1,5)	55(0.1,100)	0.35(0.1,0.4)	0(0,0)	0.005(0,0.01)
19	12.119(11.969,12.406)	3.5(1,4)	5(2.75,5.75)	55(0.1,100)	0.35(0.1,0.4)	0(0,0)	0.01(0.003,0.01)
20	12.907(12.488,13.128)	4(4,4)	1(1,2.5)	100(100,100)	0.4(0.4,0.4)	0(0,0)	0(0,0)
21	12.375(11.957,12.644)	5.5(1.75,8)	4.5(1,5)	55(2.575,100)	0.35(0.15,0.4)	0(0,0)	0.005(0,0.01)
22	12.103(11.374,12.573)	6(4,11.75)	5.5(3.5,6)	100(32.5,100)	0.4(0.325,0.4)	0(0,0.01)	0.01(0,0.01)
23	12.54(12.245,12.898)	3(1,9.25)	4(2.25,5)	5.5(0.1,77.5)	0.25(0.1,0.375)	0(0,0.008)	0.005(0,0.01)
24	12.126(11.979,12.277)	4(1.5,4)	3(1.25,5)	100(2.575,100)	0.4(0.15,0.4)	0(0,0)	0.005(0,0.01)
25	12.533(12.308,13.031)	2.5(1,4)	2(1,5)	50.05(0.1,100)	0.25(0.1,0.4)	0(0,0)	0(0,0.01)
26	12.497(12.371,12.829)	4(2.25,7)	2.5(1,5)	100(3.25,100)	0.4(0.225,0.4)	0(0,0)	0.005(0,0.01)
27	12.713(12.599,12.976)	4(1.25,4)	3.5(1,5)	55(0.325,100)	0.35(0.125,0.4)	0(0,0)	0.005(0,0.01)
28	12.528(12.439,12.585)	4(1.75,4)	3.5(1.5,75)	100(2.575,100)	0.4(0.15,0.4)	0(0,0)	0(0,0.008)
29	12.571(12.112,12.838)	1(1,4)	5(4.25,5.75)	0.1(0.1,100)	0.1(0.1,0.4)	0(0,0)	0.01(0.01,0.01)
30	12.723(12.058,13.143)	4(1.75,4)	3(1,5)	100(25.075,100)	0.4(0.175,0.4)	0(0,0)	0.005(0,0.01)
31	12.556(12.436,12.7)	4(1.5,7)	5(1,5)	100(2.575,100)	0.4(0.15,0.4)	0(0,0)	0.01(0,0.01)
32	12.699(12.173,13.067)	4(1.75,4)	1(1,5)	100(25.075,100)	0.4(0.175,0.4)	0(0,0)	0(0,0.01)

Table 27. The median (first quantile – third quantile) of the model fit indices in 10 iterations group by the number of hybrid principal component analysis (HPCA) for frontal region of brain.

2.2.3.3 Central Region (2)

The MSPE and Correlation group by number of HPCA, Train and Test ([Table 28](#))

The User, System and Elapsed Time group by number of HPCA ([Table 29](#)): The median (first quantile – third quantile) of the Time (user-system-elapsed) in 10 iterations group by the number of hybrid principal component analysis (HPCA) for whole brain.

The Model fitting parameters group by number of HPCA ([Table 30](#)): The median (first quantile – third quantile) of the model fit indices in 10 iterations group by the number of hybrid principal component analysis (HPCA) for whole brain.

HPCA	MSPE		Correlation	
	Train	Test	Train	Test
1	0.012(0.002,0.016)	0.221(0.165,0.235)	0.994(0.992,0.999)	0.885(0.877,0.916)
2	0.002(0.001,0.009)	0.136(0.126,0.146)	0.999(0.996,1)	0.929(0.924,0.935)
3	0.06(0.037,0.081)	0.466(0.4,0.533)	0.97(0.959,0.982)	0.74(0.694,0.782)
4	0.059(0.053,0.065)	0.446(0.386,0.506)	0.97(0.967,0.973)	0.748(0.707,0.784)
5	0.262(0.206,0.262)	0.656(0.656,0.656)	0.859(0.859,0.891)	0.606(0.606,0.606)
6	0.263(0.263,0.263)	0.639(0.639,0.639)	0.859(0.859,0.859)	0.616(0.616,0.616)
7	0.138(0.138,0.138)	0.645(0.639,0.658)	0.928(0.928,0.928)	0.629(0.619,0.631)
8	0.145(0.142,0.15)	0.664(0.624,0.691)	0.924(0.922,0.926)	0.618(0.595,0.637)
9	0.25(0.21,0.364)	0.69(0.666,0.713)	0.866(0.798,0.889)	0.6(0.578,0.622)
10	0.396(0.21,0.423)	0.733(0.683,0.769)	0.798(0.778,0.889)	0.555(0.5,0.6)
11	0.4(0.253,0.445)	0.83(0.774,0.92)	0.785(0.764,0.864)	0.465(0.371,0.541)
12	0.386(0.28,0.444)	0.776(0.754,0.853)	0.804(0.758,0.853)	0.503(0.418,0.549)
13	0.459(0.384,0.471)	0.815(0.79,0.903)	0.735(0.727,0.796)	0.485(0.37,0.51)
14	0.464(0.384,0.516)	0.812(0.763,0.893)	0.736(0.696,0.796)	0.49(0.372,0.527)
15	0.502(0.416,0.562)	0.814(0.756,0.863)	0.705(0.661,0.777)	0.466(0.39,0.514)
16	0.538(0.429,0.586)	0.818(0.738,0.899)	0.687(0.643,0.758)	0.459(0.376,0.536)
17	0.541(0.495,0.58)	0.835(0.756,0.892)	0.676(0.648,0.71)	0.444(0.393,0.513)
18	0.494(0.353,0.603)	0.912(0.879,0.979)	0.725(0.638,0.818)	0.335(0.318,0.363)
19	0.509(0.416,0.621)	0.896(0.87,0.909)	0.71(0.623,0.776)	0.365(0.36,0.393)
20	0.545(0.496,0.612)	0.824(0.791,0.903)	0.677(0.629,0.712)	0.444(0.343,0.467)
21	0.584(0.495,0.621)	0.824(0.754,0.86)	0.644(0.623,0.71)	0.456(0.377,0.503)
22	0.503(0.392,0.579)	0.808(0.735,0.944)	0.711(0.648,0.79)	0.443(0.302,0.539)
23	0.43(0.31,0.558)	0.867(0.809,0.928)	0.76(0.676,0.846)	0.376(0.329,0.456)
24	0.497(0.349,0.562)	0.869(0.752,0.932)	0.709(0.661,0.821)	0.449(0.334,0.515)
25	0.543(0.441,0.639)	0.862(0.815,0.873)	0.675(0.614,0.755)	0.391(0.376,0.45)
26	0.425(0.369,0.544)	0.896(0.72,0.981)	0.763(0.674,0.808)	0.364(0.308,0.55)
27	0.518(0.5,0.609)	0.863(0.806,0.909)	0.694(0.63,0.711)	0.404(0.342,0.455)
28	0.586(0.516,0.634)	0.827(0.754,0.846)	0.643(0.618,0.695)	0.44(0.416,0.502)
29	0.581(0.428,0.644)	0.862(0.817,0.94)	0.661(0.611,0.767)	0.378(0.326,0.439)
30	0.555(0.496,0.635)	0.899(0.846,0.929)	0.673(0.616,0.719)	0.362(0.342,0.4)
31	0.587(0.498,0.647)	0.856(0.807,0.87)	0.65(0.609,0.708)	0.403(0.375,0.458)
32	0.502(0.355,0.562)	0.9(0.844,0.925)	0.708(0.675,0.817)	0.356(0.332,0.441)

Table 28. The median (first quantile – third quantile) of the Mean Square Prediction Error (MSPE) and Correlation in 10 iterations group by Train and Test and the number of hybrid principal component analysis (HPCA) for central region of the brain.

HPCA	Time		
	User	System	Elapsed
1	191.625(189.765,193.765)	0.11(0.095,0.143)	192.17(190.075,194.162)
2	186.92(184.365,188.205)	0.105(0.09,0.16)	187.26(184.72,188.573)
3	261.955(259.61,264.145)	0.095(0.082,0.12)	262.44(259.97,264.582)
4	269.97(268.013,272.723)	0.115(0.093,0.188)	270.475(268.373,273.14)
5	358.79(353.998,366.905)	0.165(0.132,0.19)	359.39(354.515,368.33)
6	360.3(356.403,365.733)	0.14(0.118,0.2)	360.8(358.752,366.21)
7	451.085(443.948,455)	0.185(0.162,0.257)	451.995(444.622,455.853)
8	449.625(443.995,455.312)	0.185(0.148,0.235)	450.365(445.015,456.13)
9	544.545(537.902,558.155)	0.2(0.158,0.255)	545.17(538.638,558.952)
10	542.55(539.722,546.735)	0.21(0.192,0.237)	543.3(540.74,547.525)
11	535.185(531.37,542.455)	0.22(0.192,0.273)	536(531.997,543.437)
12	540.28(531.577,546.755)	0.225(0.173,0.273)	541.05(532.725,547.515)
13	591.565(575.998,599.887)	0.22(0.197,0.262)	592.495(576.775,600.855)
14	579.385(574.537,583.605)	0.2(0.18,0.228)	580.055(575.317,584.375)
15	646.53(636.348,675.118)	0.325(0.227,0.455)	647.295(637.765,717.863)
16	649.47(641.218,673.715)	0.31(0.235,0.35)	651.555(642.777,675.642)
17	647.685(641.085,652.395)	0.315(0.285,0.33)	649.6(643.242,654.713)
18	655.995(641.885,666.055)	0.4(0.265,0.443)	657.735(644.205,668.588)
19	635.955(621.17,645.705)	0.225(0.175,0.295)	637.525(622.122,646.398)
20	625.475(614.745,629.478)	0.23(0.19,0.303)	626.515(615.825,630.46)
21	624.765(614.887,639.603)	0.26(0.203,0.315)	626.57(615.86,640.828)
22	627.415(615.98,634.055)	0.285(0.263,0.34)	628.47(617.645,635.085)
23	578.88(549.797,631.913)	0.185(0.155,0.332)	582.915(550.787,632.72)
24	538.035(525.488,608.597)	0.215(0.16,0.257)	538.68(526.16,614.497)
25	540.23(526.32,550.383)	0.195(0.16,0.228)	540.79(526.707,551.265)
26	528.695(522.322,535.735)	0.225(0.175,0.273)	529.17(522.773,536.258)
27	533.99(529.04,537.028)	0.2(0.165,0.22)	534.535(529.513,537.46)
28	533.165(529.427,539.49)	0.155(0.115,0.225)	533.605(529.913,540.048)
29	533.92(530.8,535.54)	0.18(0.15,0.218)	534.4(531.388,535.982)
30	532.51(525.675,536.183)	0.21(0.178,0.243)	532.875(526.13,536.737)
31	528.555(518.625,537.218)	0.195(0.175,0.302)	529.17(519.14,537.86)
32	538.885(532.288,546.8)	0.19(0.145,0.258)	539.5(532.828,547.45)

Table 29. The median (first quantile – third quantile) of the Time (user-system-elapsed) in 10 iterations group by the number of hybrid principal component analysis (HPCA) for central region of the brain.

HPCA	Min_Error	Optimum					
		index	K	Tau	Lambda	Eta	Kappa
1	1.364(1.283,1.62)	6(4,8)	2.5(2,5.5)	100(100,100)	0.4(0.4,0.4)	0(0,0)	0(0,0)
2	1.071(0.936,1.452)	4(4,6.25)	4(2,4)	100(10,100)	0.4(0.3,0.4)	0(0,0)	0(0,0)
3	5.06(4.721,5.552)	3.5(1,4)	2(1,2.75)	55(0.1,100)	0.35(0.1,0.4)	0(0,0)	0(0,0)
4	5.348(4.541,6.229)	3(1.25,3)	1(1,1)	5.5(0.1,10)	0.25(0.1,0.3)	0(0,0)	0(0,0)
5	7.238(6.792,7.751)	4(4,4)	1(1,1)	100(100,100)	0.4(0.4,0.4)	0(0,0)	0(0,0)
6	7.003(6.676,7.411)	4(4,4)	1(1,1)	100(100,100)	0.4(0.4,0.4)	0(0,0)	0(0,0)
7	8.178(7.705,8.556)	3(3,3.75)	2(2,2)	10(10,10)	0.3(0.3,0.3)	0(0,0)	0(0,0)
8	8.275(7.88,9.019)	4(3,6.25)	2(2,2)	10(10,77.5)	0.3(0.3,0.375)	0(0,0)	0(0,0)
9	9.921(9.187,10.366)	3(3,4)	2(2,2)	10(10,77.5)	0.3(0.3,0.375)	0(0,0)	0(0,0)
10	10.015(9.841,10.118)	3.5(3,6.25)	2(2,2)	10(10,10)	0.3(0.3,0.3)	0(0,0)	0(0,0.01)
11	10.314(10.012,10.723)	3.5(3,5.75)	2(2,2)	10(3.25,10)	0.3(0.225,0.3)	0(0,0)	0.005(0,0.01)
12	10.482(10.123,10.894)	3(3,3)	2(2,2.75)	10(10,10)	0.3(0.3,0.3)	0(0,0)	0.005(0,0.01)
13	11.425(11.119,11.709)	5.5(3,7)	1(1,2.5)	10(10,10)	0.3(0.3,0.3)	0(0,0)	0(0,0.01)
14	11.242(11.101,11.812)	4(1.5,7)	1(1,2.75)	10(2.575,77.5)	0.3(0.15,0.375)	0(0,0)	0(0,0.01)
15	12.33(12.168,12.717)	2.5(1,3.75)	1(1,2)	5.5(0.1,10)	0.25(0.1,0.3)	0(0,0)	0(0,0.01)
16	12.323(12.233,12.847)	4(3.25,8)	1.5(1,2)	100(32.5,100)	0.4(0.325,0.4)	0(0,0)	0.005(0,0.01)
17	12.635(12.543,12.904)	3.5(2,4)	1(1,1)	55(1,100)	0.35(0.2,0.4)	0(0,0)	0(0,0.01)
18	12.45(12.216,12.614)	2(1,3.75)	2.5(1,4.75)	1(0.1,77.5)	0.2(0.1,0.375)	0(0,0)	0.01(0.01,0.01)
19	12.386(11.975,12.67)	2(1.25,3)	1.5(1,2.75)	1(0.325,10)	0.2(0.125,0.3)	0(0,0)	0.01(0.01,0.01)
20	12.494(12.298,12.816)	3.5(1.25,4)	1(1,2.5)	10(0.325,100)	0.3(0.125,0.4)	0(0,0)	0.01(0,0.01)
21	12.633(12.147,12.716)	3(1,7)	1(1,1)	50.5(0.1,100)	0.3(0.1,0.4)	0(0,0)	0.005(0,0.01)
22	12.34(12.077,12.546)	4(2.5,9.25)	1.5(1,5)	100(3.25,100)	0.4(0.225,0.4)	0(0,0.008)	0.005(0,0.01)
23	12.312(12.088,12.658)	2.5(1,4)	3.5(1,5)	5.5(0.1,100)	0.25(0.1,0.4)	0(0,0)	0.01(0.01,0.01)
24	12.155(11.872,12.653)	3.5(2.25,4)	1(1,5.25)	55(3.25,100)	0.35(0.225,0.4)	0(0,0)	0(0,0.01)
25	12.277(12.012,12.368)	1.5(1,3.75)	1(1,2.5)	0.55(0.1,77.5)	0.15(0.1,0.375)	0(0,0)	0.01(0.003,0.01)
26	12.374(12.226,12.757)	4(4,10)	3(1.25,5.75)	100(100,100)	0.4(0.4,0.4)	0(0,0.008)	0.01(0,0.01)
27	12.251(12.142,12.42)	3.5(2,6.25)	1(1,1)	10(1,77.5)	0.3(0.2,0.375)	0(0,0)	0.005(0,0.01)
28	12.307(11.997,12.558)	4(3.25,8)	1(1,1.75)	100(10,100)	0.4(0.3,0.4)	0(0,0)	0.01(0,0.01)
29	12.494(12.338,12.55)	4(2.25,7.25)	2(1,4.25)	55(0.325,100)	0.35(0.125,0.4)	0(0,0)	0.01(0.01,0.01)
30	12.405(12.151,12.748)	3.5(1.25,7.75)	1(1,2)	10(0.325,77.5)	0.3(0.125,0.375)	0(0,0)	0.01(0.01,0.01)
31	12.548(12.434,12.978)	1.5(1,6.75)	1(1,1)	0.55(0.1,77.5)	0.15(0.1,0.375)	0(0,0)	0.01(0,0.01)
32	12.307(12.079,12.604)	2.5(1,6.25)	2(1,4.5)	5.5(0.1,10)	0.25(0.1,0.3)	0(0,0)	0.01(0.01,0.01)

Table 30. The median (first quantile – third quantile) of the model fit indices in 10 iterations group by the number of hybrid principal component analysis (HPCA) for central region of the brain.

2.2.3.4 Occipital Region (3)

The MSPE and Correlation group by number of HPCA, Train and Test ([Table 31](#))

The User, System and Elapsed Time group by number of HPCA ([Table 32](#)): The median (first quantile – third quantile) of the Time (user-system-elapsed) in 10 iterations group by the number of hybrid principal component analysis (HPCA) for whole brain.

The Model fitting parameters group by number of HPCA ([Table 33](#)): The median (first quantile – third quantile) of the model fit indices in 10 iterations group by the number of hybrid principal component analysis (HPCA) for whole brain.

HPCA	MSPE		Correlation	
	Train	Test	Train	Test
1	0(0,0)	0.003(0.002,0.003)	1(1,1)	0.999(0.998,0.999)
2	0.008(0.006,0.008)	0.104(0.104,0.162)	0.996(0.996,0.997)	0.947(0.918,0.947)
3	0.036(0.034,0.039)	0.091(0.088,0.091)	0.982(0.98,0.983)	0.955(0.954,0.956)
4	0.063(0.063,0.063)	0.119(0.119,0.119)	0.968(0.968,0.968)	0.939(0.939,0.939)
5	0.064(0.064,0.064)	0.106(0.106,0.106)	0.967(0.967,0.967)	0.946(0.946,0.946)
6	0.08(0.08,0.08)	0.22(0.22,0.22)	0.959(0.959,0.959)	0.887(0.887,0.887)
7	0.087(0.087,0.087)	0.223(0.223,0.223)	0.956(0.956,0.956)	0.885(0.885,0.885)
8	0.097(0.097,0.097)	0.244(0.244,0.244)	0.95(0.95,0.95)	0.873(0.873,0.873)
9	0.1(0.1,0.1)	0.246(0.246,0.246)	0.949(0.949,0.949)	0.872(0.872,0.872)
10	0.115(0.115,0.115)	0.293(0.293,0.293)	0.941(0.941,0.941)	0.846(0.846,0.846)
11	0.117(0.117,0.117)	0.291(0.291,0.291)	0.939(0.939,0.939)	0.847(0.847,0.847)
12	0.139(0.134,0.139)	0.299(0.299,0.302)	0.928(0.928,0.93)	0.839(0.837,0.839)
13	0.14(0.14,0.14)	0.301(0.301,0.301)	0.927(0.927,0.927)	0.838(0.838,0.838)
14	0.152(0.131,0.152)	0.309(0.309,0.348)	0.921(0.921,0.932)	0.833(0.811,0.833)
15	0.154(0.15,0.154)	0.31(0.31,0.312)	0.92(0.92,0.922)	0.832(0.831,0.832)
16	0.156(0.152,0.156)	0.321(0.321,0.323)	0.919(0.919,0.921)	0.826(0.825,0.826)
17	0.156(0.156,0.156)	0.321(0.321,0.321)	0.919(0.919,0.919)	0.826(0.826,0.826)
18	0.156(0.128,0.156)	0.321(0.321,0.373)	0.919(0.919,0.934)	0.826(0.796,0.826)
19	0.153(0.128,0.156)	0.322(0.321,0.372)	0.92(0.919,0.934)	0.825(0.796,0.826)
20	0.156(0.134,0.156)	0.322(0.322,0.36)	0.919(0.919,0.931)	0.825(0.803,0.825)
21	0.15(0.15,0.156)	0.323(0.321,0.323)	0.922(0.919,0.922)	0.825(0.825,0.826)
22	0.14(0.129,0.157)	0.348(0.323,0.372)	0.927(0.918,0.933)	0.81(0.796,0.825)
23	0.157(0.135,0.157)	0.322(0.322,0.36)	0.918(0.918,0.93)	0.825(0.803,0.825)
24	0.157(0.153,0.157)	0.321(0.321,0.321)	0.918(0.918,0.92)	0.825(0.825,0.825)
25	0.143(0.129,0.157)	0.334(0.321,0.371)	0.926(0.918,0.933)	0.818(0.797,0.825)
26	0.128(0.128,0.145)	0.343(0.318,0.369)	0.933(0.924,0.934)	0.812(0.798,0.827)
27	0.154(0.129,0.156)	0.318(0.318,0.368)	0.92(0.918,0.933)	0.827(0.798,0.828)
28	0.151(0.129,0.157)	0.321(0.319,0.37)	0.921(0.918,0.933)	0.826(0.798,0.827)
29	0.151(0.129,0.157)	0.321(0.319,0.37)	0.921(0.918,0.933)	0.826(0.797,0.827)
30	0.14(0.129,0.152)	0.345(0.32,0.369)	0.927(0.921,0.933)	0.812(0.798,0.826)
31	0.151(0.151,0.157)	0.321(0.319,0.321)	0.921(0.918,0.921)	0.826(0.826,0.827)
32	0.157(0.153,0.157)	0.319(0.319,0.32)	0.918(0.918,0.92)	0.827(0.826,0.827)

Table 31. The median (first quantile – third quantile) of the Mean Square Prediction Error (MSPE) and Correlation in 10 iterations group by Train and Test and the number of hybrid principal component analysis (HPCA) for occipital region of the brain.

HPCA	Time		
	User	System	Elapsed
1	108.61(107.415,110.81)	0.105(0.083,0.203)	108.9(107.507,111.078)
2	101.2(99.818,101.99)	0.09(0.07,0.135)	101.335(99.982,102.105)
3	187.765(183.86,192.588)	0.11(0.085,0.127)	188.08(184.07,192.763)
4	227.925(225.902,232.53)	0.08(0.06,0.143)	228.12(226.182,232.875)
5	269.32(264.643,273.21)	0.11(0.083,0.163)	269.525(264.968,273.545)
6	270.915(265.475,274.773)	0.1(0.08,0.11)	271.155(265.677,275.023)
7	331.53(328.902,337.395)	0.15(0.107,0.178)	331.975(329.447,337.678)
8	335.26(331.858,336.467)	0.125(0.11,0.177)	335.665(332.243,337.22)
9	335.4(331.127,337.365)	0.105(0.09,0.11)	335.67(331.51,337.662)
10	374.23(368.783,377.138)	0.14(0.068,0.163)	375.14(369.333,377.713)
11	386.7(383.84,395.887)	0.145(0.095,0.19)	387.055(384.177,396.203)
12	344.24(341.238,346.708)	0.14(0.095,0.198)	344.545(341.528,346.947)
13	370.135(363.87,381.71)	0.17(0.142,0.177)	370.565(364.2,382.045)
14	406.85(403.395,414.202)	0.12(0.102,0.165)	407.265(403.78,414.508)
15	431.905(427.718,439.005)	0.16(0.118,0.185)	432.51(428.087,439.425)
16	456.215(445.132,457.37)	0.15(0.123,0.188)	456.625(445.598,457.91)
17	458.095(448.173,461.62)	0.15(0.132,0.188)	458.7(448.553,461.985)
18	450.675(447.675,456.202)	0.17(0.162,0.19)	451.055(448.157,456.652)
19	455.445(454.062,459.948)	0.21(0.143,0.238)	455.95(454.31,460.31)
20	451.89(447.278,460.168)	0.185(0.147,0.2)	452.48(447.918,460.67)
21	454.2(447.37,458.56)	0.16(0.093,0.225)	454.545(447.787,459.295)
22	450.53(447.452,457.253)	0.17(0.14,0.208)	450.99(447.983,457.585)
23	453.42(446.497,457.823)	0.15(0.115,0.218)	453.86(446.84,458.268)
24	451.15(450.065,455.895)	0.125(0.093,0.195)	451.53(450.477,456.51)
25	454.595(444.477,461.317)	0.165(0.133,0.17)	455(444.935,461.65)
26	458.385(453.415,461.715)	0.165(0.145,0.185)	458.765(453.815,462.208)
27	454.995(450.182,459.613)	0.215(0.173,0.235)	455.37(450.818,460.055)
28	452.325(450.755,458.182)	0.18(0.153,0.215)	452.845(451.283,458.63)
29	455.895(450.095,459.145)	0.2(0.153,0.317)	456.27(450.515,459.91)
30	453.29(447.945,454.907)	0.165(0.135,0.178)	453.66(448.522,455.335)
31	453.225(446.313,461.353)	0.13(0.098,0.162)	453.6(446.71,461.765)
32	447.905(444.095,459.76)	0.235(0.16,0.257)	448.29(444.527,460.4)

Table 32. The median (first quantile – third quantile) of the Time (user-system-elapsed) in 10 iterations group by the number of hybrid principal component analysis (HPCA) for occipital region of the brain.

HPCA	Min_Error	Optimum					
		index	K	Tau	Lambda	Eta	Kappa
1	0.025(0.023,0.029)	5.5(4,8)	2(2,3.75)	100(100,100)	0.4(0.4,0.4)	0(0,0)	0(0,0)
2	0.36(0.319,0.404)	4(4,12)	1(1,2)	100(100,100)	0.4(0.4,0.4)	0(0,0.01)	0(0,0)
3	1.227(1.164,1.281)	9.5(8,12)	1(1,1)	100(100,100)	0.4(0.4,0.4)	0.005(0,0.01)	0(0,0)
4	1.83(1.706,1.882)	12(12,12)	1(1,1)	100(100,100)	0.4(0.4,0.4)	0.01(0.01,0.01)	0(0,0)
5	1.856(1.784,2.145)	12(12,12)	1(1,1)	100(100,100)	0.4(0.4,0.4)	0.01(0.01,0.01)	0(0,0)
6	2.122(2.073,2.186)	12(12,12)	1(1,1)	100(100,100)	0.4(0.4,0.4)	0.01(0.01,0.01)	0(0,0)
7	2.343(2.217,2.471)	12(12,12)	1(1,1)	100(100,100)	0.4(0.4,0.4)	0.01(0.01,0.01)	0(0,0)
8	2.603(2.465,2.647)	12(12,12)	1(1,1)	100(100,100)	0.4(0.4,0.4)	0.01(0.01,0.01)	0(0,0)
9	2.581(2.428,2.744)	12(12,12)	1(1,1)	100(100,100)	0.4(0.4,0.4)	0.01(0.01,0.01)	0(0,0)
10	2.855(2.763,2.932)	12(12,12)	1(1,1)	100(100,100)	0.4(0.4,0.4)	0.01(0.01,0.01)	0(0,0)
11	2.902(2.779,3.015)	12(12,12)	1(1,1)	100(100,100)	0.4(0.4,0.4)	0.01(0.01,0.01)	0(0,0)
12	3.704(3.481,4.033)	12(5.75,12)	1(1,1)	100(100,100)	0.4(0.4,0.4)	0.01(0.003,0.01)	0(0,0)
13	3.644(3.566,3.672)	12(12,12)	1(1,1)	100(100,100)	0.4(0.4,0.4)	0.01(0.01,0.01)	0(0,0)
14	3.958(3.761,4.066)	12(11,12)	1(1,1)	100(32.5,100)	0.4(0.325,0.4)	0.01(0.01,0.01)	0(0,0)
15	3.726(3.593,3.871)	12(11.25,12)	1(1,1)	100(100,100)	0.4(0.4,0.4)	0.01(0.01,0.01)	0(0,0)
16	3.83(3.736,3.953)	12(11.25,12)	1(1,1)	100(100,100)	0.4(0.4,0.4)	0.01(0.01,0.01)	0(0,0)
17	3.714(3.639,3.85)	12(12,12)	1(1,1)	100(100,100)	0.4(0.4,0.4)	0.01(0.01,0.01)	0(0,0)
18	3.638(3.53,3.767)	12(11,12)	1(1,1)	100(10,100)	0.4(0.3,0.4)	0.01(0.01,0.01)	0(0,0)
19	3.637(3.517,3.719)	11.5(11,12)	1(1,1)	100(10,100)	0.4(0.3,0.4)	0.01(0.01,0.01)	0(0,0)
20	3.761(3.719,3.971)	12(11,12)	1(1,1)	100(32.5,100)	0.4(0.325,0.4)	0.01(0.01,0.01)	0(0,0)
21	4.056(3.77,4.288)	8(4,12)	1(1,1)	100(100,100)	0.4(0.4,0.4)	0.005(0,0.01)	0(0,0)
22	3.639(3.518,3.826)	11(11,12)	1(1,1)	55(10,100)	0.35(0.3,0.4)	0.01(0.01,0.01)	0(0,0)
23	3.798(3.709,3.87)	12(11,12)	1(1,1)	100(32.5,100)	0.4(0.325,0.4)	0.01(0.01,0.01)	0(0,0)
24	3.781(3.539,3.929)	12(11.25,12)	1(1,1)	100(100,100)	0.4(0.4,0.4)	0.01(0.01,0.01)	0(0,0)
25	3.83(3.655,4.035)	12(11,12)	1(1,1)	100(10,100)	0.4(0.3,0.4)	0.01(0.01,0.01)	0(0,0)
26	3.856(3.758,4.229)	11.5(11,12)	1(1,1.75)	100(10,100)	0.4(0.3,0.4)	0.01(0.01,0.01)	0(0,0)
27	3.677(3.536,4.206)	11.5(11,12)	1(1,1)	100(10,100)	0.4(0.3,0.4)	0.01(0.01,0.01)	0(0,0)
28	3.865(3.649,3.963)	11(11,12)	1(1,1)	100(10,100)	0.4(0.3,0.4)	0.01(0.01,0.01)	0(0,0)
29	3.848(3.548,4.11)	11(11,12)	1(1,1)	100(10,100)	0.4(0.3,0.4)	0.01(0.01,0.01)	0(0,0)
30	3.847(3.645,3.937)	11(4,11)	1(1,1)	55(10,100)	0.35(0.3,0.4)	0.01(0,0.01)	0(0,0)
31	3.773(3.727,3.941)	7.5(4,12)	1(1,1)	100(100,100)	0.4(0.4,0.4)	0.005(0,0.01)	0(0,0)
32	3.748(3.723,3.954)	12(6,12)	1(1,1)	100(100,100)	0.4(0.4,0.4)	0.01(0.003,0.01)	0(0,0)

Table 33. The median (first quantile – third quantile) of the model fit indices in 10 iterations group by the number of hybrid principal component analysis (HPCA) for occipital region of the brain.

2.2.3.5 Parietal (4)

The MSPE and Correlation group by number of HPCA, Train and Test ([Table 34](#))

The User, System and Elapsed Time group by number of HPCA ([Table 35](#)): The median (first quantile – third quantile) of the Time (user-system-elapsed) in 10 iterations group by the number of hybrid principal component analysis (HPCA) for whole brain.

The Model fitting parameters group by number of HPCA ([Table 36](#)): The median (first quantile – third quantile) of the model fit indices in 10 iterations group by the number of hybrid principal component analysis (HPCA) for whole brain.

HPCA	MSPE		Correlation	
	Train	Test	Train	Test
1	0.007(0.002,0.017)	0.134(0.127,0.155)	0.996(0.991,0.999)	0.931(0.921,0.935)
2	0.009(0.009,0.009)	0.102(0.102,0.153)	0.995(0.995,0.995)	0.948(0.921,0.948)
3	0.08(0.047,0.092)	0.414(0.384,0.415)	0.959(0.953,0.976)	0.775(0.775,0.794)
4	0.077(0.053,0.091)	0.403(0.403,0.408)	0.961(0.953,0.973)	0.782(0.782,0.782)
5	0.191(0.145,0.191)	0.59(0.59,0.607)	0.899(0.899,0.925)	0.662(0.658,0.662)
6	0.196(0.192,0.196)	0.594(0.594,0.594)	0.896(0.896,0.899)	0.66(0.66,0.66)
7	0.115(0.11,0.152)	0.596(0.569,0.627)	0.941(0.921,0.943)	0.672(0.663,0.684)
8	0.137(0.109,0.283)	0.591(0.567,0.62)	0.929(0.846,0.944)	0.674(0.665,0.685)
9	0.191(0.164,0.287)	0.569(0.566,0.595)	0.899(0.844,0.914)	0.681(0.673,0.687)
10	0.301(0.17,0.311)	0.583(0.572,0.594)	0.84(0.83,0.911)	0.668(0.66,0.677)
11	0.311(0.202,0.371)	0.705(0.684,0.716)	0.83(0.793,0.896)	0.608(0.607,0.613)
12	0.317(0.212,0.322)	0.686(0.685,0.73)	0.826(0.823,0.887)	0.61(0.603,0.619)
13	0.332(0.277,0.337)	0.805(0.785,0.821)	0.817(0.814,0.858)	0.533(0.523,0.555)
14	0.335(0.332,0.352)	0.796(0.748,0.807)	0.815(0.805,0.817)	0.543(0.532,0.567)
15	0.389(0.376,0.438)	0.775(0.746,0.784)	0.782(0.749,0.79)	0.557(0.524,0.574)
16	0.374(0.373,0.395)	0.765(0.689,0.785)	0.791(0.778,0.791)	0.556(0.538,0.578)
17	0.388(0.373,0.46)	0.783(0.686,0.812)	0.782(0.734,0.791)	0.547(0.52,0.606)
18	0.389(0.375,0.394)	0.773(0.746,0.784)	0.782(0.778,0.79)	0.566(0.542,0.573)
19	0.393(0.376,0.395)	0.773(0.746,0.786)	0.779(0.777,0.789)	0.561(0.537,0.574)
20	0.395(0.374,0.438)	0.765(0.746,0.787)	0.777(0.749,0.791)	0.555(0.537,0.574)
21	0.384(0.373,0.395)	0.766(0.745,0.786)	0.785(0.778,0.791)	0.556(0.538,0.575)
22	0.382(0.375,0.393)	0.78(0.752,0.784)	0.786(0.779,0.79)	0.557(0.543,0.573)
23	0.378(0.373,0.392)	0.787(0.754,0.79)	0.789(0.779,0.791)	0.547(0.537,0.571)
24	0.396(0.385,0.437)	0.743(0.686,0.774)	0.777(0.75,0.784)	0.576(0.562,0.608)
25	0.395(0.39,0.395)	0.744(0.744,0.766)	0.777(0.777,0.781)	0.576(0.544,0.576)
26	0.381(0.375,0.395)	0.765(0.744,0.785)	0.786(0.777,0.79)	0.566(0.557,0.576)
27	0.387(0.372,0.394)	0.774(0.753,0.798)	0.783(0.778,0.792)	0.561(0.532,0.571)
28	0.394(0.394,0.456)	0.754(0.694,0.754)	0.778(0.737,0.778)	0.571(0.571,0.602)
29	0.376(0.372,0.381)	0.796(0.794,0.798)	0.789(0.787,0.792)	0.541(0.531,0.551)
30	0.391(0.374,0.394)	0.769(0.754,0.797)	0.78(0.778,0.791)	0.566(0.536,0.57)
31	0.373(0.373,0.39)	0.796(0.771,0.796)	0.791(0.781,0.791)	0.532(0.521,0.541)
32	0.373(0.373,0.392)	0.796(0.764,0.826)	0.791(0.78,0.791)	0.532(0.515,0.565)

Table 34. The median (first quantile – third quantile) of the Mean Square Prediction Error (MSPE) and Correlation in 10 iterations group by Train and Test and the number of hybrid principal component analysis (HPCA) for parietal region of the brain.

HPCA	Time		
	User	System	Elapsed
1	160.14(158.207,162.27)	0.105(0.063,0.157)	160.295(158.4,162.435)
2	159.07(156.555,159.705)	0.11(0.102,0.173)	159.265(156.782,160.032)
3	227.29(220.58,229.47)	0.11(0.08,0.135)	227.58(220.783,229.675)
4	225.595(223.162,228.65)	0.11(0.072,0.127)	225.775(223.45,228.958)
5	297.525(294.877,303.31)	0.12(0.085,0.138)	297.81(295.153,303.695)
6	301.585(299.55,307.025)	0.115(0.095,0.158)	301.795(299.805,307.285)
7	384.135(379.835,388.283)	0.135(0.073,0.183)	384.375(380.238,388.653)
8	386.185(380.38,390.907)	0.15(0.12,0.16)	386.515(380.815,391.227)
9	456.045(453.145,457.408)	0.17(0.14,0.263)	456.595(453.745,457.962)
10	456.2(448.053,463.75)	0.19(0.133,0.222)	456.675(448.495,464.17)
11	453.405(448.482,456.342)	0.21(0.173,0.515)	453.94(449.292,456.845)
12	450.8(446.487,453.668)	0.22(0.133,0.235)	451.11(446.995,454.04)
13	499.48(491.625,502.94)	0.235(0.197,0.267)	499.955(492.095,503.4)
14	498.59(486.395,500.81)	0.195(0.17,0.237)	499.08(486.9,501.27)
15	510.865(509.472,514.317)	0.2(0.155,0.235)	511.375(509.9,514.893)
16	514.38(506.665,520.85)	0.17(0.147,0.185)	514.815(507.085,521.468)
17	517.37(514.19,528.412)	0.18(0.132,0.22)	517.895(514.682,528.905)
18	513.385(504.965,515.657)	0.145(0.125,0.23)	513.98(505.32,516.285)
19	517.205(513.13,524.365)	0.24(0.177,0.268)	517.665(513.75,524.915)
20	516.435(513.23,523.005)	0.21(0.172,0.237)	517.02(513.688,523.563)
21	526.8(520.613,529.625)	0.17(0.14,0.213)	528.565(521,530.113)
22	517.145(514.51,523.18)	0.23(0.163,0.278)	517.67(515.003,523.69)
23	448.505(440.025,494.533)	0.185(0.16,0.21)	449.235(440.368,495.132)
24	443.705(439.673,447.177)	0.19(0.132,0.243)	444.185(440.035,447.705)
25	410.84(380.967,437.235)	0.165(0.145,0.22)	411.32(381.42,437.587)
26	382.765(377.34,389.635)	0.165(0.153,0.235)	383.115(377.612,390.027)
27	377.44(368.298,382.185)	0.18(0.14,0.198)	377.735(368.545,382.47)
28	374.44(367.972,376.772)	0.18(0.17,0.228)	374.905(368.318,377.058)
29	382.755(380.878,384.09)	0.18(0.14,0.25)	383.15(381.285,384.665)
30	377.665(375.368,382.94)	0.165(0.132,0.185)	378.025(375.697,383.255)
31	378.645(371.202,382.71)	0.16(0.15,0.198)	379.02(371.505,383.045)
32	374.225(368.89,385.467)	0.125(0.1,0.185)	374.445(369.137,385.853)

Table 35. The median (first quantile – third quantile) of the Time (user-system-elapsed) in 10 iterations group by the number of hybrid principal component analysis (HPCA) for parietal region of the brain.

HPCA	Min_Error	Optimum					
		index	K	Tau	Lambda	Eta	Kappa
1	1.064(0.94,1.236)	4(4,7)	2.5(1.25,4.75)	100(100,100)	0.4(0.4,0.4)	0(0,0)	0(0,0)
2	1.012(0.837,1.083)	4(4,4)	2(1.25,2)	100(100,100)	0.4(0.4,0.4)	0(0,0)	0(0,0)
3	3.989(3.741,4.382)	4(3.25,4)	1(1,1.75)	100(32.5,100)	0.4(0.325,0.4)	0(0,0)	0(0,0)
4	4.257(3.748,4.817)	4(4,4)	1(1,1)	100(32.5,100)	0.4(0.325,0.4)	0(0,0)	0(0,0)
5	5.349(5.186,5.726)	4(4,4.75)	1(1,1)	100(32.5,100)	0.4(0.325,0.4)	0(0,0)	0(0,0)
6	4.858(4.741,5.097)	8(4,8)	1(1,1)	100(100,100)	0.4(0.4,0.4)	0(0,0)	0(0,0)
7	7.028(5.984,7.615)	5(3,7)	2(2,2)	10(10,10)	0.3(0.3,0.3)	0(0,0)	0(0,0)
8	6.396(6.124,6.66)	4.5(3.25,7)	2(1,2)	10(10,100)	0.3(0.3,0.4)	0(0,0)	0(0,0)
9	7.389(7.087,7.643)	4(3.25,6.25)	2(1.25,2)	10(10,100)	0.3(0.3,0.4)	0(0,0)	0(0,0)
10	7.391(7.24,7.465)	7(3.75,7)	1.5(1,2)	10(10,10)	0.3(0.3,0.3)	0(0,0)	0(0,0)
11	7.853(7.768,7.973)	3.5(3,4)	1(1,2)	55(10,100)	0.35(0.3,0.4)	0(0,0)	0(0,0)
12	7.858(7.663,8.192)	9(4,11)	1(1,2)	10(10,77.5)	0.3(0.3,0.375)	0.005(0,0.01)	0(0,0)
13	8.453(8.381,8.856)	5(1.25,7.75)	1(1,1.75)	10(0.325,10)	0.3(0.125,0.3)	0(0,0)	0(0,0)
14	8.579(8.357,8.916)	2(1,10.25)	1(1,1)	5.05(0.1,10)	0.2(0.1,0.3)	0(0,0.008)	0(0,0)
15	9.762(9.243,10.313)	3.5(2.25,4.75)	1(1,1)	10(1,10)	0.3(0.2,0.3)	0(0,0)	0(0,0)
16	9.653(9.222,10.073)	3(1.25,9.25)	1(1,1)	5.5(0.325,77.5)	0.25(0.125,0.375)	0(0,0.008)	0(0,0)
17	9.45(9.274,10.086)	5(2.25,10.25)	1(1,1)	10(1,100)	0.3(0.2,0.4)	0(0,0.008)	0(0,0)
18	9.749(9.456,9.943)	7(2.25,10.25)	1(1,1)	10(3.25,10)	0.3(0.225,0.3)	0(0,0.008)	0(0,0)
19	9.659(9.512,10.035)	7(2.25,10.25)	1(1,1)	10(3.25,10)	0.3(0.225,0.3)	0(0,0.008)	0(0,0)
20	9.393(9.19,9.431)	3(1,10.25)	1(1,1)	5.5(0.1,10)	0.25(0.1,0.3)	0(0,0.008)	0(0,0)
21	9.527(9.349,9.599)	4.5(1,11)	1(1,1)	5.05(0.1,10)	0.2(0.1,0.3)	0(0,0.01)	0(0,0)
22	9.713(9.231,10.08)	3(2.25,10)	1(1,1)	10(3.25,10)	0.3(0.225,0.3)	0(0,0.008)	0(0,0)
23	9.674(9.371,9.929)	3(1,7)	1(1,1)	10(0.1,77.5)	0.3(0.1,0.375)	0(0,0)	0(0,0)
24	9.778(9.338,9.851)	8(3.25,11)	1(1,1)	10(10,77.5)	0.3(0.3,0.375)	0(0,0.01)	0(0,0)
25	9.393(9.18,9.759)	11(8,11)	1(1,1)	10(10,10)	0.3(0.3,0.3)	0.01(0.003,0.01)	0(0,0)
26	9.538(9.379,10.089)	3.5(3,11)	1(1,1)	10(10,10)	0.3(0.3,0.3)	0(0,0.01)	0(0,0)
27	9.394(9.283,9.55)	7(2,11)	1(1,1)	10(1,10)	0.3(0.2,0.3)	0.005(0,0.01)	0(0,0)
28	9.794(9.473,10.005)	11(8,11)	1(1,1)	10(10,77.5)	0.3(0.3,0.375)	0.01(0,0.01)	0(0,0)
29	9.613(9.369,9.822)	2.5(1.25,3)	1(1,1)	5.5(0.325,10)	0.25(0.125,0.3)	0(0,0)	0(0,0)
30	9.608(9.256,9.91)	7.5(2.25,11)	1(1,1)	10(3.25,10)	0.3(0.225,0.3)	0(0,0.01)	0(0,0)
31	9.516(9.42,9.634)	2(1,9.5)	1(1,1)	0.55(0.1,7.75)	0.15(0.1,0.275)	0(0,0.008)	0(0,0)
32	9.6(9.318,10.235)	3.5(2,4)	1(1,1)	10(1,100)	0.3(0.2,0.4)	0(0,0)	0(0,0)

Table 36. The median (first quantile – third quantile) of the model fit indices in 10 iterations group by the number of hybrid principal component analysis (HPCA) for parietal region of the brain.

2.2.3.6 Left Temporal (5)

The MSPE and Correlation group by number of HPCA, Train and Test ([Table 37](#))

The User, System and Elapsed Time group by number of HPCA ([Table 38](#)): The median (first quantile – third quantile) of the Time (user-system-elapsed) in 10 iterations group by the number of hybrid principal component analysis (HPCA) for whole brain.

The Model fitting parameters group by number of HPCA ([Table 39](#)): The median (first quantile – third quantile) of the model fit indices in 10 iterations group by the number of hybrid principal component analysis (HPCA) for whole brain.

HPCA	MSPE		Correlation	
	Train	Test	Train	Test
1	0(0,0)	0.001(0.001,0.001)	1(1,1)	1(1,1)
2	0.035(0.015,0.05)	0.172(0.114,0.222)	0.983(0.974,0.992)	0.91(0.884,0.941)
3	0.028(0.013,0.059)	0.178(0.116,0.224)	0.986(0.97,0.993)	0.907(0.883,0.94)
4	0.155(0.079,0.202)	0.206(0.173,0.313)	0.919(0.893,0.96)	0.891(0.831,0.909)
5	0.203(0.131,0.243)	0.174(0.151,0.222)	0.893(0.87,0.932)	0.909(0.882,0.922)
6	0.191(0.137,0.191)	0.411(0.39,0.431)	0.9(0.899,0.929)	0.769(0.76,0.791)
7	0.178(0.138,0.193)	0.41(0.388,0.429)	0.906(0.898,0.928)	0.77(0.76,0.791)
8	0.249(0.15,0.255)	0.454(0.454,0.466)	0.866(0.863,0.922)	0.739(0.731,0.739)
9	0.253(0.24,0.255)	0.436(0.393,0.456)	0.864(0.863,0.872)	0.751(0.738,0.779)
10	0.279(0.258,0.298)	0.569(0.542,0.578)	0.849(0.838,0.862)	0.657(0.651,0.684)
11	0.322(0.246,0.427)	0.513(0.495,0.533)	0.824(0.757,0.869)	0.705(0.694,0.717)
12	0.38(0.358,0.484)	0.55(0.517,0.565)	0.787(0.719,0.801)	0.671(0.66,0.696)
13	0.433(0.3,0.48)	0.546(0.493,0.546)	0.753(0.721,0.836)	0.676(0.676,0.713)
14	0.452(0.331,0.5)	0.594(0.59,0.594)	0.739(0.707,0.818)	0.641(0.639,0.641)
15	0.411(0.381,0.429)	0.578(0.571,0.595)	0.768(0.756,0.787)	0.651(0.639,0.656)
16	0.428(0.348,0.494)	0.611(0.578,0.63)	0.757(0.711,0.807)	0.629(0.614,0.652)
17	0.425(0.4,0.454)	0.624(0.582,0.645)	0.758(0.739,0.775)	0.619(0.599,0.65)
18	0.472(0.427,0.52)	0.625(0.58,0.642)	0.726(0.693,0.757)	0.618(0.603,0.65)
19	0.434(0.4,0.503)	0.632(0.602,0.643)	0.752(0.705,0.775)	0.608(0.602,0.631)
20	0.486(0.427,0.522)	0.638(0.591,0.64)	0.717(0.691,0.757)	0.608(0.603,0.642)
21	0.424(0.401,0.484)	0.642(0.579,0.649)	0.759(0.747,0.774)	0.603(0.596,0.651)
22	0.43(0.428,0.444)	0.607(0.584,0.639)	0.755(0.745,0.756)	0.629(0.604,0.647)
23	0.437(0.428,0.522)	0.62(0.579,0.648)	0.75(0.701,0.756)	0.619(0.597,0.651)
24	0.43(0.425,0.454)	0.602(0.575,0.637)	0.755(0.739,0.758)	0.631(0.606,0.655)
25	0.43(0.403,0.443)	0.594(0.577,0.605)	0.755(0.746,0.772)	0.642(0.631,0.654)
26	0.428(0.422,0.499)	0.611(0.571,0.638)	0.756(0.707,0.76)	0.626(0.606,0.659)
27	0.475(0.428,0.522)	0.633(0.571,0.633)	0.724(0.691,0.756)	0.609(0.609,0.658)
28	0.427(0.422,0.5)	0.612(0.574,0.641)	0.757(0.707,0.76)	0.624(0.602,0.656)
29	0.423(0.42,0.44)	0.574(0.57,0.605)	0.759(0.749,0.761)	0.654(0.631,0.657)
30	0.426(0.422,0.496)	0.6(0.573,0.633)	0.757(0.709,0.76)	0.634(0.61,0.657)
31	0.423(0.405,0.497)	0.614(0.574,0.634)	0.759(0.708,0.772)	0.625(0.608,0.656)
32	0.474(0.427,0.521)	0.607(0.573,0.634)	0.724(0.692,0.757)	0.631(0.608,0.657)

Table 37. The median (first quantile – third quantile) of the Mean Square Prediction Error (MSPE) and Correlation in 10 iterations group by Train and Test and the number of hybrid principal component analysis (HPCA) for left temporal region of the brain.

HPCA	Time		
	User	System	Elapsed
1	85.22(82.24,88.103)	0.085(0.072,0.18)	85.285(82.385,88.298)
2	109.025(107.6,109.975)	0.05(0.033,0.1)	109.12(107.698,110.093)
3	152.06(148.175,154.522)	0.085(0.06,0.118)	152.39(148.273,154.65)
4	159.19(157.465,162.08)	0.075(0.06,0.088)	159.3(157.608,162.18)
5	145.71(144.542,148.22)	0.08(0.063,0.097)	145.815(144.602,148.353)
6	211.98(209.11,214.64)	0.115(0.083,0.158)	212.335(209.238,214.767)
7	197.115(195.108,202.04)	0.13(0.065,0.168)	197.225(195.288,202.237)
8	260.785(255.61,262.813)	0.115(0.085,0.15)	260.945(255.775,262.975)
9	239.12(235.985,245.053)	0.12(0.095,0.137)	239.295(236.23,245.58)
10	294.78(293.9,300.085)	0.135(0.11,0.165)	295.025(294.45,300.273)
11	284.495(279.705,287.048)	0.12(0.093,0.165)	284.75(279.882,287.257)
12	340.275(336.577,342.192)	0.14(0.132,0.155)	340.51(336.818,342.448)
13	328.79(325.352,333.43)	0.13(0.1,0.165)	329.065(325.553,333.665)
14	343.705(339.407,348.03)	0.145(0.103,0.168)	343.935(339.817,348.188)
15	343.96(341.745,346.987)	0.12(0.103,0.137)	344.195(341.932,347.185)
16	368.495(365.69,371.485)	0.115(0.09,0.168)	368.755(365.975,371.655)
17	372.245(368.833,377.332)	0.15(0.118,0.167)	372.48(369.087,377.605)
18	371.535(369.907,374.832)	0.135(0.113,0.16)	371.755(370.12,375.06)
19	371.565(366.975,376.44)	0.14(0.1,0.158)	371.755(367.173,376.758)
20	371.87(365.462,375.325)	0.15(0.112,0.16)	372.07(365.8,375.518)
21	369.815(367.247,378.235)	0.12(0.11,0.155)	370.055(367.9,378.382)
22	371.575(370.975,373.905)	0.155(0.14,0.21)	371.86(371.255,374.35)
23	366.97(361.977,371.255)	0.135(0.123,0.14)	367.155(362.295,371.45)
24	366.125(363.695,370.655)	0.17(0.143,0.242)	366.335(363.942,370.962)
25	369.765(367.602,373.958)	0.14(0.125,0.17)	370.06(367.815,374.215)
26	379.255(372.052,384.143)	0.12(0.087,0.137)	379.455(372.287,384.585)
27	369.96(368.803,373.07)	0.16(0.123,0.17)	370.125(369.037,373.43)
28	379.885(374.61,382.557)	0.17(0.13,0.197)	380.115(374.832,383.03)
29	378.04(373.005,380.638)	0.16(0.132,0.178)	378.335(373.4,380.917)
30	377.445(366.725,381.855)	0.175(0.137,0.205)	377.785(367.048,382.265)
31	374.415(369.205,379.86)	0.13(0.12,0.195)	374.705(369.527,380.072)
32	371.06(366.777,378.42)	0.16(0.125,0.178)	371.315(367.035,378.672)

Table 38. The median (first quantile – third quantile) of the Time (user-system-elapsed) in 10 iterations group by the number of hybrid principal component analysis (HPCA) for left temporal region of the brain.

HPCA	Min_Error	Optimum					
		index	K	Tau	Lambda	Eta	Kappa
1	0.006(0.005,0.007)	5(4,8)	2(2,3)	100(25.075,100)	0.4(0.175,0.4)	0(0,0)	0(0,0)
2	1.415(1.343,1.713)	9(4,12)	1(1,1)	100(10,100)	0.4(0.3,0.4)	0.005(0,0.01)	0(0,0)
3	1.677(1.494,1.932)	3.5(3,12)	1(1,1)	55(10,100)	0.35(0.3,0.4)	0(0,0.01)	0(0,0)
4	7.154(6.448,7.419)	4(4,4)	1.5(1,2.75)	100(100,100)	0.4(0.4,0.4)	0(0,0)	0(0,0)
5	5.96(5.559,7.008)	8(4,12)	1(1,1.75)	100(100,100)	0.4(0.4,0.4)	0.005(0,0.01)	0(0,0)
6	7.504(7.116,8.419)	4(4,4)	2(2,3)	100(100,100)	0.4(0.4,0.4)	0(0,0)	0(0,0)
7	8.101(7.478,8.806)	4(4,10)	2.5(2,3)	100(100,100)	0.4(0.4,0.4)	0(0,0.008)	0(0,0)
8	9.462(8.715,10.392)	4(1.25,4)	2(2,2)	100(0.325,100)	0.4(0.125,0.4)	0(0,0)	0(0,0)
9	9.851(9.118,10.531)	4(4,12)	2(2,2.75)	100(100,100)	0.4(0.4,0.4)	0(0,0.01)	0(0,0)
10	9.943(9.804,10.184)	8(4,12)	3(3,3)	100(100,100)	0.4(0.4,0.4)	0.005(0,0.01)	0(0,0)
11	10.151(9.916,10.718)	4(4,10)	2(1,3)	100(100,100)	0.4(0.4,0.4)	0(0,0.008)	0(0,0)
12	11.566(11.277,11.967)	4(1.75,4)	1(1,2)	100(25.075,100)	0.4(0.175,0.4)	0(0,0)	0(0,0)
13	11.326(10.992,12.048)	4(4,4)	1(1,2.75)	100(100,100)	0.4(0.4,0.4)	0(0,0)	0(0,0)
14	11.241(10.954,11.54)	4(4,4)	1(1,2)	100(100,100)	0.4(0.4,0.4)	0(0,0)	0(0,0)
15	11.455(11.111,11.86)	4(3.25,4)	1.5(1,2)	100(32.5,100)	0.4(0.325,0.4)	0(0,0)	0(0,0)
16	12.005(11.844,12.357)	4(1,4)	1(1,2.5)	100(0.1,100)	0.4(0.1,0.4)	0(0,0)	0(0,0)
17	11.657(11.452,12.376)	4(1.75,4)	1(1,2)	100(2.575,100)	0.4(0.15,0.4)	0(0,0)	0(0,0)
18	12.144(11.743,12.628)	4(1.25,4)	1(1,1)	100(0.325,100)	0.4(0.125,0.4)	0(0,0)	0(0,0)
19	12.109(11.637,12.573)	4(3,7)	1(1,2.5)	100(10,100)	0.4(0.3,0.4)	0(0,0)	0(0,0)
20	12.622(11.901,12.717)	4(2.5,4)	1(1,1)	100(25.75,100)	0.4(0.25,0.4)	0(0,0)	0(0,0)
21	12.871(12.405,13.751)	3(1.25,4)	1.5(1,2)	50.5(0.325,100)	0.3(0.125,0.4)	0(0,0)	0(0,0)
22	12.097(12.048,12.454)	3.5(1.5,4)	1(1,1.75)	55(2.575,100)	0.35(0.15,0.4)	0(0,0)	0(0,0)
23	12.474(11.857,13.081)	3.5(1,4)	1(1,1.75)	55(0.1,100)	0.35(0.1,0.4)	0(0,0)	0(0,0)
24	11.861(11.441,12.459)	3.5(1,4)	1(1,1.75)	55(0.1,100)	0.35(0.1,0.4)	0(0,0)	0(0,0)
25	12.064(11.699,12.139)	3.5(1.25,4)	1(1,2)	55(0.325,100)	0.35(0.125,0.4)	0(0,0)	0(0,0)
26	12.856(12.322,13.023)	3.5(1,4)	1(1,1.75)	55(0.1,100)	0.35(0.1,0.4)	0(0,0)	0(0,0)
27	12.602(11.965,12.783)	4(1,4)	1(1,1)	100(0.1,100)	0.4(0.1,0.4)	0(0,0)	0(0,0)
28	12.137(11.7,13.283)	3.5(1.25,4)	1(1,1.75)	55(0.325,100)	0.35(0.125,0.4)	0(0,0)	0(0,0)
29	11.8(11.454,12.45)	1.5(1,3.75)	1(1,1)	0.55(0.1,77.5)	0.15(0.1,0.375)	0(0,0)	0(0,0)
30	12.408(12.038,12.576)	3.5(1,4)	1(1,1)	55(0.1,100)	0.35(0.1,0.4)	0(0,0)	0(0,0)
31	12.383(12.078,12.646)	4(1.25,4)	1(1,1.75)	100(0.325,100)	0.4(0.125,0.4)	0(0,0)	0(0,0)
32	12.327(11.821,12.582)	3(1,4)	1(1,1)	50.5(0.1,100)	0.3(0.1,0.4)	0(0,0)	0(0,0)

Table 39. The median (first quantile – third quantile) of the model fit indices in 10 iterations group by the number of hybrid principal component analysis (HPCA) for left temporal region of the brain.

2.2.3.7 Right Temporal (6)

The MSPE and Correlation group by number of HPCA, Train and Test ([Table 40](#))

The User, System and Elapsed Time group by number of HPCA ([Table 41](#)): The median (first quantile – third quantile) of the Time (user-system-elapsed) in 10 iterations group by the number of hybrid principal component analysis (HPCA) for whole brain.

The Model fitting parameters group by number of HPCA ([Table 42](#)): The median (first quantile – third quantile) of the model fit indices in 10 iterations group by the number of hybrid principal component analysis (HPCA) for whole brain.

HPCA	MSPE		Correlation	
	Train	Test	Train	Test
1	0.016(0.005,0.025)	0.193(0.128,0.209)	0.992(0.988,0.998)	0.899(0.89,0.935)
2	0.016(0.016,0.075)	0.378(0.209,0.401)	0.992(0.962,0.992)	0.801(0.789,0.893)
3	0.133(0.104,0.152)	0.577(0.532,0.675)	0.931(0.921,0.947)	0.678(0.619,0.705)
4	0.263(0.263,0.263)	0.351(0.351,0.351)	0.858(0.858,0.858)	0.81(0.81,0.81)
5	0.395(0.378,0.395)	0.468(0.468,0.525)	0.778(0.778,0.789)	0.733(0.7,0.733)
6	0.4(0.392,0.411)	0.567(0.566,0.574)	0.774(0.767,0.78)	0.674(0.67,0.676)
7	0.387(0.378,0.477)	0.835(0.76,0.879)	0.783(0.727,0.789)	0.443(0.425,0.541)
8	0.348(0.342,0.359)	0.856(0.751,0.891)	0.807(0.801,0.811)	0.468(0.447,0.551)
9	0.48(0.427,0.54)	0.774(0.651,0.86)	0.72(0.678,0.757)	0.498(0.473,0.615)
10	0.443(0.434,0.53)	0.832(0.7,0.849)	0.746(0.698,0.752)	0.468(0.467,0.563)
11	0.524(0.462,0.57)	0.779(0.718,0.927)	0.693(0.655,0.733)	0.502(0.447,0.564)
12	0.482(0.437,0.502)	0.793(0.749,0.842)	0.727(0.723,0.751)	0.496(0.447,0.545)
13	0.633(0.515,0.633)	0.944(0.929,0.952)	0.615(0.615,0.704)	0.33(0.318,0.351)
14	0.616(0.608,0.631)	0.795(0.774,0.916)	0.619(0.615,0.629)	0.511(0.339,0.522)
15	0.676(0.634,0.75)	0.979(0.815,0.993)	0.577(0.51,0.607)	0.393(0.299,0.439)
16	0.625(0.558,0.648)	0.979(0.85,1.003)	0.619(0.6,0.664)	0.394(0.392,0.411)
17	0.62(0.562,0.746)	0.985(0.832,0.985)	0.616(0.514,0.661)	0.4(0.4,0.423)
18	0.619(0.562,0.626)	0.908(0.79,0.986)	0.616(0.611,0.662)	0.408(0.396,0.512)
19	0.563(0.557,0.563)	0.987(0.987,1.005)	0.661(0.661,0.665)	0.397(0.387,0.4)
20	0.62(0.563,0.667)	0.976(0.8,0.988)	0.615(0.583,0.66)	0.398(0.363,0.49)
21	0.641(0.629,0.679)	0.904(0.792,0.985)	0.603(0.575,0.609)	0.411(0.333,0.495)
22	0.591(0.558,0.66)	0.988(0.846,0.99)	0.638(0.589,0.664)	0.396(0.362,0.418)
23	0.619(0.563,0.665)	0.975(0.797,0.982)	0.616(0.584,0.66)	0.402(0.397,0.492)
24	0.621(0.577,0.665)	0.907(0.793,0.983)	0.615(0.584,0.65)	0.41(0.396,0.51)
25	0.562(0.557,0.621)	0.983(0.788,1.004)	0.662(0.615,0.665)	0.402(0.397,0.513)
26	0.65(0.627,0.719)	0.837(0.781,0.952)	0.595(0.538,0.61)	0.42(0.404,0.518)
27	0.66(0.633,0.721)	0.866(0.84,1.007)	0.591(0.536,0.615)	0.396(0.297,0.417)
28	0.558(0.547,0.558)	1.022(1.006,1.038)	0.664(0.664,0.673)	0.387(0.371,0.391)
29	0.59(0.554,0.663)	0.999(0.879,1.028)	0.64(0.586,0.667)	0.387(0.344,0.408)
30	0.618(0.559,0.674)	1.005(0.993,1.031)	0.617(0.579,0.664)	0.378(0.301,0.391)
31	0.588(0.559,0.642)	1.009(0.999,1.031)	0.641(0.605,0.663)	0.386(0.317,0.389)
32	0.643(0.617,0.676)	0.996(0.803,1.015)	0.6(0.577,0.618)	0.344(0.293,0.505)

Table 40. The median (first quantile – third quantile) of the Mean Square Prediction Error (MSPE) and Correlation in 10 iterations group by Train and Test and the number of hybrid principal component analysis (HPCA) for right temporal region of the brain.

HPCA	Time		
	User	System	Elapsed
1	129.065(127.987,129.665)	0.105(0.08,0.148)	129.24(128.157,129.772)
2	131.62(129.708,133.14)	0.075(0.043,0.108)	131.7(129.818,133.313)
3	188.73(187.417,190.437)	0.095(0.053,0.118)	188.795(187.573,190.62)
4	185.265(182.867,185.865)	0.095(0.065,0.107)	185.405(183.045,186.02)
5	242.205(239.267,246.69)	0.085(0.065,0.105)	242.405(239.465,247.047)
6	257.465(254.97,260.955)	0.11(0.075,0.132)	257.69(255.147,261.188)
7	283.29(280.975,286.67)	0.1(0.06,0.12)	283.525(281.355,286.877)
8	281.58(276.673,282.965)	0.135(0.09,0.155)	281.795(276.958,283.393)
9	319.79(316.832,322.405)	0.145(0.11,0.212)	320.04(317.158,322.702)
10	321.48(317.448,322.82)	0.145(0.132,0.182)	321.75(317.905,323.117)
11	326.51(324.083,327.66)	0.16(0.103,0.185)	326.7(324.445,327.955)
12	326.705(323.975,329.02)	0.14(0.12,0.17)	326.975(324.205,329.268)
13	361.53(355.472,364.057)	0.14(0.13,0.155)	361.78(355.842,364.308)
14	354.99(350.935,358.965)	0.145(0.115,0.175)	355.265(351.178,359.255)
15	377.985(375.518,382.927)	0.135(0.11,0.14)	378.175(375.838,383.157)
16	376.9(372.323,378.96)	0.155(0.117,0.17)	377.15(372.575,379.31)
17	371.96(370.3,376.8)	0.17(0.145,0.197)	372.39(370.755,377.015)
18	373.455(369.725,377.683)	0.16(0.142,0.168)	373.705(369.98,378.09)
19	374.11(370.278,375.537)	0.165(0.142,0.253)	374.39(370.553,376.165)
20	372.82(367.46,391.758)	0.17(0.118,0.208)	373.38(367.85,392.828)
21	375.65(371.742,382.527)	0.16(0.132,0.185)	376.06(372.17,383.33)
22	404.245(401.195,411.692)	0.205(0.155,0.257)	405.55(402.545,413.245)
23	404.135(397.807,409.91)	0.195(0.172,0.27)	405.455(399.332,411.695)
24	425(422.528,429.973)	0.265(0.25,0.295)	427.77(424.772,432.645)
25	428.22(418.313,430.28)	0.3(0.28,0.317)	430.76(420.25,432.757)
26	425.335(420.537,429.565)	0.265(0.212,0.297)	427.515(422.198,432.053)
27	426.23(420.87,434.488)	0.195(0.173,0.207)	428.8(422.722,436.47)
28	436.95(432.378,441.75)	0.295(0.24,0.308)	439.425(434.625,443.858)
29	439.225(437.29,443.38)	0.3(0.275,0.325)	441.79(439.798,445.598)
30	440.795(435.11,444.865)	0.315(0.222,0.405)	443.65(437.67,447.873)
31	438.175(434.988,449.245)	0.275(0.215,0.347)	440.525(436.997,452.593)
32	413.435(405.858,421.507)	0.315(0.275,0.358)	415.93(408.315,424.407)

Table 41. The median (first quantile – third quantile) of the Time (user-system-elapsed) in 10 iterations group by the number of hybrid principal component analysis (HPCA) for right temporal region of the brain

HPCA	Min.Error	Optimum					
		index	K	Tau	Lambda	Eta	Kappa
1	2.163(1.784,2.666)	4(4,7)	3(2,5)	100(100,100)	0.4(0.4,0.4)	0(0,0)	0(0,0)
2	3.493(3.206,3.895)	8(5,9)	1(1,1)	0.1(0.1,100)	0.1(0.1,0.4)	0(0,0.01)	0(0,0)
3	7.92(7.159,8.081)	4(3.25,4)	1(1,1)	100(10,100)	0.4(0.3,0.4)	0(0,0)	0(0,0)
4	7.044(6.865,7.283)	12(12,12)	1(1,1)	100(100,100)	0.4(0.4,0.4)	0.01(0.01,0.01)	0(0,0)
5	10.212(9.271,10.466)	12(12,12)	1(1,1.75)	100(100,100)	0.4(0.4,0.4)	0.01(0.01,0.01)	0(0,0)
6	9.244(8.908,9.742)	12(8,12)	1(1,1)	100(100,100)	0.4(0.4,0.4)	0.01(0,0.01)	0(0,0)
7	10.368(9.767,10.924)	11(11,12)	1(1,1.75)	10(10,100)	0.3(0.3,0.4)	0.01(0.01,0.01)	0(0,0)
8	9.538(9.466,9.615)	7.5(3.25,10)	1(1,1)	10(3.25,77.5)	0.3(0.225,0.375)	0(0,0.01)	0(0,0)
9	10.974(10.683,11.17)	7.5(3.25,11.75)	1(1,1)	55(10,100)	0.35(0.3,0.4)	0.005(0,0.01)	0(0,0)
10	10.918(10.814,11.297)	5(3,8)	1(1,1.75)	10(10,77.5)	0.3(0.3,0.375)	0(0,0)	0(0,0.008)
11	11.343(11.184,11.869)	9.5(3,12)	1(1,1)	55(10,100)	0.35(0.3,0.4)	0.005(0,0.01)	0(0,0)
12	11.105(10.876,11.748)	9.5(4.75,11.75)	2(1,2)	55(10,100)	0.35(0.3,0.4)	0.005(0,0.01)	0(0,0.008)
13	12.44(12.28,12.601)	1.5(1,2.75)	1(1,1)	0.55(0.1,7.75)	0.15(0.1,0.275)	0(0,0)	0.01(0.01,0.01)
14	12.457(12.298,12.644)	6.5(4.25,11)	1(1,1)	100(25.075,100)	0.4(0.175,0.4)	0(0,0.008)	0(0,0.01)
15	12.903(12.622,13.16)	9.5(5,11)	1(1,1)	55(10,100)	0.35(0.3,0.4)	0.005(0,0.01)	0.01(0.003,0.01)
16	12.606(12.482,12.743)	7.5(7,10.25)	1(1,1)	10(10,100)	0.3(0.3,0.4)	0(0,0.008)	0(0,0.01)
17	12.842(12.67,13.169)	11(8.25,11)	1(1,1)	55(10,100)	0.35(0.3,0.4)	0.01(0.003,0.01)	0(0,0.01)
18	12.482(12.211,12.73)	7.5(4,11)	1(1,1)	55(10,100)	0.35(0.3,0.4)	0(0,0.01)	0(0,0)
19	12.458(12.098,12.665)	11(7,11)	1(1,1)	10(10,10)	0.3(0.3,0.3)	0.01(0,0.01)	0(0,0.008)
20	12.547(12.351,12.934)	11(5.75,11)	1(1,1)	10(10,100)	0.3(0.3,0.4)	0.01(0.003,0.01)	0(0,0.008)
21	12.585(12.276,12.861)	11.5(11,12)	1(1,1)	55(10,100)	0.35(0.3,0.4)	0.01(0.01,0.01)	0.005(0,0.01)
22	12.882(12.423,13.175)	5.5(4,10)	1(1,1)	55(10,100)	0.35(0.3,0.4)	0(0,0.008)	0(0,0.01)
23	12.717(12.582,12.8)	11(4.75,11)	1(1,1)	10(10,100)	0.3(0.3,0.4)	0.01(0,0.01)	0(0,0.008)
24	12.738(12.112,12.913)	8(4.75,11)	1(1,1)	100(10,100)	0.4(0.3,0.4)	0(0,0.01)	0(0,0.008)
25	12.371(12.063,12.813)	9.5(7,11)	1(1,1)	10(10,100)	0.3(0.3,0.4)	0.005(0,0.01)	0(0,0)
26	12.879(12.612,13.23)	11.5(7.25,12)	1(1,1)	100(32.5,100)	0.4(0.325,0.4)	0.01(0,0.01)	0.005(0,0.01)
27	12.676(12.547,13.261)	8(4.75,10.75)	1(1,1)	100(10,100)	0.4(0.3,0.4)	0(0,0.01)	0.01(0.01,0.01)
28	12.516(11.852,12.686)	9.5(7.25,11)	1(1,1)	10(10,10)	0.3(0.3,0.3)	0.005(0,0.01)	0(0,0)
29	12.722(12.062,13.04)	11(8.75,11.75)	1(1,1)	10(10,100)	0.3(0.3,0.4)	0.01(0.003,0.01)	0(0,0.01)
30	12.605(12.376,12.915)	8.5(7,11)	1(1,1)	10(10,10)	0.3(0.3,0.3)	0.005(0,0.01)	0(0,0.01)
31	12.343(12.085,12.642)	7(4,11)	1(1,1)	10(10,10)	0.3(0.3,0.3)	0(0,0.01)	0(0,0.008)
32	12.809(12.416,13.032)	9(4,11)	1(1,1)	55(10,100)	0.35(0.3,0.4)	0.005(0,0.01)	0.005(0,0.01)

Table 42. The median (first quantile – third quantile) of the model fit indices in 10 iterations group by the number of hybrid principal component analysis (HPCA) for right temporal region of the brain.

2.2.3.8 Figures

The MSPE over Number of HPCA group by Regions: **Figure 30**

The Correlation over Number of HPCA group by Regions: **Figure 31**

The Elapsed Time over Number of HPCA group by Regions: **Figure 32**

The MSPE over Number of HPCA group by Regions

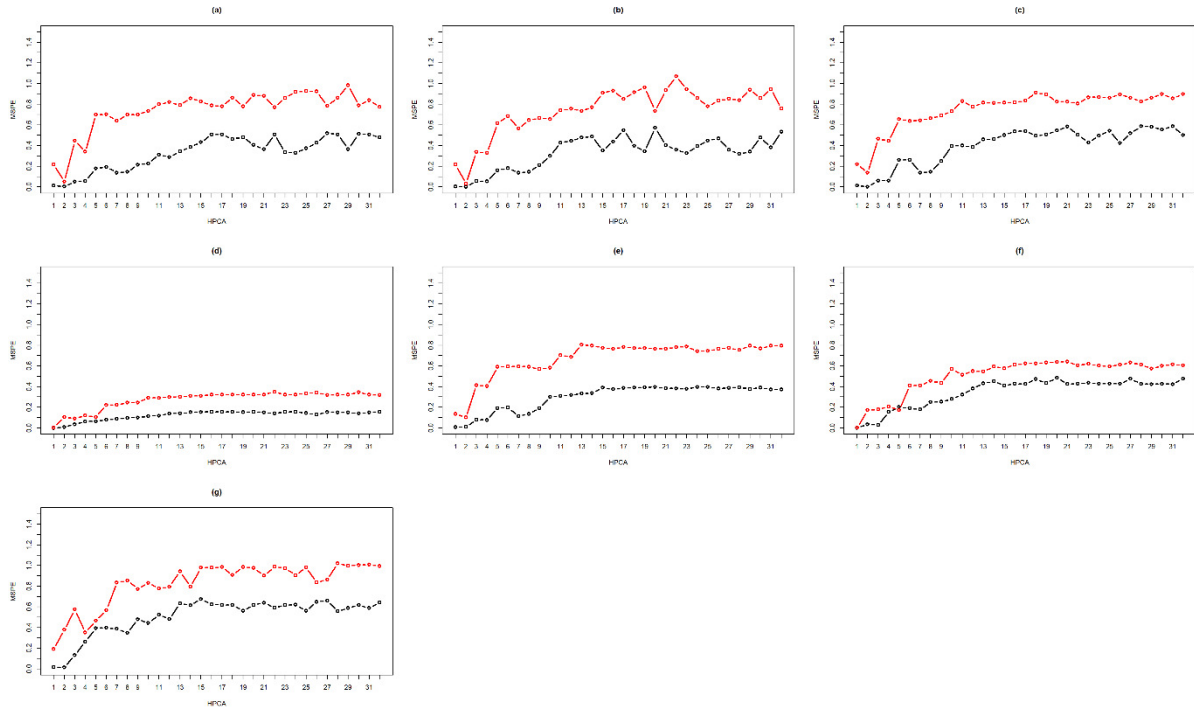


Figure 30. (a-g) Median of mean square prediction error (MSPE) for train (black solid line) and test (red solid line) in 10 iterations of (a) whole, (b) frontal, (c) central, (d) occipital, (e) parietal, (f) left temporal and (g) right temporal regions, respectively.

The Correlation over Number of HPCA group by Regions

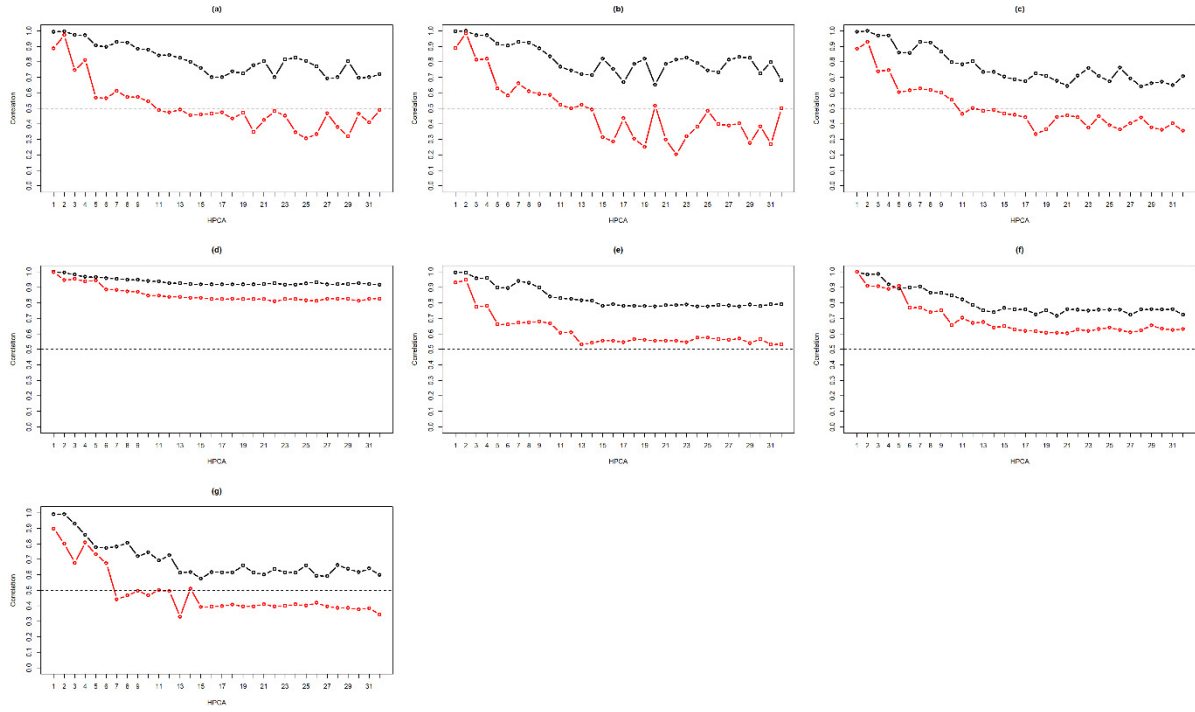


Figure 31. (a-g) Median of correlation for train (black solid line) and test (red solid line) in 10 iterations of (a) whole, (b) frontal, (c) central, (d) occipital, (e) parietal, (f) left temporal and (g) right temporal regions, respectively.

The Elapsed Time over Number of HPCA group by Regions

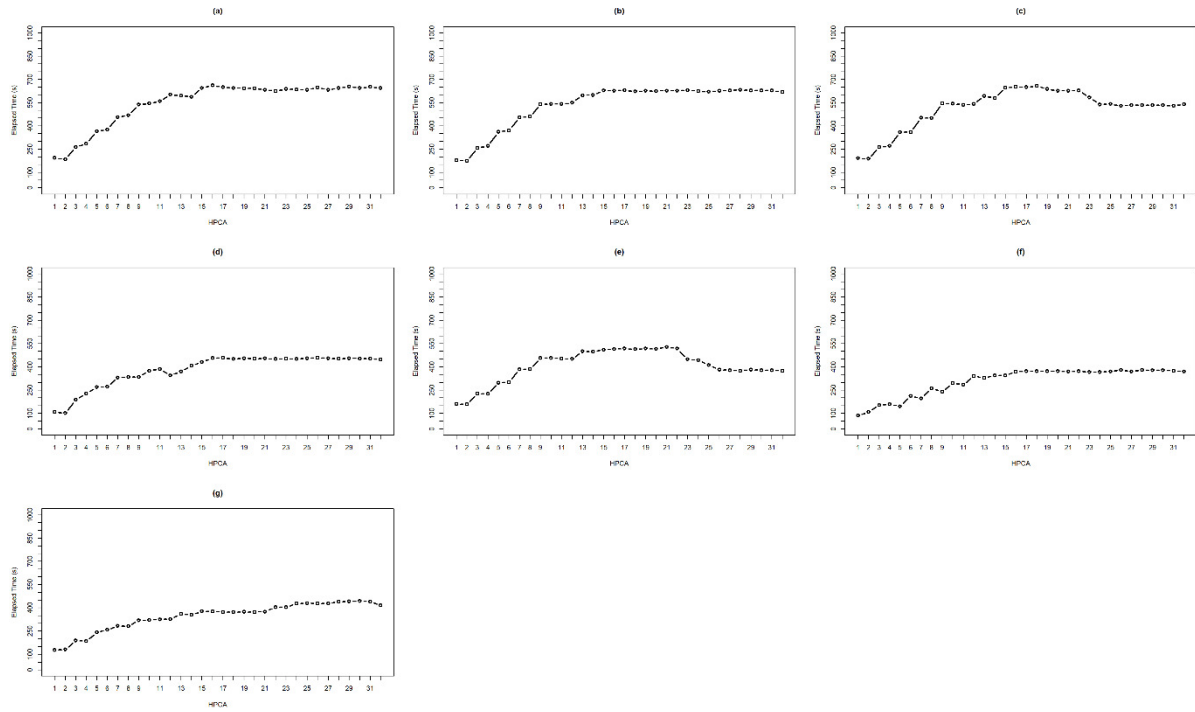


Figure 32. (a-g) Median of elapsed time in 10 iterations of (a) whole, (b) frontal, (c) central, (d) occipital, (e) parietal, (f) left temporal and (g) right temporal regions, respectively.

2.2.4. Conclusion .

According to the [Figure 30](#), the MSPE increased by adding HPCA from 1 to KLM in all regions. It also indicates that the difference between training and testing exists. The [Figure 31](#) shows the correlation. The highest correlation is in the first HPCA in all regions fro both training and testing datasets. The elapsed time is also increasing by adding HPCA. ([Figure 32](#))

We conclude that the first,second or third HPCA has the best performance based on the MSPE, correlation and elapsed times in all region of the brain. It shows that in some regions the second HPCA has the best performance than others.

3. The Visual Odd Ball Task

3.1. Standard Stimuli

3.1.1. The HPCA Decomposition

3.1.1.1 The Functional dimension

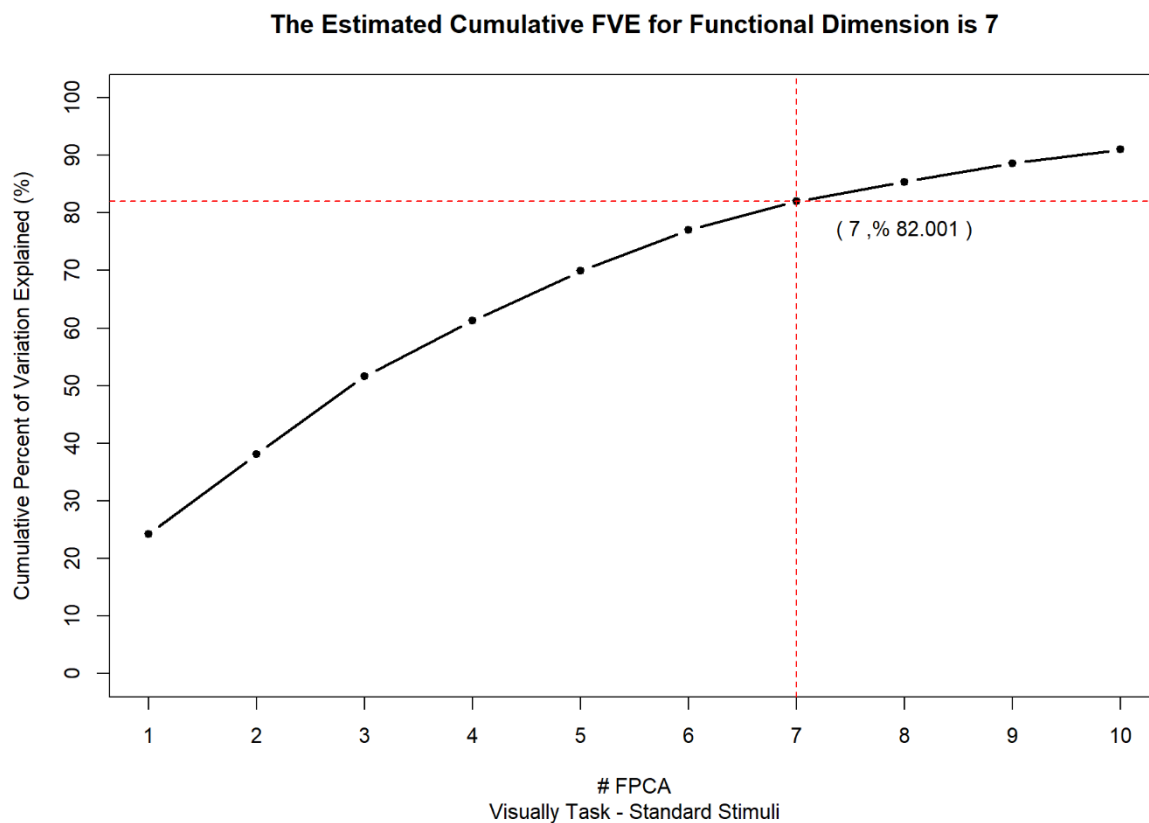


Figure 33. The estimated cumulative fraction of variations (FVE) for functional dimension.

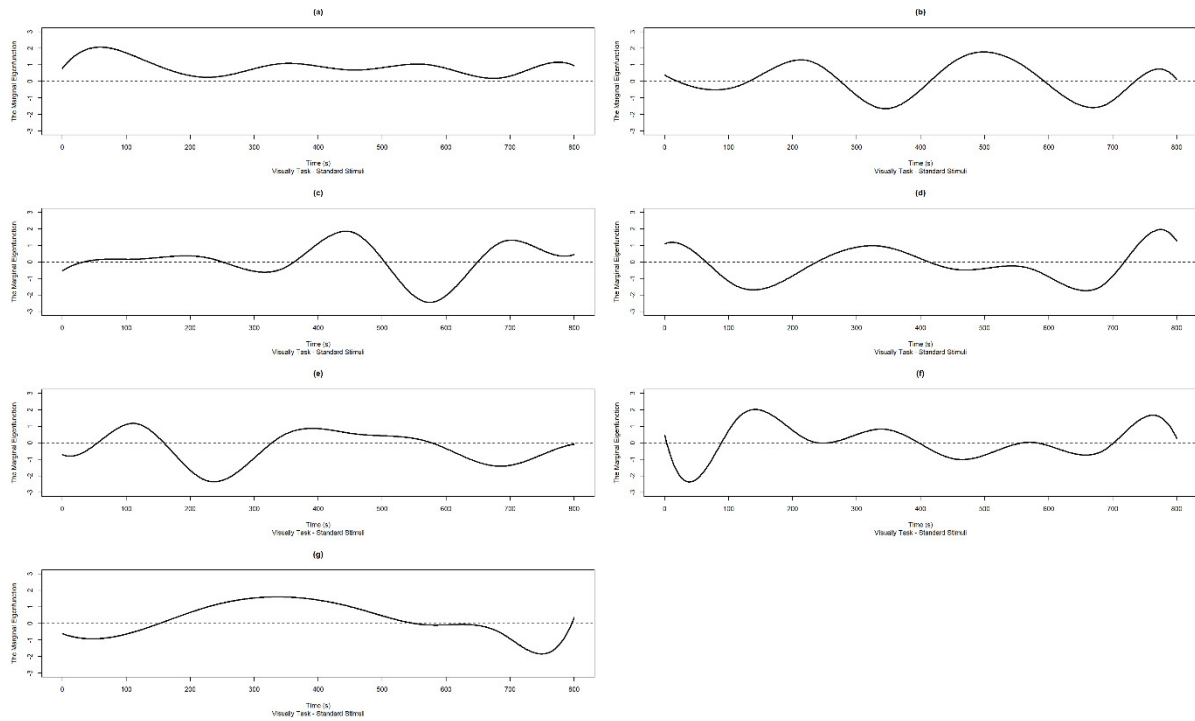


Figure 34. (a, g) Estimated first to seven leading functional marginal eigenfunctions.

According to the [Figure 33](#), the first seven marginal functional principal component analysis (FPCA) capture at least 85% of variation in this dimension. The [Figure 34](#) plots estimated first to seven leading functional marginal eigenfunctions on the x-axis (time).

3.1.1.2 The Longitudinal dimension

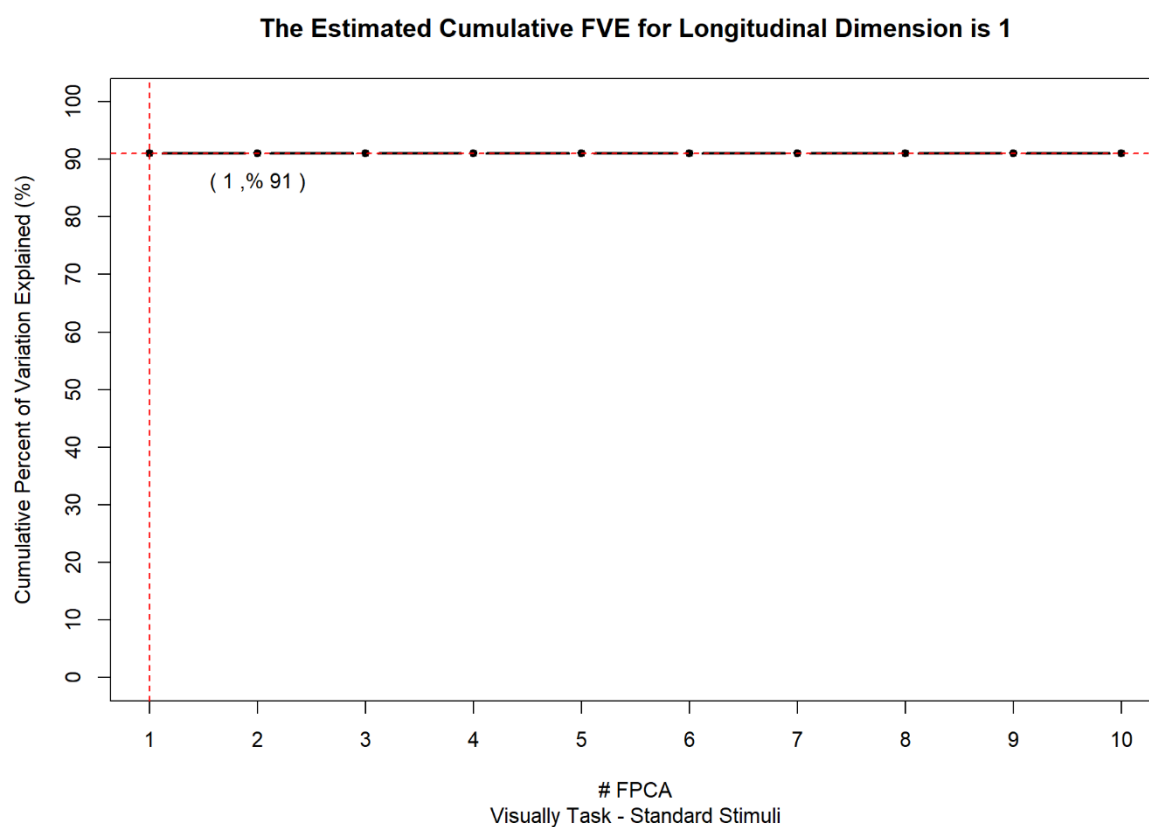


Figure 35. The estimated cumulative fraction of variations (FVE) for longitudinal dimension.

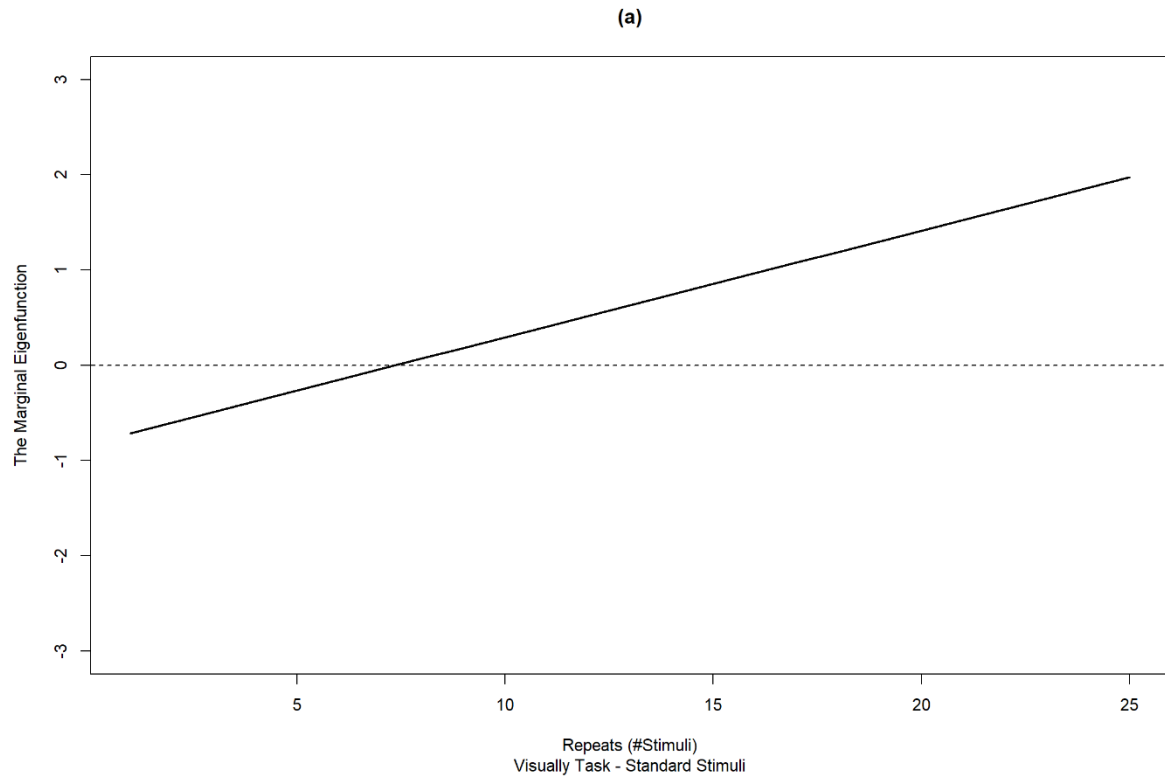


Figure 36. Estimated first leading longitudinal marginal eigenfunctions.

According to the [Figure 35](#), the first marginal functional principal component analysis (FPCA) capture at least 90% of variation in this dimension. The [Figure 36](#) plots estimated first leading longitudinal marginal eigenfunctions on the receptions.

3.1.1.3 The Regional dimension

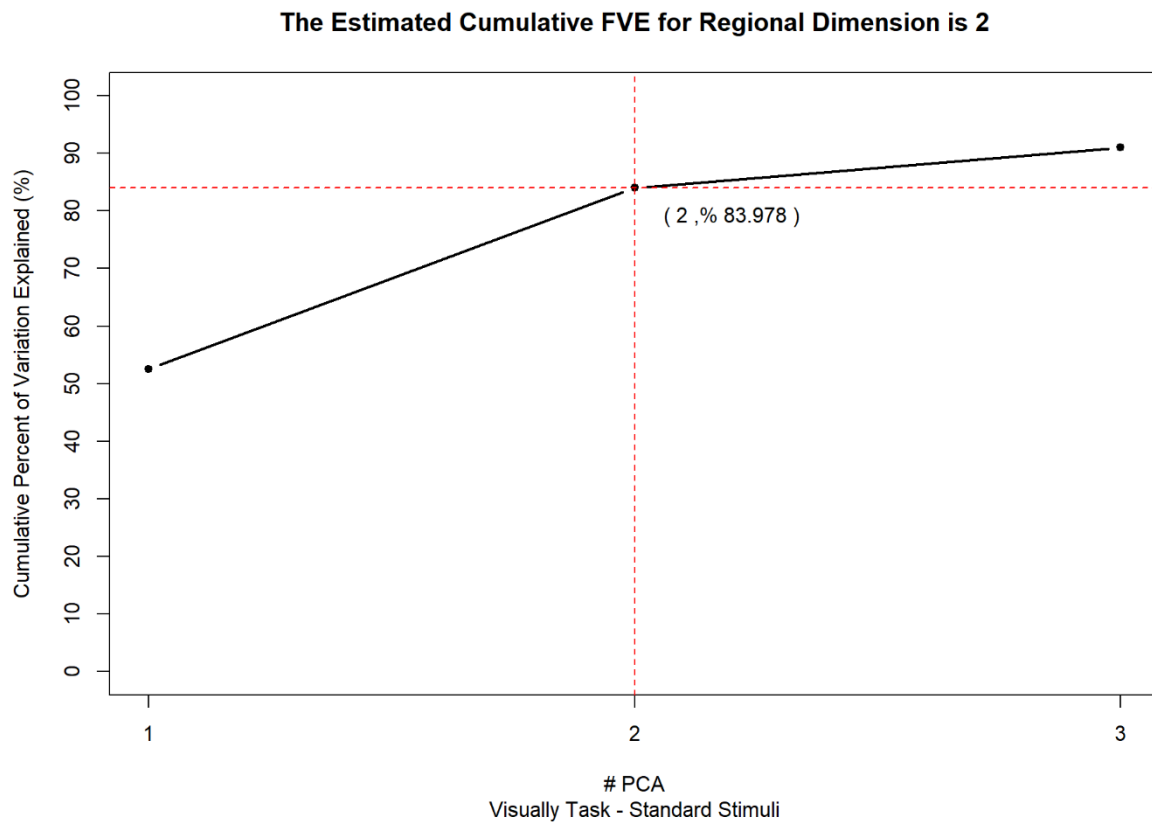


Figure 37. The estimated cumulative fraction of variations (FVE) for regional dimension.

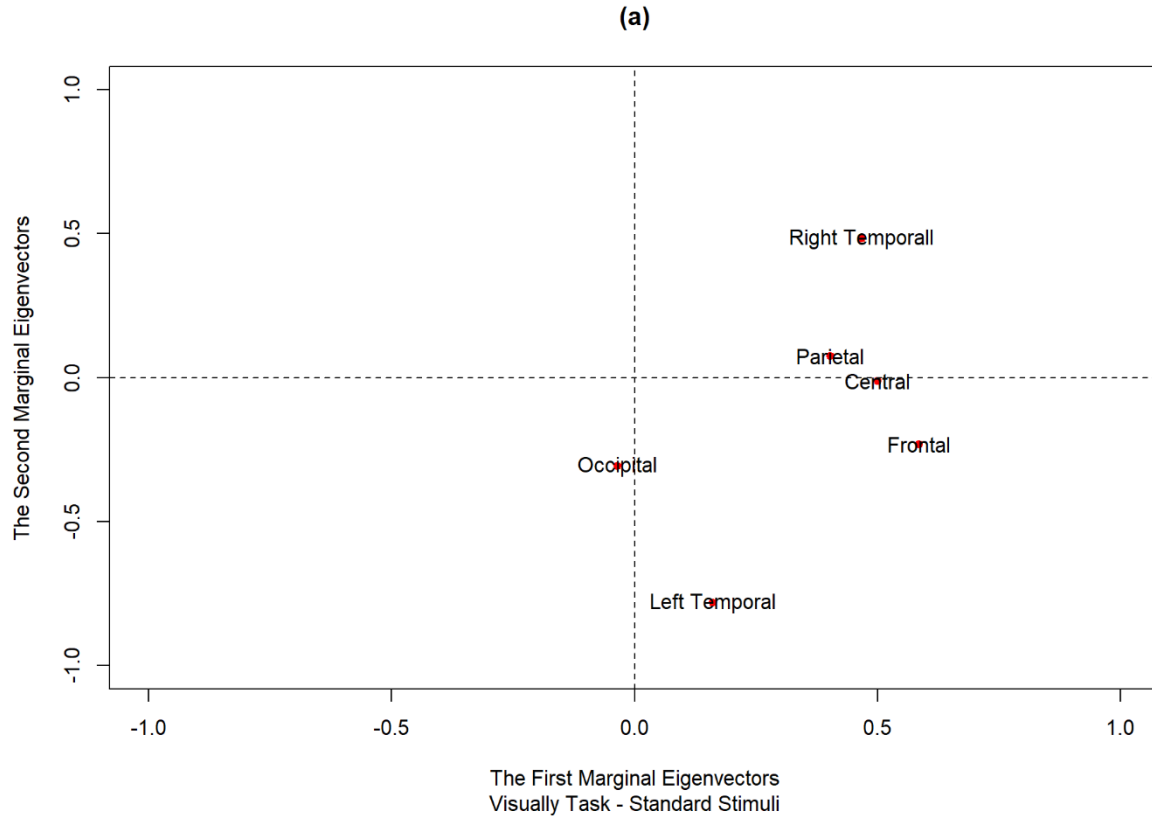


Figure 38. Estimated first and second leading regional marginal eigenvectors.

According to the Figure 37, the first and second marginal principal component analysis (PCA) capture at least nearly 85% of variation in this dimension. The Figure 38 plots estimated first and second leading regional marginal eigenvectors.

3.1.2. The Reconstructed of the Response .

The fof regression were conducted group by the number of HPCA and the regions. In the following part, we show the reconstructed ERPs for each subject (in gray) based on the number of HPCA. In this trial, we have 14 HPCA. We also show the point-wise mean and mean $\pm$ standard deviation. (Whole Brain: **Figure 39**, Frontal region (1): **Figure 40**, Central Region (2): **Figure 41**, Occipital Region (3): **Figure 42**, Parietal Region (4): **Figure 43**, Left Temporal (5): **Figure 44**, Right Temporal (6): **Figure 45**).

3.1.2.1 Whole Brain

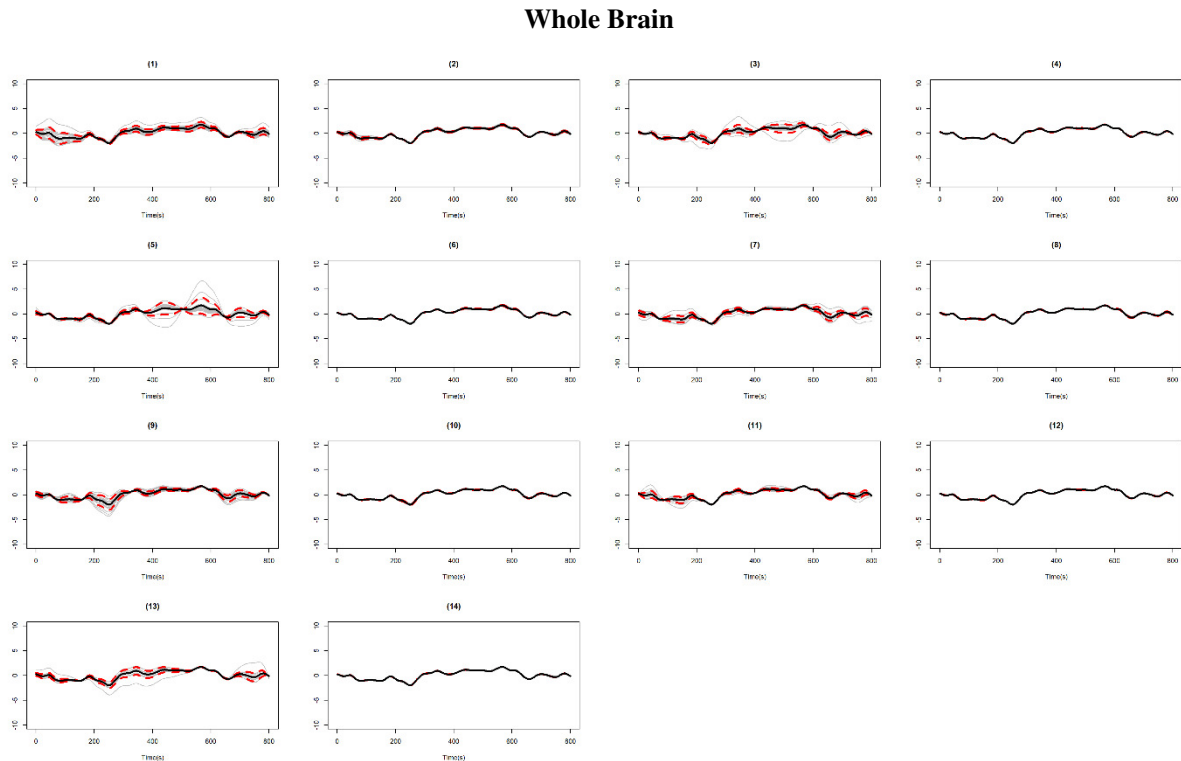


Figure 39. (1-14) Reconstructed ERPs for each subject (Gray line), point-wise mean (solid black line), and mean $\pm$ standard deviation (dotted red line) for whole region of the Brain.

3.1.2.2 Frontal Region (1)

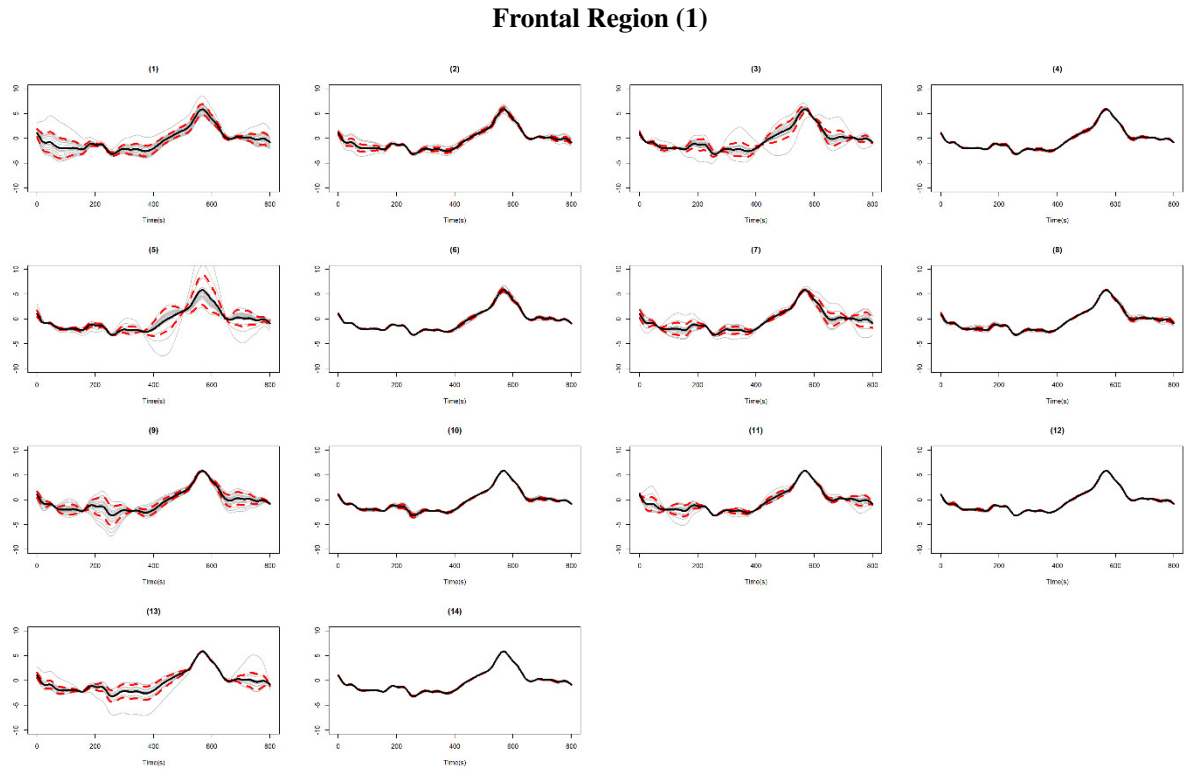


Figure 40. (1-14) Reconstructed ERPs for each subject (Gray line), point-wise mean (solid black line), and mean $\pm$ standard deviation (dotted red line) for frontal region of the Brain.

3.1.2.3 Central Region (2)

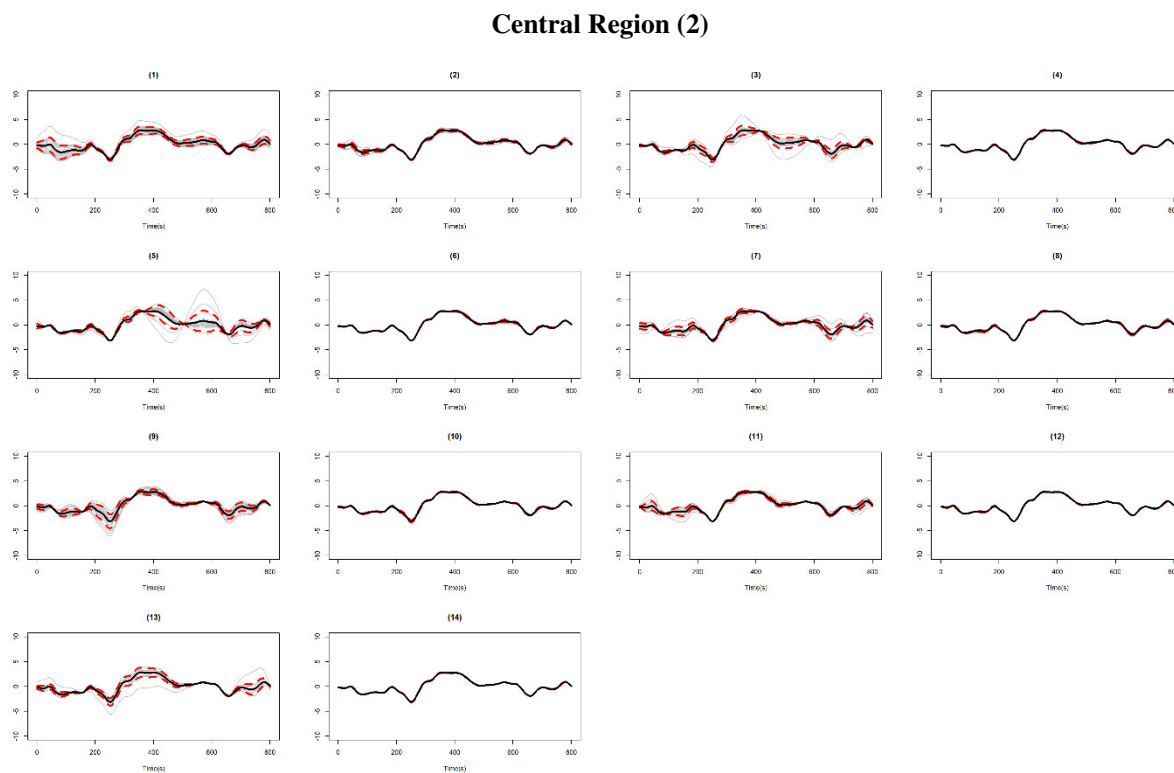


Figure 41. (1-14) Reconstructed ERPs for each subject (Gray line), point-wise mean (solid black line), and mean $\pm$ standard deviation (dotted red line) for central region of the Brain.

3.1.2.4 Occipital Region (3)

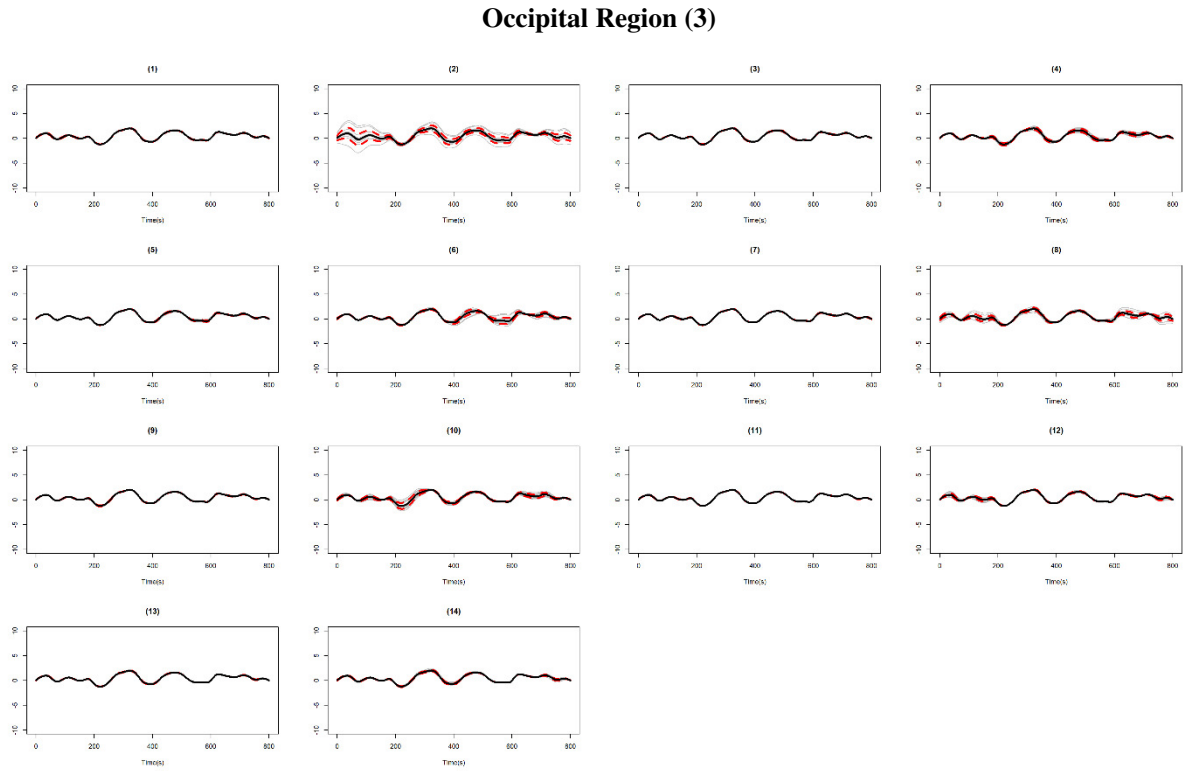


Figure 42. (1-14) Reconstructed ERPs for each subject (Gray line), point-wise mean (solid black line), and mean $\pm$ standard deviation (dotted red line) for occipital region of the Brain.

3.1.2.5 Parietal Region (4)

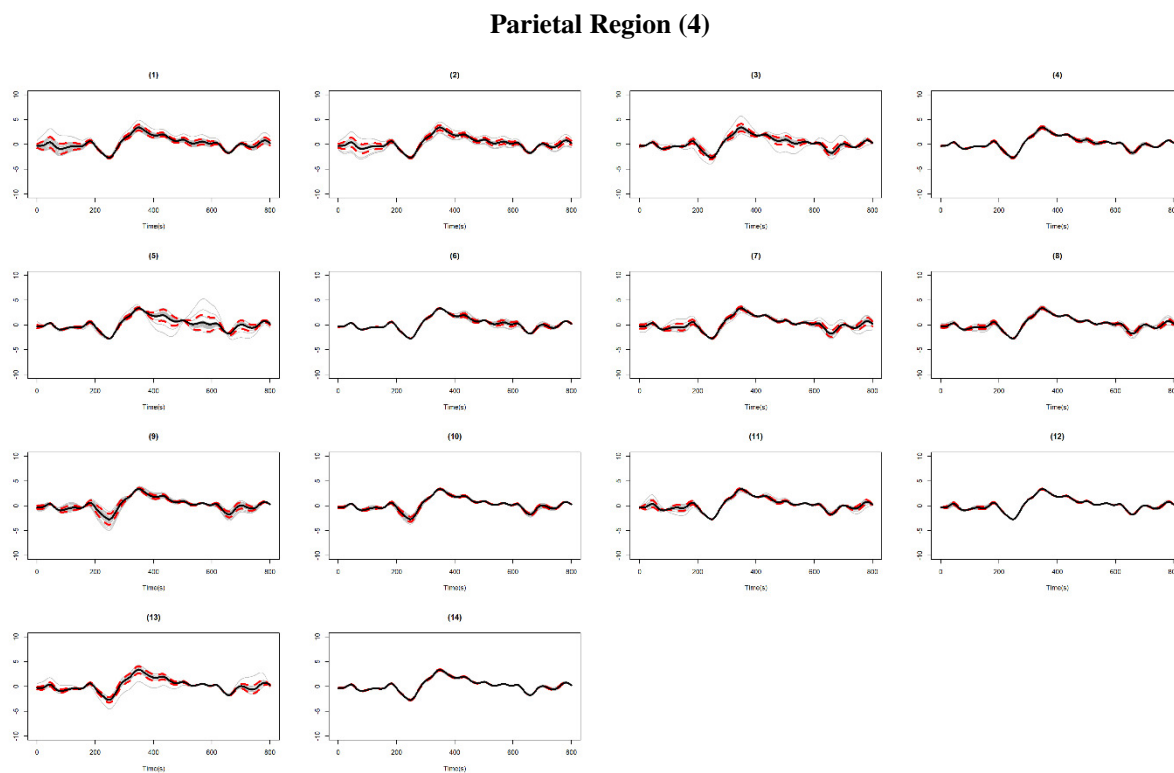


Figure 43. (1-14) Reconstructed ERPs for each subject (Gray line), point-wise mean (solid black line), and mean $\pm$ standard deviation (dotted red line) for parietal region of the Brain.

3.1.2.6 Left Temporal (5)

Left Temporal (5)

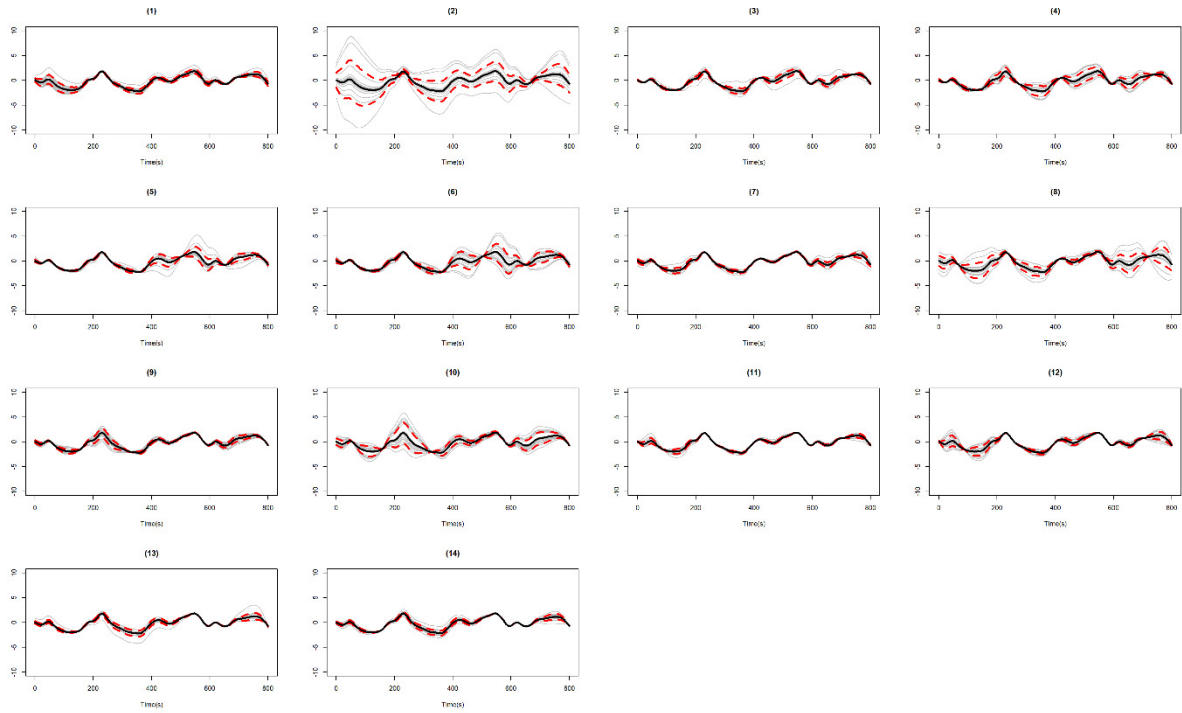


Figure 44. (1-14) Reconstructed ERPs for each subject (Gray line), point-wise mean (solid black line), and mean $\pm$ standard deviation (dotted red line) for left temporal region of the Brain.

3.1.2.7 Right Temporal (6)

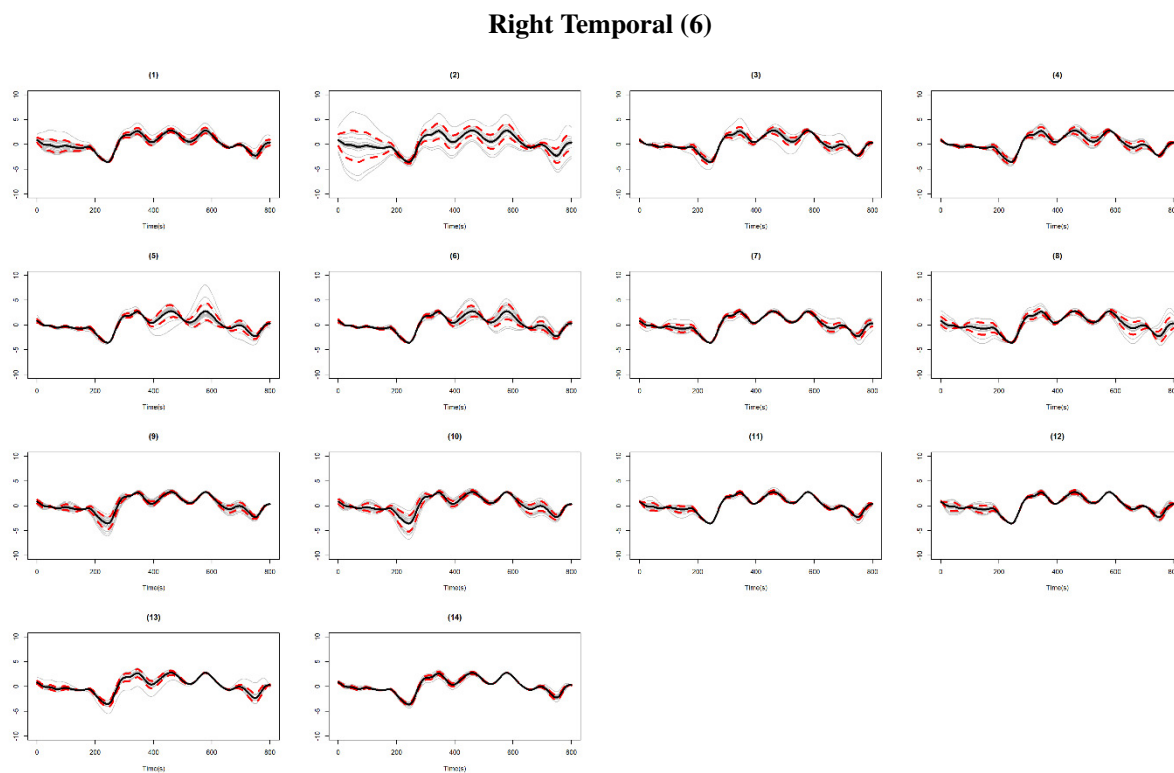


Figure 45. (1-14) Reconstructed ERPs for each subject (Gray line), point-wise mean (solid black line), and mean $\pm$ standard deviation (dotted red line) for right temporal region of the Brain.

3.1.3. *The function-on-function result* .

The fof regression result group by the region of the brain and the number of hybrid principal component analysis (HPCA) were reported.

3.1.3.1 **Whole Brain**

The MSPE and Correlation group by number of HPCA, Train and Test ([Table 43](#))

The User, System and Elapsed Time group by number of HPCA ([Table 44](#)): The median (first quantile – third quantile) of the Time (user-system-elapsed) in 10 iterations group by the number of hybrid principal component analysis (HPCA) for whole brain.

The Model fitting parameters group by number of HPCA ([Table 45](#)): The median (first quantile – third quantile) of the model fit indices in 10 iterations group by the number of hybrid principal component analysis (HPCA) for whole brain.

HPCA	MSPE		Correlation	
	Train	Test	Train	Test
1	0(0,0.002)	0.154(0.151,0.154)	1(0.999,1)	0.92(0.92,0.922)
2	0.002(0.001,0.007)	0.121(0.117,0.143)	0.999(0.997,0.999)	0.938(0.926,0.94)
3	0.052(0.052,0.052)	0.202(0.199,0.29)	0.974(0.974,0.974)	0.893(0.844,0.895)
4	0.05(0.05,0.05)	0.305(0.253,0.359)	0.975(0.975,0.975)	0.837(0.806,0.865)
5	0.106(0.106,0.116)	0.514(0.512,0.567)	0.946(0.94,0.946)	0.711(0.68,0.712)
6	0.109(0.102,0.115)	0.663(0.595,0.663)	0.944(0.941,0.948)	0.622(0.622,0.663)
7	0.069(0.056,0.104)	0.716(0.623,0.751)	0.965(0.946,0.971)	0.598(0.579,0.655)
8	0.113(0.101,0.127)	0.619(0.573,0.656)	0.942(0.934,0.948)	0.643(0.622,0.67)
9	0.25(0.242,0.25)	0.804(0.719,0.804)	0.866(0.866,0.871)	0.533(0.533,0.579)
10	0.249(0.241,0.25)	0.795(0.722,0.795)	0.866(0.866,0.871)	0.539(0.539,0.59)
11	0.274(0.274,0.274)	0.821(0.77,0.821)	0.852(0.852,0.852)	0.519(0.519,0.546)
12	0.274(0.259,0.274)	0.754(0.753,0.819)	0.852(0.852,0.861)	0.553(0.519,0.554)
13	0.282(0.282,0.353)	0.784(0.597,0.821)	0.847(0.804,0.847)	0.537(0.516,0.648)
14	0.28(0.275,0.334)	0.769(0.627,0.829)	0.848(0.815,0.851)	0.534(0.511,0.632)

Table 43. The median (first quantile – third quantile) of the Mean Square Prediction Error (MSPE) and Correlation in 10 iterations group by Train and Test and the number of hybrid principal component analysis (HPCA) for occipital region of the brain.

HPCA	Time (s)		
	User	System	Elapsed
1	132.735(129.608,137.208)	0.155(0.13,0.17)	133.845(129.99,138.443)
2	129.18(126.805,136.423)	0.075(0.037,0.08)	129.35(127.315,138.178)
3	275.04(261.73,283.003)	0.17(0.105,0.188)	277.265(266.14,283.955)
4	279.52(276.978,283.388)	0.17(0.145,0.208)	280.24(277.713,284.248)
5	356.87(354.003,358.885)	0.23(0.22,0.273)	357.73(355.052,359.998)
6	346.875(342.578,348.95)	0.15(0.14,0.18)	347.785(343.438,349.96)
7	442.385(434.89,452.09)	0.235(0.16,0.288)	443.43(435.905,453.085)
8	440.895(439.738,448.192)	0.235(0.175,0.258)	442.035(440.887,449.04)
9	447.46(441.603,453.878)	0.215(0.2,0.253)	448.535(442.63,455.027)
10	452.43(449.957,456.183)	0.23(0.205,0.247)	453.4(450.81,457.12)
11	492.1(486.892,494.73)	0.235(0.195,0.268)	493.41(487.787,496.688)
12	494.22(485.17,498)	0.24(0.23,0.27)	495.19(485.887,499.638)
13	519.22(514.678,527.563)	0.18(0.16,0.215)	519.84(515.342,528.217)
14	526.27(518.33,534.045)	0.245(0.18,0.3)	526.855(518.963,534.763)

Table 44. The median (first quantile – third quantile) of the Time (user-system-elapsed) in 10 iterations group by the number of hybrid principal component analysis (HPCA) for whole brain.

HPCA	Min.Error	Optimum					
		index	K	Tau	Lambda	Eta	Kappa
1	1.468(1.141,1.744)	9(9,9)	2.5(2,3)	0.1(0.1,0.1)	0.1(0.1,0.1)	0.01(0.01,0.01)	0(0,0)
2	1.218(0.96,1.349)	10.5(9,11.75)	2(1.25,3)	10(0.325,100)	0.3(0.125,0.4)	0.01(0.01,0.01)	0(0,0)
3	2.705(2.324,3.328)	9(5,9)	1(1,1)	0.1(0.1,0.1)	0.1(0.1,0.1)	0.01(0,0.01)	0(0,0)
4	2.841(2.383,3.361)	7.5(5,9)	1(1,1)	0.1(0.1,0.775)	0.1(0.1,0.175)	0.005(0,0.01)	0(0,0)
5	4.426(4.294,4.522)	9.5(6,10.75)	1(1,1)	1(1,7.75)	0.2(0.2,0.275)	0.01(0,0.01)	0(0,0)
6	4.499(4.335,4.877)	11(7.25,11)	1(1,1)	10(3.25,10)	0.3(0.225,0.3)	0.01(0.003,0.01)	0(0,0)
7	5.141(4.969,5.298)	11(7.25,11.75)	3(2.25,4.5)	10(3.25,77.5)	0.3(0.225,0.375)	0.01(0.003,0.01)	0(0,0)
8	5.381(5.299,5.494)	11(9.5,12)	2(2,3)	10(10,100)	0.3(0.3,0.4)	0.01(0.01,0.01)	0(0,0)
9	6.879(6.715,7.202)	11(11,11)	1(1,1)	10(10,77.5)	0.3(0.3,0.375)	0.01(0.01,0.01)	0(0,0)
10	6.904(6.446,7.004)	11(11,11)	1(1,1)	10(10,10)	0.3(0.3,0.3)	0.01(0.01,0.01)	0(0,0)
11	7.236(6.851,7.601)	11(11,11)	1(1,1)	10(10,10)	0.3(0.3,0.3)	0.01(0.01,0.01)	0(0,0)
12	7.202(7.122,7.689)	11.5(11,12)	1(1,1.75)	55(10,100)	0.35(0.3,0.4)	0.01(0.01,0.01)	0(0,0)
13	7.866(7.614,8.175)	11(11,12)	1(1,1)	10(10,100)	0.3(0.3,0.4)	0.01(0.01,0.01)	0(0,0)
14	8.11(7.845,8.289)	11.5(11,12)	1(1,1)	55(10,100)	0.35(0.3,0.4)	0.01(0.01,0.01)	0(0,0)

Table 45. The median (first quantile – third quantile) of the model fit indices in 10 iterations group by the number of hybrid principal component analysis (HPCA) for whole brain.

3.1.3.2 Frontal Region (1)

The MSPE and Correlation group by number of HPCA, Train and Test (Table 46)

The User, System and Elapsed Time group by number of HPCA (Table 47): The median (first quantile – third quantile) of the Time (user-system-elapsed) in 10 iterations group by the number of hybrid principal component analysis (HPCA) for frontal region of brain.

The Model fitting parameters group by number of HPCA (Table 48): The median (first quantile – third quantile) of the model fit indices in 10 iterations group by the number of hybrid principal component analysis (HPCA) for whole brain.

HPCA	MSPE		Correlation	
	Train	Test	Train	Test
1	0(0,0.001)	0.079(0.077,0.082)	1(1,1)	0.96(0.958,0.961)
2	0.001(0.001,0.002)	0.077(0.076,0.09)	1(0.999,1)	0.961(0.954,0.961)
3	0.028(0.006,0.028)	0.234(0.229,0.238)	0.986(0.986,0.997)	0.876(0.873,0.878)
4	0.026(0.026,0.026)	0.311(0.307,0.311)	0.987(0.987,0.987)	0.833(0.833,0.834)
5	0.058(0.058,0.064)	0.307(0.307,0.346)	0.971(0.967,0.971)	0.835(0.814,0.835)
6	0.058(0.053,0.063)	0.407(0.369,0.438)	0.971(0.968,0.973)	0.779(0.763,0.801)
7	0.049(0.047,0.055)	0.437(0.386,0.451)	0.975(0.972,0.976)	0.767(0.756,0.793)
8	0.065(0.059,0.072)	0.454(0.419,0.483)	0.967(0.963,0.97)	0.75(0.735,0.768)
9	0.171(0.168,0.171)	0.638(0.616,0.638)	0.91(0.91,0.912)	0.647(0.647,0.659)
10	0.167(0.155,0.173)	0.617(0.557,0.632)	0.912(0.909,0.919)	0.658(0.649,0.687)
11	0.186(0.171,0.189)	0.637(0.547,0.656)	0.902(0.9,0.911)	0.643(0.635,0.694)
12	0.181(0.17,0.186)	0.613(0.571,0.647)	0.905(0.902,0.911)	0.662(0.639,0.677)
13	0.189(0.181,0.202)	0.576(0.48,0.639)	0.9(0.893,0.905)	0.681(0.642,0.732)
14	0.22(0.202,0.243)	0.577(0.52,0.635)	0.883(0.87,0.893)	0.679(0.649,0.71)

Table 46. The median (first quantile – third quantile) of the Mean Square Prediction Error (MSPE) and Correlation in 10 iterations group by Train and Test and the number of hybrid principal component analysis (HPCA) for frontal region brain.

HPCA	Time (s)		
	User	System	Elapsed
1	166.085(164.56,167.723)	0.125(0.093,0.22)	166.365(164.805,168.063)
2	167.16(163.678,169.053)	0.09(0.08,0.11)	167.42(164.305,169.347)
3	274.84(269.237,280.23)	0.12(0.103,0.365)	275.13(269.59,280.962)
4	277.455(273.77,279.71)	0.11(0.075,0.147)	277.815(274.035,280.075)
5	343.7(339.947,346.453)	0.145(0.112,0.188)	344.12(340.29,346.872)
6	338.175(335.905,343.098)	0.135(0.112,0.195)	338.675(336.355,343.65)
7	401.455(393.633,409.298)	0.17(0.16,0.188)	402.095(394.068,409.857)
8	399.575(395.168,402.768)	0.195(0.17,0.238)	400.25(395.653,403.48)
9	438.85(431.328,453.083)	0.275(0.188,0.362)	439.475(432.115,453.77)
10	439.19(436.31,441.21)	0.25(0.188,0.27)	440.385(438.233,442.22)
11	477.085(472.262,490.36)	0.195(0.163,0.243)	478.12(475.555,491.237)
12	468.9(465.2,471.443)	0.225(0.192,0.325)	470.45(466.73,472.423)
13	488.875(480.19,491.64)	0.19(0.17,0.205)	489.73(480.883,492.357)
14	486.085(478.745,494.245)	0.195(0.157,0.22)	486.865(479.295,495.055)

Table 47. The median (first quantile – third quantile) of the Time (user-system-elapsed) in 10 iterations group by the number of hybrid principal component analysis (HPCA) for frontal region of brain.

HPCA	Min_Error	Optimum					
		index	K	Tau	Lambda	Eta	Kappa
1	0.731(0.705,0.85)	9(6,9)	3(1,3)	0.1(0.1,0.1)	0.1(0.1,0.1)	0.01(0.003,0.01)	0(0,0)
2	0.54(0.459,0.582)	9(9,11)	2(1,2)	10(0.1,77.5)	0.3(0.1,0.375)	0.01(0.01,0.01)	0(0,0)
3	2.399(2.192,2.865)	5(5,5)	1(1,2)	0.1(0.1,0.1)	0.1(0.1,0.1)	0(0,0)	0(0,0)
4	2.09(1.981,2.662)	5(5,5)	1(1,1)	0.1(0.1,0.1)	0.1(0.1,0.1)	0(0,0)	0(0,0)
5	2.776(2.563,2.918)	6(6,9.75)	1(1,1)	1(1,7.75)	0.2(0.2,0.275)	0(0,0.008)	0(0,0)
6	2.478(2.191,2.9)	11(6.75,11)	1(1,1)	10(1,10)	0.3(0.2,0.3)	0.01(0.003,0.01)	0(0,0)
7	3.223(3.098,3.377)	11(7,11.75)	2(2,3.75)	10(1,77.5)	0.3(0.2,0.375)	0.01(0.003,0.01)	0(0,0)
8	3.064(2.823,3.314)	11(9.5,11.75)	2(2,2.75)	10(10,100)	0.3(0.3,0.4)	0.01(0.01,0.01)	0(0,0)
9	4.563(4.456,4.623)	11(11,11)	1(1,1)	10(10,10)	0.3(0.3,0.3)	0.01(0.01,0.01)	0(0,0)
10	4.726(4.524,4.888)	9(4.75,11)	1(1,1.75)	10(10,100)	0.3(0.3,0.4)	0.005(0,0.01)	0(0,0)
11	5.222(5.025,5.397)	11.5(8,12)	1(1,1.75)	55(10,100)	0.35(0.3,0.4)	0.01(0.003,0.01)	0(0,0)
12	5.112(4.989,5.26)	8(3.25,9.75)	1(1,1.75)	10(3.25,100)	0.3(0.225,0.4)	0(0,0.01)	0(0,0)
13	5.291(5.243,5.412)	8.5(5,11)	1(1,1.75)	55(3.25,100)	0.35(0.225,0.4)	0.005(0,0.01)	0(0,0)
14	5.273(5.063,5.521)	11.5(11,12)	1(1,1)	100(32.5,100)	0.4(0.325,0.4)	0.01(0.01,0.01)	0(0,0)

Table 48. The median (first quantile – third quantile) of the model fit indices in 10 iterations group by the number of hybrid principal component analysis (HPCA) for frontal region of brain.

3.1.3.3 Central Region (2)

The MSPE and Correlation group by number of HPCA, Train and Test (Table 49)

The User, System and Elapsed Time group by number of HPCA (Table 50): The median (first quantile – third quantile) of the Time (user-system-elapsed) in 10 iterations group by the number of hybrid principal component analysis (HPCA) for whole brain.

The Model fitting parameters group by number of HPCA (Table 51): The median (first quantile – third quantile) of the model fit indices in 10 iterations group by the number of hybrid principal component analysis (HPCA) for whole brain.

HPCA	MSPE		Correlation	
	Train	Test	Train	Test
1	0.001(0,0.001)	0.092(0.089,0.092)	1(1,1)	0.953(0.953,0.955)
2	0(0,0)	0.141(0.124,0.145)	1(1,1)	0.929(0.926,0.937)
3	0.048(0.048,0.048)	0.233(0.196,0.233)	0.975(0.975,0.975)	0.877(0.877,0.897)
4	0.051(0.051,0.051)	0.121(0.121,0.162)	0.974(0.974,0.974)	0.937(0.915,0.937)
5	0.086(0.049,0.086)	0.4(0.38,0.417)	0.956(0.956,0.975)	0.781(0.771,0.793)
6	0.092(0.09,0.101)	0.359(0.348,0.366)	0.953(0.948,0.954)	0.802(0.799,0.81)
7	0.068(0.057,0.143)	0.46(0.432,0.483)	0.965(0.926,0.971)	0.745(0.729,0.762)
8	0.155(0.147,0.155)	0.41(0.41,0.415)	0.919(0.919,0.924)	0.77(0.77,0.77)
9	0.191(0.191,0.191)	0.489(0.489,0.489)	0.899(0.899,0.899)	0.726(0.726,0.726)
10	0.192(0.192,0.192)	0.509(0.509,0.509)	0.899(0.899,0.899)	0.713(0.713,0.713)
11	0.208(0.208,0.208)	0.598(0.597,0.598)	0.89(0.89,0.89)	0.661(0.661,0.661)
12	0.209(0.2,0.209)	0.601(0.601,0.644)	0.889(0.889,0.894)	0.66(0.64,0.66)
13	0.216(0.213,0.275)	0.567(0.445,0.586)	0.885(0.851,0.887)	0.677(0.667,0.748)
14	0.218(0.215,0.218)	0.562(0.545,0.58)	0.884(0.884,0.886)	0.68(0.67,0.689)

Table 49. The median (first quantile – third quantile) of the Mean Square Prediction Error (MSPE) and Correlation in 10 iterations group by Train and Test and the number of hybrid principal component analysis (HPCA) for occipital region of the brain.

HPCA	Time (s)		
	User	System	Elapsed
1	162.555(160.223,164.527)	0.12(0.095,0.175)	162.87(160.62,164.865)
2	163.9(162.222,165.41)	0.075(0.045,0.115)	164.14(162.355,165.612)
3	275.92(273.293,278.887)	0.13(0.095,0.155)	276.385(273.51,279.178)
4	278.11(271.488,283.51)	0.155(0.115,0.21)	278.825(271.795,284.1)
5	354.905(352.413,358.658)	0.175(0.118,0.215)	355.835(353.87,359.615)
6	356.675(351.792,359.213)	0.21(0.157,0.22)	357.32(352.435,359.88)
7	443.325(436.328,451.31)	0.16(0.15,0.168)	443.78(436.838,451.868)
8	436.39(428.53,438.917)	0.205(0.143,0.3)	436.855(429.007,439.357)
9	506.645(500.028,512.225)	0.2(0.153,0.245)	507.29(500.59,512.75)
10	515.82(507.658,525.483)	0.21(0.16,0.265)	516.505(508.327,525.87)
11	507.61(500.242,515.095)	0.205(0.173,0.23)	508.17(500.915,515.702)
12	500.73(497.435,502.777)	0.17(0.16,0.198)	501.185(497.878,503.408)
13	537.455(530.072,541.39)	0.2(0.162,0.28)	538.275(530.612,542.215)
14	542.41(536.01,549.055)	0.23(0.17,0.397)	543.1(536.728,549.872)

Table 50. The median (first quantile – third quantile) of the Time (user-system-elapsed) in 10 iterations group by the number of hybrid principal component analysis (HPCA) for central region of the brain.

HPCA	Min.Error	Optimum					
		index	K	Tau	Lambda	Eta	Kappa
1	0.882(0.804,1.046)	9(9,9)	2(1,3)	0.1(0.1,0.1)	0.1(0.1,0.1)	0.01(0.01,0.01)	0(0,0)
2	1.147(1.084,1.225)	9(2,9)	3(2,3)	0.1(0.1,0.1)	0.1(0.1,0.1)	0.01(0,0.01)	0(0,0)
3	2.256(2.085,2.488)	5(5,8)	1(1,1)	0.1(0.1,0.1)	0.1(0.1,0.1)	0(0,0.008)	0(0,0)
4	2.287(1.93,2.425)	9(6,9)	1(1,1)	0.1(0.1,0.1)	0.1(0.1,0.1)	0.01(0.003,0.01)	0(0,0)
5	4.14(4.074,4.318)	6(5.25,9.75)	1(1,1.75)	1(0.325,1)	0.2(0.125,0.2)	0(0,0.01)	0(0,0)
6	4.149(3.875,4.279)	8.5(6,11)	1(1,1)	10(1,10)	0.3(0.2,0.3)	0.005(0,0.01)	0(0,0)
7	5.158(4.975,5.377)	10.5(6,11)	2.5(1,3)	5.5(1,10)	0.25(0.2,0.3)	0.01(0,0.01)	0(0,0)
8	5.275(5.139,5.623)	11(11,11)	1(1,1)	10(10,10)	0.3(0.3,0.3)	0.01(0.01,0.01)	0(0,0)
9	5.827(5.635,5.986)	11(11,11)	1(1,1)	10(10,10)	0.3(0.3,0.3)	0.01(0.01,0.01)	0(0,0)
10	5.623(5.605,5.778)	11(11,11)	1(1,1)	10(10,10)	0.3(0.3,0.3)	0.01(0.01,0.01)	0(0,0)
11	6.109(5.937,6.243)	11(11,11)	1(1,1)	10(10,10)	0.3(0.3,0.3)	0.01(0.01,0.01)	0(0,0)
12	6.121(5.991,6.344)	11(10.25,11)	1(1,1)	10(3.25,10)	0.3(0.225,0.3)	0.01(0.01,0.01)	0(0,0)
13	6.921(6.576,7.054)	11(8,11.75)	1(1,1)	10(10,77.5)	0.3(0.3,0.375)	0.01(0.003,0.01)	0(0,0)
14	6.748(6.51,6.934)	11(8,11)	1(1,1)	10(10,10)	0.3(0.3,0.3)	0.01(0.003,0.01)	0(0,0)

Table 51. The median (first quantile – third quantile) of the model fit indices in 10 iterations group by the number of hybrid principal component analysis (HPCA) for central region of the brain.

3.1.3.4 Occipital Region (3)

The MSPE and Correlation group by number of HPCA, Train and Test ([Table 52](#))

The User, System and Elapsed Time group by number of HPCA ([Table 53](#)): The median (first quantile – third quantile) of the Time (user-system-elapsed) in 10 iterations group by the number of hybrid principal component analysis (HPCA) for whole brain.

The Model fitting parameters group by number of HPCA ([Table 54](#)): The median (first quantile – third quantile) of the model fit indices in 10 iterations group by the number of hybrid principal component analysis (HPCA) for whole brain.

HPCA	MSPE		Correlation	
	Train	Test	Train	Test
1	0(0,0)	0.002(0.002,0.002)	1(1,1)	0.999(0.999,0.999)
2	0.002(0.001,0.002)	0.322(0.322,0.326)	0.999(0.999,1)	0.836(0.836,0.837)
3	0.004(0.004,0.004)	0.326(0.326,0.326)	0.998(0.998,0.998)	0.834(0.834,0.835)
4	0.001(0,0.032)	0.638(0.544,0.647)	0.999(0.984,1)	0.683(0.677,0.717)
5	0.007(0,0.007)	0.659(0.651,0.663)	0.997(0.997,1)	0.667(0.665,0.675)
6	0.072(0.072,0.072)	0.563(0.563,0.563)	0.963(0.963,0.963)	0.698(0.698,0.698)
7	0.001(0,0.055)	0.759(0.619,0.759)	1(0.972,1)	0.605(0.605,0.669)
8	0.05(0.006,0.095)	0.628(0.55,0.706)	0.974(0.951,0.997)	0.663(0.631,0.695)
9	0.1(0.019,0.1)	0.638(0.549,0.711)	0.948(0.948,0.99)	0.648(0.623,0.696)
10	0.224(0.147,0.224)	0.742(0.655,0.742)	0.881(0.881,0.923)	0.586(0.586,0.638)
11	0.22(0.148,0.224)	0.736(0.622,0.739)	0.883(0.88,0.923)	0.588(0.587,0.651)
12	0.229(0.222,0.229)	0.787(0.785,0.787)	0.878(0.878,0.882)	0.559(0.559,0.56)
13	0.228(0.139,0.232)	0.789(0.656,0.791)	0.879(0.876,0.928)	0.557(0.556,0.63)
14	0.24(0.237,0.24)	0.787(0.787,0.789)	0.872(0.872,0.873)	0.555(0.554,0.555)

Table 52. The median (first quantile – third quantile) of the Mean Square Prediction Error (MSPE) and Correlation in 10 iterations group by Train and Test and the number of hybrid principal component analysis (HPCA) for occipital region of the brain.

HPCA	Time (s)		
	User	System	Elapsed
1	129.64(128.415,130.41)	0.1(0.058,0.125)	129.81(128.705,130.555)
2	175.47(173.245,178.508)	0.09(0.042,0.118)	175.855(173.44,178.717)
3	241.895(237.55,246.548)	0.095(0.082,0.125)	242.25(237.877,246.823)
4	283.435(277.077,286.468)	0.135(0.095,0.165)	283.74(277.515,286.76)
5	335.115(326.84,339.542)	0.125(0.088,0.177)	335.59(327.575,340.162)
6	372.7(360.67,377.63)	0.175(0.162,0.19)	373.26(361.44,378.335)
7	352.57(348.503,355.042)	0.185(0.14,0.205)	352.965(348.925,355.487)
8	418.205(412.29,427.415)	0.165(0.12,0.193)	418.81(412.86,428.153)
9	405.76(400.345,410.537)	0.14(0.117,0.185)	406.33(400.82,411.18)
10	497.015(493.42,500.673)	0.17(0.132,0.218)	497.745(494.16,501.418)
11	470.325(461.04,477.205)	0.215(0.2,0.265)	470.995(461.877,477.938)
12	571.86(557.247,579.693)	0.245(0.23,0.307)	572.5(557.992,581.095)
13	539.38(532.263,550.772)	0.2(0.172,0.22)	539.97(532.902,551.52)
14	not calculated	not calculated	not calculated

Table 53. The median (first quantile – third quantile) of the Time (user-system-elapsed) in 10 iterations group by the number of hybrid principal component analysis (HPCA) for occipital region of the brain.

HPCA	Min.Error	Optimum					
		index	K	Tau	Lambda	Eta	Kappa
1	0.017(0.015,0.021)	9(9,10.5)	2(1,3.75)	0.1(0.1,77.5)	0.1(0.1,0.375)	0.01(0.01,0.01)	0(0,0)
2	1.346(1.228,1.416)	1(1,1)	1(1,2.5)	0.1(0.1,0.1)	0.1(0.1,0.1)	0(0,0)	0(0,0)
3	1.292(1.14,1.422)	1(1,1.75)	1(1,1)	0.1(0.1,0.1)	0.1(0.1,0.1)	0(0,0)	0(0,0)
4	1.807(1.593,1.856)	2(2,3)	3(1,4)	1(1,10)	0.2(0.2,0.3)	0(0,0)	0(0,0)
5	1.716(1.524,1.932)	1.5(1,2)	3.5(2,5)	0.55(0.1,1)	0.15(0.1,0.2)	0(0,0)	0(0,0)
6	2.853(2.71,3.36)	9(9,9)	1(1,1)	0.1(0.1,0.1)	0.1(0.1,0.1)	0.01(0.01,0.01)	0(0,0)
7	2.928(2.75,3.142)	1(1,1)	4(1.75,4.75)	0.1(0.1,0.1)	0.1(0.1,0.1)	0(0,0)	0(0,0)
8	5.398(5.252,6.001)	1(1,1)	2.5(1,4)	0.1(0.1,0.1)	0.1(0.1,0.1)	0(0,0)	0(0,0)
9	5.361(5.045,5.489)	1(1,1)	1(1,3.5)	0.1(0.1,0.1)	0.1(0.1,0.1)	0(0,0)	0(0,0)
10	5.22(5.144,5.442)	12(5.25,12)	1(1,1)	100(25.075,100)	0.4(0.175,0.4)	0.01(0.003,0.01)	0(0,0)
11	5.222(5.002,5.583)	8(1.75,12)	1(1,1)	100(25.075,100)	0.4(0.175,0.4)	0.005(0,0.01)	0(0,0)
12	5.535(5.387,5.828)	12(5,12)	1(1,1)	100(100,100)	0.4(0.4,0.4)	0.01(0,0.01)	0(0,0)
13	5.689(5.339,6.061)	10.5(1.75,12)	1(1,1)	100(0.1,100)	0.4(0.1,0.4)	0.01(0,0.01)	0(0,0)
14	5.981(5.911,6.086)	12(8,12)	1(1,1)	100(100,100)	0.4(0.4,0.4)	0.01(0,0.01)	0(0,0)

Table 54. The median (first quantile – third quantile) of the model fit indices in 10 iterations group by the number of hybrid principal component analysis (HPCA) for occipital region of the brain.

3.1.3.5 Parietal (4)

The MSPE and Correlation group by number of HPCA, Train and Test (Table 55)

The User, System and Elapsed Time group by number of HPCA (Table 56): The median (first quantile – third quantile) of the Time (user-system-elapsed) in 10 iterations group by the number of hybrid principal component analysis (HPCA) for whole brain.

The Model fitting parameters group by number of HPCA (Table 57): The median (first quantile – third quantile) of the model fit indices in 10 iterations group by the number of hybrid principal component analysis (HPCA) for whole brain.

HPCA	MSPE		Correlation	
	Train	Test	Train	Test
1	0(0,0)	0.062(0.062,0.062)	1(1,1)	0.968(0.968,0.969)
2	0(0,0)	0.218(0.158,0.28)	1(1,1)	0.89(0.86,0.919)
3	0.052(0.052,0.052)	0.293(0.292,0.293)	0.974(0.974,0.974)	0.847(0.847,0.847)
4	0.065(0.021,0.066)	0.152(0.121,0.295)	0.967(0.967,0.99)	0.922(0.854,0.938)
5	0.096(0.096,0.096)	0.295(0.275,0.321)	0.951(0.951,0.951)	0.841(0.826,0.852)
6	0.107(0.107,0.117)	0.212(0.177,0.255)	0.945(0.939,0.945)	0.888(0.863,0.909)
7	0.156(0.105,0.162)	0.296(0.264,0.451)	0.919(0.915,0.946)	0.839(0.755,0.858)
8	0.176(0.176,0.176)	0.236(0.236,0.236)	0.908(0.908,0.908)	0.875(0.875,0.875)
9	0.192(0.192,0.192)	0.263(0.263,0.263)	0.899(0.899,0.899)	0.858(0.858,0.858)
10	0.195(0.192,0.195)	0.343(0.343,0.343)	0.897(0.897,0.899)	0.811(0.811,0.811)
11	0.209(0.205,0.209)	0.416(0.416,0.416)	0.889(0.889,0.891)	0.767(0.767,0.767)
12	0.21(0.206,0.21)	0.408(0.386,0.412)	0.889(0.888,0.891)	0.772(0.77,0.784)
13	0.217(0.215,0.258)	0.383(0.354,0.384)	0.885(0.861,0.886)	0.786(0.786,0.804)
14	0.223(0.217,0.223)	0.378(0.358,0.378)	0.881(0.881,0.885)	0.79(0.79,0.801)

Table 55. The median (first quantile – third quantile) of the Mean Square Prediction Error (MSPE) and Correlation in 10 iterations group by Train and Test and the number of hybrid principal component analysis (HPCA) for parietal region of the brain.

HPCA	Time (s)		
	User	System	Elapsed
1	166.93(165.84,168.605)	0.09(0.09,0.122)	167.61(166.298,169.408)
2	186.96(185.335,191.493)	0.14(0.088,0.19)	187.505(186.075,192.173)
3	305.285(301.555,306.425)	0.15(0.13,0.218)	306.18(302.185,307.85)
4	308.05(305.672,314.672)	0.13(0.095,0.17)	308.745(306.16,314.965)
5	362.675(358.795,366.98)	0.18(0.145,0.22)	363.43(359.375,367.575)
6	377.175(369.793,383.8)	0.17(0.135,0.215)	377.74(370.928,384.35)
7	456.545(412.535,462.275)	0.205(0.175,0.238)	457.14(413.547,463.5)
8	387.355(382.09,389.03)	0.175(0.123,0.19)	387.86(382.545,389.725)
9	428.75(423.345,432.682)	0.18(0.163,0.197)	429.17(423.718,433.07)
10	437.535(434.617,444.57)	0.215(0.153,0.247)	437.885(435.063,445.193)
11	494.575(490.223,503.015)	0.2(0.14,0.303)	495.11(490.58,503.48)
12	440.58(436.94,447.995)	0.13(0.065,0.185)	440.97(437.268,448.352)
13	462.525(456.09,470.052)	0.175(0.13,0.215)	463.28(456.415,470.478)
14	472.345(470.677,475.735)	0.22(0.197,0.243)	472.82(471.042,476.163)

Table 56. The median (first quantile – third quantile) of the Time (user-system-elapsed) in 10 iterations group by the number of hybrid principal component analysis (HPCA) for parietal region of the brain.

HPCA	Min_Error	Optimum					
		index	K	Tau	Lambda	Eta	Kappa
1	0.591(0.45,0.645)	9(9,9)	3(3,3)	0.1(0.1,0.1)	0.1(0.1,0.1)	0.01(0.01,0.01)	0(0,0)
2	1.625(1.354,1.72)	5(1,9)	3(3,3)	0.1(0.1,0.1)	0.1(0.1,0.1)	0.005(0,0.01)	0(0,0)
3	2.124(2.022,2.447)	9(9,9)	1(1,1)	0.1(0.1,0.1)	0.1(0.1,0.1)	0.01(0.01,0.01)	0(0,0)
4	2.436(2.341,2.624)	9(9,10)	1(1,1.75)	0.1(0.1,1)	0.1(0.1,0.2)	0.01(0.01,0.01)	0(0,0)
5	4.195(3.929,4.427)	7(6,9)	1(1,1)	1(0.1,1)	0.2(0.1,0.2)	0(0,0.01)	0(0,0)
6	3.862(3.699,4.218)	9(6.25,11)	1(1,1)	5.5(0.325,10)	0.25(0.125,0.3)	0.01(0,0.01)	0(0,0)
7	4.894(4.821,5.035)	10.5(7.25,11)	1(1,2)	10(3.25,10)	0.3(0.225,0.3)	0.01(0,0.01)	0(0,0)
8	4.929(4.665,5.188)	11(11,11)	1(1,1)	10(10,10)	0.3(0.3,0.3)	0.01(0.01,0.01)	0(0,0)
9	5.124(4.999,5.286)	11(11,11)	1(1,1)	10(10,10)	0.3(0.3,0.3)	0.01(0.01,0.01)	0(0,0)
10	5.329(5.133,5.71)	11(7.75,11)	1(1,1)	10(10,10)	0.3(0.3,0.3)	0.01(0.003,0.01)	0(0,0)
11	5.449(5.17,5.716)	11(5,11)	1(1,1)	10(10,10)	0.3(0.3,0.3)	0.01(0.003,0.01)	0(0,0)
12	5.513(5.403,5.604)	11(7.5,11)	1(1,1)	10(10,10)	0.3(0.3,0.3)	0.01(0.003,0.01)	0(0,0)
13	6.351(5.922,6.701)	11(7.25,11)	1(1,1)	10(10,77.5)	0.3(0.3,0.375)	0.01(0,0.01)	0(0,0)
14	6.291(6.215,6.352)	9.5(6.25,11)	1(1,1)	10(10,10)	0.3(0.3,0.3)	0.005(0,0.01)	0(0,0)

Table 57. The median (first quantile – third quantile) of the Mean Square Prediction Error (MSPE) and Correlation in 10 iterations group by Train and Test and the number of hybrid principal component analysis (HPCA) for occipital region of the brain.

3.1.3.6 Left Temporal (5)

The MSPE and Correlation group by number of HPCA, Train and Test ([Table 58](#))

The User, System and Elapsed Time group by number of HPCA ([Table 59](#)): The median (first quantile – third quantile) of the Time (user-system-elapsed) in 10 iterations group by the number of hybrid principal component analysis (HPCA) for whole brain.

The Model fitting parameters group by number of HPCA ([Table 60](#)): The median (first quantile – third quantile) of the model fit indices in 10 iterations group by the number of hybrid principal component analysis (HPCA) for whole brain.

HPCA	MSPE		Correlation	
	Train	Test	Train	Test
1	0.001(0,0.001)	0.052(0.05,0.059)	1(1,1)	0.974(0.97,0.975)
2	0.013(0.013,0.013)	0.653(0.653,0.653)	0.993(0.993,0.993)	0.657(0.657,0.657)
3	0.087(0.087,0.087)	0.698(0.698,0.703)	0.955(0.955,0.955)	0.613(0.606,0.613)
4	0.127(0.041,0.127)	0.841(0.831,1.111)	0.934(0.934,0.979)	0.514(0.406,0.523)
5	0.216(0.086,0.216)	0.841(0.841,1.105)	0.885(0.885,0.955)	0.484(0.348,0.485)
6	0.185(0.018,0.278)	0.962(0.952,1.194)	0.901(0.849,0.991)	0.367(0.258,0.38)
7	0.006(0.001,0.122)	1.177(0.935,1.185)	0.997(0.937,1)	0.268(0.264,0.389)
8	0.026(0.026,0.122)	1.139(1.128,1.14)	0.987(0.936,0.987)	0.258(0.257,0.261)
9	0.349(0.213,0.756)	0.969(0.862,1.06)	0.806(0.447,0.886)	0.259(0.202,0.418)
10	0.36(0.36,0.36)	0.853(0.793,0.947)	0.8(0.8,0.8)	0.425(0.352,0.469)
11	0.718(0.372,0.902)	0.93(0.917,0.991)	0.507(0.311,0.792)	0.272(0.261,0.37)
12	0.39(0.39,0.768)	0.968(0.888,0.97)	0.781(0.439,0.781)	0.333(0.219,0.389)
13	0.408(0.408,0.872)	0.963(0.845,1.001)	0.769(0.355,0.769)	0.329(0.168,0.42)
14	0.425(0.425,0.425)	1.036(0.886,1.059)	0.758(0.758,0.758)	0.275(0.128,0.384)

Table 58. - The median (first quantile – third quantile) of the Mean Square Prediction Error (MSPE) and Correlation in 10 iterations group by Train and Test and the number of hybrid principal component analysis (HPCA) for left temporal region of the brain.

HPCA	Time (s)		
	User	System	Elapsed
1	135.275(133.348,136.925)	0.1(0.068,0.125)	135.49(133.415,137.345)
2	170.325(169.16,173.732)	0.085(0.055,0.145)	170.455(169.347,175.115)
3	228.65(227.853,231.705)	0.115(0.09,0.16)	229.24(228.117,232.06)
4	252.56(250.083,254.155)	0.11(0.078,0.263)	252.9(250.3,254.41)
5	327.765(321.878,334.377)	0.12(0.09,0.13)	328.09(322.153,334.64)
6	333.8(327.013,334.78)	0.135(0.113,0.15)	334.08(327.37,335.1)
7	369.76(368.825,371.26)	0.13(0.12,0.14)	370.08(369.095,371.635)
8	383.12(379.718,388.095)	0.135(0.113,0.148)	383.52(380.072,388.362)
9	368.225(365.783,372.327)	0.15(0.12,0.17)	368.595(366.083,372.698)
10	440.295(434.132,449.418)	0.19(0.147,0.208)	440.685(434.632,449.998)
11	423.33(421.53,433.58)	0.175(0.145,0.197)	423.66(421.86,434.078)
12	491.27(488.068,501.69)	0.17(0.147,0.218)	491.68(488.525,502.177)
13	473.115(470.413,480.36)	0.205(0.19,0.25)	473.55(470.75,480.76)
14	478.865(469.862,484.09)	0.15(0.14,0.22)	479.44(470.245,484.76)

Table 59. The median (first quantile – third quantile) of the Time (user-system-elapsed) in 10 iterations group by the number of hybrid principal component analysis (HPCA) for left temporal region of the brain.

HPCA	Min_Error	Optimum					
		index	K	Tau	Lambda	Eta	Kappa
1	0.55(0.508,0.641)	9(8,9)	1(1,3)	0.1(0.1,75.025)	0.1(0.1,0.325)	0.01(0,0.01)	0(0,0)
2	3.326(3.138,3.519)	2(2,2)	1(1,1)	1(1,1)	0.2(0.2,0.2)	0(0,0)	0(0,0)
3	4.716(4.349,5.151)	2(2,2)	1(1,1)	1(1,1)	0.2(0.2,0.2)	0(0,0)	0(0,0)
4	5.775(5.277,6.333)	1.5(1,2)	1(1,1.75)	0.55(0.1,1)	0.15(0.1,0.2)	0(0,0)	0(0,0)
5	7.017(6.789,7.296)	2(2,3)	1(1,2.5)	1(1,10)	0.2(0.2,0.3)	0(0,0)	0(0,0)
6	9.47(9.174,9.885)	3(1.25,5)	1.5(1,4.5)	0.1(0.1,0.775)	0.1(0.1,0.175)	0(0,0)	0(0,0)
7	9.356(8.903,9.871)	1(1,2.75)	5.5(2,6)	0.1(0.1,0.775)	0.1(0.1,0.175)	0(0,0)	0(0,0)
8	12.046(11.794,12.189)	1(1,1)	4(4,4)	0.1(0.1,0.1)	0.1(0.1,0.1)	0(0,0)	0(0,0)
9	12.465(11.742,12.926)	1(1,1)	2(1,3.5)	0.1(0.1,0.1)	0.1(0.1,0.1)	0(0,0)	0(0,75)
10	11.09(10.846,12.297)	1(1,8)	1(1,1)	0.1(0.1,0.1)	0.1(0.1,0.1)	0(0,0.008)	0(0,0)
11	12.491(11.899,12.73)	8.5(3.5,9.75)	1.5(1,2)	0.55(0.1,7.75)	0.15(0.1,0.275)	0.005(0,0.01)	50(0,100)
12	12.352(11.812,12.904)	4.5(1,9)	1(1,2.5)	0.1(0.1,0.1)	0.1(0.1,0.1)	0(0,0.01)	0(0,75)
13	12.693(12.128,12.918)	1.5(1,4)	1(1,6.75)	0.1(0.1,0.775)	0.1(0.1,0.175)	0(0,0)	0(0,75.25)
14	12.482(11.919,12.914)	1(1,9)	1(1,3.25)	0.1(0.1,0.1)	0.1(0.1,0.1)	0(0,0.01)	0(0,0)

Table 60. The median (first quantile – third quantile) of the model fit indices in 10 iterations group by the number of hybrid principal component analysis (HPCA) for left temporal region of the brain.

3.1.3.7 Right Temporal (6)

The MSPE and Correlation group by number of HPCA, Train and Test ([Table 61](#))

The User, System and Elapsed Time group by number of HPCA ([Table 62](#)): The median (first quantile – third quantile) of the Time (user-system-elapsed) in 10 iterations group by the number of hybrid principal component analysis (HPCA) for whole brain.

The Model fitting parameters group by number of HPCA ([Table 63](#)): The median (first quantile – third quantile) of the model fit indices in 10 iterations group by the number of hybrid principal component analysis (HPCA) for whole brain.

HPCA	MSPE		Correlation	
	Train	Test	Train	Test
1	0(0,0)	0.051(0.05,0.051)	1(1,1)	0.974(0.974,0.975)
2	0.004(0,0.008)	0.612(0.367,0.863)	0.998(0.996,1)	0.701(0.584,0.814)
3	0.057(0.057,0.057)	0.716(0.491,0.943)	0.971(0.971,0.971)	0.637(0.531,0.743)
4	0.179(0.037,0.183)	0.378(0.378,0.476)	0.906(0.904,0.981)	0.797(0.751,0.797)
5	0.052(0.014,0.14)	0.62(0.449,1.007)	0.973(0.927,0.993)	0.658(0.468,0.753)
6	0.208(0.092,0.264)	0.438(0.434,0.474)	0.89(0.858,0.953)	0.757(0.752,0.759)
7	0.245(0.104,0.281)	0.498(0.462,0.534)	0.869(0.848,0.947)	0.717(0.694,0.745)
8	0.34(0.339,0.34)	0.491(0.49,0.491)	0.812(0.812,0.813)	0.72(0.72,0.721)
9	0.348(0.348,0.348)	0.451(0.451,0.451)	0.807(0.807,0.807)	0.744(0.744,0.744)
10	0.372(0.372,0.372)	0.604(0.604,0.604)	0.792(0.792,0.792)	0.644(0.644,0.644)
11	0.377(0.377,0.377)	0.624(0.624,0.625)	0.789(0.789,0.789)	0.629(0.629,0.629)
12	0.39(0.39,0.39)	0.607(0.607,0.608)	0.781(0.781,0.781)	0.639(0.638,0.639)
13	0.41(0.41,0.41)	0.576(0.576,0.576)	0.768(0.768,0.768)	0.657(0.657,0.657)
14	0.411(0.411,0.411)	0.626(0.626,0.626)	0.767(0.767,0.767)	0.627(0.627,0.627)

Table 61. The median (first quantile – third quantile) of the Mean Square Prediction Error (MSPE) and Correlation in 10 iterations group by Train and Test and the number of hybrid principal component analysis (HPCA) for right temporal region of the brain.

HPCA	Time (s)		
	User	System	Elapsed
1	134.655(133.883,135.99)	0.055(0.05,0.083)	134.765(134.035,136.087)
2	162.19(159.59,163.907)	0.075(0.063,0.11)	162.345(159.685,164.372)
3	246.31(242.053,250.637)	0.125(0.073,0.148)	246.55(242.32,250.93)
4	261.48(255.098,266.955)	0.105(0.085,0.185)	261.845(255.382,267.31)
5	337.975(335.01,341.485)	0.145(0.133,0.157)	338.405(335.375,341.86)
6	377.94(374.205,383.617)	0.12(0.087,0.16)	378.23(374.632,384.005)
7	426.21(416.545,430.733)	0.135(0.112,0.198)	426.79(416.918,431.165)
8	395.57(389.53,399.938)	0.16(0.12,0.205)	396.005(389.818,400.46)
9	422.66(416.485,425.012)	0.175(0.125,0.21)	423.02(416.872,425.427)
10	439.865(434.787,446.15)	0.165(0.118,0.193)	440.275(435.172,446.623)
11	491.8(487.195,495.978)	0.17(0.143,0.23)	492.225(487.715,496.43)
12	441.275(434.273,446.172)	0.18(0.113,0.208)	441.705(434.72,446.542)
13	463.785(459.97,470.732)	0.14(0.12,0.185)	464.235(460.348,471.085)
14	480.595(473.235,483.88)	0.18(0.12,0.223)	481.07(473.688,485.953)

Table 62. The median (first quantile – third quantile) of the Mean Square Prediction Error (MSPE) and Correlation in 10 iterations group by Train and Test and the number of hybrid principal component analysis (HPCA) for occipital region of the brain.

HPCA	Min_Error	Optimum					
		index	K	Tau	Lambda	Eta	Kappa
1	0.455(0.386,0.494)	9(9,9)	2.5(2,3)	0.1(0.1,0.1)	0.1(0.1,0.1)	0.01(0.01,0.01)	0(0,0)
2	3.141(2.705,3.767)	5(1,9)	1.5(1,2.75)	0.1(0.1,0.1)	0.1(0.1,0.1)	0.005(0,0.01)	0(0,0)
3	3.948(3.611,4.378)	9(1,9)	1(1,1)	0.1(0.1,0.1)	0.1(0.1,0.1)	0.01(0,0.01)	0(0,0)
4	4.017(3.777,4.201)	10.5(8.25,12)	1(1,2.5)	100(3.25,100)	0.4(0.225,0.4)	0.01(0.003,0.01)	0(0,0)
5	5.3(5.109,5.62)	9(1.75,10.5)	2.5(2,3)	0.1(0.1,77.5)	0.1(0.1,0.375)	0.01(0,0.01)	0(0,0)
6	5.743(5.654,5.838)	10(1,12)	1.5(1,3.5)	100(0.1,100)	0.4(0.1,0.4)	0.005(0,0.01)	0(0,0)
7	6.179(5.969,6.349)	4.5(1,11.25)	1(1,3.5)	0.1(0.1,100)	0.1(0.1,0.4)	0(0,0.01)	0(0,0)
8	7.555(7.325,7.715)	12(9,12)	1(1,1)	100(100,100)	0.4(0.4,0.4)	0.01(0.003,0.01)	0(0,0)
9	8.236(7.706,8.756)	12(12,12)	1(1,1)	100(100,100)	0.4(0.4,0.4)	0.01(0.01,0.01)	0(0,0)
10	8.356(8.035,8.634)	12(12,12)	1(1,1)	100(100,100)	0.4(0.4,0.4)	0.01(0.01,0.01)	0(0,0)
11	8.474(8.241,8.638)	12(12,12)	1(1,1)	100(100,100)	0.4(0.4,0.4)	0.01(0.01,0.01)	0(0,0)
12	9.009(8.102,9.14)	12(12,12)	1(1,1)	100(100,100)	0.4(0.4,0.4)	0.01(0.01,0.01)	0(0,0)
13	8.861(8.574,9.42)	12(12,12)	1(1,1)	100(100,100)	0.4(0.4,0.4)	0.01(0.01,0.01)	0(0,0)
14	8.952(8.858,9.638)	12(12,12)	1(1,1)	100(100,100)	0.4(0.4,0.4)	0.01(0.01,0.01)	0(0,0)

Table 63. The median (first quantile – third quantile) of the model fit indices in 10 iterations group by the number of hybrid principal component analysis (HPCA) for right temporal region of the brain.

3.1.3.8 Figures

The MSPE over Number of HPCA group by Regions: **Figure 46**

The Correlation over Number of HPCA group by Regions: **Figure 47**

The Elapsed Time over Number of HPCA group by Regions: **Figure 48**

The MSPE over Number of HPCA group by Regions

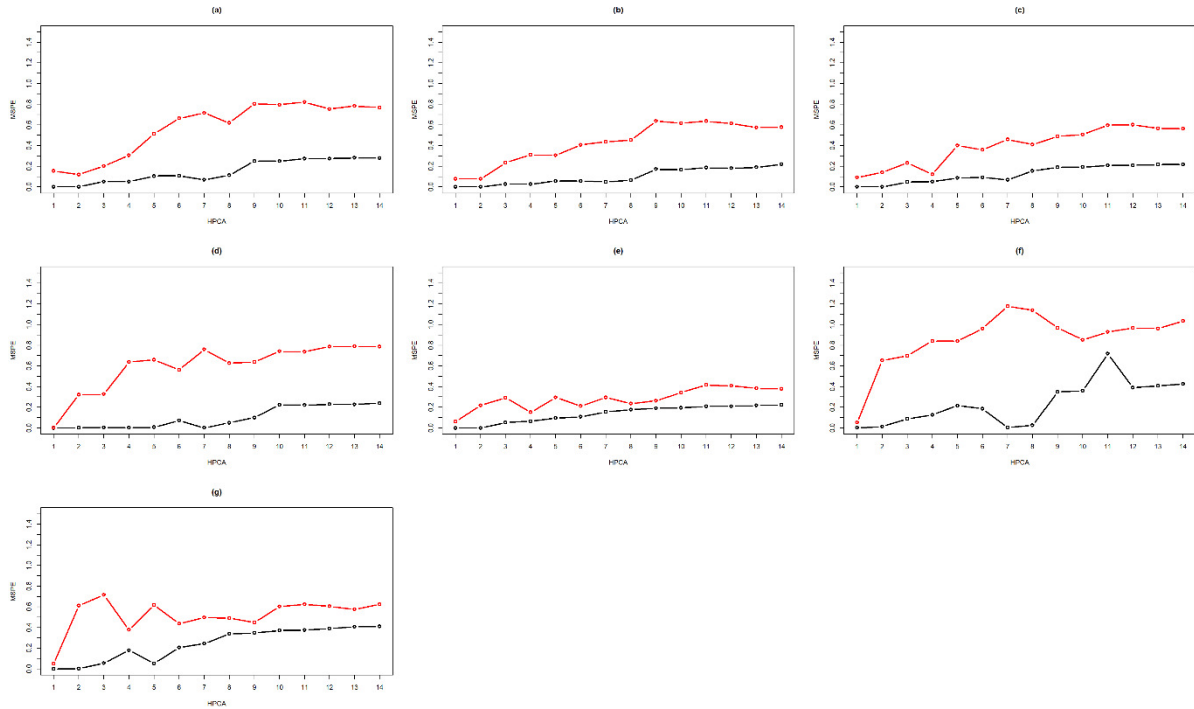


Figure 46. (a-g) Median of mean square prediction error (MSPE) for train (black solid line) and test (red solid line) in 10 iterations of (a) whole, (b) frontal, (c) central, (d) occipital, (e) parietal, (f) left temporal and (g) right temporal regions, respectively.

The Correlation over Number of HPCA group by Regions

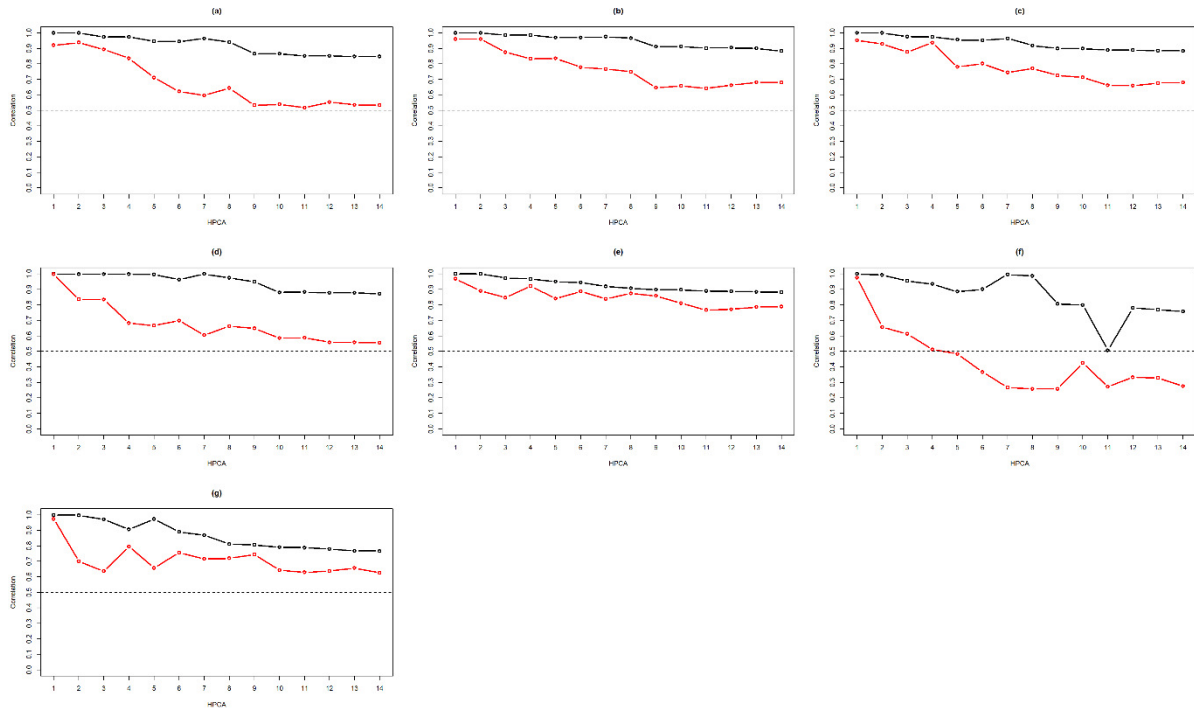


Figure 47. (a-g) Median of correlation for train (black solid line) and test (red solid line) in 10 iterations of (a) whole, (b) frontal, (c) central, (d) occipital, (e) parietal, (f) left temporal and (g) right temporal regions, respectively.

The Elapsed Time over Number of HPCA group by Regions

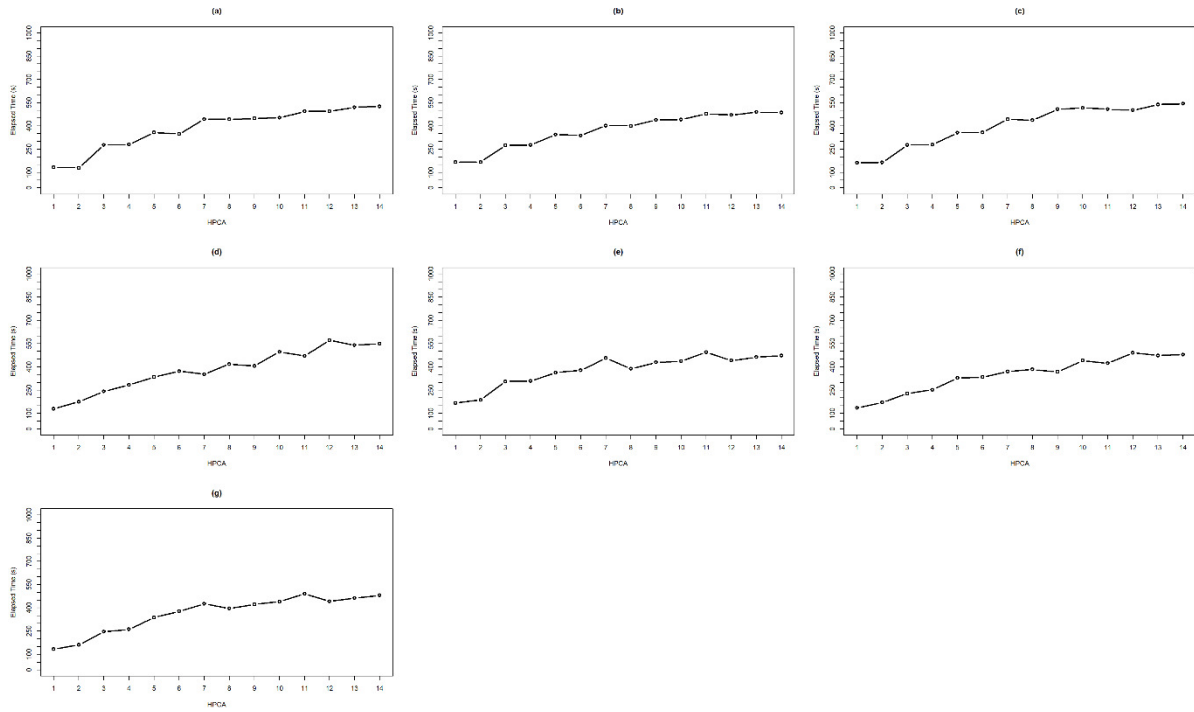


Figure 48. (a-g) Median of elapsed time in 10 iterations of (a) whole, (b) frontal, (c) central, (d) occipital, (e) parietal, (f) left temporal and (g) right temporal regions, respectively.

3.1.4. Conclusion .

According to the [Figure 46](#), the MSPE increased by adding HPCA from 1 to KLM in all regions. It also indicates that the difference between training and testing exists. The [Figure 47](#) shows the correlation. The highest correlation is in the first HPCA in all regions fro both training and testing datasets. The elapsed time is also increasing by adding HPCA. ([Figure 48](#))

We conclude that the first HPCA has the best performance based on the MSPE, correlation and elapsed times in all region of the brain.

3.2. Target Stimuli

3.2.1. The HPCA Decomposition

3.2.1.1 The Functional dimension

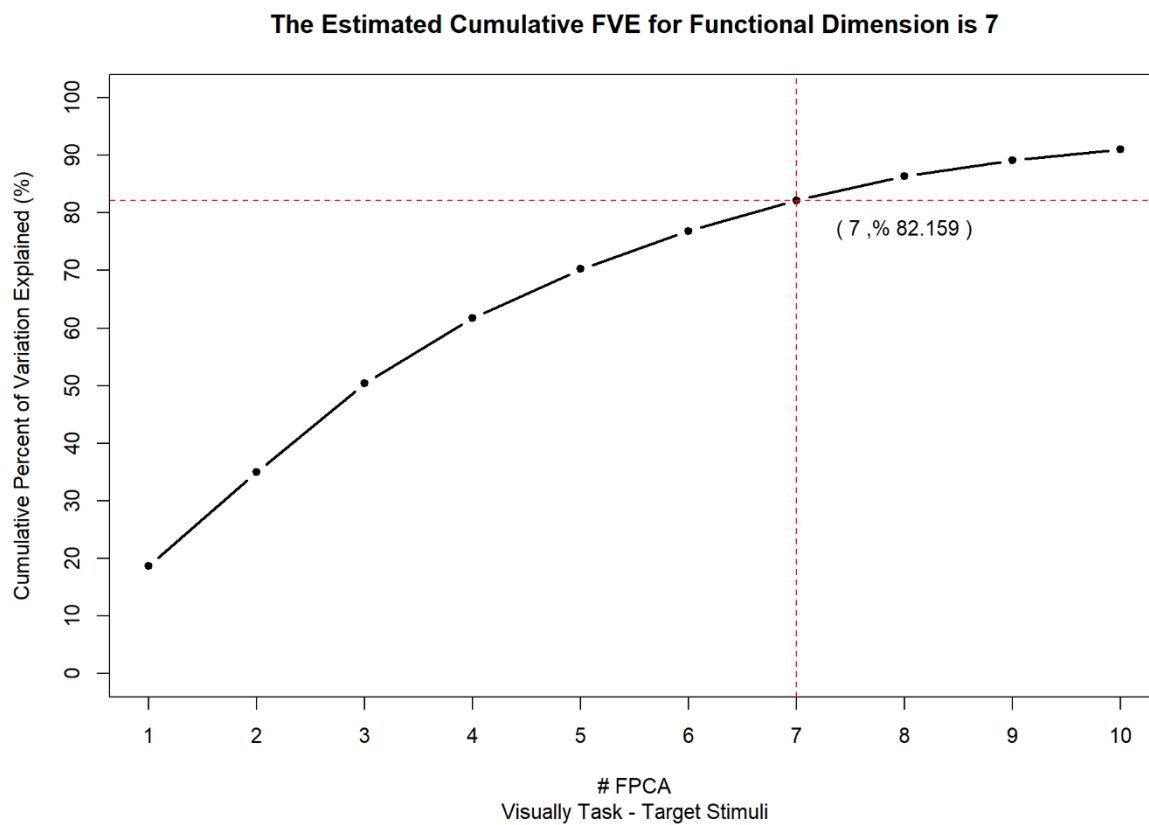


Figure 49. The estimated cumulative fraction of variations (FVE) for functional dimension.

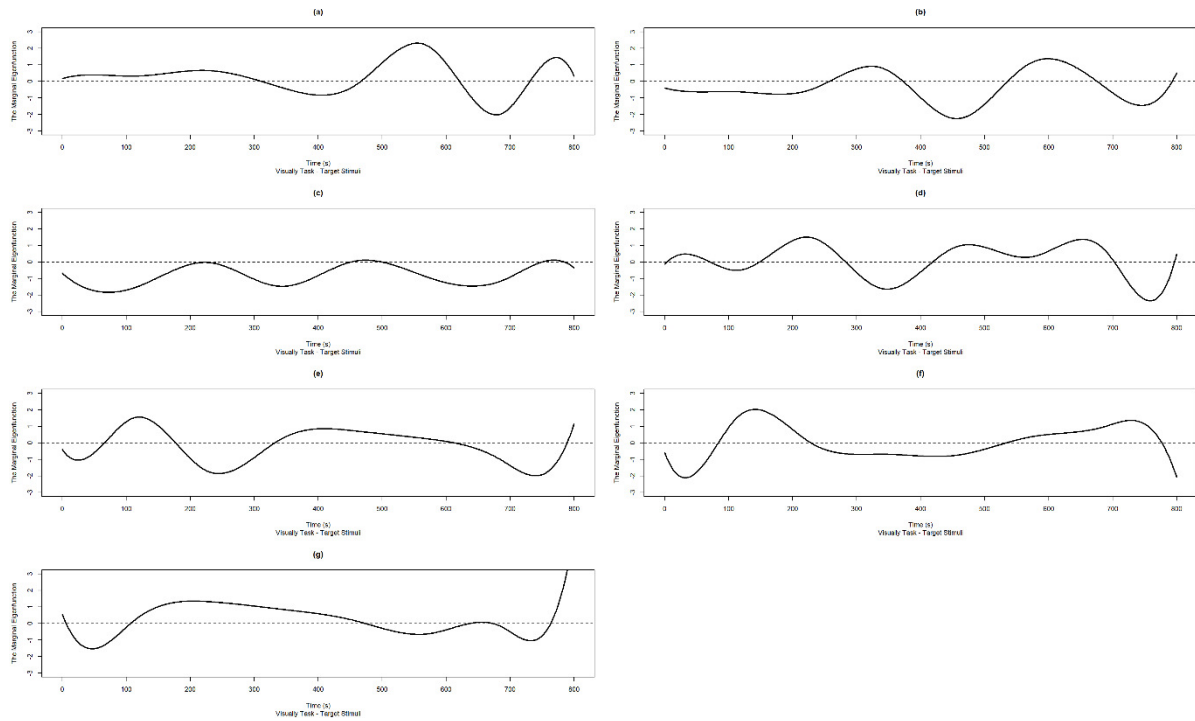


Figure 50. (a, g) Estimated first to seven leading functional marginal eigenfunctions.

According to the [Figure 49](#), the first seven marginal functional principal component analysis (FPCA) capture at least 85% of variation in this dimension. The [Figure 50](#) plots estimated first to seven leading functional marginal eigenfunctions on the x-axis (time).

3.2.1.2 The Longitudinal dimension

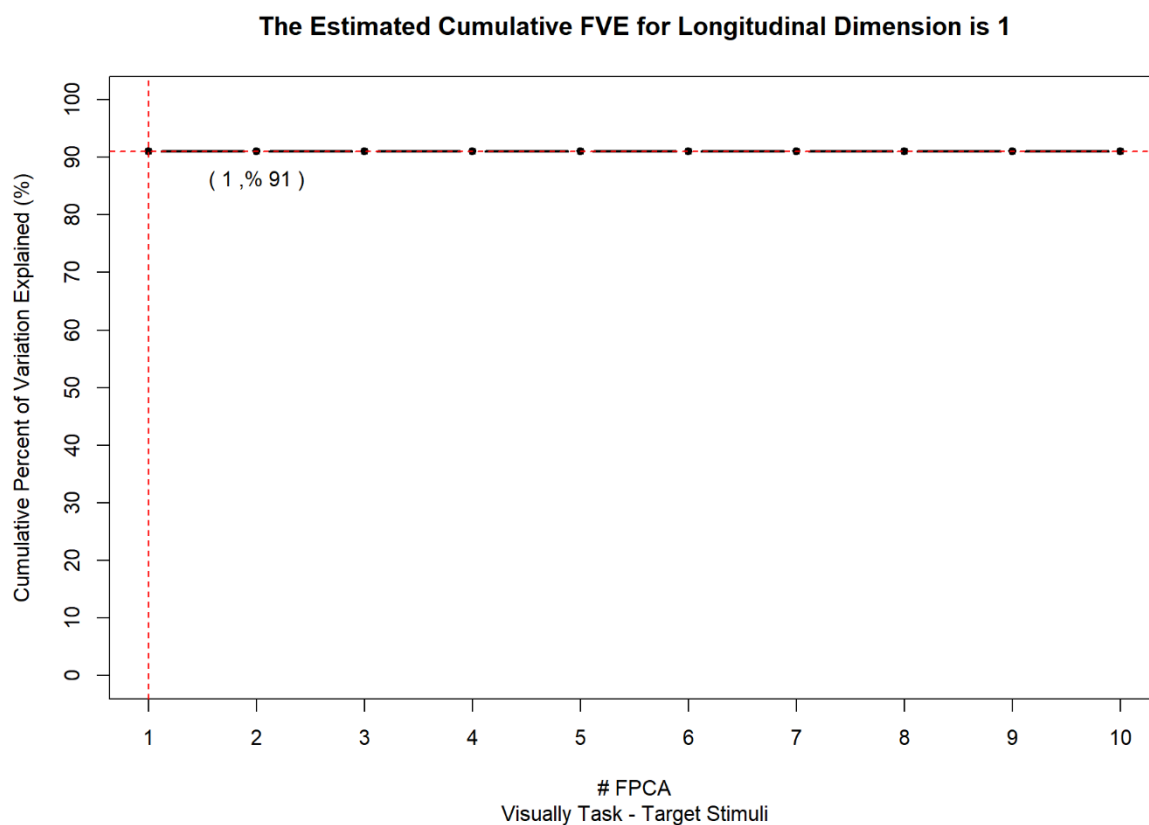


Figure 51. The estimated cumulative fraction of variations (FVE) for longitudinal dimension.

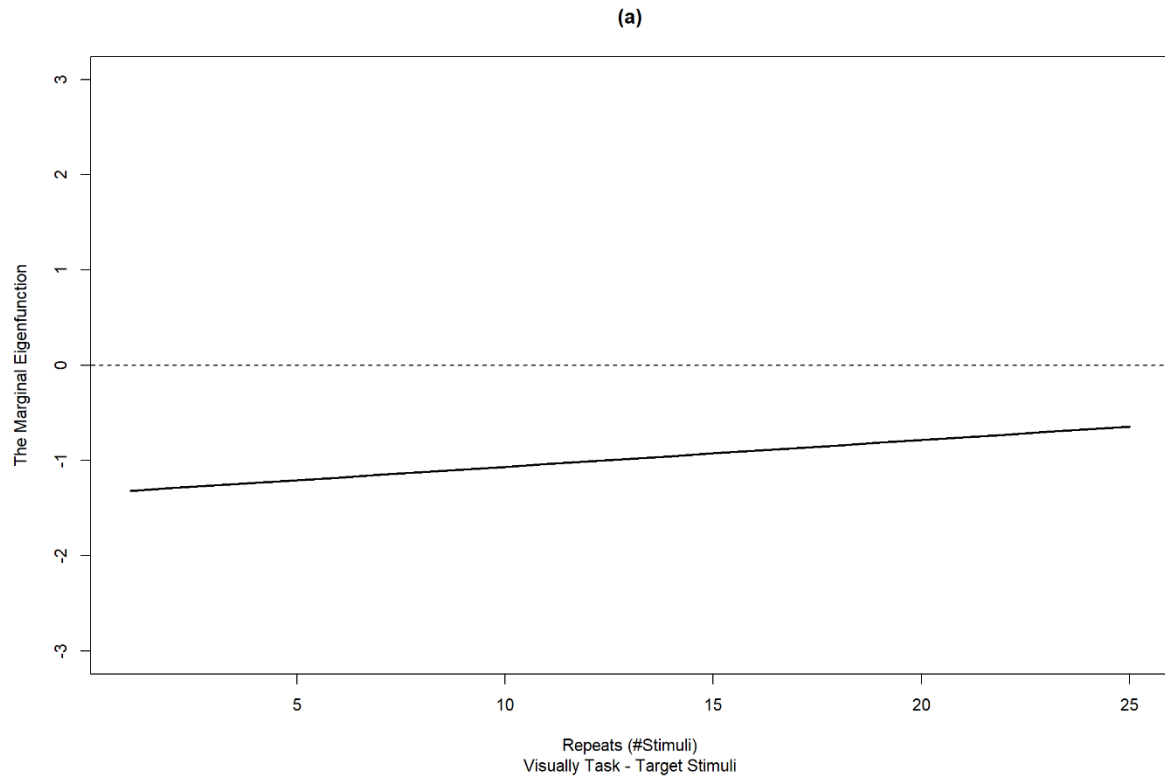


Figure 52. Estimated first leading longitudinal marginal eigenfunctions.

According to the [Figure 51](#), the first marginal functional principal component analysis (FPCA) capture at least 90% of variation in this dimension. The [Figure 52](#) plots estimated first leading longitudinal marginal eigenfunctions on the receptions.

3.2.1.3 The Regional dimension

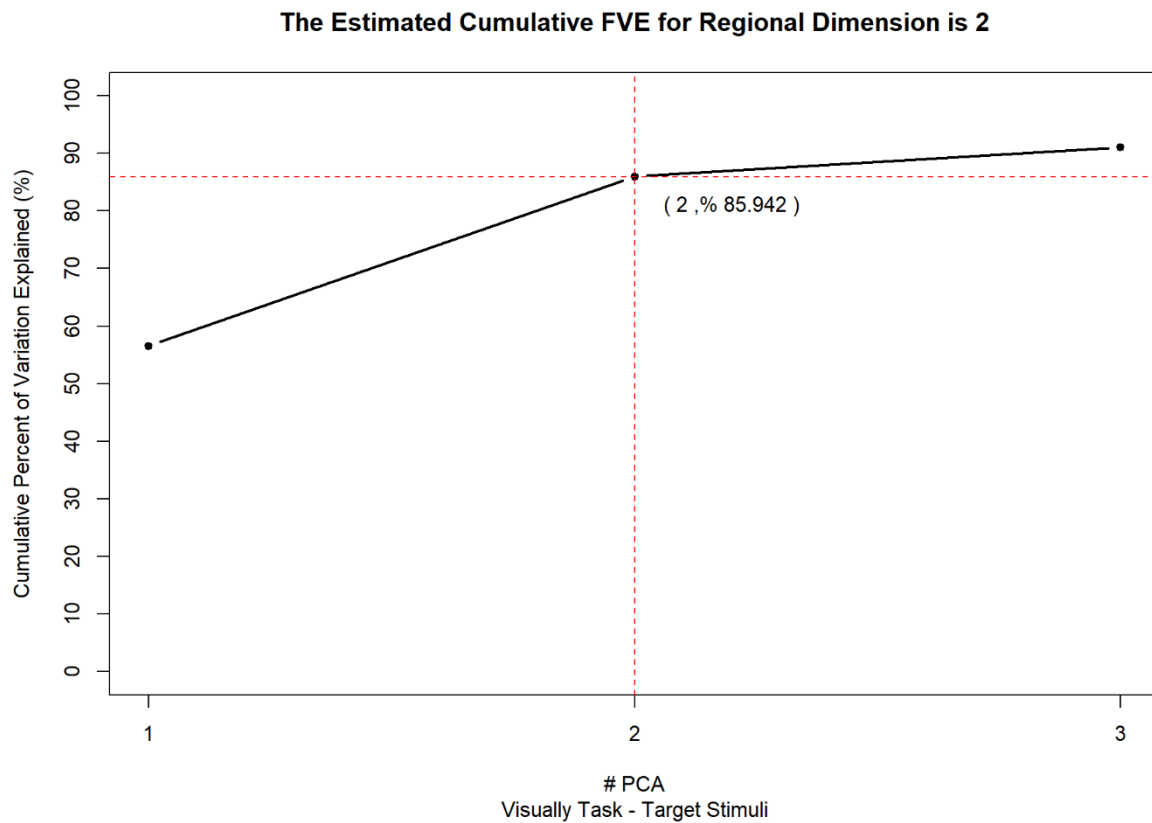


Figure 53. The estimated cumulative fraction of variations (FVE) for regional dimension.



Figure 54. Estimated first and second leading regional marginal eigenvectors.

According to the Figure 53, the first and second marginal principal component analysis (PCA) capture at least nearly 85% of variation in this dimension. The Figure 54 plots estimated first and second leading regional marginal eigenvectors.

3.2.2. The Reconstructed of the Response .

The fof regression were conducted group by the number of HPCA and the regions. In the following part, we show the reconstructed ERPs for each subject (in gray) based on the number of HPCA. In this trial, we have 14 HPCA. We also show the point-wise mean and mean $\pm$ standard deviation. (Whole Brain: **Figure 55**, Frontal region (1): **Figure 56**, Central Region (2): **Figure 57**, Occipital Region (3): **Figure 58**, Parietal Region (4): **Figure 59**, Left Temporal (5): **Figure 60**, Right Temporal (6): **Figure 61**).

3.2.2.1 Whole Brain

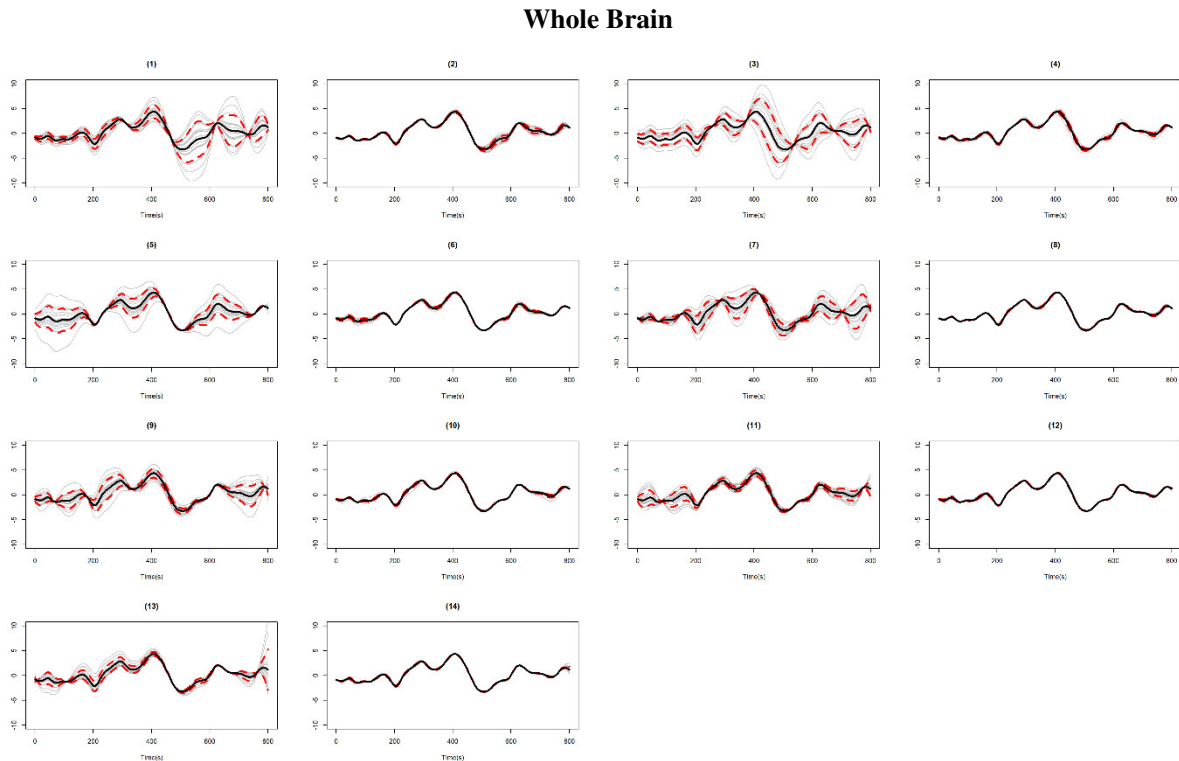


Figure 55. (1-14) Reconstructed ERPs for each subject (Gray line), point-wise mean (solid black line), and mean $\pm$ standard deviation (dotted red line) for whole Brain.

3.2.2.2 Frontal Region (1)

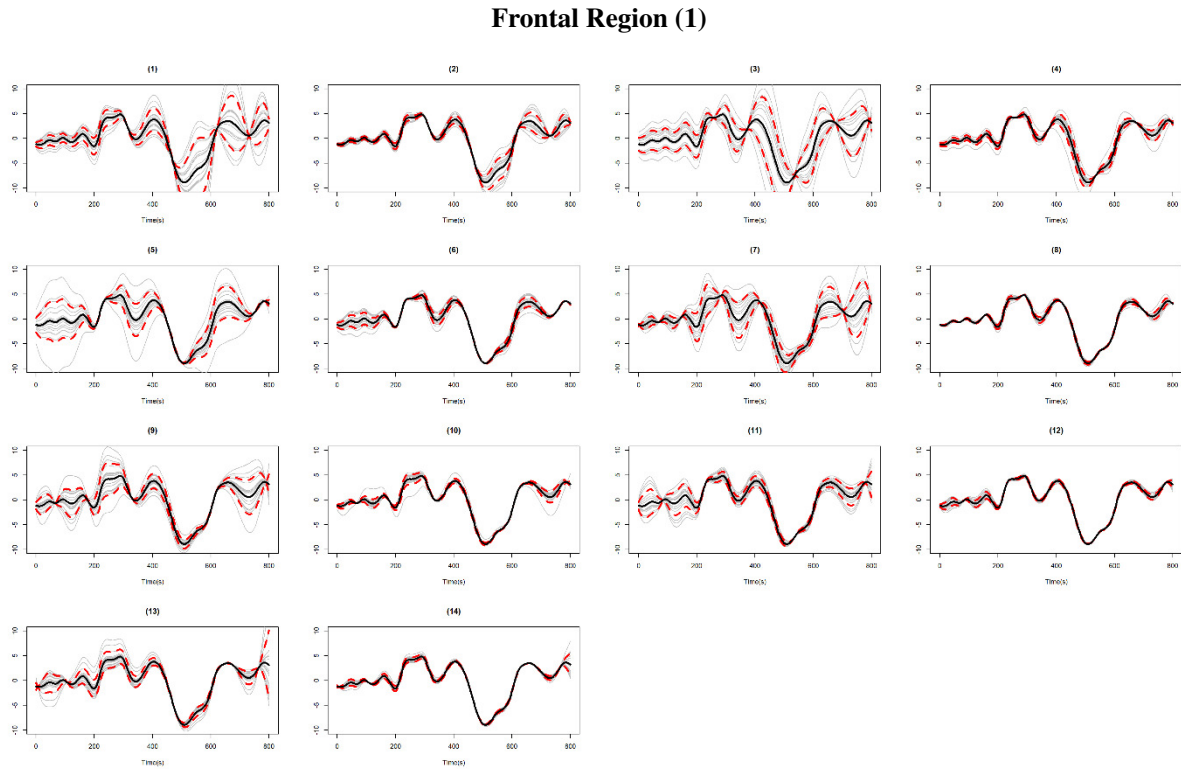


Figure 56. (1-14) Reconstructed ERPs for each subject (Gray line), point-wise mean (solid black line), and mean $\pm$ standard deviation (dotted red line) for frontal region of the Brain.

3.2.2.3 Central Region (2)

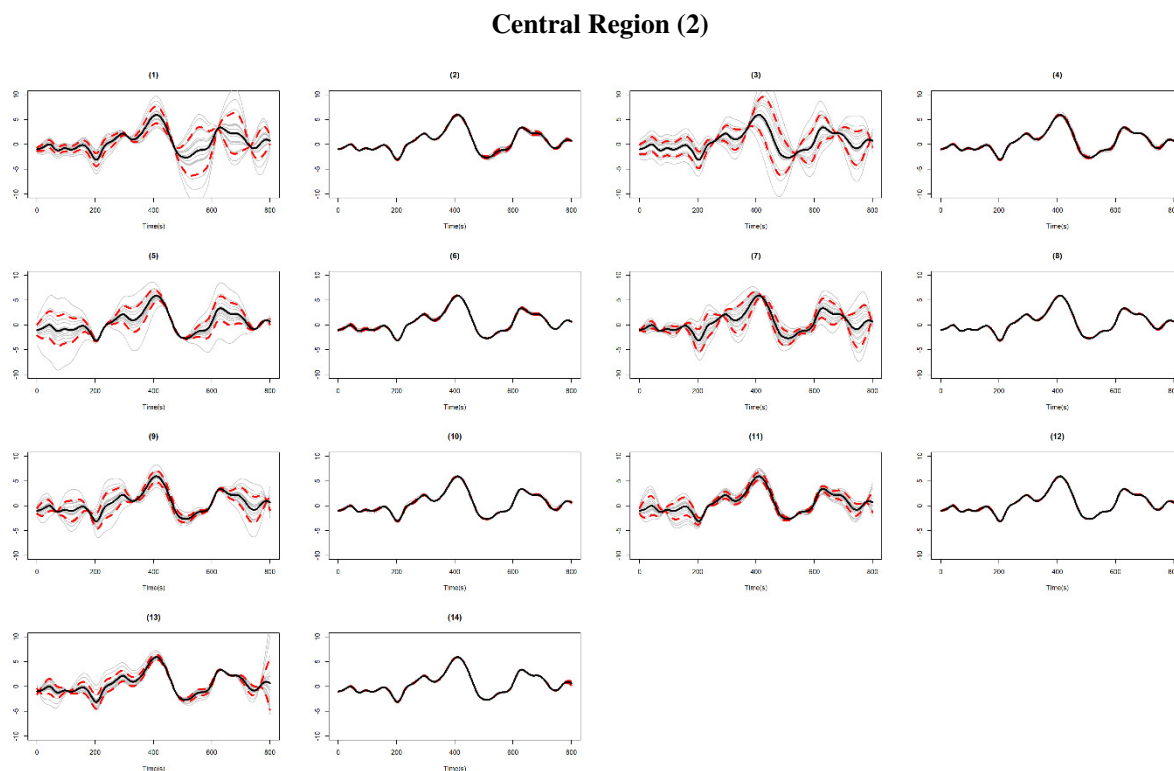


Figure 57. (1-14) Reconstructed ERPs for each subject (Gray line), point-wise mean (solid black line), and mean $\pm$ standard deviation (dotted red line) for central region of the Brain.

3.2.2.4 Occipital Region (3)

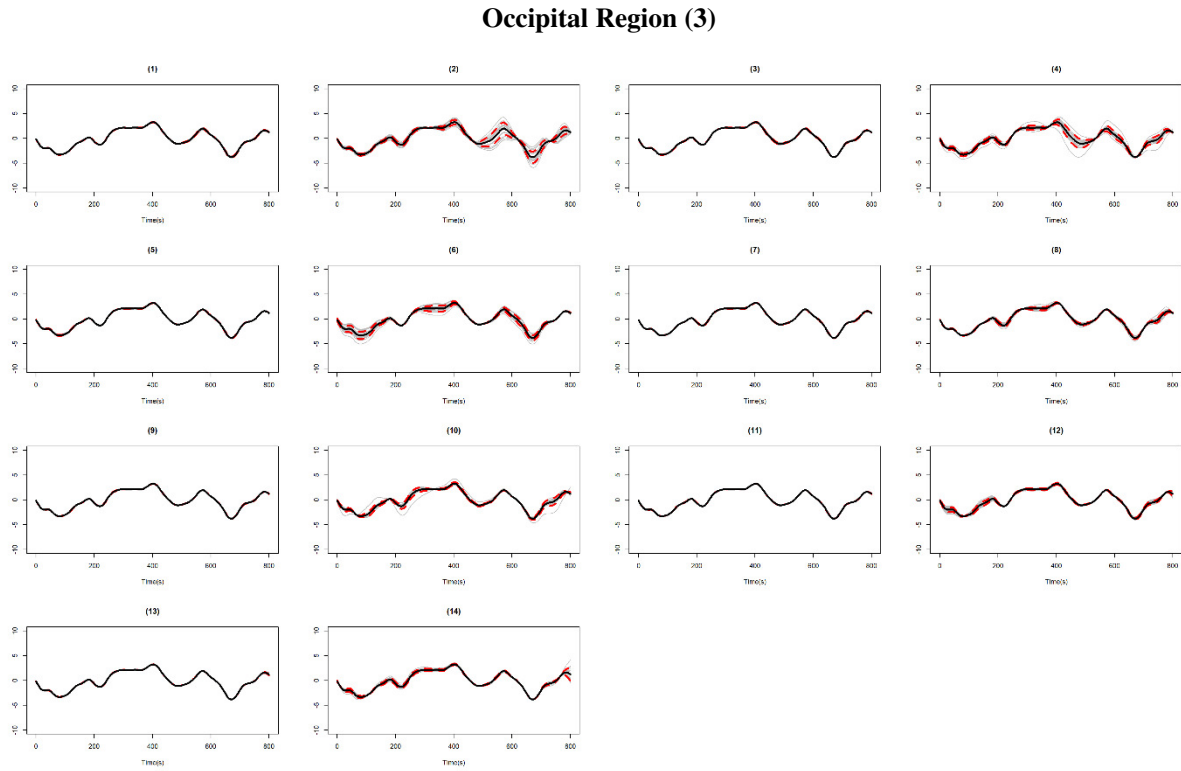


Figure 58. (1-14) Reconstructed ERPs for each subject (Gray line), point-wise mean (solid black line), and mean $\pm$ standard deviation (dotted red line) for occipital region of the Brain.

3.2.2.5 Parietal Region (4)

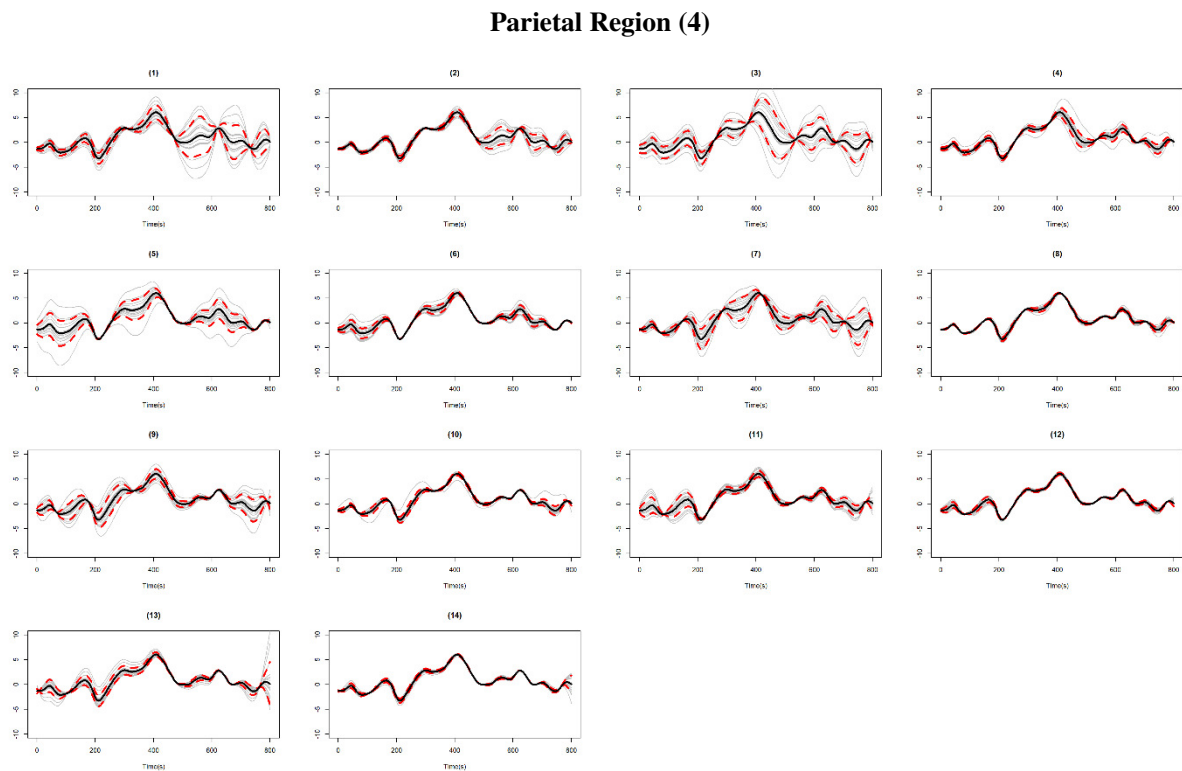


Figure 59. (1-14) Reconstructed ERPs for each subject (Gray line), point-wise mean (solid black line), and mean $\pm$ standard deviation (dotted red line) for parietal region of the Brain.

3.2.2.6 Left Temporal (5)

Left Temporal (5)

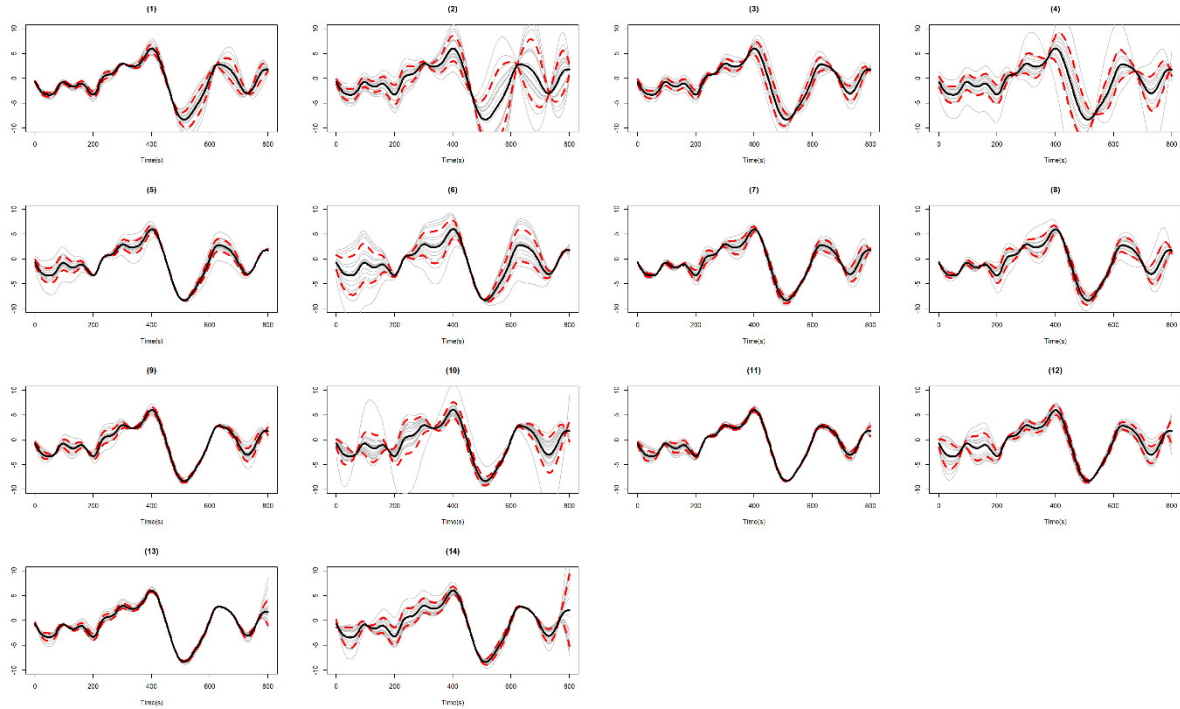


Figure 60. (1-14) Reconstructed ERPs for each subject (Gray line), point-wise mean (solid black line), and mean $\pm$ standard deviation (dotted red line) for left temporal region of the Brain.

3.2.2.7 Right Temporal (6)

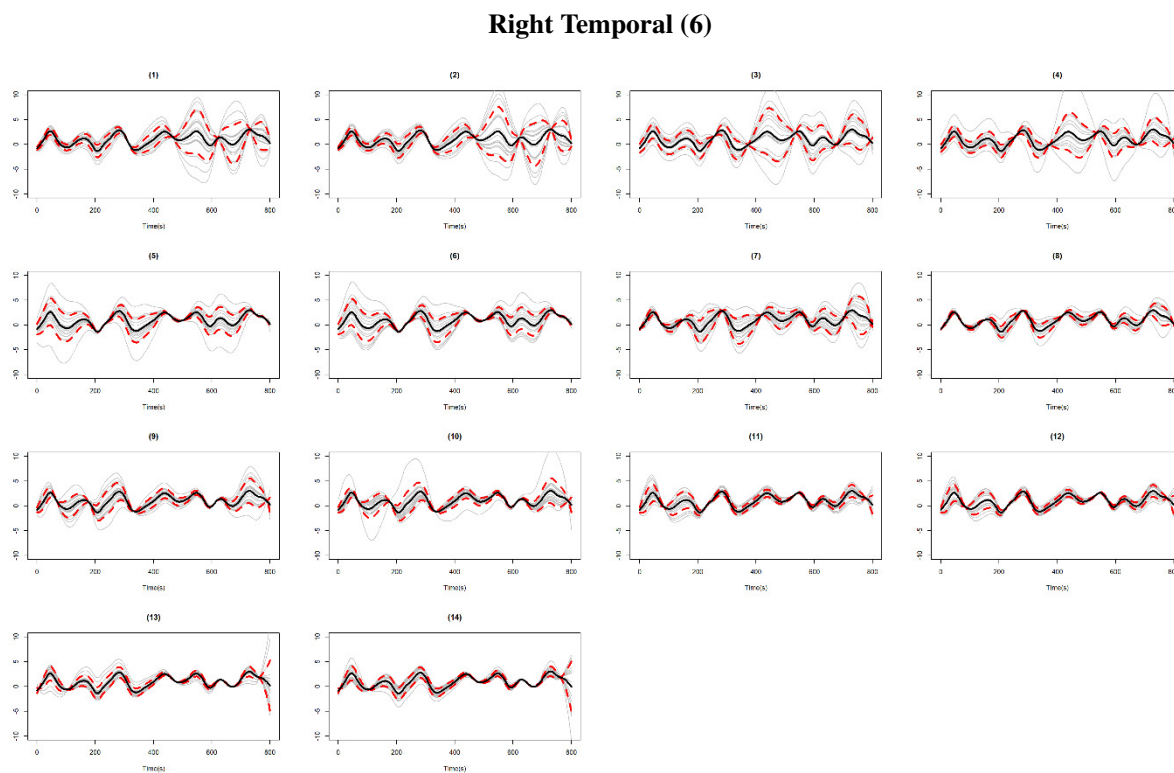


Figure 61. (1-14) Reconstructed ERPs for each subject (Gray line), point-wise mean (solid black line), and mean $\pm$ standard deviation (dotted red line) for right temporal region of the Brain.

3.2.3. *The function-on-function result* .

The fof regression result group by the region of the brain and the number of hybrid principal component analysis (HPCA) were reported.

3.2.3.1 **Whole Brain**

The MSPE and Correlation group by number of HPCA, Train and Test ([Table 64](#))

The User, System and Elapsed Time group by number of HPCA ([Table 65](#)): The median (first quantile – third quantile) of the Time (user-system-elapsed) in 10 iterations group by the number of hybrid principal component analysis (HPCA) for whole brain.

The Model fitting parameters group by number of HPCA ([Table 66](#)): The median (first quantile – third quantile) of the model fit indices in 10 iterations group by the number of hybrid principal component analysis (HPCA) for whole brain.

HPCA	MSPE		Correlation	
	Train	Test	Train	Test
1	0.444(0.258,0.497)	0.545(0.438,0.599)	0.773(0.731,0.869)	0.692(0.645,0.755)
2	0.316(0.192,0.437)	0.588(0.543,0.64)	0.84(0.778,0.899)	0.66(0.612,0.703)
3	0.331(0.32,0.332)	0.505(0.505,0.508)	0.817(0.817,0.824)	0.71(0.709,0.71)
4	0.328(0.324,0.328)	0.487(0.487,0.525)	0.819(0.819,0.822)	0.722(0.697,0.722)
5	0.341(0.341,0.341)	0.559(0.559,0.559)	0.811(0.811,0.811)	0.679(0.679,0.679)
6	0.338(0.338,0.338)	0.599(0.599,0.6)	0.813(0.813,0.813)	0.655(0.655,0.655)
7	0.482(0.472,0.482)	0.637(0.637,0.637)	0.719(0.719,0.726)	0.612(0.612,0.612)
8	0.634(0.487,0.653)	0.689(0.655,0.737)	0.633(0.614,0.717)	0.572(0.519,0.594)
9	0.633(0.556,0.666)	0.694(0.615,0.71)	0.63(0.598,0.671)	0.56(0.546,0.623)
10	0.631(0.557,0.666)	0.707(0.612,0.708)	0.63(0.597,0.671)	0.547(0.544,0.624)
11	0.47(0.47,0.613)	0.683(0.678,0.741)	0.728(0.636,0.728)	0.576(0.51,0.584)
12	0.599(0.468,0.683)	0.71(0.687,0.759)	0.642(0.576,0.729)	0.546(0.491,0.578)
13	0.636(0.582,0.7)	0.855(0.798,0.893)	0.621(0.56,0.65)	0.392(0.355,0.48)
14	0.633(0.604,0.691)	0.854(0.777,0.896)	0.623(0.568,0.642)	0.393(0.353,0.471)

Table 64. The median (first quantile – third quantile) of the Mean Square Prediction Error (MSPE) and Correlation in 10 iterations group by Train and Test and the number of hybrid principal component analysis (HPCA) for whole brain.

HPCA	Time (s)		
	User	System	Elapsed
1	168.715(161.143,176.38)	0.16(0.088,0.23)	169.94(161.608,178.215)
2	166.94(154.008,183.305)	0.135(0.09,0.158)	168.51(154.563,184.663)
3	327.46(323.39,330.028)	0.175(0.16,0.33)	328.37(324.238,333.67)
4	318.93(315.188,320.623)	0.165(0.15,0.225)	320.045(316.515,321.395)
5	356.46(350.34,360.425)	0.19(0.163,0.238)	357.45(351.248,361.522)
6	354.865(349.84,357.45)	0.19(0.137,0.2)	355.7(350.555,358.248)
7	438.385(431.875,448.308)	0.23(0.215,0.28)	439.745(433.153,449.355)
8	449.37(444.8,451.5)	0.21(0.175,0.243)	450.26(445.908,452.617)
9	466.72(460.81,470.72)	0.215(0.173,0.245)	467.845(461.83,471.713)
10	464.585(459.84,471.187)	0.22(0.167,0.23)	465.46(460.665,472.24)
11	510.275(506.03,520.41)	0.235(0.192,0.285)	511.365(506.805,521.725)
12	500.905(499.87,519.165)	0.225(0.175,0.257)	501.735(500.65,520.31)
13	564.01(561.85,567.015)	0.24(0.193,0.302)	564.64(562.49,567.8)
14	570.475(560.6,575.29)	0.23(0.2,0.3)	571.065(561.118,576.287)

Table 65. The median (first quantile – third quantile) of the Time (user-system-elapsed) in 10 iterations group by the number of hybrid principal component analysis (HPCA) for whole brain.

HPCA	Min_Error	Optimum					
		index	K	Tau	Lambda	Eta	Kappa
1	10.039(9.127,10.515)	12(9.75,12)	1(1,1)	100(100,100)	0.4(0.4,0.4)	0.01(0.01,0.01)	0.01(0.003,0.01)
2	9.766(9.35,10.066)	12(5,12)	1(1,2)	100(100,100)	0.4(0.4,0.4)	0.01(0,0.01)	0.005(0,0.01)
3	7.869(7.433,9.153)	12(11.25,12)	1(1,1)	100(100,100)	0.4(0.4,0.4)	0.01(0.01,0.01)	0(0,0)
4	8.452(7.974,8.956)	12(11.25,12)	1(1,1)	100(100,100)	0.4(0.4,0.4)	0.01(0.01,0.01)	0(0,0)
5	8.428(8.07,9.118)	12(12,12)	1(1,1)	100(100,100)	0.4(0.4,0.4)	0.01(0.01,0.01)	0(0,0)
6	8.746(8.18,9.326)	12(12,12)	1(1,1)	100(100,100)	0.4(0.4,0.4)	0.01(0.01,0.01)	0(0,0)
7	11.103(10.188,11.416)	12(11.25,12)	1(1,1)	100(100,100)	0.4(0.4,0.4)	0.01(0.01,0.01)	0(0,0)
8	12.229(12.058,12.913)	12(12,12)	1(1,1.75)	100(100,100)	0.4(0.4,0.4)	0.01(0.01,0.01)	0.01(0,0.01)
9	12.245(11.899,12.781)	12(12,12)	1.5(1,2)	100(100,100)	0.4(0.4,0.4)	0.01(0.01,0.01)	0.01(0.003,0.01)
10	12.743(11.594,12.998)	12(12,12)	1(1,2)	100(100,100)	0.4(0.4,0.4)	0.01(0.01,0.01)	0.01(0.003,0.01)
11	12.236(11.833,12.85)	12(12,12)	2(2,2)	100(100,100)	0.4(0.4,0.4)	0.01(0.01,0.01)	0(0,0.008)
12	12.824(12.698,12.923)	12(12,12)	2(1.25,2)	100(100,100)	0.4(0.4,0.4)	0.01(0.01,0.01)	0.005(0,0.01)
13	12.887(12.341,13.112)	8(8,12)	1.5(1,2)	100(100,100)	0.4(0.4,0.4)	0(0,0.01)	0.01(0.003,0.01)
14	12.94(12.677,13.172)	12(6.75,12)	1(1,2)	100(100,100)	0.4(0.4,0.4)	0.01(0.003,0.01)	0.01(0.01,0.01)

Table 66. The median (first quantile – third quantile) of the model fit indices in 10 iterations group by the number of hybrid principal component analysis (HPCA) for whole brain.

3.2.3.2 Frontal Region (1)

The MSPE and Correlation group by number of HPCA, Train and Test ([Table 67](#))

The User, System and Elapsed Time group by number of HPCA ([Table 68](#)): The median (first quantile – third quantile) of the Time (user-system-elapsed) in 10 iterations group by the number of hybrid principal component analysis (HPCA) for frontal region of brain.

The Model fitting parameters group by number of HPCA ([Table 69](#)): The median (first quantile – third quantile) of the model fit indices in 10 iterations group by the number of hybrid principal component analysis (HPCA) for whole brain.

HPCA	MSPE		Correlation	
	Train	Test	Train	Test
1	0.175(0.166,0.175)	0.329(0.326,0.329)	0.909(0.909,0.913)	0.819(0.819,0.823)
2	0.152(0.055,0.152)	0.522(0.514,0.522)	0.921(0.921,0.972)	0.719(0.719,0.719)
3	0.287(0.287,0.287)	0.506(0.506,0.517)	0.844(0.844,0.844)	0.715(0.706,0.715)
4	0.282(0.278,0.282)	0.474(0.474,0.479)	0.847(0.847,0.849)	0.736(0.733,0.736)
5	0.294(0.288,0.294)	0.575(0.575,0.599)	0.84(0.84,0.843)	0.677(0.639,0.677)
6	0.292(0.255,0.292)	0.647(0.542,0.662)	0.841(0.841,0.863)	0.625(0.625,0.69)
7	0.428(0.424,0.537)	0.676(0.675,0.712)	0.756(0.692,0.759)	0.587(0.536,0.594)
8	0.434(0.427,0.569)	0.691(0.689,0.727)	0.752(0.675,0.756)	0.575(0.522,0.583)
9	0.427(0.427,0.486)	0.66(0.66,0.689)	0.757(0.717,0.757)	0.598(0.574,0.598)
10	0.564(0.421,0.564)	0.73(0.675,0.734)	0.677(0.677,0.761)	0.519(0.515,0.594)
11	0.425(0.425,0.519)	0.71(0.71,0.723)	0.758(0.703,0.758)	0.567(0.535,0.567)
12	0.422(0.422,0.422)	0.725(0.725,0.725)	0.76(0.76,0.76)	0.558(0.558,0.558)
13	0.565(0.565,0.565)	0.876(0.876,0.876)	0.673(0.673,0.673)	0.394(0.394,0.394)
14	0.556(0.458,0.562)	0.821(0.793,0.873)	0.679(0.675,0.736)	0.461(0.398,0.493)

Table 67. The median (first quantile – third quantile) of the Mean Square Prediction Error (MSPE) and Correlation in 10 iterations group by Train and Test and the number of hybrid principal component analysis (HPCA) for frontal region brain.

HPCA	Time (s)		
	User	System	Elapsed
1	218.345(213.175,221.848)	0.105(0.082,0.135)	218.645(213.56,222.15)
2	205.185(203.98,206.122)	0.105(0.053,0.135)	205.46(204.257,206.502)
3	316.81(312.365,321.803)	0.095(0.08,0.145)	317.185(312.778,322.167)
4	301.97(298.912,305.875)	0.145(0.1,0.198)	302.425(299.19,306.237)
5	351.81(351.195,353.833)	0.125(0.08,0.165)	352.3(351.478,354.388)
6	358.74(357.285,365.42)	0.145(0.13,0.168)	359.3(357.662,365.765)
7	434.1(431.98,439.97)	0.175(0.16,0.235)	434.775(432.523,440.43)
8	433.56(432.743,446.1)	0.26(0.213,0.32)	434.57(433.793,447.135)
9	499.855(496.142,504.325)	0.275(0.227,0.3)	500.8(497.74,505.27)
10	504.96(497.02,513.663)	0.255(0.22,0.278)	506.335(498.852,514.852)
11	505.04(501.072,517.07)	0.23(0.19,0.27)	505.835(502.165,518.5)
12	497.58(488.878,509.483)	0.23(0.205,0.337)	498.485(489.512,511.962)
13	526.46(521.37,533.782)	0.22(0.175,0.245)	527.06(522.022,534.465)
14	533.31(520.935,534.908)	0.175(0.147,0.225)	534.245(521.575,537.73)

Table 68. The median (first quantile – third quantile) of the Time (user-system-elapsed) in 10 iterations group by the number of hybrid principal component analysis (HPCA) for frontal region of brain.

HPCA	Min.Error	Optimum					
		index	K	Tau	Lambda	Eta	Kappa
1	7.418(7.079,7.597)	12(12,12)	1(1,1)	100(100,100)	0.4(0.4,0.4)	0.01(0.01,0.01)	0(0,0)
2	5.707(5.21,6.397)	12(6,12)	1(1,1)	100(100,100)	0.4(0.4,0.4)	0.01(0.003,0.01)	0(0,0)
3	7.916(7.208,8.489)	12(11.25,12)	1(1,1)	100(100,100)	0.4(0.4,0.4)	0.01(0.01,0.01)	0(0,0)
4	6.844(6.491,7.541)	12(11.25,12)	1(1,1)	100(100,100)	0.4(0.4,0.4)	0.01(0.01,0.01)	0(0,0)
5	7.273(6.85,7.583)	12(11.25,12)	1(1,1)	100(100,100)	0.4(0.4,0.4)	0.01(0.01,0.01)	0(0,0)
6	7.796(7.039,8.073)	12(12,12)	1(1,2)	100(100,100)	0.4(0.4,0.4)	0.01(0.01,0.01)	0(0,0)
7	10.169(9.678,10.927)	12(12,12)	1(1,1)	100(100,100)	0.4(0.4,0.4)	0.01(0.01,0.01)	0(0,0.008)
8	11.137(10.292,11.783)	12(12,12)	1(1,1)	100(100,100)	0.4(0.4,0.4)	0.01(0.01,0.01)	0(0,0.01)
9	10.957(10.578,11.486)	12(12,12)	2(1.25,2)	100(100,100)	0.4(0.4,0.4)	0.01(0.01,0.01)	0(0,0)
10	10.943(10.672,11.133)	12(12,12)	2(2,2)	100(100,100)	0.4(0.4,0.4)	0.01(0.01,0.01)	0.01(0,0.01)
11	11.248(10.841,11.516)	12(12,12)	2(2,2)	100(100,100)	0.4(0.4,0.4)	0.01(0.01,0.01)	0(0,0.008)
12	10.984(10.415,11.417)	12(12,12)	2(2,2)	100(100,100)	0.4(0.4,0.4)	0.01(0.01,0.01)	0(0,0)
13	11.151(10.975,11.545)	12(12,12)	2(2,2)	100(100,100)	0.4(0.4,0.4)	0.01(0.01,0.01)	0.01(0.01,0.01)
14	11.194(10.958,11.755)	12(9,12)	2(1.25,2)	100(100,100)	0.4(0.4,0.4)	0.01(0.003,0.01)	0.01(0,0.01)

Table 69. The median (first quantile – third quantile) of the model fit indices in 10 iterations group by the number of hybrid principal component analysis (HPCA) for frontal region of brain.

3.2.3.3 Central Region (2)

The MSPE and Correlation group by number of HPCA, Train and Test ([Table 70](#))

The User, System and Elapsed Time group by number of HPCA ([Table 71](#)): The median (first quantile – third quantile) of the Time (user-system-elapsed) in 10 iterations group by the number of hybrid principal component analysis (HPCA) for whole brain.

The Model fitting parameters group by number of HPCA ([Table 72](#)): The median (first quantile – third quantile) of the model fit indices in 10 iterations group by the number of hybrid principal component analysis (HPCA) for whole brain.

HPCA	MSPE		Correlation	
	Train	Test	Train	Test
1	0.473(0.28,0.533)	0.638(0.445,0.648)	0.762(0.712,0.857)	0.617(0.612,0.757)
2	0.533(0.284,0.533)	0.618(0.468,0.618)	0.714(0.714,0.855)	0.649(0.649,0.74)
3	0.348(0.337,0.355)	0.478(0.47,0.488)	0.807(0.803,0.814)	0.725(0.719,0.73)
4	0.343(0.31,0.349)	0.503(0.496,0.683)	0.81(0.807,0.831)	0.709(0.586,0.714)
5	0.367(0.325,0.367)	0.551(0.551,0.568)	0.795(0.795,0.821)	0.679(0.676,0.69)
6	0.369(0.369,0.547)	0.538(0.535,0.603)	0.794(0.709,0.794)	0.688(0.662,0.689)
7	0.692(0.532,0.692)	0.716(0.578,0.716)	0.586(0.586,0.684)	0.553(0.553,0.65)
8	0.692(0.565,0.882)	0.711(0.593,0.843)	0.586(0.329,0.664)	0.559(0.466,0.641)
9	0.653(0.595,0.705)	0.65(0.547,0.662)	0.62(0.568,0.643)	0.617(0.61,0.678)
10	0.705(0.664,0.705)	0.664(0.655,0.664)	0.568(0.568,0.61)	0.607(0.607,0.613)
11	0.87(0.656,0.924)	0.884(0.731,0.892)	0.387(0.283,0.611)	0.379(0.344,0.525)
12	0.657(0.585,0.725)	0.7(0.553,0.727)	0.61(0.542,0.644)	0.563(0.53,0.668)
13	0.911(0.739,0.931)	0.901(0.734,0.907)	0.31(0.27,0.53)	0.337(0.315,0.527)
14	0.734(0.662,0.869)	0.808(0.735,0.909)	0.534(0.394,0.605)	0.437(0.311,0.525)

Table 70. The median (first quantile – third quantile) of the Mean Square Prediction Error (MSPE) and Correlation in 10 iterations group by Train and Test and the number of hybrid principal component analysis (HPCA) for central region of the brain.

HPCA	Time (s)		
	User	System	Elapsed
1	220.365(214.725,221.885)	0.17(0.133,0.212)	220.91(215.255,222.393)
2	223.38(219.653,228.685)	0.085(0.065,0.15)	224.445(219.945,229.237)
3	322.7(321.922,325.2)	0.15(0.133,0.215)	323.45(322.463,325.658)
4	323.415(319.345,325.37)	0.14(0.122,0.177)	323.725(319.862,325.617)
5	361.27(360.655,366.165)	0.14(0.085,0.2)	361.715(361.277,366.628)
6	365.8(361.9,368.033)	0.17(0.118,0.19)	366.255(362.355,368.48)
7	445.965(442.86,449.15)	0.16(0.145,0.17)	446.255(443.355,449.63)
8	456.06(453.188,462.875)	0.19(0.175,0.243)	456.56(453.692,463.493)
9	517.325(508.728,525.685)	0.2(0.19,0.215)	517.85(509.223,526.208)
10	469.56(460.123,478.21)	0.165(0.145,0.177)	470.15(460.795,478.643)
11	505.39(502.915,512.243)	0.165(0.143,0.198)	505.89(503.663,512.798)
12	503.285(498.54,506.713)	0.19(0.155,0.208)	503.935(499.068,507.493)
13	561.45(551.643,567.152)	0.24(0.163,0.348)	562.48(552.598,567.783)
14	560.96(553.13,566.22)	0.21(0.192,0.243)	561.845(554.095,567.1)

Table 71. The median (first quantile – third quantile) of the Time (user-system-elapsed) in 10 iterations group by the number of hybrid principal component analysis (HPCA) for central region of the brain.

HPCA	Min_Error	Optimum					
		index	K	Tau	Lambda	Eta	Kappa
1	10.771(10.508,11.298)	12(9,12)	1(1,2)	100(100,100)	0.4(0.4,0.4)	0.01(0.003,0.01)	0.01(0.003,0.01)
2	10.748(10.075,11.169)	12(12,12)	1(1,1)	100(100,100)	0.4(0.4,0.4)	0.01(0.01,0.01)	0.01(0.003,0.01)
3	9.034(8.397,9.873)	10(5,12)	1(1,1)	100(100,100)	0.4(0.4,0.4)	0.005(0,0.01)	0(0,0)
4	8.492(8.096,8.919)	11(8,11)	1(1,1)	100(10,100)	0.4(0.3,0.4)	0.01(0,0.01)	0(0,0)
5	9.688(8.459,10.539)	11.5(5.75,12)	1(1,1)	100(32.5,100)	0.4(0.325,0.4)	0.01(0.003,0.01)	0(0,0)
6	9.985(9.031,10.811)	12(12,12)	1(1,1.75)	100(100,100)	0.4(0.4,0.4)	0.01(0.01,0.01)	0(0,0.01)
7	12.777(12.536,12.974)	12(12,12)	1(1,1)	100(100,100)	0.4(0.4,0.4)	0.01(0.01,0.01)	0.01(0,0.01)
8	12.495(12.035,13.358)	12(12,12)	1(1,1)	100(100,100)	0.4(0.4,0.4)	0.01(0.01,0.01)	0.01(0.003,0.753)
9	12.647(12.069,13.058)	12(12,12)	1(1,2)	100(100,100)	0.4(0.4,0.4)	0.01(0.01,0.01)	0.01(0.003,0.01)
10	13.201(12.706,13.449)	12(12,12)	1(1,1)	100(100,100)	0.4(0.4,0.4)	0.01(0.01,0.01)	0.01(0.01,0.01)
11	13.462(13.145,13.498)	10(5,12)	2(1.25,2)	100(100,100)	0.4(0.4,0.4)	0.005(0,0.01)	1(0.01,1)
12	12.826(12.29,13.071)	12(9,12)	1(1,2)	100(100,100)	0.4(0.4,0.4)	0.01(0.003,0.01)	0.01(0,0.01)
13	13.438(13.352,13.53)	11.5(8,12)	1(1,2)	100(100,100)	0.4(0.4,0.4)	0.01(0,0.01)	1(0.01,1)
14	13.382(13.177,13.479)	10(5,12)	2(1,2)	100(100,100)	0.4(0.4,0.4)	0.005(0,0.01)	0.01(0.01,1)

Table 72. The median (first quantile – third quantile) of the model fit indices in 10 iterations group by the number of hybrid principal component analysis (HPCA) for central region of the brain.

3.2.3.4 Occipital Region (3)

The MSPE and Correlation group by number of HPCA, Train and Test ([Table 73](#))

The User, System and Elapsed Time group by number of HPCA ([Table 74](#)): The median (first quantile – third quantile) of the Time (user-system-elapsed) in 10 iterations group by the number of hybrid principal component analysis (HPCA) for whole brain.

The Model fitting parameters group by number of HPCA ([Table 75](#)): The median (first quantile – third quantile) of the model fit indices in 10 iterations group by the number of hybrid principal component analysis (HPCA) for whole brain.

HPCA	MSPE		Correlation	
	Train	Test	Train	Test
1	0.001(0,0.001)	0.002(0.001,0.002)	1(1,1)	0.999(0.999,0.999)
2	0.002(0.002,0.002)	0.18(0.18,0.18)	0.999(0.999,0.999)	0.908(0.908,0.908)
3	0.003(0.003,0.024)	0.182(0.182,0.208)	0.998(0.988,0.998)	0.907(0.895,0.907)
4	0.053(0.053,0.053)	0.194(0.194,0.194)	0.973(0.973,0.973)	0.9(0.9,0.9)
5	0.053(0.053,0.053)	0.19(0.19,0.19)	0.973(0.973,0.973)	0.902(0.902,0.902)
6	0.066(0.023,0.066)	0.172(0.172,0.206)	0.967(0.967,0.988)	0.911(0.894,0.911)
7	0.066(0.023,0.066)	0.173(0.173,0.209)	0.967(0.967,0.988)	0.911(0.892,0.911)
8	0.071(0.071,0.071)	0.205(0.205,0.205)	0.964(0.964,0.964)	0.894(0.894,0.894)
9	0.071(0.071,0.071)	0.205(0.205,0.205)	0.964(0.964,0.964)	0.894(0.894,0.894)
10	0.095(0.095,0.095)	0.2(0.2,0.2)	0.951(0.951,0.951)	0.896(0.896,0.896)
11	0.095(0.095,0.095)	0.201(0.201,0.201)	0.951(0.951,0.951)	0.895(0.895,0.895)
12	0.102(0.102,0.102)	0.219(0.219,0.219)	0.948(0.948,0.948)	0.886(0.886,0.886)
13	0.102(0.102,0.102)	0.22(0.22,0.22)	0.948(0.948,0.948)	0.885(0.885,0.885)
14	0.104(0.104,0.104)	0.213(0.213,0.213)	0.946(0.946,0.946)	0.889(0.889,0.889)

Table 73. The median (first quantile – third quantile) of the Mean Square Prediction Error (MSPE) and Correlation in 10 iterations group by Train and Test and the number of hybrid principal component analysis (HPCA) for occipital region of the brain.

HPCA	Time (s)		
	User	System	Elapsed
1	143.385(141.523,144.563)	0.095(0.06,0.138)	143.685(141.695,144.755)
2	186.47(183.667,189.717)	0.075(0.07,0.112)	186.875(183.997,190.005)
3	219.855(217.753,222.37)	0.1(0.063,0.108)	220.125(218.038,222.59)
4	323(315.808,325.338)	0.11(0.102,0.14)	323.405(316.158,326.018)
5	304.89(301.05,307.197)	0.155(0.132,0.2)	305.41(301.477,308.17)
6	369.33(364.565,374.962)	0.17(0.115,0.257)	369.97(365.073,375.595)
7	342.45(337.37,350.225)	0.145(0.09,0.26)	342.97(338.15,350.755)
8	409.84(399.265,411.34)	0.15(0.117,0.178)	410.295(399.753,411.9)
9	431.045(425.51,435.938)	0.18(0.12,0.227)	431.555(426.555,436.908)
10	429.645(426.66,440.167)	0.18(0.162,0.208)	430.205(427.167,440.88)
11	434.305(430.528,437.292)	0.2(0.135,0.235)	435.005(431.117,437.805)
12	502.24(496.112,527.575)	0.21(0.175,0.417)	503.79(496.77,608.355)
13	515.18(513.065,520.54)	0.275(0.218,0.315)	517.12(514.697,522.903)
14	510.005(507.388,524.177)	0.245(0.205,0.308)	510.615(508.185,526.305)

Table 74. The median (first quantile – third quantile) of the Time (user-system-elapsed) in 10 iterations group by the number of hybrid principal component analysis (HPCA) for occipital region of the brain.

HPCA	Min.Error	Optimum					
		index	K	Tau	Lambda	Eta	Kappa
1	0.027(0.025,0.028)	12(12,12)	1.5(1,2)	100(100,100)	0.4(0.4,0.4)	0.01(0.01,0.01)	0(0,0)
2	1.917(1.817,2.1)	1(1,1)	1(1,1)	0.1(0.1,0.1)	0.1(0.1,0.1)	0(0,0)	0(0,0)
3	1.983(1.784,2.21)	1(1,3.25)	1(1,1)	0.1(0.1,75.025)	0.1(0.1,0.325)	0(0,0)	0(0,0)
4	2.267(2.005,2.435)	1(1,1)	1(1,1)	0.1(0.1,0.1)	0.1(0.1,0.1)	0(0,0)	0(0,0)
5	2.364(2.226,2.428)	1(1,1)	1(1,1)	0.1(0.1,0.1)	0.1(0.1,0.1)	0(0,0)	0(0,0)
6	2.999(2.931,3.086)	1(1,1)	1(1,1.75)	0.1(0.1,0.1)	0.1(0.1,0.1)	0(0,0)	0(0,0)
7	2.795(2.772,3.035)	1(1,1)	1(1,1.75)	0.1(0.1,0.1)	0.1(0.1,0.1)	0(0,0)	0(0,0)
8	2.893(2.758,2.984)	1(1,1)	1(1,1)	0.1(0.1,0.1)	0.1(0.1,0.1)	0(0,0)	0(0,0)
9	2.897(2.739,3.035)	1(1,1)	1(1,1)	0.1(0.1,0.1)	0.1(0.1,0.1)	0(0,0)	0(0,0)
10	3.265(3.129,3.34)	1(1,1)	1(1,1)	0.1(0.1,0.1)	0.1(0.1,0.1)	0(0,0)	0(0,0)
11	3.508(3.351,3.553)	1(1,1)	1(1,1)	0.1(0.1,0.1)	0.1(0.1,0.1)	0(0,0)	0(0,0)
12	3.699(3.45,3.825)	1(1,1)	1(1,1)	0.1(0.1,0.1)	0.1(0.1,0.1)	0(0,0)	0(0,0)
13	3.607(3.42,3.647)	1(1,1)	1(1,1)	0.1(0.1,0.1)	0.1(0.1,0.1)	0(0,0)	0(0,0)
14	3.244(3.156,3.437)	1(1,1)	1(1,1)	0.1(0.1,0.1)	0.1(0.1,0.1)	0(0,0)	0(0,0)

Table 75. The median (first quantile – third quantile) of the Mean Square Prediction Error (MSPE) and Correlation in 10 iterations group by Train and Test and the number of hybrid principal component analysis (HPCA) for occipital region of the brain.

3.2.3.5 Parietal (4)

The MSPE and Correlation group by number of HPCA, Train and Test (Table 76)

The User, System and Elapsed Time group by number of HPCA (Table 77): The median (first quantile – third quantile) of the Time (user-system-elapsed) in 10 iterations group by the number of hybrid principal component analysis (HPCA) for whole brain.

The Model fitting parameters group by number of HPCA (Table 78): The median (first quantile – third quantile) of the model fit indices in 10 iterations group by the number of hybrid principal component analysis (HPCA) for whole brain.

HPCA	MSPE		Correlation	
	Train	Test	Train	Test
1	0.329(0.174,0.392)	0.402(0.4,0.472)	0.831(0.788,0.909)	0.785(0.729,0.791)
2	0.402(0.361,0.41)	0.362(0.345,0.378)	0.783(0.777,0.811)	0.817(0.802,0.832)
3	0.309(0.28,0.311)	0.292(0.292,0.298)	0.831(0.83,0.848)	0.842(0.838,0.842)
4	0.447(0.349,0.448)	0.512(0.394,0.563)	0.758(0.758,0.81)	0.709(0.668,0.782)
5	0.328(0.316,0.34)	0.399(0.395,0.402)	0.819(0.812,0.827)	0.776(0.775,0.778)
6	0.392(0.353,0.475)	0.398(0.378,0.457)	0.787(0.736,0.804)	0.787(0.758,0.788)
7	0.597(0.57,0.616)	0.541(0.5,0.566)	0.645(0.629,0.669)	0.7(0.669,0.742)
8	0.612(0.508,0.612)	0.526(0.414,0.526)	0.631(0.631,0.702)	0.711(0.711,0.772)
9	0.635(0.616,0.64)	0.495(0.483,0.498)	0.611(0.607,0.628)	0.741(0.74,0.748)
10	0.588(0.518,0.636)	0.475(0.356,0.503)	0.654(0.611,0.694)	0.764(0.734,0.807)
11	0.663(0.612,0.755)	0.553(0.537,0.707)	0.586(0.521,0.632)	0.691(0.6,0.709)
12	0.786(0.786,0.786)	0.775(0.771,0.775)	0.499(0.499,0.499)	0.54(0.539,0.54)
13	0.782(0.782,0.782)	0.796(0.796,0.796)	0.506(0.506,0.506)	0.497(0.497,0.497)
14	0.786(0.783,0.8)	0.793(0.789,0.794)	0.504(0.485,0.507)	0.498(0.498,0.51)

Table 76. The median (first quantile – third quantile) of the Mean Square Prediction Error (MSPE) and Correlation in 10 iterations group by Train and Test and the number of hybrid principal component analysis (HPCA) for parietal region of the brain.

HPCA	Time (s)		
	User	System	Elapsed
1	212.225(210.46,213.86)	0.115(0.065,0.17)	212.565(210.907,214.235)
2	209.245(206.64,209.845)	0.11(0.102,0.203)	209.645(207.033,210.28)
3	334.46(329.627,335.858)	0.15(0.118,0.188)	335.32(330.417,336.377)
4	339.975(336.457,344.645)	0.16(0.133,0.185)	340.555(336.87,345.205)
5	339.14(330.145,344.805)	0.135(0.13,0.185)	339.425(330.617,345.6)
6	335.505(330.57,338.158)	0.15(0.132,0.178)	335.775(330.96,338.677)
7	395.98(391.73,401.565)	0.15(0.143,0.34)	396.415(392.085,401.935)
8	393.225(388.58,398.12)	0.165(0.132,0.212)	393.65(388.902,398.615)
9	393.86(387.64,401.985)	0.155(0.125,0.17)	394.15(387.93,402.495)
10	393.455(387.81,399.482)	0.15(0.14,0.19)	393.745(388.227,401.177)
11	446.54(438.563,449.64)	0.19(0.145,0.218)	447.16(439.078,450.178)
12	452.215(444.308,455.963)	0.12(0.112,0.167)	452.65(444.698,456.382)
13	488.355(484.348,492.717)	0.18(0.17,0.22)	488.935(484.968,494.722)
14	487.125(474.545,491.555)	0.18(0.153,0.228)	487.54(474.993,492)

Table 77. The median (first quantile – third quantile) of the Time (user-system-elapsed) in 10 iterations group by the number of hybrid principal component analysis (HPCA) for parietal region of the brain.

HPCA	Min.Error	Optimum					
		index	K	Tau	Lambda	Eta	Kappa
1	8.529(7.209,8.8)	12(5,12)	1(1,1.75)	100(100,100)	0.4(0.4,0.4)	0.01(0,0.01)	0.01(0,0.01)
2	9.035(8.486,9.458)	12(9.75,12)	1(1,2)	100(100,100)	0.4(0.4,0.4)	0.01(0.01,0.01)	0.01(0.01,0.01)
3	7.538(7.124,8.19)	12(8.75,12)	1(1,1)	100(100,100)	0.4(0.4,0.4)	0.01(0.003,0.01)	0(0,0)
4	9.222(8.665,9.843)	12(1,12)	1(1,1)	100(0.1,100)	0.4(0.1,0.4)	0.01(0,0.01)	0.01(0.003,0.01)
5	8.546(8.341,8.881)	12(4,12)	1(1,1.75)	100(100,100)	0.4(0.4,0.4)	0.01(0,0.01)	0(0,0)
6	9.415(8.649,9.722)	12(12,12)	1(1,2)	100(100,100)	0.4(0.4,0.4)	0.01(0.01,0.01)	0.005(0,0.01)
7	11.5(10.916,11.627)	12(11.25,12)	1(1,1)	100(100,100)	0.4(0.4,0.4)	0.01(0.01,0.01)	0.01(0.01,0.01)
8	10.987(10.728,11.561)	12(12,12)	1(1,1)	100(100,100)	0.4(0.4,0.4)	0.01(0.01,0.01)	0.01(0.003,0.01)
9	11.628(11.144,11.726)	10(8,12)	1(1,1)	100(100,100)	0.4(0.4,0.4)	0.005(0,0.01)	0.01(0.01,0.01)
10	11.507(10.94,12.059)	12(12,12)	1(1,1.75)	100(100,100)	0.4(0.4,0.4)	0.01(0.01,0.01)	0.01(0,0.01)
11	11.95(11.825,12.124)	12(8,12)	1(1,1)	100(100,100)	0.4(0.4,0.4)	0.01(0,0.01)	0.01(0.01,0.753)
12	12.109(12.025,12.148)	1.5(1,6.25)	7(4.75,7)	0.55(0.1,77.5)	0.15(0.1,0.375)	0(0,0)	1(1,1)
13	12.089(12.068,12.12)	1(1,3.25)	8(8,8)	0.1(0.1,75.025)	0.1(0.1,0.325)	0(0,0)	1(1,1)
14	12.194(12.101,12.223)	4.5(1,11)	5.5(2.5,7.75)	50.05(0.1,100)	0.25(0.1,0.4)	0(0,0.008)	1(1,1)

Table 78. The median (first quantile – third quantile) of the model fit indices in 10 iterations group by the number of hybrid principal component analysis (HPCA) for parietal region of the brain.

3.2.3.6 Left Temporal (5)

The MSPE and Correlation group by number of HPCA, Train and Test ([Table 79](#))

The User, System and Elapsed Time group by number of HPCA ([Table 80](#)): The median (first quantile – third quantile) of the Time (user-system-elapsed) in 10 iterations group by the number of hybrid principal component analysis (HPCA) for whole brain.

The Model fitting parameters group by number of HPCA ([Table 81](#)): The median (first quantile – third quantile) of the model fit indices in 10 iterations group by the number of hybrid principal component analysis (HPCA) for whole brain.

HPCA	MSPE		Correlation	
	Train	Test	Train	Test
1	0.03(0.024,0.03)	0.071(0.071,0.071)	0.985(0.985,0.988)	0.964(0.964,0.964)
2	0.057(0.037,0.129)	0.711(0.698,0.738)	0.971(0.934,0.981)	0.595(0.583,0.606)
3	0.147(0.106,0.189)	0.694(0.679,0.749)	0.924(0.901,0.946)	0.595(0.548,0.601)
4	0.232(0.197,0.267)	0.775(0.739,0.776)	0.876(0.856,0.896)	0.55(0.55,0.563)
5	0.216(0.185,0.275)	0.762(0.751,0.938)	0.885(0.852,0.903)	0.546(0.449,0.546)
6	0.256(0.256,0.263)	0.616(0.616,0.756)	0.862(0.859,0.862)	0.627(0.536,0.627)
7	0.291(0.291,0.291)	0.597(0.597,0.597)	0.842(0.842,0.842)	0.637(0.637,0.637)
8	0.316(0.316,0.316)	0.805(0.805,0.832)	0.827(0.827,0.827)	0.482(0.465,0.482)
9	0.336(0.336,0.336)	0.803(0.803,0.825)	0.815(0.815,0.815)	0.48(0.467,0.48)
10	0.371(0.371,0.376)	0.723(0.722,0.749)	0.793(0.79,0.793)	0.533(0.511,0.536)
11	0.383(0.376,0.383)	0.732(0.73,0.757)	0.786(0.786,0.789)	0.524(0.508,0.528)
12	0.402(0.399,0.405)	0.721(0.72,0.722)	0.773(0.771,0.775)	0.532(0.53,0.534)
13	0.4(0.394,0.4)	0.767(0.761,0.767)	0.774(0.774,0.778)	0.493(0.493,0.503)
14	0.424(0.424,0.424)	0.728(0.728,0.728)	0.759(0.759,0.759)	0.525(0.525,0.525)

Table 79. The median (first quantile – third quantile) of the Mean Square Prediction Error (MSPE) and Correlation in 10 iterations group by Train and Test and the number of hybrid principal component analysis (HPCA) for left temporal region of the brain.

HPCA	Time (s)		
	User	System	Elapsed
1	148.545(147.735,148.967)	0.08(0.065,0.135)	148.775(147.89,149.105)
2	175.77(172.21,178.328)	0.105(0.085,0.138)	175.96(172.39,178.51)
3	235.465(232.122,239.198)	0.115(0.077,0.22)	235.775(232.405,239.46)
4	263.25(261.567,266.112)	0.13(0.088,0.167)	263.63(261.783,266.348)
5	296.825(293.233,300.13)	0.125(0.102,0.137)	297.06(293.493,300.55)
6	327.705(326.283,333.48)	0.125(0.102,0.15)	327.935(326.665,333.915)
7	392.24(385.547,395.955)	0.16(0.12,0.178)	392.65(385.89,396.38)
8	383.52(377.318,386.302)	0.165(0.118,0.19)	383.95(377.673,386.742)
9	371.84(367.83,373.81)	0.135(0.095,0.14)	372.05(368.193,374.555)
10	384.705(379.76,388.762)	0.145(0.115,0.197)	384.98(380.118,389.575)
11	413.915(411.235,417.465)	0.175(0.13,0.198)	414.295(411.713,417.77)
12	445.94(443.623,450.29)	0.185(0.135,0.2)	446.405(443.928,450.69)
13	460.235(456.767,462.132)	0.165(0.142,0.21)	460.74(457.005,462.51)
14	481.795(467.27,496.51)	0.17(0.14,0.238)	482.225(467.615,497.065)

Table 80. The median (first quantile – third quantile) of the Time (user-system-elapsed) in 10 iterations group by the number of hybrid principal component analysis (HPCA) for left temporal region of the brain.

HPCA	Min_Error	Optimum					
		index	K	Tau	Lambda	Eta	Kappa
1	1.662(1.554,1.801)	12(12,12)	1(1,1)	100(100,100)	0.4(0.4,0.4)	0.01(0.01,0.01)	0(0,0)
2	6.441(5.614,6.696)	2.5(1,4)	1(1,1)	50.05(0.1,100)	0.25(0.1,0.4)	0(0,0)	0(0,0)
3	7.156(6.884,8.136)	4(1.5,4)	1(1,1.75)	100(2.575,100)	0.4(0.15,0.4)	0(0,0)	0(0,0)
4	7.761(7.635,7.919)	4(1.75,4)	1(1,1)	100(25.075,100)	0.4(0.175,0.4)	0(0,0)	0(0,0)
5	7.56(7.243,8.29)	4(3.25,4)	1(1,2.5)	100(32.5,100)	0.4(0.325,0.4)	0(0,0)	0(0,0)
6	9.175(8.545,9.325)	1(1,3.25)	1(1,1)	0.1(0.1,75.025)	0.1(0.1,0.325)	0(0,0)	0(0,0)
7	9.65(9.459,10.059)	1(1,1)	1(1,1)	0.1(0.1,0.1)	0.1(0.1,0.1)	0(0,0)	0(0,0)
8	9.94(9.634,10.193)	1(1,1.75)	1(1,1)	0.1(0.1,0.775)	0.1(0.1,0.175)	0(0,0)	0(0,0)
9	10.092(9.759,10.39)	1(1,1)	1(1,1)	0.1(0.1,0.1)	0.1(0.1,0.1)	0(0,0)	0(0,0)
10	10.867(10.67,10.957)	1(1,1)	1(1,1)	0.1(0.1,0.1)	0.1(0.1,0.1)	0(0,0)	0(0,0)
11	11.145(10.444,11.266)	1(1,1)	1(1,1)	0.1(0.1,0.1)	0.1(0.1,0.1)	0(0,0)	0(0,0)
12	10.81(10.421,11.711)	1(1,1)	1(1,1)	0.1(0.1,0.1)	0.1(0.1,0.1)	0(0,0)	0(0,0)
13	11.053(10.597,11.403)	1(1,1)	1(1,1)	0.1(0.1,0.1)	0.1(0.1,0.1)	0(0,0)	0(0,0)
14	11.896(11.599,12.262)	1(1,1)	1(1,1)	0.1(0.1,0.1)	0.1(0.1,0.1)	0(0,0)	0(0,0)

Table 81. The median (first quantile – third quantile) of the model fit indices in 10 iterations group by the number of hybrid principal component analysis (HPCA) for left temporal region of the brain.

3.2.3.7 Right Temporal (6)

The MSPE and Correlation group by number of HPCA, Train and Test (Table 82)

The User, System and Elapsed Time group by number of HPCA (Table 83): The median (first quantile – third quantile) of the Time (user-system-elapsed) in 10 iterations group by the number of hybrid principal component analysis (HPCA) for whole brain.

The Model fitting parameters group by number of HPCA (Table 84): The median (first quantile – third quantile) of the model fit indices in 10 iterations group by the number of hybrid principal component analysis (HPCA) for whole brain.

HPCA	MSPE		Correlation	
	Train	Test	Train	Test
1	0.94(0.409,0.942)	0.936(0.542,0.938)	0.245(0.242,0.719)	0.261(0.253,0.642)
2	0.57(0.203,0.947)	1.023(0.987,1.057)	0.571(0.228,0.893)	0.153(0.134,0.377)
3	0.53(0.52,0.844)	0.833(0.72,0.981)	0.692(0.363,0.712)	0.407(0.166,0.548)
4	0.967(0.958,0.975)	0.977(0.968,0.978)	0.178(0.154,0.209)	0.151(0.149,0.179)
5	0.98(0.98,0.98)	0.962(0.96,0.968)	0.136(0.136,0.136)	0.206(0.199,0.213)
6	0.978(0.959,0.979)	0.983(0.982,0.992)	0.144(0.141,0.2)	0.126(0.111,0.128)
7	0.982(0.982,0.983)	0.987(0.986,0.987)	0.129(0.127,0.13)	0.114(0.11,0.116)
8	0.984(0.984,0.984)	0.98(0.98,0.981)	0.123(0.123,0.123)	0.138(0.133,0.139)
9	0.986(0.981,0.986)	0.989(0.986,0.989)	0.114(0.112,0.132)	0.098(0.098,0.113)
10	0.986(0.971,0.987)	0.989(0.989,0.999)	0.112(0.106,0.167)	0.099(0.078,0.099)
11	0.968(0.967,0.983)	1.008(0.988,1.017)	0.177(0.125,0.178)	0.088(0.067,0.107)
12	0.976(0.968,0.982)	0.989(0.984,1.004)	0.15(0.129,0.176)	0.101(0.084,0.123)
13	0.979(0.976,0.985)	0.981(0.98,0.984)	0.141(0.119,0.15)	0.137(0.131,0.139)
14	0.989(0.981,0.989)	0.985(0.984,0.987)	0.101(0.101,0.132)	0.121(0.115,0.124)

Table 82. The median (first quantile – third quantile) of the Mean Square Prediction Error (MSPE) and Correlation in 10 iterations group by Train and Test and the number of hybrid principal component analysis (HPCA) for right temporal region of the brain.

HPCA	Time (s)		
	User	System	Elapsed
1	179.165(177.64,181.667)	0.095(0.058,0.14)	179.375(177.805,181.818)
2	193.11(191.085,197.035)	0.075(0.047,0.102)	193.295(191.245,197.272)
3	299.895(289.92,304.407)	0.115(0.102,0.155)	300.145(290.225,304.74)
4	288.595(285.35,295.878)	0.1(0.093,0.118)	288.815(285.642,296.093)
5	309.11(305.035,310.622)	0.12(0.087,0.207)	309.585(305.352,310.99)
6	313.925(313.377,316.307)	0.11(0.09,0.17)	314.19(313.628,316.625)
7	360.065(357.067,362.037)	0.135(0.105,0.193)	360.38(357.598,362.325)
8	352.785(351.688,356.355)	0.115(0.102,0.155)	353.01(352.045,356.623)
9	393.765(389.928,398.733)	0.165(0.135,0.19)	394.14(390.705,399.075)
10	389.105(384.775,397.482)	0.15(0.14,0.178)	389.47(385.157,397.832)
11	447.41(439.542,451.202)	0.17(0.14,0.22)	447.825(439.94,451.715)
12	433.06(428.905,437.673)	0.16(0.11,0.248)	433.295(429.29,438.31)
13	483.605(441.205,494.715)	0.155(0.15,0.208)	484.065(441.463,495.143)
14	417.03(414.825,422.892)	0.18(0.17,0.215)	417.39(415.175,423.237)

Table 83. The median (first quantile – third quantile) of the Time (user-system-elapsed) in 10 iterations group by the number of hybrid principal component analysis (HPCA) for right temporal region of the brain.

HPCA	Min.Error	Optimum					
		index	K	Tau	Lambda	Eta	Kappa
1	13.443(13.07,13.457)	4(4,5.75)	1(1,1.75)	55(0.325,100)	0.35(0.125,0.4)	0(0,0)	1(0.25,1)
2	13.367(10.934,13.412)	12(10.5,12)	1.5(1,2)	100(25.75,100)	0.4(0.25,0.4)	0.01(0.01,0.01)	0.5(0,75.25)
3	12.68(12.381,13.509)	11.5(11,12)	2(1,3)	100(32.5,100)	0.4(0.325,0.4)	0.01(0.01,0.01)	0.005(0,0.753)
4	13.761(13.745,13.771)	2.5(1.25,9.75)	1(1,1.75)	5.5(0.325,77.5)	0.25(0.125,0.375)	0(0,0.008)	100(1,100)
5	13.783(13.761,13.82)	6(1.25,9)	1(1,1)	0.1(0.1,0.1)	0.1(0.1,0.1)	0.005(0,0.01)	100(100,100)
6	13.731(13.711,13.749)	5.5(3,9)	1.5(1,6)	5.5(0.325,10)	0.25(0.125,0.3)	0(0,0.01)	100(100,100)
7	13.852(13.809,13.869)	4(1.25,7.75)	1(1,1)	55(0.325,100)	0.35(0.125,0.4)	0(0,0)	100(100,100)
8	13.885(13.871,13.888)	3.5(2,6.5)	1(1,1)	5.5(0.325,77.5)	0.25(0.125,0.375)	0(0,0)	100(100,100)
9	13.887(13.878,13.897)	2.5(1,8)	1(1,1)	0.55(0.1,100)	0.15(0.1,0.4)	0(0,0)	100(100,100)
10	13.879(13.841,13.926)	3(2.25,4)	1(1,4.75)	5.5(0.1,10)	0.25(0.1,0.3)	0(0,0)	100(100,100)
11	13.858(13.826,13.894)	8(4,9)	6.5(2.25,8)	10(0.325,77.5)	0.3(0.125,0.375)	0.005(0,0.01)	100(100,100)
12	13.825(13.788,13.849)	4(3.25,9.25)	3.5(1.25,5.5)	10(10,77.5)	0.3(0.3,0.375)	0(0,0.008)	100(100,100)
13	13.836(13.805,13.861)	3(1,3.75)	3(2,4)	0.1(0.1,10)	0.1(0.1,0.3)	0(0,0)	100(100,100)
14	13.854(13.842,13.869)	4(2,8.25)	1(1,3.25)	0.55(0.1,1)	0.15(0.1,0.2)	0(0,0.008)	100(100,100)

Table 84. The median (first quantile – third quantile) of the model fit indices in 10 iterations group by the number of hybrid principal component analysis (HPCA) for right temporal region of the brain.

3.2.3.8 Figures

The MSPE over Number of HPCA group by Regions: [Figure 62](#)

The Correlation over Number of HPCA group by Regions: [Figure 63](#)

The Elapsed Time over Number of HPCA group by Regions: [Figure 64](#)

The MSPE over Number of HPCA group by Regions

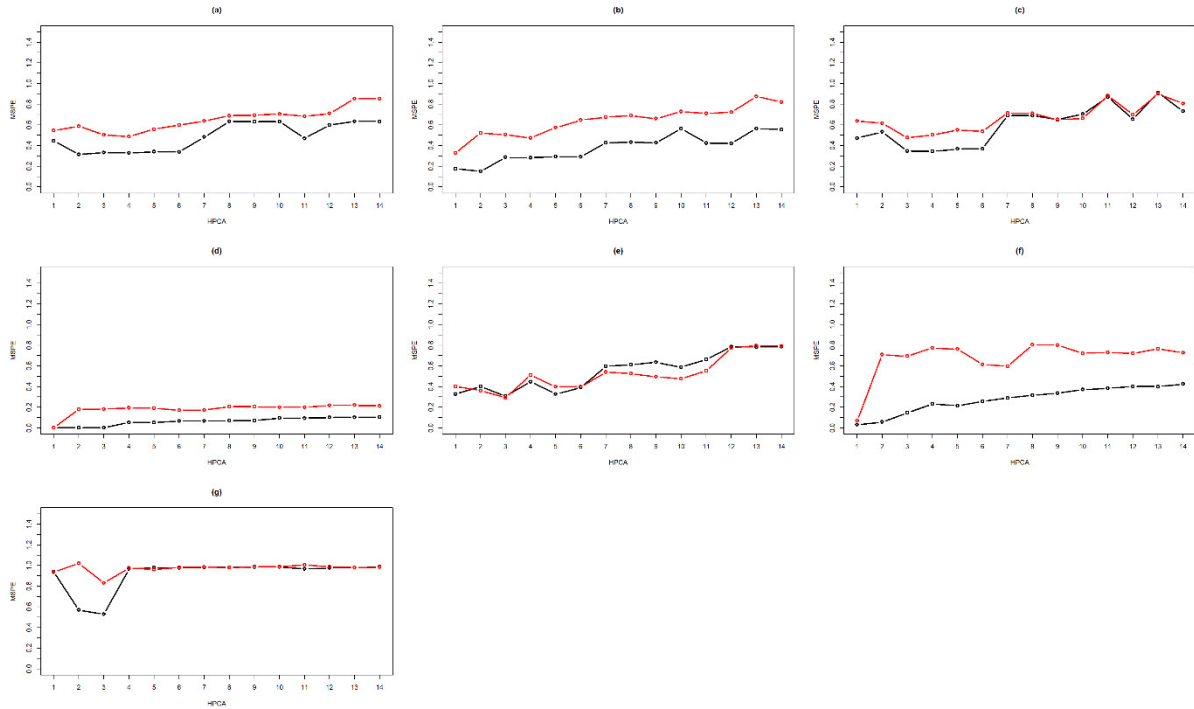


Figure 62. (a-g) Median of mean square prediction error (MSPE) for train (black solid line) and test (red solid line) in 10 iterations of (a) whole, (b) frontal, (c) central, (d) occipital, (e) parietal, (f) left temporal and (g) right temporal regions, respectively.

The Correlation over Number of HPCA group by Regions

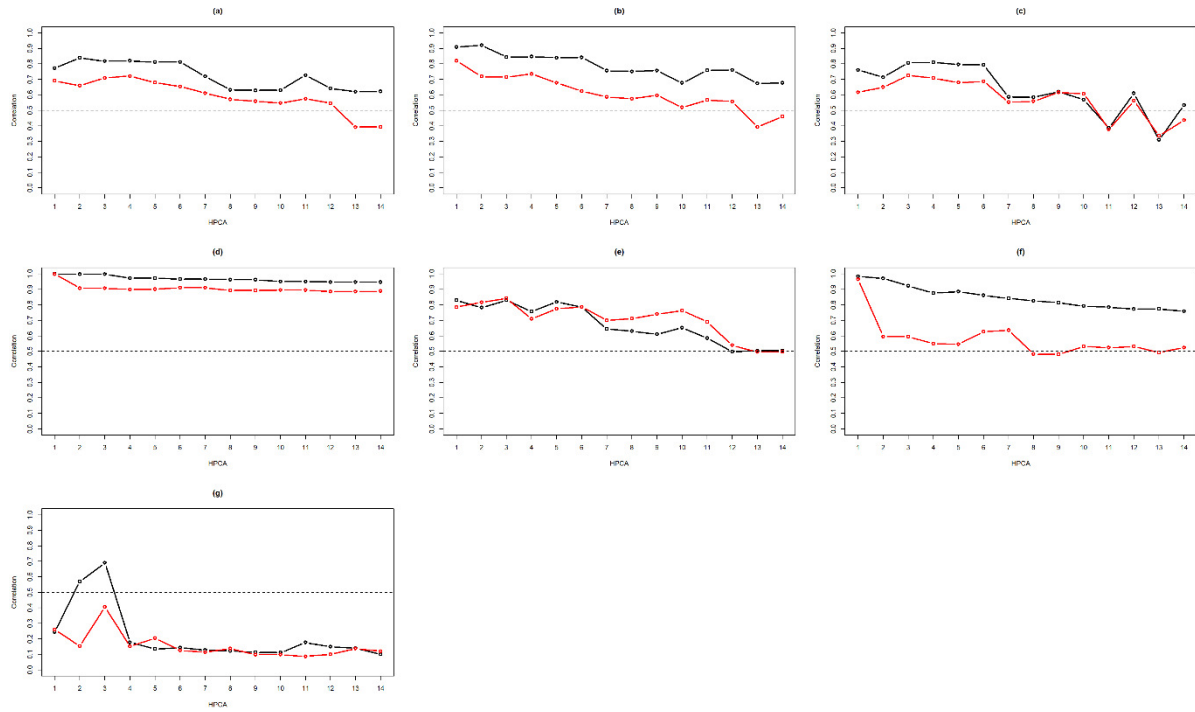


Figure 63. (a-g) Median of correlation for train (black solid line) and test (red solid line) in 10 iterations of (a) whole, (b) frontal, (c) central, (d) occipital, (e) parietal, (f) left temporal and (g) right temporal regions, respectively.

The Elapsed Time over Number of HPCA group by Regions

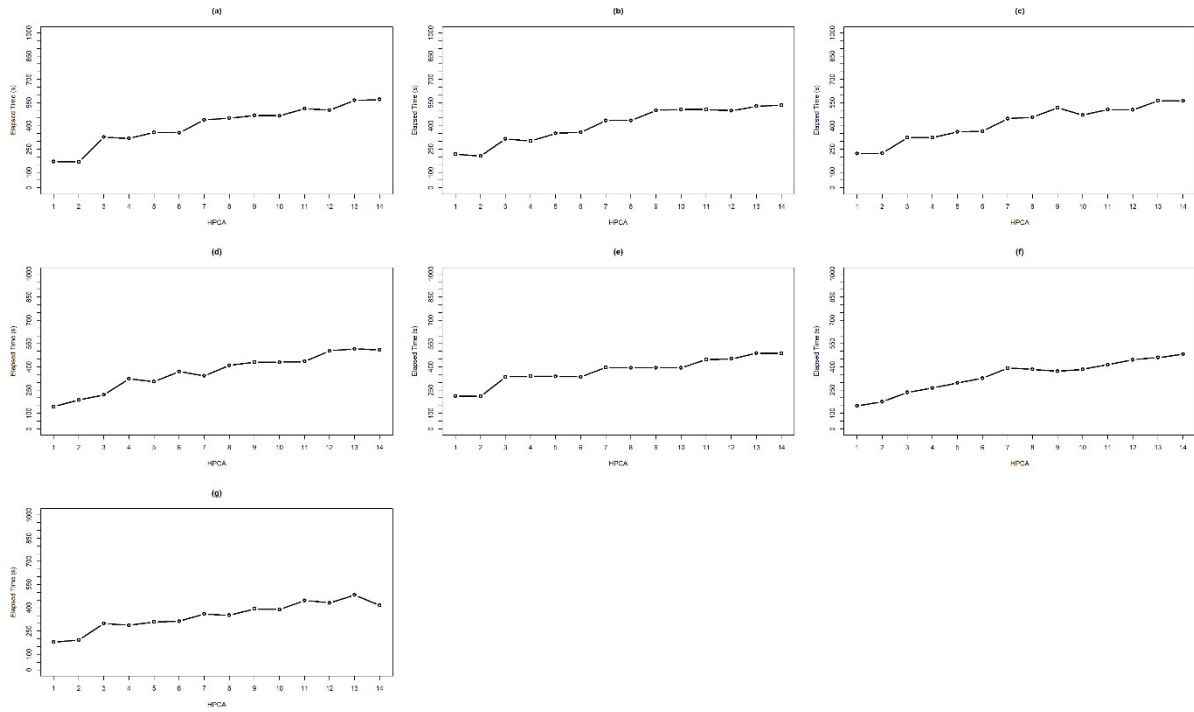


Figure 64. (a-g) Median of elapsed time in 10 iterations of (a) whole, (b) frontal, (c) central, (d) occipital, (e) parietal, (f) left temporal and (g) right temporal regions, respectively.

3.2.4. Conclusion .

According to the **Figure 46**, the MSPE is not changing in many HPCAs but some of them have lowest MSPE in training and testing datasets. For example, in the right temporal (g), the minimum MSPE is obtained with first three HPCA, but in the left temporal, the minimum MSPE is obtained with the first HPCA.

The **Figure 47** shows the correlation. The highest correlation are obtained in different HPCA in different regions for both training and testing datasets.

But the elapsed time is increasing by adding HPCA for all regions. (**Figure 48**)

We conclude that the different number of HPCA has the best performance based on the MSPE, correlation and elapsed times. It depends based on the datasets. But the elapsed time is increasing for all regions.

4. Bayesian Simulation

4.1. Without HPCA

Without HPCA (Table A)							
#HPCA	Deviance	DIC	LPML	MSPE (Mean)	Correlation (Mean)	Time (user)	Time (Elapsed)
1	-11112 (-11157.45,-11065.91)	-10646.7 (-10689.65,-10599.83)	17.03 (15.33,18.74)	0.00038 (0.00033,0.00044)	0.9993 (0.9991,0.9995)	7.93 (7.81,8.13)	49 (48.59,49.63)
2	-11129.6 (-11175.2,-11078.12)	-10672.05 (-10718.45,-10623.8)	17.14 (15.73,19.59)	0.00044 (0.00036,0.00051)	0.9993 (0.9991,0.9994)	8.27 (8.14,8.42)	49.89 (49.11,51.37)
3	-11133.69 (-11175.57,-11088.06)	-10706.25 (-10764.33,-10652.8)	16.81 (14.6,19.39)	0.0004 (0.00035,0.00047)	0.9994 (0.9992,0.9995)	15.59 (8.53,15.98)	56.63 (50.5,58.01)
4	-11142.12 (-11178.3,-11089.92)	-10723.85 (-10798.4,-10675.18)	14.91 (12.96,16.64)	0.00038 (0.00033,0.00044)	0.9994 (0.9992,0.9996)	9.32 (8.94,11.48)	51.79 (51.12,55.78)
5	-11153.12 (-11186.71,-11095.58)	-10754.8 (-10828.25,-10696.43)	13.91 (11.68,16.35)	0.00034 (0.0003,0.00039)	0.9995 (0.9994,0.9996)	9.31 (9.9,9.5)	52.42 (51.77,55.85)
6	-11161.37 (-11204.75,-11099.88)	-10762.6 (-10849.83,-10709.3)	18.19 (15.85,21.02)	0.0003 (0.00027,0.00035)	0.9996 (0.9995,0.9997)	9.96 (9.42,18.44)	54.33 (52.9,62.11)
7	-11166.77 (-11204.95,-11101.98)	-10779.5 (-10857.2,-10724.48)	17.51 (15.62,18.95)	0.00029 (0.00025,0.00033)	0.9997 (0.9995,0.9997)	9.9 (9.68,10.19)	54.2 (53.71,55.38)
8	-11167.97 (-11207.04,-11107.69)	-10796.6 (-10865.3,-10732)	18.37 (16.62,20.2)	0.00027 (0.00024,0.00031)	0.9997 (0.9996,0.9998)	18.62 (9.91,20.92)	63.64 (55.79,66.27)
9	-11172.8 (-11214.64,-11113.6)	-10819.05 (-10895.08,-10752.4)	20.12 (18.65,21.28)	0.00025 (0.00022,0.00028)	0.9997 (0.9996,0.9998)	21.16 (11.1,34.9)	67.61 (58.86,80.88)
10	-11172.68 (-11213.23,-11113.14)	-10835.3 (-10938.6,-10771.85)	18.38 (17.39,19.98)	0.00024 (0.00021,0.00027)	0.9997 (0.9997,0.9998)	20.32 (11.19,37.6)	67.88 (58.17,84.97)

Table 85. The median (first quantile – third quantile) of the model fit indices.(100 iterations)

4.2. First HPCA

First HPCA (Table B)							
#HPCA	Deviance	DIC	LPML	MSPE (Mean)	Correlation (Mean)	Time (user)	Time (Elapsed)
1	-20131.37 (-20135.31,-20127.99)	-19588.6 (-19592.63,-19585.28)	29.12 (28.16,30.08)	0.00015 (0.00014,0.00016)	0.999996 (0.999992,0.999997)	21.04 (10.49,35.67)	62.43 (51.34,77.27)
2	-20135.13 (-20137.96,-20130.74)	-19876.9 (-20240.3,-19751.13)	32.19 (31.57,32.95)	0.00014 (0.00013,0.00015)	0.999998 (0.999997,0.999998)	20.23 (8.28,33.97)	61.44 (49.65,75.33)
3	-20139.53 (-20143.07,-20136.35)	-20460.95 (-20958.68,-20170.68)	33.28 (32.33,34.33)	0.00014 (0.00013,0.00015)	0.999998 (0.999997,0.999998)	8.66 (8.25,27.33)	51.32 (50.26,69.98)
4	-20138.93 (-20141.34,-20134.87)	-20609.8 (-20801.55,-20390.98)	28.83 (27.62,29.78)	0.00014 (0.00013,0.00015)	0.999998 (0.999998,0.999998)	8.61 (8.27,17.7)	51.53 (50.64,59.94)
5	-20141.05 (-20145.43,-20137.68)	-21110.45 (-21398,-20944.03)	29.17 (27.9,30.7)	0.00013 (0.00012,0.00014)	0.999998 (0.999998,0.999999)	9.01 (8.63,18.06)	52.49 (51.63,60.49)
6	-20143.45 (-20146.14,-20140.02)	-21428.4 (-21641.08,-21174.48)	31.16 (29.65,32.3)	0.00013 (0.00012,0.00014)	0.999999 (0.999998,0.999999)	10.08 (8.79,19.48)	55.03 (52.52,63.02)
7	-20143.42 (-20146.81,-20140.06)	-21404.25 (-21666.03,-21188.78)	29.57 (28.33,31.13)	0.00013 (0.00012,0.00013)	0.999999 (0.999999,0.999999)	14.51 (9.08,23.75)	59.44 (53.35,68.77)
8	-20147.25 (-20150.92,-20143.65)	-21165.15 (-21296.8,-21012.43)	29.88 (28.02,31.33)	0.00012 (0.00012,0.00013)	0.999999 (0.999998,0.999999)	15.71 (8.77,26.89)	60.96 (54.38,72.56)
9	-20147.86 (-20151.19,-20144.29)	-21414.9 (-21560.13,-21242.43)	32.39 (30.82,33.63)	0.00012 (0.00011,0.00013)	0.999999 (0.999999,0.999999)	15.9 (8.85,27.47)	62.43 (55.22,73.67)
10	-20145.8 (-20149.02,-20142.56)	-21416.45 (-21632.78,-21149.28)	29.88 (28.26,31.32)	0.00012 (0.00011,0.00013)	0.999999 (0.999999,0.999999)	16.1 (8.44,30.86)	63.49 (55.98,78.22)

Table 86. The median (first quantile – third quantile) of the model fit indices.(100 iterations)

4.3. All HPCA

All HPCA (Table C)							
#HPCA	Deviance	DIC	LPML	MSPE (Mean)	Correlation (Mean)	Time (user)	Time (Elapsed)
1	-19937.82 (-19948.3,-19929.46)	-19396.45 (-19407.48,-19388.33)	-37.71 (-42.66,9.45)	0.13847 (0.00088,0.16171)	0.3917 (0.28117,0.9978)	15.52 (8.27,30.54)	56.26 (49.67,72.09)
2	-19962.29 (-20000.84,-19937.26)	-19424.95 (-19468.63,-19404.8)	21.26 (16.52,25.09)	0.00045 (0.00034,0.0006)	0.99913 (0.99862,0.99938)	9.39 (8.28,17.4)	52.37 (49.73,61.48)
3	-20023.1 (-20044.4,-19985.27)	-19489.35 (-19528.45,-19464.85)	25.14 (22.4,27.53)	0.00033 (0.00029,0.0004)	0.99945 (0.99921,0.9996)	9.3 (8.27,17.24)	51.77 (50.24,59.81)
4	-20129.8 (-20133.02,-20127.03)	-19590.25 (-19593.4,-19586.98)	28.79 (27.91,29.55)	0.00016 (0.00015,0.00017)	0.99997 (0.99994,0.99999)	17.82 (8.76,24.41)	60.24 (51.55,68.1)
5	-20131.72 (-20135.6,-20127.7)	-20294.05 (-20858.55,-20067.8)	25.74 (24.44,27.03)	0.00016 (0.00014,0.00017)	0.99997 (0.99995,0.99999)	15.48 (8.35,25.53)	58.4 (51.94,68.29)
6	-20131.2 (-20135.39,-20127.29)	-20434.5 (-20663.78,-20188.93)	29.03 (27.56,30.63)	0.00015 (0.00014,0.00016)	0.99997 (0.99995,0.99999)	15.34 (8.37,26.37)	58.91 (52.46,70.08)
7	-20135.81 (-20138.76,-20131.51)	-20792.2 (-21114.08,-20551.7)	28.35 (27.02,29.9)	0.00015 (0.00014,0.00016)	0.99998 (0.99996,0.99999)	13.29 (8.26,25.32)	58.72 (52.59,70)
8	-20139.32 (-20142.71,-20135.15)	-20876.3 (-21079.25,-20697.53)	27.63 (25.5,28.84)	0.00014 (0.00013,0.00015)	0.99998 (0.99996,0.99999)	14.29 (8.33,19.65)	59.59 (53.82,65.13)
9	-20140.15 (-20143.75,-20135.61)	-20983.65 (-21189.83,-20796.15)	30.76 (29.14,32.12)	0.00014 (0.00013,0.00014)	0.99998 (0.99996,0.99999)	15.86 (12.52,23.74)	62.36 (58.6,70.75)
10	-20138.89 (-20141.58,-20135.02)	-20952.45 (-21105.6,-20787.9)	28.17 (26.36,29.72)	0.00014 (0.00013,0.00015)	0.99998 (0.99996,0.99999)	19.17 (12.81,27.24)	67.27 (60.08,75.2)

Table 87. The median (first quantile – third quantile) of the model fit indices. (100 iterations)

REFERENCES

1. Walz, J.M., et al., *Simultaneous EEG-fMRI reveals a temporal cascade of task-related and default-mode activations during a simple target detection task.*, Neuroimage, 2014. 102: p. 229-239.